# Windows Server get started documentation

Windows Server builds connected app, network, and web service infrastructure from workgroup to data center, bridging on-premises with Azure.

## Overview

OVERVIEW

What is Windows Server?

What's new in Windows Server 2025

Windows Server Insiders Preview

CONCEPT

Editions feature comparison

Hardware requirements

Features removed or no longer developed in Windows Server

## Plan

CONCEPT

Install, upgrade, or migrate

Server Core vs Server with Desktop Experience

Servicing channels comparison

CONCEPT

Microsoft server applications compatibility

Azure Hybrid Benefit for Windows Server

## Install & upgrade

DEPLOY

Install Windows Server from installation media

Plan your Windows Server upgrade

Perform an in-place upgrade

HOW-TO GUIDE

Convert editions and license types

Install Server Core App Compatibility

## Activation

CONCEPT

Automatic VM Activation

KMS activation planning

HOW-TO GUIDE

Create KMS host

KMS client activation

Troubleshooting Windows volume activation

## Azure Arc enabled servers

CONCEPT

Azure Connected Machine agent deployment options

HOW-TO GUIDE

Connect Windows Server to Azure Arc

Get Extended Security Updates

Deliver Extended Security Updates with Azure Arc

Enable Hotpatch for Azure Arc-enabled servers

# What is Windows Server?

08/01/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

> **Tip**
>
> Already familiar with Windows Server? Jump to **What's new in Windows Server 2025** to learn about the latest features and enhancements.

Windows Server is Microsoft's enterprise server platform that enables organizations to run and secure applications, services, and workloads across on-premises, hybrid, and cloud environments.

Whether you're running a small business or managing enterprise infrastructure, Windows Server provides the foundation for secure, scalable, and high-performance computing that grows with your organization. Built on decades of Windows innovation, it serves as the backbone for millions of organizations worldwide, powering everything from file servers and web applications to complex enterprise workloads and AI-driven solutions.

https://learn-video.azurefd.net/vod/player?id=715f723f-0644-4b9d-b7df-6e708da43242&locale=en-us&embedUrl=%2Fwindows-server%2Fget-started%2Foverview

## Why choose Windows Server?

Windows Server delivers comprehensive capabilities that help organizations modernize their infrastructure while maintaining security, performance, and reliability.

Use Windows Server for:

- **Mixed environments**: Unified management of Windows, Linux, and container workloads.
- **Enterprise integration**: Seamless integration with SQL Server, System Center, Exchange, SharePoint, Project, and existing Windows infrastructure.
- **Commercial support**: Enterprise-grade support with an active tech community.
- **Compliance scenarios**: Built-in features for regulated industries (healthcare, finance, government).
- **Flexible licensing**: Choose between pay-as-you-go, perpetual, or subscription-based licensing.

Windows Server builds on this foundation with new features that enhance security, performance, and cloud integration. The following sections highlight some of the key benefits and features of Windows Server.

## Advanced, multi-layered security

Improved protections against ever-evolving threats, multilayered safeguards, and improved resilience.

- **Advanced identity protection**: Enterprise directory services (Active Directory) that provide centralized authentication, authorization, and policy management across Windows, Linux, and Mac systems.
- **Virtualization-based security**: Enhanced Hyper-V security with VBS enclaves and key protection for confidential computing.
- **Secure networking**: TLS 1.3 encryption, SMB over QUIC, and advanced micro-segmentation to prevent lateral attack spread.

## Cloud agility anywhere

Get the power of Azure delivered to your edge and hybrid cloud environment with Azure Arc. Deploy with complete flexibility - in the cloud, in your datacenter, or hybrid at the edge.

- **Deployment flexibility**: Consistent Windows Server experience whether you deploy on-premises, hybrid, or in Azure with unified management.
- **Hotpatching**: Apply security updates without reboots - reducing from 12 required reboots per year to just 4 for enhanced productivity.
- **Azure Arc integration**: Azure Arc extends Azure's management capabilities to your on-premises servers, giving you cloud-style governance, security policies, and monitoring without moving your data whether they're in Azure, your datacenter, or other clouds.
- **Pay-as-you-go licensing**: Flexible, subscription-based pricing with Azure Arc.

## High-performance, future-ready infrastructure

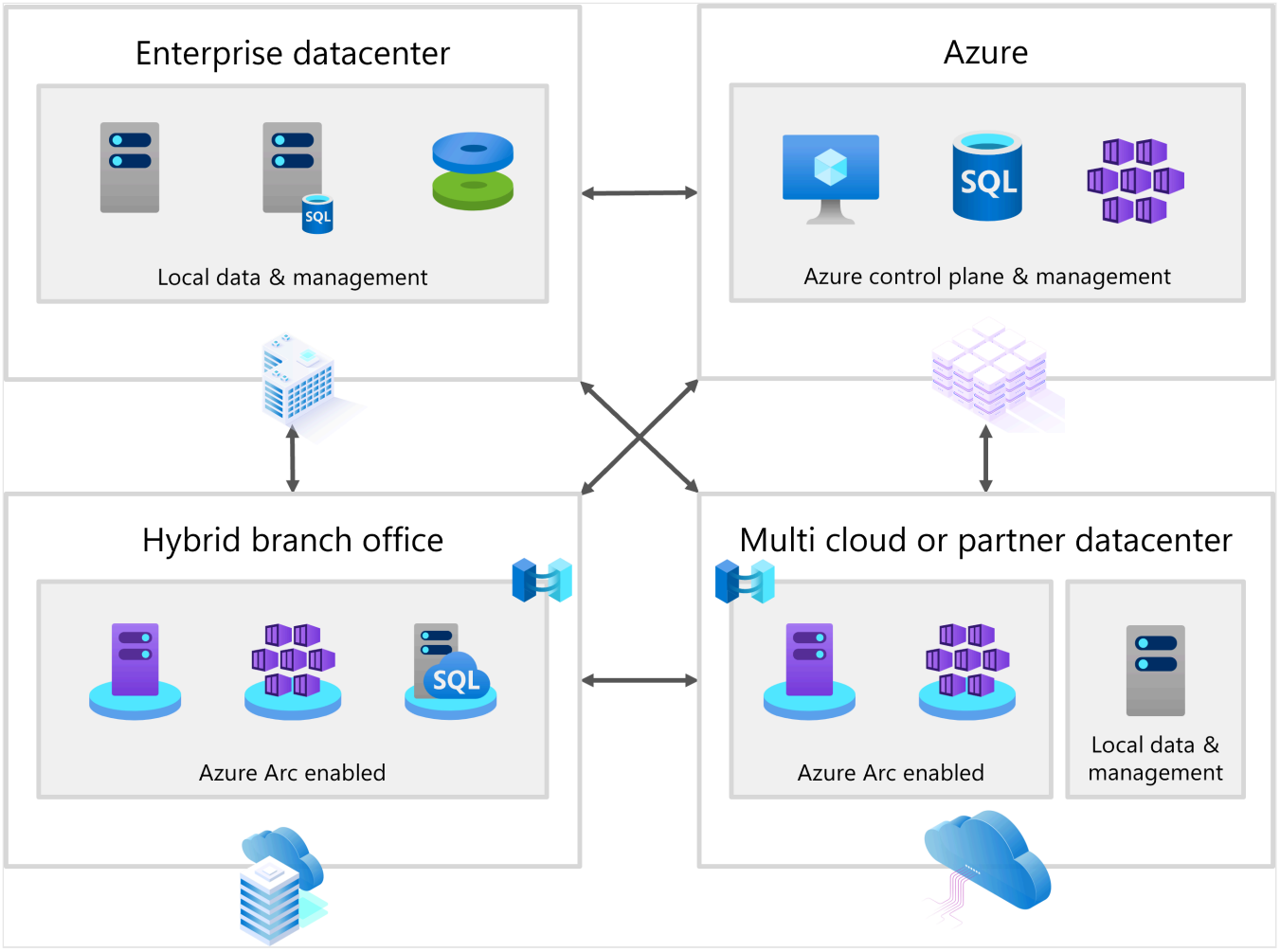
Leaps ahead in scalability, performance, storage, and capabilities to support AI and ML.

- **AI and ML ready**: GPU partitioning across VMs with live migration support and enhanced performance for AI workloads and edge inferencing.
- **Massive scale**: Use Hyper-V to virtualize your workloads with support for up to 240-TB RAM and 2,048 virtual processors per VM for the most demanding workloads.
- **Storage optimization**: Enhanced ReFS with optimized deduplication and compression for software-defined storage.

For complete details on new features, see What's new in Windows Server 2025.

# Where can you deploy Windows Server?

Windows Server gives you the flexibility to choose where your applications and data live based on your business needs. Each deployment option offers unique advantages to help power your organization's growth and innovation. You can deploy Windows Server in the cloud, in your datacenter, or as a hybrid solution at the edge across different environments.




Deploy Windows Server:

- **In the cloud**

  Deploy Windows Server virtual machines in Azure for maximum scalability and global reach. Perfect for new applications, development environments, or when you need to scale quickly.

  **Best for**: New projects, variable workloads, global applications, and disaster recovery scenarios.

- **In your datacenter**

Upgrade your existing infrastructure to Windows Server for complete control over your environment. Ideal when you have specific compliance requirements, existing hardware investments, or need air-gapped environments.

**Best for**: Regulated industries, existing infrastructure, specialized hardware requirements, and complete data control.

- 🌐 **Hybrid at the edge**

  Extend your datacenter to Azure with Azure Arc while keeping workloads running locally. Get the best of both worlds - local performance with cloud management and services.

  Keep sensitive workloads on-premises for compliance, performance, or cost reasons while gaining cloud management, monitoring, and services through Azure Arc. Perfect for organizations that need cloud benefits without full cloud commitment or data residency concerns.

  **Best for**: Gradual cloud adoption, distributed locations, edge computing scenarios, and using existing investments while gaining cloud capabilities.

# Editions

Windows Server offers two editions designed for different scales and virtualization needs:

Expand table

| Edition | Standard | Datacenter |
|---|---|---|
| **Environments** | Physical servers or environments with limited virtualization needs | Highly virtualized environments and cloud scenarios |
| **Virtualization rights** | 2 virtual machines plus one Hyper-V host per license | Unlimited virtual machines plus one Hyper-V host per license |
| **Best for** | Small to medium deployments, specific workloads, cost-conscious environments | Large datacenters, cloud deployments, extensive virtualization |

Both editions include many of the same features and capabilites. For a detailed comparison, see:

- Comparison of Windows Server editions
- Comparison of locks and limits in Windows Server

# Deployment approaches

Choose the right approach for implementing Windows Server based on your current environment and requirements:

Expand table

| Approach | Description | Best For |
|---|---|---|
| **Clean install** | Install on new hardware or completely replace existing OS | New deployments, maximum performance, complex configurations |
| **In-place upgrade** | Upgrade while keeping settings, roles, and data | VMs, stable configurations, minimal downtime |
| **Migration** | Gradually move roles and features to new installations | Complex environments, mission-critical systems |
| **Cluster rolling upgrade** | Upgrade cluster nodes one at a time | Hyper-V clusters, Scale-Out File Server, strict SLAs |

For detailed step-by-step guidance on each approach, see Installation, upgrade, and migration options.

# Installation options

When installing Windows Server, choose between two options based on how you plan to manage your server:

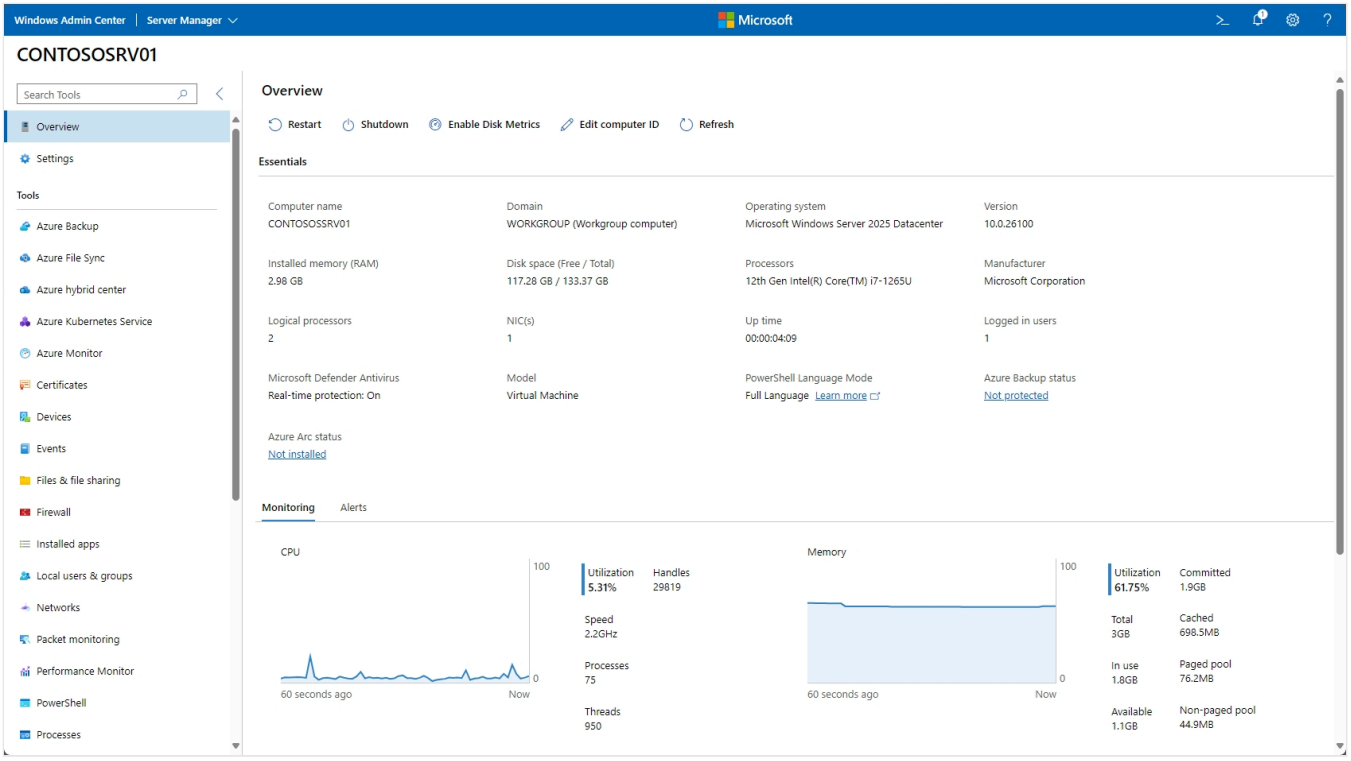
Expand table

| Installation Option | Server Core | Server with Desktop Experience |
|---|---|---|
| **Description** | Minimal installation without graphical interface. Managed via PowerShell, SConfig, or remote tools. Offers smaller footprint, reduced attack surface, and better performance. | Full installation with standard Windows desktop and all GUI management tools. Provides familiar interface and local graphical management. |
| **Best For** | Most server workloads, especially when using remote management | Servers requiring local GUI access or desktop-dependent applications |

Learn more about the differences and detailed guidance at Server Core vs Server with Desktop Experience.

# Manage Windows Server

Windows Server provides multiple management tools that scale from single servers to enterprise datacenters.




You can manage Windows Server using one or more of the following approaches:

🌐 **Windows Admin Center**: Modern, browser-based management for day-to-day server administration. Works locally or integrates with Azure portal for hybrid scenarios.

☁ **Azure Arc**: Cloud-based management that extends Azure governance, security, and monitoring to servers anywhere - in your datacenter, other clouds, or edge locations.

🔧 **Local tools**: Traditional management including Server Manager (GUI administration), PowerShell (object-oriented command shell and scripting language), and Remote Server Administration Tools for direct server access and automation.

🏢 **System Center**: Enterprise-scale management for large datacenters requiring comprehensive monitoring, deployment, and configuration management.

Choose your management approach based on your environment size and cloud strategy. For detailed guidance, see Windows Server management overview.

# Windows Insider Program

The Windows Insider Program for Windows Server provides preview builds of Windows Server allowing you early access to learn, test, and help shape the future of Windows Server. To learn more, you can get started with the Windows Insider Program for Windows Server and participate in the Windows Server Insiders Community.

# Get started with Windows Server

Ready to begin your Windows Server journey? Start with a free evaluation to test in your environment:

- Download Windows Server 2025 evaluation
- View system requirements
- Get upgrade guidance
- Browse learning paths for Windows Server to help learn new skills and accelerate your deployment with step-by-step guidance.

For the latest news on Windows Server, visit the Windows Server blog to stay up to date on announcements, features, events, and other information from the Windows Server engineering teams. You can also visit the Windows Server Community to share best practices, get the latest news, and learn from experts about Windows Server.

ⓘ **Note:** This article contains content created with AI. Learn more

# What's new in Windows Server 2025

Applies to: ✅ Windows Server 2025

This article describes some of the newest developments in Windows Server 2025, which boasts advanced features that improve security, performance, and flexibility. With faster storage options and the ability to integrate with hybrid cloud environments, managing your infrastructure is now more streamlined. Windows Server 2025 builds on the strong foundation of its predecessor and introduces a range of innovative enhancements to adapt to your needs.

## Desktop experience and upgrade

Explore upgrade options and the desktop experience.

### In-place upgrade from Windows Server 2012 R2

With Windows Server 2025, you can upgrade up to four versions at a time. You can upgrade directly to Windows Server 2025 from Windows Server 2012 R2 and later.

### Desktop shell

When you sign in for the first time, the desktop shell experience conforms to the style and appearance of Windows 11.

### Bluetooth

You can now connect mice, keyboards, headsets, audio devices, and more via Bluetooth in Windows Server 2025.

### DTrace

Windows Server 2025 comes equipped with `dtrace` as a native tool. DTrace is a command-line utility that enables users to monitor and troubleshoot their system's performance in real time. With DTrace, you can dynamically instrument both the kernel and user-space code without any need to modify the code itself. This versatile tool supports a range of data collection and analysis techniques, such as aggregations, histograms, and tracing of user-level events. To learn more, see DTrace for command-line help and DTrace on Windows for other capabilities.

### Email and accounts

You can now add the following types of accounts in Windows **Settings** under **Accounts** > **Email & accounts** for Windows Server 2025:

- Microsoft Entra ID
- Microsoft account
- Work or school account

Domain join is still required for most situations.

## Feedback Hub

To submit feedback or report problems that you encounter when you use Windows Server 2025, use the Windows Feedback Hub. Include screenshots or recordings of the process that caused the issue to help us understand your situation and share suggestions to enhance your Windows experience. To learn more, see Explore the Feedback Hub.

## File compression

Windows Server 2025 has a new compression feature. To compress an item, right-click and select **Compress to**. This feature supports **ZIP**, **7z**, and **TAR** compression formats with specific compression methods for each one.

## Pinned apps

Pinning your most-used apps is now available through the **Start** menu and is customizable to suit your needs. The default pinned apps are currently:

- Azure Arc Setup
- Feedback Hub
- File Explorer
- Microsoft Edge
- Server Manager
- Settings
- Terminal
- Windows PowerShell

## Task Manager

Windows Server 2025 uses the modern Task Manager app with Mica material that conforms to the style of Windows 11.

## Wi-Fi

It's now easier to enable wireless capabilities because the Wireless LAN Service feature is now installed by default. The wireless startup service is set to manual. To enable it, run `net start wlansvc` in the command prompt, Windows Terminal, or PowerShell.

## Windows Terminal

The Windows Terminal, a powerful and efficient multishell application for command-line users, is available in Windows Server 2025. Search for **Terminal** on the search bar.

## WinGet

WinGet is installed by default, which is a command-line Windows Package Manager tool that provides comprehensive package manager solutions for installing applications on Windows devices. To learn more, see Use the WinGet tool to install and manage applications.

# Advanced multilayer security

Learn about security in Windows 2025.

## Hotpatch (preview)

Hotpatch is now available for Windows Server 2025 machines connected to Azure Arc after Hotpatch is enabled in the Azure Arc portal. You can use Hotpatch to apply OS security updates without restarting your machine. To learn more, see Hotpatch.

> **Important**
>
> Azure Arc-enabled Hotpatch is currently in preview. See the **Supplemental Terms of Use for Microsoft Azure Previews** for legal terms that apply to Azure features that are in beta, preview, or otherwise not yet released into general availability.

## Credential Guard

Starting with Windows Server 2025, Credential Guard is now enabled by default on devices that meet the requirements. For more information about Credential Guard, see Configure Credential Guard.

## Active Directory Domain Services

The latest enhancements to Active Directory Domain Services (AD DS) and Active Directory Lightweight Domain Services (AD LDS) introduce a range of new functionalities and capabilities aimed at optimizing your domain management experience:

- **32k database page size optional feature**: Active Directory uses an Extensible Storage Engine (ESE) database since its introduction in Windows 2000 that uses an 8k database page size. The 8k architectural design decision resulted in limitations throughout Active Directory that are documented in Active Directory maximum limits: Scalability. An example of this limitation is a single record Active Directory object, which can't exceed 8k-byte size. Moving to a 32k database page format offers a huge improvement in areas affected by legacy restrictions. Multivalued attributes can now hold up to approximately 3,200 values, which is an increase by a factor of 2.6.

  You can install new domain controllers (DCs) with a 32k-page database that uses 64-bit Long Value IDs (LIDs) and run in 8k-page mode for compatibility with previous versions. An upgraded DC continues to use its current database format and 8k pages. Moving to a 32k-page database is done on a forestwide basis and requires that all DCs in the forest have a 32k-page capable database.

- **Active Directory schema updates**: Three new log database files are introduced that extend the Active Directory schema: `sch89.ldf`, `sch90.ldf`, and `sch91.ldf`. The AD LDS equivalent schema updates are in `MS-ADAM-Upgrade3.ldf`. To learn more about previous schema updates, see Windows Server Active Directory schema updates.

- **Active Directory object repair**: Enterprise administrators can now repair objects with the missing core attributes `SamAccountType` and `ObjectCategory`. Enterprise administrators can reset the `LastLogonTimeStamp` attribute on an object to the current time. These operations are achieved through a new RootDSE modify operation feature on the affected object called `fixupObjectState`.

- **Channel binding audit support**: You can now enable events 3074 and 3075 for Lightweight Directory Access Protocol (LDAP) channel binding. When the channel binding policy is modified to a more secure setting, an administrator can identify devices in the environment that don't support or fail channel binding. These audit events are also available in Windows Server 2022 and later via KB4520412.

- **DC-location algorithm improvements**: The DC discovery algorithm provides new functionality with improvements to mapping of short NetBIOS-style domain names to DNS-style domain names. To learn more, see Locating domain controllers in Windows and Windows Server.

> **Note**
>
> Windows doesn't use mailslots during DC discovery operations because Microsoft has announced the deprecation of WINS and mailslots for these legacy technologies.

- **Forest and domain functional levels**: The new functional level is used for general supportability and is required for the new 32k database page size feature. The new functional level maps to the value of `DomainLevel 10` and `ForestLevel 10` for unattended installations. Microsoft has no plans to retrofit functional levels for Windows Server 2019 and Windows Server 2022. To perform an unattended promotion and demotion of a DC, see DCPROMO answer file syntax for unattended promotion and demotion of domain controllers.

  The DsGetDcName API also supports the new flag `DS_DIRECTORY_SERVICE_13_REQUIRED` that enables location of DCs running Windows Server 2025. You can learn more about functional levels in the following articles:
  - AD DS functional levels
  - Raise the domain functional level
  - Raise the forest functional level

  > **Note**
  >
  > New Active Directory forests or AD LDS configuration sets are required to have a functional level of Windows Server 2016 or later. Promotion of an Active Directory or AD LDS replica requires that the existing domain or configuration set is already running with a functional level of Windows Server 2016 or later.
  >
  > Microsoft recommends that all customers begin planning now to upgrade their Active Directory and AD LDS servers to Windows Server 2022 in preparation for the next release.

- **Improved algorithms for Name/SID Lookups**: Local security authority (LSA) Name and SID Lookup forwarding between machine accounts no longer uses the legacy Netlogon secure channel. Kerberos authentication and the DC Locator algorithm are used instead. To maintain compatibility with legacy operating systems, it's still possible to use the Netlogon secure channel as a fallback option.

- **Improved security for confidential attributes**: DCs and AD LDS instances only allow LDAP to add, search, and modify operations that involve confidential attributes when the connection is encrypted.

- **Improved security for default machine account passwords**: Active Directory now uses default computer account passwords that are randomly generated. Windows 2025 DCs block setting computer account passwords to the default password of the computer account name.

  To control this behavior, enable the Group Policy Object (GPO) setting **Domain controller: Refuse setting default machine account password** located in *Computer Configuration\Windows Settings\Security Settings\Local Policies\Security Options*.

  Utilities like Active Directory Administrative Center (ADAC), Active Directory Users and Computers (ADUC), `net computer`, and `dsmod` also honor this new behavior. Both ADAC and ADUC no longer allow creation of a pre-Windows 2000 account.

- **Kerberos PKINIT support for cryptographic agility**: The Kerberos Public Key Cryptography for Initial Authentication in Kerberos (PKINIT) protocol implementation is updated to allow for cryptographic agility by supporting more algorithms and removing hardcoded algorithms.

- **Kerberos changes for Algorithms used for Ticket Granting Tickets**: The Kerberos Distribution Center will no longer issue Ticket Granting Tickets using RC4 encryption, such as RC4-HMAC(NT).

- **Kerberos changes for supported encryption type configuration**: Kerberos no longer honors the legacy registry key REG_DWORD `SupportedEncryptionTypes` found in the path `HKEY_LOCAL_MACHINE\CurrentControlSet\Control\Lsa\Kerberos\Parameters`, Microsoft recommends using group policy instead. To learn more about the group policy settings, see Network security: Configure encryption types allowed for Kerberos.

- **LAN Manager GPO setting**: The GPO setting **Network security: Don't store LAN Manager hash value on next password change** is no longer present and doesn't apply to new versions of Windows.

- **LDAP encryption by default**: All new Active Directory deployments require LDAP signing (sealing) by default for all LDAP client communication after a Simple Authentication and Security Layer (SASL) bind. To learn more about the signing behavior, see LDAP signing for Active Directory Domain Services.

- **LDAP support for Transport Layer Security (TLS) 1.3**: LDAP uses the latest SCHANNEL implementation and supports TLS 1.3 for LDAP over TLS connections. Using TLS 1.3 eliminates obsolete cryptographic algorithms and enhances security over older versions. TLS 1.3 aims to encrypt as much of the handshake as possible. To learn more, see Protocols in TLS/SSL (Schannel SSP) and TLS Cipher Suites in Windows Server 2022.

- **Legacy Security Account Manager (SAM) remote procedure call (RPC) password change behavior**: Secure protocols such as Kerberos are the preferred way to change domain user passwords. On DCs, the latest SAM RPC password change method SamrUnicodeChangePasswordUser4 by using Advanced Encryption Standard (AES) is accepted by default when it's called remotely. The following legacy SAM RPC methods are blocked by default when they're called remotely:
  - SamrChangePasswordUser
  - SamrOemChangePasswordUser2
  - SamrUnicodeChangePasswordUser2

  For domain users that are members of the Protected Users group and for local accounts on domain member computers, all remote password changes through the legacy SAM RPC interface are blocked by default, including `SamrUnicodeChangePasswordUser4`.

  To control this behavior, use the following GPO setting:

  **Computer Configuration** > **Administrative Templates** > **System** > **Security Account Manager** > **Configure SAM change password RPC methods policy**

- **Non-uniform Memory Access (NUMA) support**: AD DS now takes advantage of NUMA-capable hardware by using CPUs in all processor groups. Previously, Active Directory would only use CPUs in group 0. Active Directory can expand beyond 64 cores.

- **Performance counters**: Monitoring and troubleshooting the performance of the following counters are now available:
  - **DC Locator**: Counters specific to clients and DCs are available.
  - **LSA Lookups**: Name and SID Lookups through the `LsaLookupNames`, `LsaLookupSids`, and equivalent APIs. These counters are available on both client and server versions.
  - **LDAP client**: Available in Windows Server 2022 and later via the KB 5029250 update.

- **Replication priority order**: Administrators can now increase the system-calculated replication priority with a particular replication partner for a particular naming context. This feature allows more flexibility in configuring the replication order to address specific scenarios.

## Delegated Managed Service Account

This new type of account enables migration from a service account to a delegated Managed Service Account (dMSA). This account type comes with managed and fully randomized keys to ensure minimal application changes while the original service account passwords are disabled. To learn more, see Delegated Managed Service Accounts overview.

# Windows Local Administrator Password Solution

Windows Local Administrator Password Solution (LAPS) helps organizations manage local administrator passwords on their domain-joined computers. It automatically generates unique passwords for each computer's local administrator account. It then stores them securely in Active Directory and updates them regularly. Automatically generated passwords help to improve security. They reduce the risk of attackers gaining access to sensitive systems by using compromised or easily guessable passwords.

Several features new to Microsoft LAPS introduce the following improvements:

- **New automatic account management**: IT admins can now create a managed local account with ease. With this feature, you can customize the account name and enable or disable the account. You can even randomize the account name for enhanced security. The update also includes improved integration with existing local account management policies from Microsoft. To learn more about this feature, see Windows LAPS account management modes.

- **New image rollback detection**: Windows LAPS now detects when an image rollback occurs. If a rollback does happen, the password stored in Active Directory might no longer match the password stored locally on the device. Rollbacks can result in a *torn state*. In this case, the IT admin is unable to sign in to the device by using the persisted Windows LAPS password.

  To address this issue, a new feature was added that includes an Active Directory attribute called `msLAPS-CurrentPasswordVersion`. This attribute contains a random globally unique identifier (GUID) written by Windows LAPS every time a new password is persisted in Active Directory and saved locally. During every processing cycle, the GUID stored in `msLAPS-CurrentPasswordVersion` is queried and compared to the locally persisted copy. If they're different, the password is immediately rotated.

  To enable this feature, run the latest version of the `Update-LapsADSchema` cmdlet. Windows LAPS then recognizes the new attribute and begins to use it. If you don't run the updated version of the `Update-LapsADSchema` cmdlet, Windows LAPS logs a 10108 warning event in the event log but continues to function normally in all other respects.

  No policy settings are used to enable or configure this feature. The feature is always enabled after the new schema attribute is added.

- **New passphrase**: IT admins can now use a new feature in Windows LAPS that enables the generation of less-complex passphrases. An example is a passphrase such as **EatYummyCaramelCandy**. This phrase is easier to read, remember, and type compared to a traditional password like **V3r_b4tim#963?**.

With this new feature, you can configure the `PasswordComplexity` policy setting to select one of three different word lists for passphrases. All of the lists are included in Windows and don't require a separate download. A new policy setting called `PassphraseLength` controls the number of words used in the passphrase.

When you create a passphrase, the specified number of words are randomly selected from the chosen word list and concatenated. The first letter of each word is capitalized to enhance readability. This feature also fully supports backing up passwords to either Active Directory or Microsoft Entra ID.

The passphrase word lists used in the three new `PasswordComplexity` passphrase settings are sourced from the Electronic Frontier Foundation article Deep Dive: EFF's New Wordlists for Random Passphrases. The Windows LAPS Passphrase Word Lists is licensed under the CC-BY-3.0 attribution license and is available for download.

> ⓘ **Note**
>
> Windows LAPS doesn't allow for customization of the built-in word lists or the use of customer-configured word lists.

- **Improved readability password dictionary**: Windows LAPS introduces a new `PasswordComplexity` setting that enables IT admins to create less complex passwords. You can use this feature to customize LAPS to use all four character categories (uppercase letters, lowercase letters, numbers, and special characters) like the existing complexity setting of `4`. With the new setting of `5`, the more complex characters are excluded to enhance password readability and minimize confusion. For example, the numeral **1** and the letter **l** are never used with the new setting.

  When `PasswordComplexity` is configured to `5`, the following changes are made to the default password dictionary character set:
  - **Don't use:** The letters **I**, **O**, **Q**, **l**, **o**
  - **Don't use:** The numbers **0**, **1**
  - **Don't use:** The special characters **,**, **.**, **&**, **{**, **}**, **[**, **]**, **(**, **)**, **;**
  - **Use:** The special characters **:**, **=**, **?**, *****

  The ADUC snap-in (via Microsoft Management Console) now features an improved Windows LAPS tab. The Windows LAPS password now appears in a new font that enhances its readability when it appears in plain text.

- **Post-authentication action support for terminating individual processes**: A new option is added to the **Post-authentication actions** (PAA) Group Policy setting, `Reset the password, sign out the managed account, and terminate any remaining processes`, which

is located in **Computer Configuration** > **Administrative Templates** > **System** > **LAPS** > **Post-authentication actions**.

This new option is an extension of the previous option, `Reset the password and log off the managed account`. After configuration, the PAA notifies and then terminates any interactive sign-in sessions. It enumerates and terminates any remaining processes that are still running under the local account identity managed by Windows LAPS. No notification precedes this termination.

The expansion of logging events during the execution of PAA provides deeper insights into the operation.

To learn more about Windows LAPS, see What is Windows LAPS?.

## OpenSSH

In earlier versions of Windows Server, the OpenSSH connectivity tool required manual installation before use. The OpenSSH server-side component is installed by default in Windows Server 2025. The Server Manager UI also includes a one-step option under **Remote SSH Access** that enables or disables the `sshd.exe` service. Also, you can add users to the **OpenSSH Users** group to allow or restrict access to your devices. To learn more, see OpenSSH for Windows overview.

## Security baseline

By implementing a customized security baseline, you can establish security measures right from the beginning for your device or VM role based on the recommended security posture. This baseline comes equipped with more than 350 preconfigured Windows security settings. You can use the settings to apply and enforce specific security settings that align with the best practices recommended by Microsoft and industry standards. To learn more, see OSConfig overview.

## Virtualization-based security enclaves

A virtualization-based security (VBS) enclave is a software-based trusted execution environment inside the address space of a host application. VBS enclaves use underlying VBS technology to isolate the sensitive portion of an application in a secure partition of memory. VBS enclaves enable isolation of sensitive workloads from both the host application and the rest of the system.

VBS enclaves enable applications to protect their secrets by removing the need to trust admins and hardening against malicious attackers. For more information, read the VBS enclaves Win32

reference.

## Virtualization-based security key protection

VBS key protection enables Windows developers to secure cryptographic keys by using VBS. VBS uses the virtualization extension capability of the CPU to create an isolated runtime outside of the normal OS.

When in use, VBS keys are isolated in a secure process. Key operations can occur without exposing the private key material outside of this space. At rest, a TPM key encrypts private key material, which binds VBS keys to the device. Keys protected in this way can't be dumped from process memory or exported in plain text from a user's machine.

VBS key protection helps prevent exfiltration attacks by any admin-level attacker. VBS must be enabled to use key protection. For information about how to enable VBS, see Enable memory integrity.

## Secured connectivity

The following sections discuss security for connections.

### Secure certificate management

Searching or retrieving certificates on Windows now supports SHA-256 hashes, as described in the functions CertFindCertificateInStore and CertGetCertificateContextProperty. TLS server authentication is more secure across Windows and now requires a minimum RSA key length of 2,048 bits. For more information, read TLS server authentication: Deprecation of weak RSA certificates.

### SMB over QUIC

The SMB over QUIC server feature, which was only available in Windows Server Azure Edition, is now available in both Windows Server Standard and Windows Server Datacenter versions. SMB over QUIC adds the benefits of the QUIC, which provides low-latency, encrypted connections over the internet.

#### SMB over QUIC enablement policy

Administrators can disable the SMB over QUIC client through Group Policy and PowerShell. To disable SMB over QUIC by using Group Policy, set the **Enable SMB over QUIC** policy in the following paths to **Disabled**:

- *Computer Configuration\Administrative Templates\Network\Lanman Workstation*
- *Computer Configuration\Administrative Templates\Network\Lanman Server*

To disable SMB over QUIC by using PowerShell, run this command in an elevated PowerShell prompt:

```PowerShell
Set-SmbClientConfiguration -EnableSMBQUIC $false
```

### SMB signing and encryption auditing

Administrators can enable auditing of the SMB server and client for support of SMB signing and encryption. If a non-Microsoft client or server lacks support for SMB encryption or signing, it can be detected. When a non-Microsoft device or software states it supports SMB 3.1.1, but fails to support SMB signing, it violates the SMB 3.1.1 Pre-authentication integrity protocol requirement.

You can configure SMB signing and encryption auditing settings by using Group Policy or PowerShell. You can change these policies in the following Group Policy paths:

- *Computer Configuration\Administrative Templates\Network\Lanman Server\Audit client does not support encryption*
- *Computer Configuration\Administrative Templates\Network\Lanman Server\Audit client does not support signing*
- *Computer Configuration\Administrative Templates\Network\Lanman Workstation\Audit server does not support encryption*
- *Computer Configuration\Administrative Templates\Network\Lanman Workstation\Audit server does not support signing*

To perform these changes by using PowerShell, run these commands in an elevated prompt where `$true` enables and `$false` disables these settings:

```PowerShell
Set-SmbServerConfiguration -AuditClientDoesNotSupportEncryption $true
Set-SmbServerConfiguration -AuditClientDoesNotSupportSigning $true

Set-SmbClientConfiguration -AuditServerDoesNotSupportEncryption $true
Set-SmbClientConfiguration -AuditServerDoesNotSupportSigning $true
```

Event logs for these changes are stored in the following Event Viewer paths with their specific Event ID.

Expand table

| Path | Event ID |
| --- | --- |
| Applications and Services Logs\Microsoft\Windows\SMBClient\Audit | 31998<br>31999 |
| Applications and Services Logs\Microsoft\Windows\SMBServer\Audit | 3021<br>3022 |

### SMB over QUIC auditing

SMB over QUIC client connection auditing captures events that are written to an event log to include the QUIC transport in the Event Viewer. These logs are stored in the following paths with their specific Event ID.

Expand table

| Path | Event ID |
| --- | --- |
| Applications and Services Logs\Microsoft\Windows\SMBClient\Connectivity | 30832 |
| Applications and Services Logs\Microsoft\Windows\SMBServer\Connectivity | 1913 |

### SMB over QUIC client access control

Windows Server 2025 includes client access control for SMB over QUIC. SMB over QUIC is an alternative to TCP and RDMA that supplies secure connectivity to edge file servers over untrusted networks. Client access control introduces more controls to restrict access to your data by using certificates. To learn more, see How client access control works.

## SMB alternative ports

You can use the SMB client to connect to alternative TCP, QUIC, and RDMA ports instead of their IANA/IETF defaults of 445, 5445, and 443. You can configure alternative ports via Group Policy or PowerShell. Previously, the SMB server in Windows mandated inbound connections to use the IANA-registered port TCP/445 while the SMB TCP client allowed only outbound connections to that same TCP port. Now, SMB over QUIC allows for SMB alternative ports where QUIC-mandated UDP/443 ports are available for both server and client devices. To learn more, see Configure alternative SMB ports.

## SMB firewall rule hardening

Previously, when a share was created, the SMB firewall rules were automatically configured to enable the **File and Printer Sharing** group for the relevant firewall profiles. Now the creation of an SMB share in Windows results in the automatic configuration of the new **File and Printer Sharing (Restrictive)** group, which no longer permits inbound NetBIOS ports 137-139. To learn more, see Updated firewall rules.

## SMB encryption

Enforce SMB encryption is enabled for all outbound SMB client connections. With this update, administrators can set a mandate that all destination servers support SMB 3.x and encryption. If a server lacks these capabilities, the client is unable to establish a connection.

## SMB authentication rate limiter

The SMB authentication rate limiter limits the number of authentication attempts within a certain time period. The SMB authentication rate limiter helps combat brute-force authentication attacks. The service for the SMB server uses the authentication rate limiter to implement a delay between each failed NTLM- or PKU2U-based authentication attempt. The service is enabled by default. To learn more, see How SMB authentication rate limiter works.

## Disable SMB NTLM

Beginning with Windows Server 2025, the SMB client supports NTLM blocking for remote outbound connections. Previously, the Windows Simple and Protected GSSAPI Negotiation Mechanism (SPNEGO) negotiated Kerberos, NTLM, and other mechanisms with the destination server to determine a supported security package. To learn more, see Block NTLM connections on SMB.

## SMB dialect control

You can now manage SMB dialects in Windows. When configured, the SMB server determines which SMB 2 and SMB 3 dialects it negotiates compared to the previous behavior and matches only the highest dialect.

## SMB signing

SMB signing is now required by default for all SMB outbound connections. Previously, it was required only when you connected to shares named **SYSVOL** and **NETLOGON** on Active Directory DCs. To learn more, see How signing works.

### Remote Mailslot

The Remote Mailslot protocol is disabled by default for SMB and for DC Locator protocol use with Active Directory. Remote Mailslot might be removed in a later release. To learn more, see Features removed or no longer developed in Windows Server.

### Routing and Remote Access Services hardening

By default, new Routing and Remote Access Services (RRAS) installations don't accept VPN connections based on PPTP and L2TP. You can still enable these protocols, if necessary. VPN connections based on SSTP and IKEv2 are still accepted without any change.

Existing configurations retain their behavior. For example, if you run Windows Server 2019 and accept PPTP and L2TP connections, and you upgrade to Windows Server 2025 by using an in-place upgrade, connections based on L2TP and PPTP are still accepted. This change doesn't affect Windows client operating systems. To learn more about how to reenable PPTP and L2TP, see Configure VPN protocols.

## IPsec default keying protocol change

The default keying modules have been changed to IKEv1 and IKEv2 for IPsec connections authenticated with machine certificates. For other authentication methods, the default AuthIP and IKEv1 remain. This applies to both Windows Server 2025 and Windows 11 24H2 clients. In the registry path `HKLM:\SYSTEM\CurrentControlSet\Services\MpsSvc\Parameters`, the `IpsecRestoreLegacyKeyMod` entry with a value of **0** utilizes the new sequence, IKEv2 and IKEv1. A value of **1** utilizes the previous sequence, AuthIP and IKEv1. To revert to the previous behavior, add the following registry key on systems using the new default keying protocol sequence. A reboot is required for changes to take effect.

PowerShell

```
New-ItemProperty -Path "HKLM:\SYSTEM\CurrentControlSet\Services\MpsSvc\Parameters" -
Name "IpsecRestoreLegacyKeyMod" -PropertyType "DWORD" -Value 1
```

# Hyper-V, AI, and performance

The following sections discuss Hyper-V, AI, and performance.

## Accelerated Networking (preview)

Accelerated Networking (AccelNet) simplifies the management of single root I/O virtualization (SR-IOV) for virtual machines (VMs) hosted on Windows Server 2025 clusters. This feature uses the high-performance SR-IOV data path to reduce latency, jitter, and CPU utilization. AccelNet also includes a management layer that handles prerequisite checking, host configuration, and VM performance settings. To learn more, see Accelerated Networking (preview).

## Dynamic processor compatibility

The dynamic processor compatibility mode is updated to take advantage of new processor capabilities in a clustered environment. Dynamic processor compatibility uses the maximum number of processor features available across all servers in a cluster. The mode improves performance compared to the previous version of processor compatibility.

You can also use dynamic processor compatibility to save its state between virtualization hosts that use different generations of processors. The processor compatibility mode now provides enhanced, dynamic capabilities on processors capable of second-level address translation. To learn more about the updated compatibility mode, see Processor compatibility for Hyper-V virtual machines.

## Hyper-V Manager

When you create a new VM through Hyper-V Manager, **Generation 2** is now set as the default option in the **New Virtual Machine Wizard**.

## Hypervisor-enforced paging translation

Hypervisor-enforced paging translation (HVPT) is a security enhancement to enforce the integrity of linear address translations. HVPT protects critical system data from write-what-where attacks, where the attacker writes an arbitrary value to an arbitrary location, often as the result of a buffer overflow. HVPT guards page tables that configure critical system data structures. HVPT includes everything already secured with hypervisor-protected code integrity (HVCI). HVPT is enabled by default, where hardware support is available. HVPT isn't enabled when Windows Server runs as a guest in a VM.

## GPU partitioning

You can share a physical GPU device with multiple VMs by using GPU partitioning. Instead of allocating the entire GPU to a single VM, GPU partitioning (GPU-P) assigns dedicated fractions of the GPU to each VM. With Hyper-V GPU-P high availability, a GPU-P VM is automatically enabled on another cluster node if there is unplanned downtime.

GPU-P Live Migration provides a solution to move a VM (for planned downtime or load balancing) with GPU-P to another node whether it's standalone or clustered. To learn more about GPU partitioning, see GPU partitioning.

## Network ATC

Network ATC streamlines the deployment and management of network configurations for Windows Server 2025 clusters. Network ATC uses an intent-based approach, where users specify their desired intents, such as management, compute, or storage for a network adapter. The deployment is automated based on the intended configuration.

This approach reduces the time, complexity, and errors associated with host networking deployment. It ensures configuration consistency across the cluster and also eliminates configuration drift. To learn more, see Deploy host networking with Network ATC.

## Scalability

With Windows Server 2025, Hyper-V now supports up to 4 petabytes of memory and 2,048 logical processors per host. This increase allows for greater scalability and performance for virtualized workloads.

Windows Server 2025 also supports up to 240 TB of memory and 2,048 virtual processors for generation 2 VMs, providing increased flexibility for running large workloads. To learn more, see Plan for Hyper-V scalability in Windows Server.

## Workgroup clusters

Hyper-V workgroup clusters are a special type of Windows Server Failover Cluster where the Hyper-V cluster nodes aren't members of an Active Directory domain with the ability to live migrate VMs in a workgroup cluster.

# Storage

The following sections describe storage updates.

## Block cloning support

Dev Drive now supports block cloning starting with Windows 11 24H2 and Windows Server 2025. Because Dev Drive uses the Resilient File System (ReFS) format, block cloning support provides significant performance benefits when you copy files. With block cloning, the file

system can copy a range of file bytes for an application as a low-cost metadata operation instead of performing expensive read-and-write operations to the underlying physical data.

The result is faster completion of file copying, reduced I/O to the underlying storage, and improved storage capacity by enabling multiple files to share the same logical clusters. To learn more, see Block cloning on ReFS.

## Dev Drive

Dev Drive is a storage volume that aims to enhance the performance of crucial developer workloads. Dev Drive uses ReFS technology and incorporates specific file system optimizations to offer greater control over storage volume settings and security. Administrators now have the ability to designate trust, configure antivirus settings, and exercise administrative control over attached filters. To learn more, see Set up a Dev Drive on Windows 11.

## NVMe

NVMe is a new standard for fast solid-state drives. NVMe storage performance is optimized in Windows Server 2025. The result is improved performance with an increase in IOPS and a decrease in CPU utilization.

## Storage Replica compression

Storage Replica compression reduces the amount of data transferred over the network during replication. To learn more about compression in Storage Replica, see Storage Replica overview.

## Storage Replica Enhanced Log

Storage Replica Enhanced Log helps with log implementation to eliminate the performance costs associated with file system abstractions. Block replication performance is improved. To learn more, see Storage Replica Enhanced Log.

## ReFS native storage deduplication and compression

ReFS native storage deduplication and compression are techniques used to optimize storage efficiency for both static and active workloads, such as file servers or virtual desktops. To learn more about ReFS deduplication and compression, see Optimize storage with ReFS deduplication and compression in Azure Local.

## Thin provisioned volumes

Thin provisioned volumes with Storage Spaces Direct are a way to allocate storage resources more efficiently and avoid costly overallocation by allocating from the pool only when needed in a cluster. You can also convert fixed to thin provisioned volumes. Converting from fixed to thin provisioned volumes returns any unused storage back to the pool for other volumes to use. To learn more about thin provisioned volumes, see Storage thin provisioning.

### Server Message Block

Server Message Block (SMB) is one of the most widely used protocols in networking. SMB provides a reliable way to share files and other resources between devices on your network. Windows Server 2025 includes SMB compression support for the industry standard LZ4 compression algorithm. LZ4 is in addition to SMB's existing support for XPRESS (LZ77), XPRESS Huffman (LZ77+Huffman), LZNT1, and PATTERN_V1.

## Azure Arc and hybrid

The following sections discuss Azure Arc and hybrid configurations.

### Simplified Azure Arc Setup

Azure Arc Setup is a Feature on Demand, so it's installed by default. A user-friendly wizard interface and a system tray icon in the taskbar help to facilitate the process of adding servers to Azure Arc. Azure Arc extends the capabilities of the Azure platform so that you can create applications and services that can operate in diverse environments. These environments include datacenters, the edge, and multicloud environments and provide increased flexibility. To learn more, see Connect Windows Server machines to Azure through Azure Arc Setup.

### Pay-as-you-go licensing

The Azure Arc pay-as-you-go subscription licensing option is an alternative to the conventional perpetual licensing for Windows Server 2025. With the pay-as-you-go option, you can deploy a Windows Server device, license it, and pay for only as much as you use. This feature is facilitated through Azure Arc and billed via your Azure subscription. To learn more, see Azure Arc pay-as-you-go licensing.

### Windows Server Management enabled by Azure Arc

Windows Server Management enabled by Azure Arc offers new benefits to customers with Windows Server licenses that have active Software Assurance or Windows Server licenses that are active subscription licenses. Windows Server 2025 has the following key benefits:

- **Windows Admin Center in Azure Arc**: Integrates Azure Arc with Windows Admin Center so that you can manage your Windows Server instances from the Azure Arc portal. This integration provides a unified management experience for your Windows Server instances, whether they're running on-premises, in the cloud, or at the edge.
- **Remote Support**: Offers customers with professional support the ability to grant just-in-time access with detailed execution transcripts and revocation rights.
- **Best Practices Assessment**: Collection and analysis of server data generates issues and remediation guidance and performance improvements.
- **Azure Site Recovery Configuration**: Configuration of Azure Site Recovery ensures business continuity and provides replication and data resilience for critical workloads.

To learn more about Windows Server Management enabled by Azure Arc and the available benefits, see Windows Server Management enabled by Azure Arc.

# Software-Defined Networking

Software-Defined Networking (SDN) is an approach to networking that network administrators can use to manage network services through abstraction of lower-level functionality. SDN enables the separation of the network control plane, which manages the network, from the data plane, which handles the traffic. This separation allows for increased flexibility and programmability in network management. SDN provides the following benefits in Windows Server 2025:

- **Network Controller**: This control plane for SDN is now hosted directly as Failover Cluster services on the physical host machines. Using a cluster role eliminates the need to deploy VMs, which simplifies deployment and management and conserves resources.
- **Tag-based segmentation**: Administrators can use custom service tags to associate network security groups (NSGs) and VMs for access control. Instead of specifying IP ranges, administrators can now use simple, self-explanatory labels to tag workload VMs and apply security policies based on these tags. Tags simplify the process of managing network security and eliminate the need to remember and retype IP ranges. To learn more, see Configure network security groups with tags in Windows Admin Center.
- **Default network policies in Windows Server 2025**: These policies bring Azure-like protection options to NSGs for workloads deployed through Windows Admin Center. The default policy denies all inbound access, allowing selective opening of well-known inbound ports while permitting full outbound access from workload VMs. Default network policies ensure that workload VMs are secured from the point of creation. To learn more, see Use default network access policies on virtual machines on Azure Local.
- **SDN Multisite**: This feature provides native layer 2 and layer 3 connectivity between applications across two locations without any extra components. With SDN Multisite, applications can move seamlessly without the need to reconfigure the application or

networks. It also offers unified network policy management for workloads so that you don't need to update policies when a workload VM moves from one location to another. To learn more, see What is SDN Multisite?.

- **Enhanced performance of SDN layer 3 gateways**: Layer 3 gateways achieve higher throughput and reduced CPU cycles. These improvements are enabled by default. Users automatically experience better performance when an SDN gateway layer 3 connection is configured through PowerShell or Windows Admin Center.

## Windows Containers portability

Portability is a crucial aspect of container management and has the ability to simplify upgrades by applying enhanced flexibility and compatibility of containers in Windows.

Portability is a Windows Server feature that users can employ to move container images, and their associated data, between different hosts or environments without requiring any modifications. Users can create a container image on one host and then deploy it on another host without having to worry about compatibility issues. To learn more, see Portability for containers.

## Windows Server Insider Program

The Windows Server Insider Program provides early access to the latest Windows OS releases for a community of enthusiasts. As a member, you're among the first to try out new ideas and concepts that Microsoft is developing. After you register as a member, you can participate in different release channels. Go to **Start** > **Settings** > **Windows Update** > **Windows Insider Program**.

## Related content

Windows Server Insider Community discussions

Last updated on 01/15/2026

# What's new in Windows Server 2022

07/03/2025Applies to: ✅ Windows Server 2022

This article describes some of the new features in Windows Server 2022. Windows Server 2022 is built on the strong foundation of Windows Server 2019 and brings many innovations on three key themes: security, Azure hybrid integration and management, and the application platform.

## Azure Edition

Windows Server 2022 Datacenter: Azure Edition helps you use the benefits of the cloud to keep your VMs up to date while minimizing downtime. This section describes some of the new features in Windows Server 2022 Datacenter: Azure Edition. Learn more about how Azure Automanage for Windows Server brings these new capabilities to Windows Server Azure Edition in the Azure Automanage for Windows Server services article.

Windows Server 2022 Datacenter: Azure Edition builds on Datacenter Edition to deliver a VM-only operating system that helps to use the benefits of the cloud, with advanced features like SMB over QUIC, Hotpatch, and Azure Extended Networking. This section describes some of these new features.

Compare the differences in the editions in Windows Server 2022. You can also learn more about how Azure Automanage for Windows Server brings these new capabilities to Windows Server Azure Edition in the Azure Automanage for Windows Server services article.

### April 2023

#### Hotpatch

Windows Server 2022 Datacenter: Azure Edition Hotpatch is now public preview for the Desktop Experience both in Azure and as a supported guest VM on Azure Local version 22H2.

### September 2022

This section lists the features and improvements that are now available in Windows Server Datacenter: Azure Edition beginning with the 2022-09 Cumulative Update for Microsoft server operating system version 21H2 for x64-based Systems (KB5017381 ). After you've install the Cumulative Update, the OS build number will be 20348.1070 or higher.

## Storage Replica compression for data transfer

This update includes Storage Replica compression for data transferred between the source and destination servers. This new functionality compresses the replication data at the source system, sent over the network and decompressed and saved on the destination. The compression results in fewer network packets to transfer the same amount of data, allowing for more throughput and less network utilization. Higher data throughput should also result in reduced synchronization time when you need it most, for example in a disaster recovery scenario.

New Storage Replica PowerShell parameters are available for existing commands. Review the StorageReplica PowerShell reference to learn more. For more information about Storage Replica, see the Storage Replica overview.

## Support for Azure Local

With this release, you can run Windows Server 2022 Datacenter: Azure Edition as a supported guest VM on Azure Local, version 22H2. With Azure Edition running on Azure Local, you'll be able to use all the existing features, including Hotpatch for Server Core and SMB over QUIC at your datacenter and edge locations.

Begin deploying Windows Server 2022 Datacenter: Azure Edition by using the Deploy from Azure Marketplace on Arc-enabled Azure Local (preview) or by using an ISO. You can download the ISO from here:

- Windows Server 2022 Datacenter: Azure Edition (EN-US) ISO
- Windows Server 2022 Datacenter: Azure Edition (ZH-CN) ISO

Your Azure subscription permits you to use Windows Server Datacenter: Azure Edition on any virtual machine instances running on Azure Local. For more information, see your product terms.

Learn more about the latest Azure Local features in What's new in Azure Local, version 22H2.

## Deploy from Azure Marketplace on Arc-enabled Azure Local (preview)

Windows Server 2022 Datacenter: Azure Edition images are available in the Azure Marketplace for Arc-enabled Azure Local, making it easy to try, buy, and deploy using Azure certified images.

Learn more about the Azure Marketplace integration for Azure Arc-enabled Azure Local features in the What's new in Azure Local article.

## Azure Edition (initial release)

This section lists the features and improvements available in Windows Server Datacenter: Azure Edition with the release in September 2021.

### Azure Automanage - Hotpatch

Hotpatch, part of Azure Automanage, is a new way to install updates on new Windows Server Azure Edition virtual machines (VMs) that doesn't require a reboot after installation. For more information, see Hotpatch for Windows Server.

### SMB over QUIC

SMB over QUIC updates the SMB 3.1.1 protocol to use the QUIC protocol instead of TCP in Windows Server 2022 Datacenter: Azure Edition, Windows 11 and later, and third-party clients if they support it. By using SMB over QUIC along with TLS 1.3, users and applications can securely and reliably access data from edge file servers running in Azure. Mobile and telecommuter users no longer need a VPN to access their file servers over SMB when on Windows. For more information, see SMB over QUIC and SMB over QUIC management with Automanage machine best practices.

To learn more about QUIC, see RFC 9000.

### Extended network for Azure

Azure Extended Network enables you to stretch an on-premises subnet into Azure to let on-premises virtual machines keep their original on-premises private IP addresses when you migrate to Azure. To learn more, see Azure Extended Network.

# All editions

This section describes some of the new features in Windows Server 2022 across all editions. To learn more about the different editions, see Comparison of Windows Server editions.

## Security

The new security capabilities in Windows Server 2022 combine other security capabilities in Windows Server across multiple areas to provide defense-in-depth protection against advanced threats. Advanced multi-layer security in Windows Server 2022 provides the comprehensive protection that servers need today.

## Secured-core server

Certified Secured-core server hardware from an OEM partner provides more security protections that are useful against sophisticated attacks. Certified Secured-core server hardware can provide increased assurance when handling mission-critical data in some of the most data sensitive industries. A Secured-core server uses hardware, firmware, and driver capabilities to enable advanced Windows Server security features. Many of these features are available in Windows Secured-core PCs and are now also available with Secured-core server hardware and Windows Server 2022. For more information, see Secured-core server.

### Hardware root-of-trust

Used by features such as BitLocker drive encryption, Trusted Platform Module 2.0 (TPM 2.0) secure crypto-processor chips provide a secure, hardware-based store for sensitive cryptographic keys and data, including systems integrity measurements. TPM 2.0 can verify that the server starts with legitimate code and is trusted by subsequent code execution, known as a hardware root-of-trust.

### Firmware protection

Firmware runs with high privileges and is often invisible to traditional anti-virus solutions. This has led to a rise in the number of firmware-based attacks. Secured-core servers measure and verify boot processes with Dynamic Root of Trust for Measurement (DRTM) technology. Secured-core servers can also isolate driver access to memory with Direct Memory Access (DMA) protection.

### UEFI secure boot

UEFI secure boot is a security standard that protects your servers from malicious rootkits. Secure boot ensures that the server boots only firmware and software that's trusted by the hardware manufacturer. When the server is started, the firmware checks the signature of each boot component, including firmware drivers and the OS. If the signatures are valid, the server boots and the firmware gives control to the OS.

### Virtualization-based security (VBS)

Secured-core servers support virtualization-based security (VBS) and hypervisor-based code integrity (HVCI). VBS uses hardware virtualization features to create and isolate a secure region of memory from the normal operating system, protecting against an entire class of vulnerabilities used in cryptocurrency mining attacks. VBS also allows for the use of Credential

Guard, where user credentials and secrets are stored in a virtual container that the operating system can't access directly.

HVCI uses VBS to significantly strengthen code integrity policy enforcement. Kernel mode integrity prevents unsigned kernel mode drivers or system files from being loaded into system memory.

Kernel Data Protection (KDP) provides read-only memory protection of kernel memory containing non-executable data where memory pages are protected by Hypervisor. KDP protects key structures in the Windows Defender System Guard runtime from being tampered with.

## Secure connectivity

### Transport: HTTPS and TLS 1.3 enabled by default on Windows Server 2022

Secure connections are at the heart of today's interconnected systems. Transport Layer Security (TLS) 1.3 is the latest version of the internet's most deployed security protocol, which encrypts data to provide a secure communication channel between two endpoints. HTTPS and TLS 1.3 is now enabled by default on Windows Server 2022, protecting the data of clients connecting to the server. It eliminates obsolete cryptographic algorithms, enhances security over older versions, and aims to encrypt as much of the handshake as possible. Learn more about supported TLS versions and about supported cipher suites.

Although TLS 1.3 in the protocol layer is now enabled by default, applications and services also need to actively support it. The Microsoft Security blog has more detail in the post Taking Transport Layer Security (TLS) to the next level with TLS 1.3 .

### Secure DNS: Encrypted DNS name resolution requests with DNS-over-HTTPS

DNS Client in Windows Server 2022 now supports DNS-over-HTTPS (DoH), which encrypts DNS queries using the HTTPS protocol. DoH helps keep your traffic as private as possible by preventing eavesdropping and your DNS data from being manipulated. Learn more about configuring the DNS client to use DoH.

### Server Message Block (SMB): SMB AES-256 encryption for the most security conscious

Windows Server now supports AES-256-GCM and AES-256-CCM cryptographic suites for SMB encryption. Windows automatically negotiates a more advanced cipher method when connecting to another computer that also supports it, and it can also be mandated through Group Policy. Windows Server still supports AES-128 for down-level compatibility. AES-128-GMAC signing now also accelerates signing performance.

### SMB: East-west SMB encryption controls for internal cluster communications

Windows Server failover clusters now support granular control of encrypting and signing intra-node storage communications for Cluster Shared Volumes (CSV) and the storage bus layer (SBL). When using Storage Spaces Direct, you can now decide to encrypt or sign east-west communications within the cluster itself for higher security.

### SMB Direct and RDMA encryption

SMB Direct and RDMA supply a high bandwidth, low latency networking fabric for workloads like Storage Spaces Direct, Storage Replica, Hyper-V, Scale-Out File Server, and SQL Server. SMB Direct in Windows Server 2022 supports encryption. Previously, enabling SMB encryption disabled direct data placement. This was intentional, but it seriously impacted performance. Now data is encrypted before data placement, leading to far less performance degradation while adding AES-128 and AES-256 protected packet privacy.

For more information on SMB encryption, signing acceleration, secure RDMA, and cluster support, see SMB security enhancements.

## Azure hybrid capabilities

You can increase your efficiency and agility with built-in hybrid capabilities in Windows Server 2022 that allow you to extend your datacenters to Azure more easily than ever before.

### Azure Arc enabled Windows Servers

Azure Arc-enabled servers with Windows Server 2022 brings on-premises and multicloud Windows Servers to Azure with Azure Arc. This management experience is designed to be consistent with how you manage native Azure virtual machines. When a hybrid machine is connected to Azure, it becomes a connected machine and is treated as a resource in Azure. More information can be found in the Azure Arc-enabled servers documentation.

### Add Windows Server machines

As of the KB5031364 update, you can add Windows Server machines with an easy process.

To add new Windows Server machines, go to the Azure Arc icon in the bottom-right corner of the taskbar and launch the Azure Arc Setup program to install and configure an Azure Connected Machine Agent. After it's installed, you can use the Azure Connected Machine Agent at no extra charge to your Azure account. After you enable Azure Arc on your server, you can see the status information in the taskbar icon.

To learn more, see Connect Windows Server machines to Azure through Azure Arc Setup.

### Windows Admin Center

Improvements to Windows Admin Center to manage Windows Server 2022 include capabilities to report on the current state of the Secured-core features mentioned, and, where applicable, to allow you to enable the features. For more information on these and many more improvements to Windows Admin Center, see Windows Admin Center documentation.

## Application platform

There are several platform improvements for Windows containers, including application compatibility and the Windows container experience with Kubernetes.

Some of the new features are:

- Reduced Windows container image size by up to 40%, which leads to a 30% faster startup time and better performance.
- Applications can now use Microsoft Entra ID with group Managed Service Account (gMSA) without domain-joining the container host. Windows containers now also support Microsoft Distributed Transaction Coordinator (DTC) and Microsoft Message Queuing (MSMQ).
- Simple buses can now be assigned to process-isolated Windows Server containers. Applications running in containers that need to communicate over SPI, I2C, GPIO, and UART/COM are now able to do so.
- We enabled support for hardware acceleration of DirectX APIs in Windows containers to support scenarios such as machine learning inference using local graphical processing unit (GPU) hardware. For more information, see the Bringing GPU acceleration to Windows containers blog post.
- There are several other enhancements that simplify the Windows container experience with Kubernetes. These enhancements include support for host-process containers for

node configuration, IPv6, and consistent network policy implementation with Calico.

- Windows Admin Center is updated to make it easy to containerize .NET applications. After the application is in a container, you can host it on Azure Container Registry to then deploy it to other Azure services, including Azure Kubernetes Service.
- With support for Intel Ice Lake processors, Windows Server 2022 supports business-critical and large-scale applications that require up to 48 TB of memory and 2,048 logical cores running on 64 physical sockets. Confidential computing with Intel Secured Guard Extension (SGX) on Intel Ice Lake improves application security by isolating applications from each other with protected memory.

## Other key features

### Remote Desktop IP Virtualization

As of the KB5030216 update, you can use Remote Desktop IP Virtualization.

Remote Desktop IP Virtualization simulates a single-user desktop by supporting per-session and per-program Remote Desktop IP Virtualization for Winsock applications. To learn more, see Remote Desktop IP Virtualization in Windows Server.

### Task Scheduler and Hyper-V Manager for Server Core installations

We added two management tools to the App Compatibility Feature on Demand feature package in this version: Task Scheduler (taskschd.msc) and Hyper-V Manager (virtmgmt.msc). For more information, see Server Core App Compatibility Feature on Demand (FOD).

### Nested virtualization for AMD processors

Nested virtualization is a feature that allows you to run Hyper-V inside of a Hyper-V virtual machine (VM). Windows Server 2022 brings support for nested virtualization using AMD processors, giving more choices of hardware for your environments. For more information, see the nested virtualization documentation.

### Microsoft Edge browser

Microsoft Edge is included with Windows Server 2022, replacing Internet Explorer. It's built on Chromium open source and backed by Microsoft security and innovation. It can be used with the Server with Desktop Experience installation options. For more information, see Microsoft Edge Enterprise documentation. Microsoft Edge, unlike the rest of Windows Server, follows the

Modern Lifecycle for its support lifecycle. For details, see Microsoft Edge lifecycle documentation.

## Networking performance

### UDP performance improvements

UDP is becoming a popular protocol carrying more network traffic due to the increasing popularity of RTP and custom (UDP) streaming and gaming protocols. The QUIC protocol, built on top of UDP, brings the performance of UDP to a level on par with TCP. Significantly, Windows Server 2022 includes UDP Segmentation Offload (USO). USO moves most of the work required to send UDP packets from the CPU to the network adapter's specialized hardware. Complimenting USO is UDP Receive Side Coalescing (UDP RSC), which coalesces packets and reduces CPU usage for UDP processing. In addition, hundreds of improvements to the UDP data path both transmit and receive. Windows Server 2022 and Windows 11 both have this new capability.

### TCP performance improvements

Windows Server 2022 uses TCP HyStart++ to reduce packet loss during connection start-up (especially in high-speed networks) and RACK to reduce Retransmit TimeOuts (RTO). These features are enabled in the transport stack by default and provide a smoother network data flow with better performance at high speeds. Windows Server 2022 and Windows 11 both have this new capability.

### Hyper-V virtual switch improvements

Virtual switches in Hyper-V are enhanced with updated Receive Segment Coalescing (RSC). RSC allows the hypervisor network to coalesce packets and process as one larger segment. CPU cycles are reduced and segments remain coalesced across the entire data path until processed by the intended application. RSC results in improved performance for both network traffic from an external host, received by a virtual NIC, and from a virtual NIC to another virtual NIC on the same host.

In vSwitch, RSC can also coalesce multiple TCP segments into a larger segment before data traversing the vSwitch. This change also improves networking performance for virtual workloads. RSC is enabled on external virtual switches by default.

## System Insights disk anomaly detection

System Insights has another capability via Windows Admin Center: disk anomaly detection.

Disk anomaly detection is a new capability that highlights when disks are behaving *differently* than usual. Although different isn't necessarily a bad thing, seeing these anomalous moments can be helpful when you troubleshoot issues on your systems. This capability is also available for servers running Windows Server 2019.

### Windows Update rollback improvements

Servers can now automatically recover from startup failures by removing updates if the startup failure was introduced after the installation of recent driver or quality Windows Updates. When a device is unable to start up properly after the recent installation of quality of driver updates, Windows now automatically uninstalls the updates to get the device back up and running normally.

This functionality requires the server to use the Server Core installation option with a Windows Recovery Environment partition.

## Storage

Windows Server 2022 includes the following storage updates. Storage is also affected by the updates to System Insights disk anomaly detection and Windows Admin Center.

### Storage Migration Service

Enhancements to Storage Migration Service in Windows Server 2022 make it easier to migrate storage to Windows Server or to Azure from more source locations. Here are the features that are available when running the Storage Migration Server orchestrator on Windows Server 2022:

- Migrate local users and groups to the new server.
- Migrate storage from failover clusters, migrate to failover clusters, and migrate between standalone servers and failover clusters.
- Migrate storage from a Linux server that uses Samba.
- More easily synchronize migrated shares into Azure by using Azure File Sync.
- Migrate to new networks such as Azure.
- Migrate NetApp CIFS servers from NetApp FAS arrays to Windows servers and clusters.

### Adjustable storage repair speed

User-adjustable storage repair speed is a new feature in Storage Spaces Direct that offers more control over the data resync process. Adjustable storage repair speed enables you to allocate

resources to either repair data copies (resiliency) or to run active workloads (performance). Controlling the repair speed helps improve availability and allows you to service your clusters more flexibly and efficiently.

### Faster repair and resynchronization

Storage repair and resynchronization after events such as node reboots and disk failures are now twice as fast. Repairs have less variance in elapsed time so you can be more sure of how long the repairs take. This improvement is achieved by adding more granularity to data tracking. Repairs now only move the data that needs to be moved, reducing the system resources used and the time it takes to make repairs.

### Storage bus cache with Storage Spaces on standalone servers

Storage bus cache is now available for standalone servers. It can significantly improve read and write performance, while maintaining storage efficiency and keeping operational costs low. Similar to its implementation for Storage Spaces Direct, this feature binds together faster media (for example, NVMe or SSD) with slower media (for example, HDD) to create tiers. A portion of the faster media tier is reserved for the cache. To learn more, see Enable storage bus cache with Storage Spaces on standalone servers.

### ReFS file-level snapshots

The Microsoft Resilient File System (ReFS) now includes the ability to snapshot files by using a quick metadata operation. Snapshots are different from ReFS block cloning in that clones are writable, whereas snapshots are read-only. This functionality is especially useful in virtual machine backup scenarios with VHD/VHDX files. ReFS snapshots are unique in that they take a constant time irrespective of file size. Support for snapshots is available in ReFSUtil or as an API.

### SMB compression

Enhancement to SMB in Windows Server 2022 and Windows 11 allows a user or application to compress files as they transfer over the network. You no longer need to manually zip files in order to transfer faster on slower or more congested networks. For details, see SMB compression.

## Containers

Windows Server 2022 includes the following changes to Windows containers.

## Server Core image size reduction

We reduced the size of Server Core images. The smaller image size allows you to deploy containerized applications faster. In Windows Server 2022, the Server Core container image release to manufacturing (RTM) layer at the time of GA clocks in at 2.76 GB uncompressed on disk. Compared to the Windows Server 2019 RTM layer at the time of GA, which clocks in at 3.47 GB uncompressed on disk, that's a 33% reduction in on-disk footprint for that layer. Although you shouldn't expect the total image size to be reduced by 33%, a smaller RTM layer size generally reduces the overall image size.

> **Note**
>
> Windows container base images ship as two layers: an RTM layer and a patch layer that contains the latest security fixes for OS libraries and binaries that's overlaid on the RTM layer. The patch layer's size changes over the life of the container image support cycle, depending on how many changes are in the binaries. When you pull a container base image onto a new host, you need to pull both layers.

## Longer support cycle for all Windows container images

Windows Server 2022 images, including Server Core, Nano Server, and Server, have five years of mainstream support and five years of extended support. This longer support cycle ensures that you have time to implement, use, and upgrade or migrate when appropriate for your organization. For more information, see Windows containers base image lifecycles and Windows Server 2022 lifecycles.

## Virtualized time zone

With Windows Server 2022, Windows containers can maintain a virtualized time zone configuration separate from the host. All configurations the host time zone typically uses are now virtualized and instanced for each container. To configure the container time zone, you can use the tzutil command utility or the Set-TimeZone PowerShell cmdlet. To learn more, see virtualized time zone.

## Scalability improvements for overlay networking support

Windows Server 2022 aggregates several performance and scale improvements that were already in four earlier Semi-Annual Channel (SAC) releases of Windows Server that hadn't been backported into Windows Server 2019:

- Fixed the issue that caused port exhaustion when using hundreds of Kubernetes services and pods on the same node.
- Improved packet forwarding performance in the Hyper-V virtual switch (vSwitch).
- Increased reliability across Container Network Interface (CNI) restarts in Kubernetes.
- Improvements in the Host Networking Service (HNS) control plane and in the data plane used by Windows Server containers and Kubernetes networking.

To learn more about the performance and scalability improvements for overlay networking support, see Kubernetes Overlay Networking for Windows.

## Direct Server Return routing for overlay and l2bridge networks

Direct Server Return (DSR) is an asymmetric network load distribution in load balanced systems that makes request and response traffic use different network paths. Using different network paths helps avoid extra hops and reduces latency, speeding up response time between the client and service and removing extra load from the load balancer. DSR transparently achieves increased network performance for applications with little to no infrastructure changes.

To learn more, see DSR in Introduction to Windows support in Kubernetes.

## gMSA improvements

You can use Group Managed Service Accounts (gMSA) with Windows containers to facilitate Active Directory authentication. When introduced in Windows Server 2019, gMSA required joining the container host to a domain to retrieve the gMSA credentials from Active Directory. In Windows Server 2022, gMSA for containers with a non-domain joined host uses a portable user identity instead of a host identity to retrieve gMSA credentials. Therefore, manually joining Windows worker nodes to a domain is no longer necessary. After authentication, Kubernetes saves the user identity as a secret. gMSA for containers with a non-domain joined host provides the flexibility of creating containers with gMSA without joining the host node to the domain.

To learn more about the gMSA improvements, see Create gMSAs for Windows containers.

## IPv6 support

Kubernetes in Windows now supports the IPv6 dual stack in L2bridge-based networks in Windows Server. IPv6 is dependent on the CNI that Kubernetes uses, and also requires Kubernetes version 1.20 or later to enable end-to-end IPv6 support. For more information, see IPv4/IPv6 in Introduction to Windows support in Kubernetes.

## Multi-subnet support for Windows worker nodes with Calico for Windows

The Host Network Service (HNS) now allows you to use more restrictive subnets, such as subnets with a longer prefix length, and also multiple subnets for each Windows worker node. Previously, HNS restricted Kubernetes container endpoint configurations to only use the prefix length of the underlying subnet. The first CNI that makes use of this functionality is Calico for Windows. For more information, see Multiple subnet support in Host Networking Service.

## HostProcess containers for node management

HostProcess containers are a new container type that runs directly on the host and extends the Windows container model to enable a wider range of Kubernetes cluster management scenarios. With HostProcess containers, you can package and distribute management operations that require host access while retaining versioning and deployment methods provided by containers. You can use Windows containers for various device plug-in, storage, and networking management scenarios in Kubernetes.

HostProcess containers have the following benefits:

- Cluster users no longer need to sign in and individually configure each Windows node for administrative tasks and management of Windows services.
- You can utilize the container model to deploy management logic to as many clusters as needed.
- You can build HostProcess containers on top of existing Windows Server 2019 or later base images, manage them by using Windows container runtime, and run as any user available in the domain of the host machine.
- HostProcess containers provide the best way to manage Windows nodes in Kubernetes.

For more information, see Windows HostProcess Containers.

## Windows Admin Center improvements

Windows Server 2022 expands on the Containers extension added to Windows Admin Center to containerize existing web applications based on ASP.NET from .NET Framework. You can use static folders or Visual Studio solutions from your developer.

Windows Admin Center includes the following enhancements:

- The Containers extension now supports Web Deploy files, which lets you extract the app and its configuration from a running server and then containerize the application.
- You can validate the image locally and then push that image to Azure Container Registry.

- Azure Container Registry and Azure Container Instances now have basic management functionality. You can now use the Windows Admin Center UI to create and delete registries, manage images, and start and stop new container instances.

## Azure Migrate: App Containerization tool

Azure Migrate App: Containerization is an end-to-end solution that containerizes and moves existing web applications to the Azure Kubernetes Service. You can assess existing web servers, create a container image, push the image to the Azure Container Registry, create a Kubernetes deployment, and finally deploy it to the Azure Kubernetes Service.

For more information about the Azure Migrate App: Containerization tool, see ASP.NET app containerization and migration to Azure Kubernetes Service and Java web app containerization and migration to Azure Kubernetes Service.

# What's new in Windows Server 2019

07/25/2025Applies to: ✅ Windows Server 2019

This article describes some of the new features in Windows Server 2019. Windows Server 2019 is built on the strong foundation of Windows Server 2016 and brings numerous innovations on four key themes: Hybrid Cloud, Security, Application Platform, and Hyper-Converged Infrastructure (HCI).

# General

## Windows Admin Center

Windows Admin Center is a locally deployed, browser-based app for managing servers, clusters, hyper-converged infrastructure, and Windows 10 PCs. It comes at no extra cost beyond Windows and is ready to use in production.

You can install Windows Admin Center on Windows Server 2019 and Windows 10 and 11, and use it to manage servers and clusters running Windows Server 2012 and later.

For more info, see Windows Admin Center.

## Desktop experience

Because Windows Server 2019 is a Long-Term Servicing Channel (LTSC) release, it includes the **Desktop Experience**. Semi-Annual Channel (SAC) releases don't include the Desktop Experience by design; they're strictly Server Core and Nano Server container image releases. As with Windows Server 2016, during setup of the operating system you can choose between Server Core installations or Server with Desktop Experience installations.

## System Insights

System Insights is a new feature available in Windows Server 2019 that brings local predictive analytics capabilities natively to Windows Server. These predictive capabilities, each backed by a machine-learning model, locally analyze Windows Server system data, such as performance counters and events. System Insights allows you to understand how your servers are functioning and helps you reduce the operational expenses associated with reactively managing issues in your Windows Server deployments.

# Hybrid Cloud

## Server Core App Compatibility Feature on Demand

Server Core App Compatibility Feature on Demand (FOD) significantly improves the app compatibility by including a subset of binaries and components from Windows Server with the Desktop Experience. Server Core is kept it as lean as possible by not adding the Windows Server Desktop Experience graphical environment itself, increasing the functionality and compatibility.

This optional feature on demand is available on a separate ISO and can be added to Windows Server Core installations and images only, using DISM.

## Windows Deployment Services (WDS) Transport Server role added to Server Core

Transport Server contains only the core networking parts of WDS. You can now use Server Core with the Transport Server role to create multicast namespaces that transmit data (including operating system images) from a standalone server. You can also use it if you want to have a PXE server that allows clients to PXE boot and download your own custom setup application.

## Remote Desktop Services integration with Azure AD

With Azure AD integration you can use Conditional Access policies, Multifactor Authentication, Integrated authentication with other SaaS Apps using Azure AD, and many more. For more information, see Integrate Azure AD Domain Services with your RDS deployment.

# Networking

We made several improvements to the core network stack, such as TCP Fast Open (TFO), Receive Window Autotuning, IPv6, and more. For more information, see the Core Network Stack feature improvement post.

## Dynamic vRSS and VMMQ

In the past, Virtual Machine Queues and Virtual Machine Multi-Queues (VMMQs) enabled much higher throughput to individual VMs as network throughputs first reached the 10GbE mark and beyond. Unfortunately, the planning, baselining, tuning, and monitoring required for success became a much larger undertaking than IT administrators anticipated.

Windows Server 2019 improves these optimizations by dynamically spreading and tuning the processing of network workloads as needed. Windows Server 2019 ensures peak efficiency and

removes the configuration burden for IT administrators. To learn more, see Host network requirements for Azure Local.

# Security

## Windows Defender Advanced Threat Protection (ATP)

ATP's deep platform sensors and response actions expose memory and kernel level attacks and respond by suppressing malicious files and terminating malicious processes.

- For more information about Windows Defender ATP, see Overview of Windows Defender ATP capabilities.
- For more information on onboarding servers, see Onboard servers to Windows Defender ATP service.

**Windows Defender ATP Exploit Guard** is a new set of host-intrusion prevention capabilities enabling you to balance security risk and productivity requirements. Windows Defender Exploit Guard is designed to lock down the device against a wide variety of attack vectors and block behaviors commonly used in malware attacks. The components are:

- Attack Surface Reduction (ASR) is a set of controls that enterprises can enable to prevent malware from getting on the machine by blocking suspicious malicious files. For example, Office files, scripts, lateral movement, ransomware behavior, and email-based threats.
- Network protection protects the endpoint against web-based threats by blocking any outbound process on the device to untrusted hosts/IP addresses through Windows Defender SmartScreen.
- Controlled folder access protects sensitive data from ransomware by blocking untrusted processes from accessing your protected folders.
- Exploit protection is a set of mitigations for vulnerability exploits (replacing EMET) that can be easily configured to protect your system and applications.
- Windows Defender Application Control (also known as Code Integrity (CI) policy) was released in Windows Server 2016. We've made deployment easier by including default CI policies. The default policy allows all Windows in-box files and Microsoft applications, such as SQL Server, and blocks known executables that can bypass CI.

## Security with Software Defined Networking (SDN)

Security with SDN delivers many features to increase customer confidence in running workloads, either on-premises, or as a service provider in the cloud.

These security enhancements are integrated into the comprehensive SDN platform introduced in Windows Server 2016.

For a complete list of what's new in SDN see, What's New in SDN for Windows Server 2019.

## Shielded Virtual Machines improvements

We made the following improvements to Shielded Virtual Machines.

### Branch office improvements

You can now run shielded virtual machines on machines with intermittent connectivity to the Host Guardian Service by using the new fallback HGS and offline mode features. Fallback HGS allows you to configure a second set of URLs for Hyper-V to try if it can't reach your primary HGS server.

Even if you can't reach the HGS, offline mode lets you continue to start up your shielded VMs. Offline mode also lets you start your VMs as long as the VM has started successfully once and the host's security configuration hasn't changed.

### Troubleshooting improvements

We also made it easier to troubleshoot your shielded VMs by enabling support for VMConnect Enhanced Session Mode and PowerShell Direct. These tools are useful when you lose network connectivity to your VM and need to update its configuration to restore access. To learn more, see Guarded fabric and shielded VMs.

You don't need to configure these features because they become automatically available when you place a shielded VM on a Hyper-V host running Windows Server version 1803 or later.

### Linux support

If you run mixed-OS environments, Windows Server 2019 now supports running Ubuntu, Red Hat Enterprise Linux, and SUSE Linux Enterprise Server inside shielded virtual machines.

## HTTP/2 for a faster and safer Web

- Improved coalescing of connections to deliver an uninterrupted and properly encrypted browsing experience.

- Upgraded HTTP/2's server-side cipher suite negotiation for automatic mitigation of connection failures and ease of deployment.
- Changed our default TCP congestion provider to Cubic to give you more throughput!

## Encrypted networks

Virtual network encryption encrypts virtual network traffic between virtual machines within subnets that have the **Encryption Enabled** label. Encrypted networks also use Datagram Transport Layer Security (DTLS) on the virtual subnet to encrypt packets. DTLS protects your data from eavesdropping, tampering, and forgery by anyone with access to the physical network.

For more information, see Encrypted networks.

## Firewall auditing

Firewall auditing is a new feature for SDN firewall that records any flow processed by SDN firewall rules and access control lists (ACLs) that have logging enabled.

## Virtual network peering

Virtual network peering lets you connect two virtual networks seamlessly. Once peered, the virtual networks appear in monitoring as one.

## Egress metering

Egress metering offers usage meters for outbound data transfers. Network Controller uses this feature to keep an allowlist of all IP ranges used within SDN per virtual network. These lists consider any packet heading to a destination not included within the listed IP ranges to be billed as outbound data transfers.

# Storage

Here are some of the changes we've made to Storage in Windows Server 2019. Storage is also affected by updates to Data deduplication, particularly its update to DataPort API for optimized ingress or egress to deduplicated volumes.

## File Server Resource Manager

It's now possible to prevent the File Server Resource Manager service from creating a change journal (also known as a USN journal) on all volumes when the service starts. Preventing the creation of the change journey can conserve space on each volume, but will disable real-time file classification. For more information, see File Server Resource Manager overview.

## SMB

- Windows Server no longer installs the SMB1 client and server by default. Additionally, the ability to authenticate as a guest in SMB2 and later is off by default. For more information, see SMBv1 is not installed by default in Windows 10 version 1709, Windows Server version 1709 and later versions.

- You can now disable oplocks in SMB2+ for legacy applications. You can also require signing or encryption on per-connection basis from a client. For more information, see SMBShare PowerShell module help.

## Storage Migration Service

Storage Migration Service makes it easier to migrate servers to a newer version of Windows Server. This graphical tool inventories data on servers, then transfers the data and configuration to newer servers. The Storage Migration Service can also move the identities of the old servers to the new servers so users don't have to reconfigure their profiles and apps. For more information, see Storage Migration Service.

Windows Admin Center version 1910 added the ability to deploy Azure virtual machines. This update integrates Azure VM deployment into Storage Migration Service. For more information, see Azure VM migration.

You can also access the following post-release-to-manufacturing (RTM) features when running the Storage Migration Server orchestrator on Windows Server 2019 with KB5001384 installed or on Windows Server 2022:

- Migrate local users and groups to the new server.
- Migrate storage from failover clusters, migrate to failover clusters, and migrate between standalone servers and failover clusters.
- Migrate storage from a Linux server that uses Samba.
- Sync migrated shares more easily into Azure by using Azure File Sync.
- Migrate to new networks such as Azure.
- Migrate NetApp Common Internet File System (CIFS) servers from NetApp Federated Authentication Service (FAS) arrays to Windows servers and clusters.

## Storage Spaces Direct

Here's what's new in Storage Spaces Direct. For more information about how to acquire validated Storage Spaces Direct systems, see Azure Local solution overview.

- Deduplication and compression for ReFS volumes. Variable-size chunk store with optional compression maximizes savings rates, while multi-threaded post-processing architecture minimizes performance impact. This feature supports volumes up to 64 TB and deduplicates the first 4 MB of each file.
- Native support for persistent memory, which lets you manage persistent memory like any other drive in PowerShell or Windows Admin Center. This feature supports Intel Optane DC PM and NVDIMM-N persistent memory modules.
- Nested resiliency for two-node hyper-converged infrastructure at the edge. With the help of a new software resiliency option based on RAID 5+1, you can now survive two hardware failures simultaneously. A two-node Storage Spaces Direct cluster provides continuously accessible storage for apps and virtual machines even if one server node goes down and another server node has a drive failure.
- Two-server clusters can now use a USB flash drive as a witness. If a server goes down and then back up, the USB drive cluster knows which server has the most up-to-date data. For more information, see our Storage Spaces Direct announcement blog post and Configure a file share witness for Failover Clustering.
- Windows Admin Center supports a dashboard that lets you manage and monitor Storage Spaces direct. You can monitor IOPS and IO latency from the overall cluster level down to individual SSDs or HDDs at no extra cost. To learn more, see Manage a hyper-converged infrastructure by using Windows Admin Center.
- Performance history is a new feature that provides effortless visibility into resource utilization and measurements. To learn more, see Performance history for Storage Spaces Direct.
- Scale up to 4 PB per cluster using a capacity of up to 64 volumes of up to 64 TB. You can also stitch multiple clusters together into a cluster set for even greater scale within a single storage namespace.
- By utilizing mirror-accelerated parity, it is possible to construct Storage Spaces Direct volumes that incorporate both mirror and parity strategies, similar to a blend of RAID-1 and RAID-5/6. Mirror-accelerated parity is now two times faster than Windows Server 2016.

- Drive latency outlier detection automatically identifies slow drives in PowerShell and Windows Admin Center with an "Abnormal Latency" status.
- Manually delimit the allocation of volumes to increase fault tolerance. For more information, see Delimit the allocation of volumes in Storage Spaces Direct.

## Storage Replica

Here's what's new in Storage Replica.

- Storage Replica is now available in Windows Server 2019 Standard Edition and Windows Server 2019 Datacenter Edition. However, with the Standard Edition, you can only replicate one volume, and that volume can only go up to 2 TB in size.
- Test failover is a new feature that allows you to temporarily mount a snapshot of the replicated storage on a destination server for testing or backup purposes. For more information, see Frequently asked questions about Storage Replica.
- Storage Replica log performance improvements, such as improved replication throughput and latency on all-flash storage and Storage Spaces Direct clusters that replicate between each other.
- Windows Admin Center support, including graphical management of replication using Server Manager for server-to-server, cluster-to-cluster, and stretch cluster replication.

## Data deduplication

Windows Server 2019 now supports the Resilient File System (ReFS). ReFS lets you store up to ten times more data on the same volume with deduplication and compression for the ReFS filesystem. The variable-size chunk store comes with an optional compression feature that can maximize savings rates, while the multi-threaded post-processing architecture keeps performance impact minimal. ReFS supports volumes up to 64 TB and deduplicates the first 4 TB of each file. To learn more, see How to turn on deduplication and compression in Windows Admin Center for a quick video demonstration.

# Failover Clustering

We added the following features to failover clustering in Windows Server 2019:

- Cluster sets group multiple clusters together into a loosely coupled grouping of multiple failover clusters that come in three types: compute, storage, and hyper-converged. This grouping increases the number of servers in a single software-defined datacenter (SDDC)

solution beyond the current limits of a cluster. With cluster sets, you can move online virtual machines between clusters within the cluster set. For more information, see Deploy a cluster set.

- Clusters are now Azure-aware by default. Azure-aware clusters automatically detect when they're running in Azure IaaS virtual machines, then optimize their configuration to achieve the highest levels of availability. These optimizations include proactive failover and logging of Azure planned maintenance events. Automated optimization makes deployment simpler by removing the need to configure the load balancer with Distributed Network Name for the cluster name.
- Cross-domain cluster migration lets failover clusters dynamically move from one Active Directory domain to another, simplifying domain consolidation and allowing hardware partners to create clusters and join them to the customer's domain at a later time.
- The USB witness feature lets you use a USB drive attached to a network switch as a witness in determining quorum for a cluster. This feature includes extended File Share Witness support for any SMB2-compliant device.
- The CSV cache is now enabled by default to boost virtual machine performance. MSDTC now supports Cluster Shared Volumes to allow deploying MSDTC workloads on Storage Spaces Direct, such as with SQL Server. Enhanced logic to detect partitioned nodes with self-healing to return nodes to cluster membership. Enhanced cluster network route detection and self-healing.
- Cluster Aware Updating (CAU) is now integrated and aware of Storage Spaces Direct, validating and ensuring data resynchronization completes on each node. Cluster Aware Updating inspects updates to intelligently restart only if necessary. This feature lets you restart all servers in the cluster for planned maintenance.
- You can now use file share witnesses in the following scenarios:
    - Absent or poor Internet access because of a remote location, preventing the use of a cloud witness.
    - Lack of shared drives for a disk witness. For example, a configuration that doesn't use shared disks, such as Storage Spaces Direct hyperconverged configuration, a SQL Server Always On Availability Groups (AG), or an Exchange Database Availability Group (DAG).
    - Lack of domain controller connection due to the cluster being behind a DMZ.
    - A workgroup or cross-domain cluster that doesn't have an Active Directory cluster name object (CNO). Windows Server now also blocks using a DFS Namespaces share

as a location. Adding a file share witness to a DFS share can cause stability issues for your cluster, and this configuration has never been supported. We added logic to detect if a share uses DFS Namespaces, and if DFS Namespaces is detected, Failover Cluster Manager blocks creation of the witness and displays an error message about not being supported.

- A cluster hardening feature has been implemented that enhances the security of intra-cluster communication over Server Message Block (SMB) for Cluster Shared Volumes and Storage Spaces Direct. This feature leverages certificates to provide the most secure platform possible. By doing so, Failover Clusters can now operate without any dependencies on NTLM, which enables security baselines to be established.

- Failover Clusters no longer use NTLM authentication. Instead, Windows Server 2019 clusters now exclusively use Kerberos and certificate-based authentication. Users don't need to make any changes or deploy anything to take advantage of this security enhancement. This change also lets you deploy failover clusters in environments where NTLM is disabled.

# Application Platform

## Linux containers on Windows

It's now possible to run Windows and Linux-based containers on the same container host, using the same docker daemon. You can now have a heterogeneous container host environment providing flexibility to application developers.

## Built-in support for Kubernetes

Windows Server 2019 continues the improvements to compute, networking, and storage from the Semi-Annual Channel releases needed to support Kubernetes on Windows. More details are available in upcoming Kubernetes releases.

- Container Networking in Windows Server 2019 greatly improves usability of Kubernetes on Windows. We've enhanced platform networking resiliency and support of container networking plugins.

- Deployed workloads on Kubernetes are able to use network security to protect both Linux and Windows services using embedded tooling.

## Container improvements

- **Improved integrated identity**

  We've made integrated Windows authentication in containers easier and more reliable, addressing several limitations from prior versions of Windows Server.

- **Better application compatibility**

  Containerizing Windows-based applications just got easier: The app compatibility for the existing **windowsservercore** image has been increased. For applications with more API dependencies, there's now a third base image: **windows**.

- **Reduced size and higher performance**

  The base container image download sizes, size on disk and startup times have been improved to speed up container workflows.

- **Management experience using Windows Admin Center (preview)**

  We've made it easier than ever to see which containers are running on your computer and manage individual containers with a new extension for Windows Admin Center. Look for the "Containers" extension in the Windows Admin Center public feed.

## Compute improvements

- **VM Start Ordering** VM Start Ordering is also improved with OS and Application awareness, bringing enhanced triggers for when a VM is considered started before starting the next.

- **Storage-class memory support for VMs** enables NTFS-formatted direct access volumes to be created on non-volatile DIMMs and exposed to Hyper-V VMs. Hyper-V VMs can now use the low-latency performance benefits of storage-class memory devices.

- **Persistent Memory support for Hyper-V VMs** To use the high throughput and low latency of persistent memory (also known as storage class memory) in virtual machines, it can now be projected directly into VMs. Persistent memory can help to drastically reduce database transaction latency or reduce recovery times for low latency in-memory databases on failure.

- **Container storage – persistent data volumes** Application containers now have persistent access to volumes. For more info, see Container Storage Support with Cluster Shared Volumes (CSV), Storage Spaces Direct (S2D), SMB Global Mapping.

- **Virtual machine configuration file format (updated)** The VM guest state file (`.vmgs`) has been added for virtual machines with a configuration version of 8.2 and higher. The VM

guest state file includes device state information that was previously part of the VM runtime state file.

## Encrypted Networks

Encrypted Networks - Virtual network encryption allows encryption of virtual network traffic between virtual machines that communicate with each other within subnets marked as **Encryption Enabled**. It also utilizes Datagram Transport Layer Security (DTLS) on the virtual subnet to encrypt packets. DTLS protects against eavesdropping, tampering, and forgery by anyone with access to the physical network.

## Network performance improvements for virtual workloads

Network performance improvements for virtual workloads maximizes the network throughput to virtual machines without requiring you to constantly tune or over-provision your host. Improved performance lowers the operations and maintenance cost while increasing the available density of your hosts. These new features are:

- Dynamic Virtual Machine Multi-Queue (d.VMMQ)
- Receive Segment Coalescing in the vSwitch

## Low Extra Delay Background Transport

Low Extra Delay Background Transport (LEDBAT) is a latency optimized, network congestion control provider designed to automatically yield bandwidth to users and applications. LEDBAT consumes bandwidth available while the network isn't in use. The technology is intended for use when deploying large, critical updates across an IT environment without impacting customer facing services and associated bandwidth.

## Windows Time Service

The Windows Time Service includes true UTC-compliant leap second support, a new time protocol called Precision Time Protocol, and end-to-end traceability.

## High performance SDN gateways

High performance SDN gateways in Windows Server 2019 greatly improves the performance for IPsec and GRE connections, providing ultra-high-performance throughput with much less CPU utilization.

## New Deployment UI and Windows Admin Center extension for SDN

Now, with Windows Server 2019, it's easy to deploy and manage through a new deployment UI and Windows Admin Center extension that enable anyone to harness the power of SDN.

## Windows Subsystem for Linux (WSL)

WSL enables server administrators to use existing tools and scripts from Linux on Windows Server. Many improvements showcased in the command line blog are now part of Windows Server, including Background tasks, DriveFS, WSLPath, and much more.

# Active Directory Federation Services

Active Directory Federation Services (AD FS) for Windows Server 2019 includes the following changes.

## Protected sign ins

Protected sign ins with AD FS now include the following updates:

- Users can now use third-party authentication products as their first factor without exposing passwords. In cases where the external authentication provider can prove two factors, it can use multifactor authentication (MFA).
- Users can now use passwords as an extra factor after using a non-password option as the first factor. This in-box support improves overall experience from AD FS 2016, which required downloading a GitHub adapter.
- Users can now build their own plug-in risk assessment modules to block certain types of requests during the preauthentication stage. This feature makes it easier to use cloud intelligence such as identity protection to block risky users or transactions. For more information, see Build Plug-ins with AD FS 2019 Risk Assessment Model.
- Improves Extranet Smart Lockout (ESL) quick-fix engineering (QFE) by adding the following capabilities:
    - You can now use audit mode while protected by classic extranet lockout functionality.
    - Users can now use independent lockout thresholds for familiar locations. This feature lets you run multiple instances of apps within a common service account to roll over passwords with minimal disruption.

## Other security improvements

AD FS 2019 includes the following security improvements:

- Remote PowerShell using SmartCard Sign-in lets users remote connect to AD FS with SmartCards by running PowerShell commands. Users can also use this method to manage all PowerShell functions including multi-node cmdlets.
- HTTP header customization lets users customize HTTP headers created during AD FS responses. Header customization includes the following types of headers:
    - HSTS, which only lets you use AD FS endpoints on HTTPS endpoints for a compliant browser to enforce.
    - X-frame-options, which lets AD FS admins allow specific relying parties to embed iFrames for AD FS interactive sign-in pages. You should only use this header on HTTPS hosts.
    - Future header. You can also configure multiple future headers.

    For more information, see Customize HTTP security response headers with AD FS 2019.

## Authentication and policy capabilities

AD FS 2019 includes the following authentication and policy capabilities:

- Users can now create rules to specify which authentication provider their deployment invokes for extra authentication. This feature helps with transitioning between authentication providers and securing specific apps that have special requirements for extra authentication providers.
- Optional restrictions for Transport Layer Security (TLS)-based device authentications so that only applications that require TLS can use them. Users can restrict client TLS-based device authentications so that only applications doing device-based conditional access can use them. This feature prevents unwanted prompts for device authentication for applications that don't require TLS-based device authentication.
- AD FS now supports redoing second-factor credentials based on the second factor credential freshness. This feature allows users to only require TFA for the first transaction, then only require the second factor on a periodic basis. You can only use this feature on applications that can provide an extra parameter in the request, since it's not a configurable setting in AD FS. Microsoft Entra ID supports this parameter if you configure the **Remember my MFA for X Days** setting to have *supportsMFA* set to **True** in the Microsoft Entra ID federated domain trust settings.

## Single sign-on improvements

AD FS 2019 also includes the following single sign-on (SSO) improvements:

- AD FS now uses a paginated UX flow and a centered user interface (UI) that provides a smoother sign-in experience for users. This change mirrors functionality offered in Azure AD. You may need to update your organization's logo and background images to suit the new UI.

- We fixed an issue that caused the MFA state to not persist when using Primary Refresh Token (PRT) authentication on Windows 10 devices. Users should now be prompted for second factor credentials less often. The experience should now be consistent when device authentication is successful on client TLS and PRT authentication.

## Support for building modern line-of-business apps

AD FS 2019 includes the following features to support building modern line-of-business (LOB) apps:

- AD FS now includes OAuth device flow profile support for signing in using devices without a UI surface area to support rich sign-in experiences. This feature lets users finish signing in on a different device. The Azure Command-Line Interface (CLI) experience in Azure Stack requires this functionality, and you can also use it in other scenarios.

- You no longer require the *Resource* parameter to use AD FS, which is in line with current OAUth specifications. Clients now only need to provide the relying party trust identifier as the scope parameter long with requested permissions.

- You can use cross-origin resource sharing (CORS) headers in AD FS responses. These new headings let users build single-page applications that allow client-side JavaScript libraries to validate the *id_token* signature by querying for the signing keys from the Open ID Connect (OIDC) discovery document on AD FS.

- AD FS includes Proof Key for Code Exchange (PKCE) support for secure auth code flow within OAuth. This extra layer of security prevents malicious actors from hijacking the code and replaying it from a different client.

- We fixed a minor issue that caused AD FS to only send the x5t claim. AD FS now also sends a kid claim to denote the key ID hint for signature verification.

## Supportability improvements

Admins can now configure AD FS to allow users to send error reports and debug logs to them as a ZIP file for troubleshooting. Admins can also configure a Simple Mail Transfer Protocol (SMTP) connection to automatically send the ZIP file to a triage email account. Another setting lets admins automatically create a ticket for their support system based on that email.

## Deployment updates

The following deployment updates are now included in AD FS 2019:

- AD FS has a similar function to its Windows Server 2016 version that makes it easier to upgrade Windows Server 2016 server farms into Windows Server 2019 server farms. A Windows Server 2019 server added to a Windows Server 2016 server farm will only behave like a Windows Server 2016 server until you're ready to upgrade. For more information, see Upgrading to AD FS in Windows Server 2016.

## SAML updates

AD FS 2019 includes the following Security Assertion Markup Language (SAML) updates:

- We fixed issues in aggregated federation support, such as InCommon, in these areas:
  - Improved scaling for many entities in the aggregated federation metadata document. Previously, scaling for these entities would be unsuccessful and return an ADMIN0017 error message.
  - You can now make queries using the *ScopeGroupID* parameter by running the `Get-AdfsRelyingPartyTrustsGroup` PowerShell cmdlet.
  - Improved handling of error conditions for duplicate *entityID* values.

## Azure AD style resource specification in scope parameter

Previously, AD FS required the desired resource and scope to be in a separate parameter in any authentication request. For example, the following example OAuth request contains a scope parameter:

HTTP

```
https://fs.contoso.com/adfs/oauth2/authorize?
response_type=code&client_id=claimsxrayclient&resource=urn:microsoft:adfs:claimsxr
ay&scope=oauth&redirect_uri=https://adfshelp.microsoft.com/
ClaimsXray/TokenResponse&prompt=login
```

With AD FS on Windows Server 2019, you can now pass the resource value embedded in the scope parameter. This change is consistent with authentication against Microsoft Entra ID.

The scope parameter can now be organized as a space-separated list that structures each entity as a resource or scope.

> **Note**
>
> You can only specify one resource in the authentication request. If you include more than one resource in the request, AD FS returns an error and authentication doesn't succeed.

# What's new in Windows Server 2016

Article • 04/08/2025 • Applies to: ✅ Windows Server 2016

This article describes some of the new features in Windows Server 2016 that are the ones most likely to have the greatest impact as you work with this release.

## Compute

The Virtualization area includes virtualization products and features for the IT professional to design, deploy, and maintain Windows Server.

### General

Physical and virtual machines benefit from greater time accuracy due to improvements in the Win32 Time and Hyper-V Time Synchronization Services. Windows Server can now host services that are compliant with upcoming regulations that require a 1ms accuracy regarding UTC.

### Hyper-V

Hyper-V network virtualization (HNV) is a fundamental building block of Microsoft's updated Software Defined Networking (SDN) solution and is fully integrated into the SDN stack. Windows Server 2016 includes the following changes for Hyper-V:

- Windows Server 2016 now includes a programmable Hyper-V switch. Microsoft's Network Controller pushes HNV policies down to a Host Agent running on each host using the Open vSwitch Database Management Protocol (OVSDB) as the SouthBound Interface (SBI). The Host Agent stores this policy using a customization of the VTEP schema and programs complex flow rules into a performant flow engine in the Hyper-V switch. The flow engine in the Hyper-V switch is the same one that Azure uses. The entire SDN stack up through the Network Controller and Network Resource provider is also consistent with Azure, making its performance comparable to the Azure public cloud. Within Microsoft's flow engine, the Hyper-V switch is equipped to handle both stateless and stateful flow rules through a simple match action mechanism that defines how packets should be processed within the switch.

- HNV now supports Virtual eXtensible Local Area Network (VXLAN) protocol encapsulation. HNV uses the VXLAN protocol in MAC distribution mode through the Microsoft Network Controller to map tenant overly network IP addresses to the physical underlay network IP addresses. The NVGRE and VXLAN Task Offloads support third-party drivers for improved performance.

- Windows Server 2016 includes a software load balancer (SLB) with full support for virtual network traffic and seamless interaction with HNV. The performant flow engine implements the SLB in the data plane v-Switch, then the Network Controller controls it for Virtual IP (VIP) or Dynamic IP (DIP) mappings.

- HNV implements correct L2 Ethernet headers to ensure interoperability with third-party virtual and physical appliances that depend on industry-standard protocols. Microsoft ensures that all transmitted packets have compliant values in all fields to guarantee interoperability. HNV requires support for Jumbo Frames (MTU > 1780) in the physical L2 network to account for packet overhead introduced by encapsulation protocols such as NVGRE and VXLAN. Jumbo Frame support ensures that guest Virtual Machines attached to an HNV Virtual Network maintain a 1514 MTU.

- Windows Container support adds performance improvements, simplified network management, and support for Windows containers on Windows 10. For more information, see our Windows Containers Documentation and Containers: Docker, Windows, and Trends.

- Hyper-V is now compatible with Connected Standby. When you install the Hyper-V role on a computer that uses the Always On/Always Connected (AOAC) power model, you can now configure it to use the Connected Standby power state.

- Discrete device assignment lets you give a virtual machine (VM) direct and exclusive access to certain PCIe hardware devices. This feature bypasses the Hyper-V virtualization stack, which results in faster access. For more information, see Discrete device assignment and Discrete Device Assignment - Description and background.

- Hyper-V now supports BitLocker drive encryption for operating system (OS) disks in generation 1 VMs. This protection method replaces virtual Trusted Platform Modules (TPMs), which are only available in generation 2 VMs. To decrypt the disk and start the VM, the Hyper-V host must either be part of an authorized guarded fabric or have the private key from one of the VM's guardians. Key storage requires a version 8 VM. For more information, see Upgrade virtual machine version in Hyper-V on Windows or Windows Server.

- Host resource protection prevents VMs from using too many system resources by tracking excessive levels of activity. When monitoring detects an unusually high activity level in a VM, it throttles the amount of resources the VM consumes. You can enable this feature by running the Set-VMProcessor cmdlet in PowerShell.

- You can now use hot add or remove to add or remove network adapters while the VM is running without downtime in generation 2 VMs running either Linux or Windows OSs. You can also adjust how much memory is assigned to a VM while it's running even if

Dynamic Memory isn't enabled on both generation 1 and 2 VMs running Windows Server 2016 and later or Windows 10 and later.

- Hyper-V Manager now supports the following features:
  - Alternate credentials, which let you use a different set of credentials in Hyper-V Manager when connecting to another Windows Server 2016 or Windows 10 remote host. You can also save these credentials to make signing in easier.
  - You can now manage Hyper-V on machines running Windows Server 2012 R2, Windows Server 2012, Windows 8.1, and Windows 8.
  - Hyper-V Manager now communicates with remote Hyper-V hosts using the WS-MAN protocol, which permits CredSSP, Kerberos, and NTLM authentication. When you use CredSSP to connect to a remote Hyper-V host, you can perform a live migration without enabling constrained delegation in Active Directory. WS-MAN also makes it easier to enable hosts for remote management. WS-MAN connects over port 80, which is open by default.
- Updates to integration services for Windows guests are now distributed through Windows Update. Service providers and private cloud hosts can give tenants who own the VMs control over applying updates. Windows tenants can now upgrade their VMs with all of the latest updates through a single method. For more information about how Linux tenants can use integration services, see Supported Linux and FreeBSD virtual machines for Hyper-V on Windows Server and Windows.

  > **ⓘ Important**
  >
  > Hyper-V for Windows Server 2016 no longer includes the vmguest.iso image file because it's no longer required.

- Linux OSs running on generation 2 VMs can now boot with the Secure Boot option enabled. The OSs that support Secure Boot on Windows Server 2016 hosts include Ubuntu 14.04 and later, SUSE Linux Enterprise Server 12 and later, Red Hat Enterprise Linux 7.0 and later, and CentOS 7.0 and later. Before you boot the VM for the first time, you must configure it to use the Microsoft UEFI Certificate Authority in either Hyper-V Manager, Virtual Machine Manager, or by running the Set-VMFirmware cmdlet in PowerShell.
- Generation 2 VMs and Hyper-V hosts can now use more memory and virtual processors. You can also configure hosts with more memory and virtual processors than previous versions. These changes support scenarios such as running large in-memory databases for online transaction processing (OLTP) and data warehousing (DW) for e-commerce. For

more information, see Windows Server 2016 Hyper-V large-scale VM performance for in-memory transaction processing. To earn more about version compatibility and supported maximum configurations, see Upgrade virtual machine version in Hyper-V on Windows or Windows Server and Plan for Hyper-V scalability in Windows Server.

- The Nested Virtualization feature lets you use a VM as a Hyper-V host and create VMs within the virtualized host. You can use this feature to build development and test environments running at least Windows Server 2016 or Windows 10 with an Intel VT-x capable processor. For more information, see What is Nested Virtualization?.

- You can now set up production checkpoints to comply with support policies for VMs running production workloads. These checkpoints run on backup technology inside the guest device instead of a saved state. Windows VMs use the Volume Snapshot Service (VSS), while Linux VMs flush file system buffers to create checkpoints that are consistent with the file system. You can still use checkpoints based on save states by using standard checkpoints instead. For more information, see Choose between standard or production checkpoints in Hyper-V.

  > **Important**
  >
  > New VMs use production checkpoints as the default.

- You can now resize shared virtual hard disks (`.vhdx` files) for guest clustering without downtime. You can also use guest clusters to protect shared virtual hard disks by using Hyper-V Replica for disaster recovery. You can only use this feature on collections in a guest cluster that you've enabled replication through Windows Management Instrumentation (WMI). For more information, see Msvm_CollectionReplicationService class and Virtual Hard Disk Sharing Overview.

  > **Note**
  >
  > Managing replication of a collection isn't possible via PowerShell cmdlets or using the WMI interface.

- When backing up a single virtual machine, we don't recommend using a VM group or snapshot collection regardless of whether the host is clustered or not. These options are intended for backing up guest clusters that use a shared vhdx. Instead, we recommend taking a snapshot using the Hyper-V WMI provider (V2).

- You can now create shielded Hyper-V VMs that include features to prevent Hyper-V admins on the host or malware from inspecting or tampering with the shielded VM state.

These features also protect against data theft from the shielded VM state. Data and state are encrypted so that Hyper-V admins can't see video output and available disks. You can also restrict the VMs to only run on hosts that a Host Guardian Server determines are healthy and trustworthy. For more information, see Guarded fabric and shielded VMs overview.

> **Note**
>
> Shielded VMs are compatible with Hyper-V Replica. To replicate a shielded virtual machine, you must authorize the host you want to replicate to run that shielded VM.

- The start order priority for clustered virtual machines feature gives you more control over which clustered VMs start or restart first. Deciding start order priority lets you start VMs that provide services before starting VMs that use those services. You can define sets, add VMs to sets, and specify dependencies using PowerShell cmdlets such as New-ClusterGroupSet, Get-ClusterGroupSet, and Add-ClusterGroupSetDependency.

- VM configuration files now use the `.vmcx` file extension format, while runtime state data files use the `.vmrs` file extension format. These new file formats are designed with more efficient reading and writing in mind. The updated formats also decrease the likelihood of data corruption if a storage failure happens.

  > **Important**
  >
  > The `.vmcx` file name extension indicates a binary file. Hyper-V for Windows Server 2016 doesn't support editing `.vmcx` or `.vmrs` files.

- We updated version compatibility with version 5 VMs. These VMs are compatible with both Windows Server 2012 R2 and Windows Server 2016. However, version 5 VMs that are compatible with Windows Server 2019 can only run on Windows Server 2016, not Windows Server 2012 R2. If you move or import a Windows Server 2012 R2 VM to a server running a later version of Windows Server, you must manually update the VM configuration to use features for the later versions of Windows Server. For more information about version compatibility and updated features, see Upgrade virtual machine version in Hyper-V on Windows or Windows Server.

- You can now use virtualization-based security features for generation 2 VMs, such as Device Guard and Credential Guard, to protect your OS against malware exploits. These features are available in VMs running version 8 or later. For more information, see Upgrade virtual machine version in Hyper-V on Windows or Windows Server.

- You can now run cmdlets using Windows PowerShell Direct to configure your VM from the host machine as an alternative to VMConnect or Remote PowerShell. You don't need to meet any networking or firewall requirements or have a special remote management configuration in order to start using it. For more information, see Manage Windows virtual machines with PowerShell Direct.

## Nano Server

Nano Server now includes an updated module for building Nano Server images. This update provides more separation of physical host and guest virtual machine functionality and adds support for different Windows Server editions. For more information, see Install Nano Server.

There are also improvements to the Recovery Console, including separation of inbound and outbound firewall rules and the ability to repair WinRM configuration.

## Shielded Virtual Machines

Windows Server 2016 provides a new Hyper-V-based Shielded Virtual Machine to protect any Generation 2 virtual machine from a compromised fabric. Some of the key features introduced in Windows Server 2016 include:

- A new **Encryption Supported** mode that offers more protections than for an ordinary virtual machine, but less than **Shielded** mode, while still supporting vTPM, disk encryption, Live Migration traffic encryption, and other features, including direct fabric administration conveniences such as virtual machine console connections and PowerShell Direct.

- Full support for converting existing non-shielded Generation 2 virtual machines to shielded virtual machines, including automated disk encryption.

- Hyper-V Virtual Machine Manager can now view the fabrics upon which a shielded virtual is authorized to run, providing a way for the fabric administrator to open a shielded virtual machine's key protector (KP) and view the fabrics it's permitted to run on.

- You can switch Attestation modes on a running Host Guardian Service. Now you can switch on the fly between the less secure but simpler Active Directory-based attestation and TPM-based attestation.

- End-to-end diagnostics tooling based on Windows PowerShell that is able to detect misconfigurations or errors in both guarded Hyper-V hosts and the Host Guardian Service.

- A recovery environment that offers a means to securely troubleshoot and repair shielded virtual machines within the fabric in which they normally run while offering the same level of protection as the shielded virtual machine itself.
- Host Guardian Service support for existing safe Active Directory – you can direct the Host Guardian Service to use an existing Active Directory forest as its Active Directory instead of creating its own Active Directory instance

For more information and instructions for working with shielded virtual machines, see Guarded Fabric and Shielded VMs.

# Identity and Access

New features in Identity improve the ability for organizations to secure Active Directory environments and help them migrate to cloud-only deployments and hybrid deployments, where some applications and services are hosted in the cloud and others are hosted on premises.

## Active Directory Certificate Services

Active Directory Certificate Services (AD CS) in Windows Server 2016 increases support for TPM key attestation: You can now use Smart Card KSP for key attestation, and devices that aren't joined to the domain can now use NDES enrollment to get certificates that can be attested for keys being in a TPM.

### Privileged access management

Privileged access management (PAM) helps mitigate security concerns in Active Directory environments caused by credential theft techniques, such as pass-the-hash, spear phishing, and so on. You can configure this new administrative access solution using Microsoft Identity Manager (MIM), and it introduces the following features:

- The bastion Active Directory forest, provisioned by MIM, has a special PAM trust with an existing forest. Bastion forests are a new type of Active Directory environment that's free of malicious activity due to being isolated from existing forests and only allowing access to privileged accounts.
- New processes in MIM to request administrative privileges, including new workflows for approving requests.
- New shadow security principals, or groups, provisioned in the bastion forest by MIM in response to administrative privilege requests. The shadow security groups have an

attribute that references the SID of an administrative group in an existing forest. This allows the shadow group to access resources in existing forests without changing any access control lists (ACLs).

- An expiring links feature that allows users to temporarily join a shadow group for a specified duration, enabling them to perform administrative tasks. The membership duration is controlled by a time-to-live (TTL) value, which also determines the Kerberos ticket's validity period.

  > **Note**
  >
  > Expiring links are available on all linked attributes. However, only the *member/memberOF* linked attribute relationship between a group and a user comes preconfigured with PAM to use the expiring links feature.

- Built-in Kerberos Domain Controller (KDC) enhancements allow Active Directory domain controllers to restrict Kerberos ticket lifetimes to the lowest possible TTL value when users have multiple limited-time memberships to administrative groups. For example, if you're a member of time-bound group **A**, then when you sign on, the Kerberos ticket-granting ticket (TGT) lifetime is equal to how much time you have left in group **A**. If you also join time-bound group **B**, which has a lower TTL than group **A**, then your TGT lifetime is equal to how much time you have left in group **B**.

- New monitoring capabilities that let you identify which users requested access, what access the administrators granted to them, and what activities they performed while signed in.

To learn more about PAM, see Privileged Access Management for Active Directory Domain Services.

## Microsoft Entra join

Microsoft Entra join enhances identity experiences for enterprise, business, and education customers, as well as including improved capabilities for corporate and personal devices.

- Modern Settings are now available on corporate-owned Windows devices. You no longer need a personal Microsoft account to use core Windows capabilities, and they new run using existing user work accounts to ensure compliance. These services work on PCs joined to an on-premises Windows domain and devices joined to Microsoft Entra. These settings include:

  - Roaming or personalization, accessibility settings, and credentials

    - Back up and restore
    - Access to the Microsoft Store with your work account
    - Live tiles and notifications
- Access organizational resources on mobile devices, such as phones and tablets, that can't be joined to a Windows Domain, whether they're corporate-owned or bring your own device (BYOD).
- Use Single-Sign On (SSO) for Office 365 and other organizational apps, websites, and resources.
- On BYOD devices, add a work account from an on-premises domain or Azure AD to a personally owned device. You can use SSO to access work resources through apps or on the web while remaining compliant with new features such as Conditional Account Control and Device Health attestation.
- Mobile device management (MDM) integration lets you autoenroll devices to your mobile device management (MDM) tool (Microsoft Intune or third-party).
- Set up kiosk mode and shared devices for multiple users in your organization.
- Developer experience lets you build apps that cater to both enterprise and personal contexts with a shared programming stack.
- The imaging option lets you choose between imaging and allowing your users to configure corporate-owned devices directly during the first-run experience.

## Windows Hello For Business

Windows Hello for Business is a key-based authentication approach for organizations and consumers that goes beyond passwords. This form of authentication relies on credentials that are resistant to breaches, theft, and phishing.

The user signs in to the device with a biometric or PIN linked to a certificate or an asymmetrical key pair. The Identity Providers (IDPs) validate the user by mapping the public key of the user to IDLocker and provides sign in information through One Time Password (OTP), by phone, or a different notification mechanism.

For more information, see Windows Hello for Business.

## Deprecation of File Replication Service (FRS) and Windows Server 2003 functional levels

Although File Replication Service (FRS) and the Windows Server 2003 functional levels were deprecated in previous versions of Windows Server, we would like to remind you that AD DS no longer supports Windows Server 2003. You should remove any domain controller that runs Windows Server 2003 from the domain. You should also raise the domain and forest functional level to at least Windows Server 2008.

At the Windows Server 2008 and higher domain functional levels, AD DS uses Distributed File Service (DFS) Replication to replicate SYSVOL folder contents between domain controllers. If you create a new domain at the Windows Server 2008 domain functional level or higher, DFS Replication automatically replicates the SYSVOL folder. If you created the domain at a lower functional level, you must migrate from using FRS to DFS replication for the SYSVOL folder. For more detailed migration steps, see Install, upgrade, or migrate to Windows Server.

For more information, see the following resources:

- Understanding Active Directory Domain Services (AD DS) Functional Levels
- How to raise Active Directory domain and forest functional levels

## Active Directory Federation Services

Active Directory Federation Services (AD FS) in Windows Server 2016 includes new features that enable you to configure AD FS to authenticate users stored in Lightweight Directory Access Protocol (LDAP) directories.

## Web Application Proxy

The latest version of Web Application Proxy focuses on new features that enable publishing and pre-authentication for more applications and improved user experience. Check out the full list of new features that includes preauthentication for rich client apps such as Exchange ActiveSync and wildcard domains for easier publishing of SharePoint apps. For more information, see Web Application Proxy in Windows Server 2016.

# Administration

The Management and Automation area focuses on tool and reference information for IT pros who want to run and manage Windows Server 2016, including Windows PowerShell.

Windows PowerShell 5.1 includes significant new features, including support for developing with classes and new security features that extend its use, improve its usability, and allow you to control and manage Windows-based environments more easily and comprehensively. See New Scenarios and Features in WMF 5.1 for details.

New additions for Windows Server 2016 include: the ability to run PowerShell.exe locally on Nano Server (no longer remote only), new Local Users & Groups cmdlets to replace the GUI, added PowerShell debugging support, and added support in Nano Server for security logging & transcription and JEA.

Here are some other new administration features:

## PowerShell Desired State Configuration (DSC) in Windows Management Framework (WMF) 5

Windows Management Framework 5 includes updates to Windows PowerShell Desired State Configuration (DSC), Windows Remote Management (WinRM), and Windows Management Instrumentation (WMI).

For more info about testing the DSC features of Windows Management Framework 5, see the series of blog posts discussed in Validate features of PowerShell DSC. To download, see Windows Management Framework 5.1.

## PackageManagement unified package management for software discovery, installation, and inventory

Windows Server 2016 and Windows 10 includes a new PackageManagement feature (formerly called OneGet) that enables IT Professionals or DevOps to automate software discovery, installation, and inventory (SDII), locally or remotely, no matter what the installer technology is and where the software is located.

For more info, see https://github.com/OneGet/oneget/wiki.

## PowerShell enhancements to assist digital forensics and help reduce security breaches

To help the team responsible for investigating compromised systems - sometimes known as the "blue team" - we've added additional PowerShell logging and other digital forensics functionality, and we've added functionality to help reduce vulnerabilities in scripts, such as constrained PowerShell, and secure CodeGeneration APIs.

For more info, see the PowerShell ♥ the Blue Team blog post.

# Networking

The Networking area addresses networking products and features for the IT professional to design, deploy, and maintain Windows Server 2016.

## Software-Defined Networking

Software-Defined Networking (SDN) is a new Software Defined Datacenter (SDDC) solution that includes the following features:

- Network Controller, which lets you automate the configuration of network infrastructure instead of performing manual configuration of network devices and services. Network Controller uses Representational State Transfer (REST) on its northbound interface with JavaScript Object Notation (JSON) payloads. The Network Controller southbound interface uses Open vSwitch Database Management Protocol (OVSDB).
- New features for Hyper-V:
    - Hyper-V Virtual Switch, which lets you create distributed switching and routing, and a policy enforcement layer that is aligned and compatible with Microsoft Azure. To learn more, see Hyper-V Virtual Switch.
    - Remote direct memory access (RDMA) and switch-embedded teaming (SET) for when you create virtual switches. You can set up RDMA on network adapters bound to a Hyper-V virtual switch regardless of whether you're already using SET. SET can give your virtual switches similar capabilities as NIC teaming. For more information, see Host network requirements for Azure Local.
    - Virtual machine multi-queues (VMMQs) improve on VMQ through put by allocating multiple hardware queues per VM. The default queue becomes a set of queues for a VM and spreads traffic between the queues.
    - Quality of service (QoS) for software-defined networks manages the default class of traffic through the virtual switch within the default class bandwidth.
- Network Function Virtualization (NFV), which lets you mirror or route network functions performed by hardware appliances to virtual appliances, such as load balancers, firewalls, routers, switches, and so on. You can also deploy and manage your entire SDN stack using System Center Virtual Machine Manager. You can manage Windows Server container networking with Docker and associate SDN policies with both virtual machines and containers.
- A Datacenter Firewall that provides granular access control lists (ACLs), enabling you to apply firewall policies at the VM interface level or the subnet level. To learn more, see What is Datacenter Firewall?.

- RAS Gateway, which lets you route traffic between virtual networks and physical networks, including site-to-site VPN connections from your cloud datacenter to your tenants' remote sites. Border Gateway Protocol (BGP) lets you deploy and provide dynamic routing between networks for all gateway scenarios, including Internet Key Exchange version 2 (IKEv2) site-to-site virtual private networks (VPNs), Layer 3 (L3) VPNs, and Generic Routing Encapsulation (GRE) gateways. Gateways now also support gateway pools and M+N redundancy. To learn more, see What is Remote Access Service (RAS) Gateway for Software Defined Networking?.

- Software Load Balancer (SLB) and Network Address Translation (NAT) enhances throughput by supporting Direct Server Return. This allows the return network traffic to bypass the Load Balancing multiplexer, and can be achieved using a north-south and east-west layer 4 load balancer and NAT. To learn more, see What is Software Load Balancer (SLB) for SDN? and Network Function Virtualization.

- Flexible encapsulation technologies that operate at the data plane and support both Virtual Extensible LAN (VxLAN) and Network Virtualization Generic Routing Encapsulation (NVGRE).

For more information, see Plan a Software Defined Network Infrastructure.

## Cloud scale fundamentals

Windows Server 2016 includes the following cloud scale fundamentals:

- Converged Network Interface Card (NIC), which lets you use a single network adapter for management, Remote Direct Memory Access (RDMA)-enabled storage, and tenant traffic. A Converged NIC reduces cost for each server in your datacenter because it requires fewer network adapters to manage different types of traffic per server.

- Packet Direct provides a high network traffic throughput and low-latency packet processing infrastructure.

- Switch Embedded Teaming (SET) is a NIC Teaming solution integrated into the Hyper-V Virtual Switch. SET allows the teaming of up to eight physical NICs into a single SET team, which improves availability and provides failover. In Windows Server 2016, you can create SET teams that are restricted to using Server Message Block (SMB) and RDMA. You can also use SET teams to distribute network traffic for Hyper-V Network Virtualization. For more information, see Host network requirements for Azure Local.

## TCP performance improvements

The default Initial Congestion Window (ICW) have increased from 4 to 10 and TCP Fast Open (TFO) is implemented. TFO reduces the amount of time required to establish a TCP connection and the increased ICW allows larger objects to be transferred in the initial burst. This combination can significantly reduce the time required to transfer an Internet object between the client and the cloud.

In order to improve TCP behavior when recovering from packet loss, TCP Tail Loss Probe (TLP) and Recent Acknowledgment (RACK) are implemented. TLP helps convert Retransmit TimeOuts (RTOs) to Fast Recoveries and RACK reduces the time required for Fast Recovery to retransmit a lost packet.

## Dynamic Host Configuration Protocol (DHCP)

The Dynamic Host Configuration Protocol (DHCP) has the following changes in Windows Server 2016:

- As of Windows 10, version 2004, when you're running a Windows client and connect to the internet using a tethered Android device, the connections are now labeled as metered. The traditional Client Vendor Name that appeared as MSFT 5.0 on certain Windows devices is now MSFT 5.0 XBOX.

- As of Windows 10, version 1803, The DHCP client can now read in and apply option 119, the Domain Search Option, from the DHCP server your system connects to. The Domain Search Option also provides Domain Name Services (DNS) suffixes for DNS lookups of short names. For more information, see RFC 3397.

- DHCP now supports option 82 (sub-option 5). You can use this option to allow DHCP proxy clients and relay agents to request an IP address for a specific subnet. If you're using a DHCP relay agent configured with DHCP option 82 (sub-option 5), the relay agent can request an IP address lease for DHCP clients from a specific IP address range. For more information, see DHCP Subnet Selection Options.

- New logging events for scenarios where DNS record registrations fail on the DNS server. For more information, see DHCP Logging Events for DNS Registrations.

- The DHCP Server role no longer supports Network Access Protection (NAP). DHCP servers don't enforce NAP policies, and DHCP scopes can't be NAP-enabled. DHCP client computers that are also NAP clients send a statement of health (SoH) with the DHCP request. If the DHCP server is running Windows Server 2016, these requests are processed as if no SoH is present. The DHCP server grants a normal DHCP lease to the client. If servers running Windows Server 2016 are Remote Authentication Dial-In User Service (RADIUS) proxies that forward authentication requests to a Network Policy Server (NPS) that supports NAP, the NPS evaluates these clients as non-NAP capable, causing NAP

processing to fail. For more information about NAP and NAP deprecation, see Features Removed or Deprecated in Windows Server 2012 R2.

## GRE tunneling

RAS Gateway now supports high availability Generic Routing Encapsulation (GRE) tunnels for site-to-site connections and M+N redundancy of gateways. GRE is a lightweight tunneling protocol that encapsulates a wide variety of network layer protocols inside virtual point-to-point links over an Internet Protocol internetwork. For more information, see GRE Tunneling in Windows Server 2016.

## IP Address Management (IPAM)

IPAM has the following updates:

- Enhanced IP address management. IPAM capabilities are improved for scenarios such as handling IPv4 /32 and IPv6 /128 subnets and finding free IP address subnets and ranges in an IP address block.
- You can now run the `Find-IpamFreeSubnet` cmdlet to find available subnets for allocation. This function doesn't allocate the subnets and only reports their availability. However, you can pipe the cmdlet output to the `Add-IpamSubnet` cmdlet to create a subnet. For more information, see Find-IpamFreeSubnet.
- You can now run the `Find-IpamFreeRange` cmdlet to find available IP address ranges within an IP block, prefix length, and number of requested subnets. This cmdlet doesn't allocate the IP address range, only reports their availability. However, you can pipe the output into the `AddIpamRange` cmdlet to create the range. For more information, see Find-IpamFreeRange.
- Enhanced DNS service management:
    - DNS resource records collection for non-DNSSEC DNS servers.
    - Configuring properties and operations on all types of non-DNSSEC Resource Records.
    - DNS zone management for both domain-joined Active Directory-integrated and file-backed DNS servers. You can manage all types of DNS zones, including Primary, Secondary, and Stub zones.
    - Trigger tasks on Secondary and Stub zones regardless of whether it's forward or reverse lookup zones.

    - Role-based access control for supported DNS configurations for records and zones.
    - Conditional forwarders
- Integrated DNS, DHCP, and IP address (DDI) management. You can now view all DNS resource records associated with an IP address in the IP Address Inventory. You can also automatically keep pointer (PTR) records of IP addresses and manage IP address lifecycles for both DNS and DHCP operations.
- Multiple Active Directory Forest support. You can use IPAM to manage the DNS and DHCP servers of multiple Active Directory forests when there's a two-way trust relationship between the forest where you installed IPAM and each of the remote forests. For more information, see Manage Resources in Multiple Active Directory Forests.
- The Purge Utilization Data feature lets you reduce IPAM database size by deleting old IP utilization data. Just specify a date and IPAM deletes all database entries older than or equal to the date you entered. For more information, see Purge Utilization Data.
- You can now use Role Based Access Control (RBAC) to define access scopes for IPAM objects in PowerShell. For more information, see Manage Role Based Access Control with Windows PowerShell and IP Address Management (IPAM) Server Cmdlets in Windows PowerShell.

For more information, see Manage IPAM.

# Security and Assurance

The Security and Assurance area Includes security solutions and features for the IT professional to deploy in your data center and cloud environment. For information about security in Windows Server 2016 generally, see Security and Assurance.

## Just Enough Administration (JEA)

JEA in Windows Server 2016 is security technology that enables delegated administration for anything that can be managed with Windows PowerShell. Capabilities include support for running under a network identity, connecting over PowerShell Direct, securely copying files to or from JEA endpoints, and configuring the PowerShell console to launch in a JEA context by default. For more information, see JEA on GitHub.

## Credential Guard

Credential Guard uses virtualization-based security to isolate secrets so that only privileged system software can access them. For more information, see Protect derived domain credentials with Credential Guard.

Credential Guard for Windows Server 2016 includes the following updates for signed-in user sessions:

- Kerberos and New Technology LAN Manager (NTLM) use virtualization-based security to protect Kerberos and NTLM secrets for signed-in user sessions.
- Credential Manager protects saved domain credentials using virtualization-based security. Signed-in credentials and saved domain credentials don't pass to remote hosts using Remote Desktop.
- You can enable Credential Guard without a Unified Extensible Firmware Interface (UEFI) lock.

## Remote Credential Guard

Credential Guard includes support for RDP sessions so that the user credentials remain on the client side and aren't exposed on the server side. This also provides Single Sign On for Remote Desktop. For more information, see Protect derived domain credentials with Windows Defender Credential Guard.

Remote Credential Guard for Windows Server 2016 includes the following updates for signed-in users:

- Remote Credential Guard keeps Kerberos and NTLM secrets for signed-in user credentials on the client device. Any authentication requests from the remote host for assessing network resources as the user require the client device to use the secrets.
- Remote Credential Guard protects supplied user credentials when using Remote Desktop.

## Domain protections

Domain protections now require an Active Directory domain.

## PKInit Freshness extension support

Kerberos clients now attempt the PKInit freshness extension for public key based sign-ons.

KDCs now support the PKInit freshness extension. However, they don't offer the PKInit freshness extension by default.

For more information, see Kerberos client and KDC support for RFC 8070 PKInit freshness extension.

## Rolling public key only user's NTLM secrets

Starting with the Windows Server 2016 domain functional level (DFL), DCs now support rolling the NTLM secrets of a public-key-only user. This feature is unavailable in lower domain functioning levels (DFLs).

> ⚠ **Warning**
>
> Adding a DC enabled before the November 8, 2016 update to a domain that supports rolling NTLM secrets can cause the DC to crash.

For new domains, this feature is enabled by default. For existing domains, you must configure it in the Active Directory Administrative Center.

From the Active Directory Administrative Center, right-click on the domain in the left pane and select **Properties**. Select the checkbox **Enable rolling of expiring NTLM secrets during sign on for users who are required to use Windows Hello for Business or smart card for interactive logon**. After that, select **OK** to apply this change.

## Allowing network NTLM when user is restricted to specific domain-joined devices

DCs can now support allowing network NTLM when a user is restricted to specific domain-joined devices in the Windows Server 2016 DFL and higher. This feature is unavailable in DFLs running an earlier OS than Windows Server 2016.

To configure this setting, in the authentication policy, select **Allow NTLM network authentication when the user is restricted to selected devices**.

For more information, see Authentication policies and authentication policy silos.

## Device Guard (Code Integrity)

Device Guard provides kernel mode code integrity (KMCI) and user mode code integrity (UMCI) by creating policies that specify what code can run on the server. See Introduction to Windows Defender Device Guard: virtualization-based security and code integrity policies.

## Windows Defender

Windows Defender Overview for Windows Server 2016. Windows Server Anti-malware is installed and enabled by default in Windows Server 2016, but the user interface for Windows Server Anti-malware isn't installed. However, Windows Server Anti-malware updates anti-malware definitions and protects the computer without the user interface. If you need the user interface for Windows Server Anti-malware, you can install it after the OS installation by using the Add Roles and Features Wizard.

## Control Flow Guard

Control Flow Guard (CFG) is a platform security feature that was created to combat memory corruption vulnerabilities. See Control Flow Guard for more information.

# Storage

Storage in Windows Server 2016 includes new features and enhancements for software-defined storage and traditional file servers.

## Storage Spaces Direct

Storage Spaces Direct enables building highly available and scalable storage using servers with local storage. It simplifies deploying and managing software-defined storage systems and lets you use new classes of disk devices, such as SATA SSDs and NVMe disk devices, that previously weren't available with clustered Storage Spaces with shared disks.

For more information, see Storage Spaces Direct.

## Storage Replica

Storage Replica enables storage-agnostic, block-level, synchronous replication between servers or clusters for disaster recovery, and lets you stretch a failover cluster between sites. Synchronous replication enables mirroring of data in physical sites with crash-consistent volumes to ensure zero data loss at the file-system level. Asynchronous replication allows site extension beyond metropolitan ranges with the possibility of data loss.

For more info, see Storage Replica.

## Storage Quality of Service (QoS)

You can now use storage quality of service (QoS) to centrally monitor end-to-end storage performance and create management policies using Hyper-V and CSV clusters in Windows Server 2016.

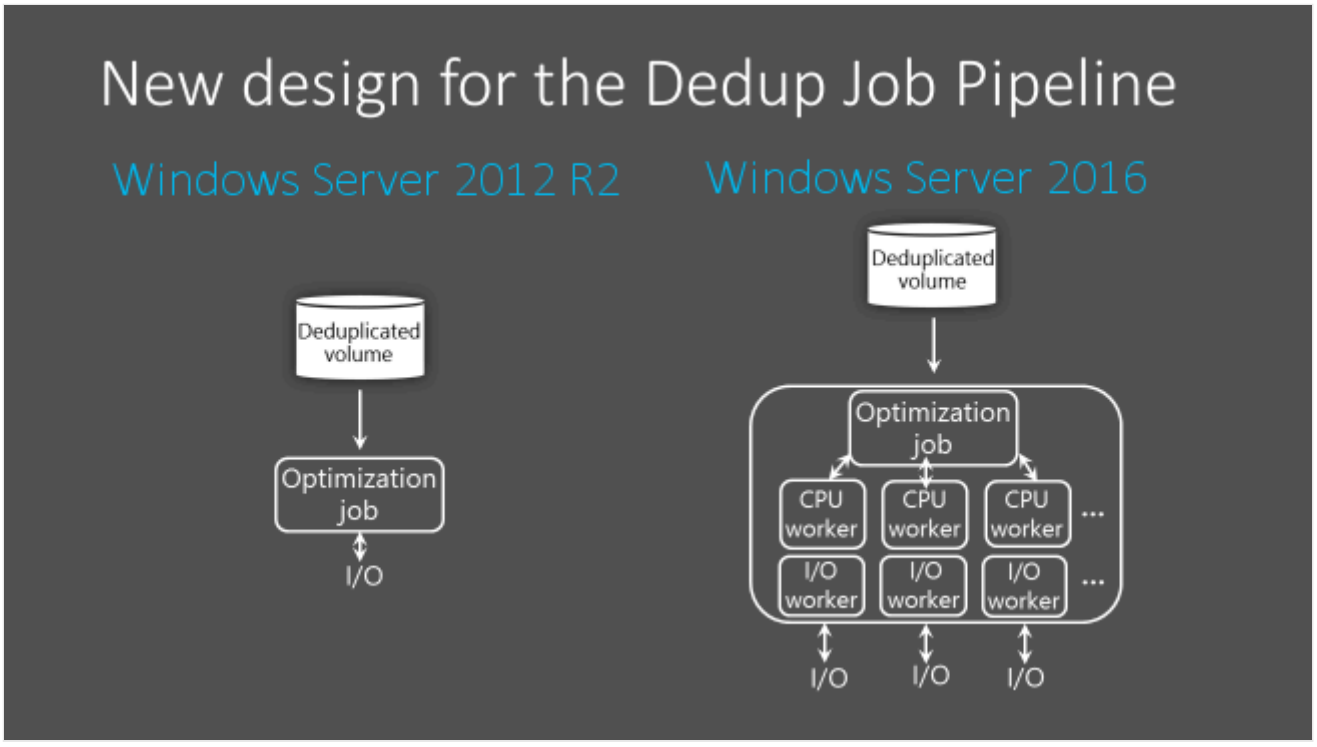
For more info, see Storage Quality of Service.

## Data deduplication

Windows Server 2016 includes the following new features for data deduplication.

### Support for large volumes

Starting with Windows Server 2016, the Data Deduplication Job pipeline can now run multiple threads in parallel using many I/O queues for each volume. This change increases performance to levels previously only possible by dividing data into several smaller volumes. These optimizations apply to all Data Deduplication Jobs, not just the Optimization Job. The following diagram demonstrates how the pipeline changed between versions of Windows Server.




Because of these performance improvements, on Windows Server 2016, Data Deduplication has high performance on volumes up to 64 TB.

### Large file support

As of Windows Server 2016, Data Deduplication uses stream map structures and other improvements to increase optimization throughput and access performance. The Deduplication Processing Pipeline can also resume optimization after failover scenarios instead of starting over from the beginning. This change improves the performance of files up to 1 TB, allowing administrators to apply deduplication savings to a larger range of workloads, such as large files associated with backup workloads.

### Support for Nano Server

Nano Server is a headless deployment option in Windows Server 2016 that requires a far smaller system resource footprint, starts up faster, and requires fewer updates and restarts than the Windows Server Core deployment option. Nano Server also fully supports Data Deduplication. For more information about Nano Server, see Container Base Images.

### Simplified configuration for Virtualized Backup Applications

Starting with Windows Server 2016, Data Deduplication for Virtualized Backup Applications scenarios is vastly simplified. This scenario is now a predefined Usage Type option. You no longer need to manually tune the deduplication settings, just enable Deduplication for a volume just like you would General Purpose File Server and Virtual Desktop Infrastructure (VDI).

### Cluster OS Rolling Upgrade support

Windows Server Failover Clusters running Data Deduplication can have a mix of nodes that run the Windows Server 2012 R2 and Windows Server 2016 versions of Data Deduplication. This mixed-mode cluster feature gives full data access to all deduplicated volumes during cluster rolling upgrades. You can now gradually roll out later versions of Data Deduplications on clusters running earlier versions of Windows Server without any downtime.

You can also now use rolling upgrades on Hyper-V. With Rolling Hyper-V Cluster upgrade, you can now add a node running Windows Server 2019 or Windows Server 2016 to a Hyper-V Cluster with nodes running Windows Server 2012 R2. After you add the node running the later version of Windows Server, you can upgrade the rest of the cluster without downtime. The cluster runs at a Windows Server 2012 R2 feature level until you upgrade all nodes in the cluster and run the Update-ClusterFunctionalLevel in PowerShell to update the cluster functioning level. For more detailed instructions for how the rolling upgrade process works, see Cluster Operating System Rolling Upgrade.

> **Note**
>
> Hyper-V on Windows 10 doesn't support failover clustering.

## SMB hardening improvements for SYSVOL and NETLOGON connections

In Windows 10 and Windows Server 2016, client connections to the Active Directory Domain Services used SYSVOL and NETLOGON shares on domain controllers by default. Now these

connections require SMB signing and mutual authentication using services such as Kerberos. If SMB signing and mutual authentication are unavailable, a Windows 10 or Windows Server 2016 computer doesn't process domain-based Group Policy and scripts. This change protects devices from adversary-in-the-middle attacks.

> **Note**
>
> The registry values for these settings aren't present by default, but the hardening rules still apply until you override them by editing Group Policy or other registry values.

For more information on these security improvements, see MS15-011: Vulnerability in Group Policy and MS15-011 & MS15-014: Hardening Group Policy.

## Work Folders

Windows Server 2016 features improved change notification when the Work Folders server is running Windows Server 2016 and the Work Folders client is Windows 10. When file changes sync to the Work Folders server, the server now immediately notifies Windows 10 clients, then syncs the file changes.

## ReFS

The next iteration of ReFS provides support for large-scale storage deployments with diverse workloads, delivering reliability, resiliency, and scalability for your data.

ReFS introduces the following improvements:

- New storage tier functionality, delivering faster performance and increased storage capacity, including:
  - Multiple resiliency types on the same virtual disk using mirroring in the performance tier and parity in the capacity tier.
  - Increased responsiveness to drifting working sets.
- Introduces block cloning to improve performance of VM operations, such as `.vhdx` checkpoint merge operations.
- A new ReFS scan tool that can help you recover leaked storage and salvage data from critical corruptions.

# Failover Clustering

Windows Server 2016 includes many new features and enhancements for multiple servers that are grouped together into a single fault-tolerant cluster using the Failover Clustering feature.

## Cluster Operating System Rolling Upgrade

Cluster Operating System Rolling Upgrade enables an administrator to upgrade the OS of the cluster nodes from Windows Server 2012 R2 to Windows Server 2016 without stopping the Hyper-V or Scale-Out File Server workloads. You can use this feature to avoid downtime penalties against Service Level Agreements (SLAs).

For more information, see Cluster Operating System Rolling Upgrade.

## Cloud witness

Cloud witness is a new type of Failover Cluster quorum witness in Windows Server 2016 that leverages Microsoft Azure as the arbitration point. The cloud witness, like any other quorum witness, gets a vote and can participate in quorum calculations. You can configure cloud witness as a quorum witness using the Configure a Cluster Quorum wizard.

For more information, see Deploy a quorum witness.

## Virtual Machine resiliency

Windows Server 2016 includes increased virtual machine (VM) compute resiliency to help reduce intra-cluster communication issues in your compute cluster. This increased resiliency includes the following updates:

- You can now configure the following options to define how the VMs should behave during transient failures:
    - **Resiliency Level** defines how your deployment should handle transient failures.
    - **Resiliency Period** defines how long all VMs are allowed to run isolated.
- Unhealthy nodes are quarantined and are no longer allowed to join the cluster. This feature prevents unhealthy nodes from negatively affecting other nodes and the overall cluster.

For more information about compute resiliency features, see Virtual Machine Compute Resiliency in Windows Server 2016.

Windows Server 2016 VMs also include new storage resiliency features for handling transient storage failures. Improved resiliency helps preserve tenant VM session states if a storage

disruption occurs. When a VM disconnects from its underlying storage, it pauses and waits for storage to recover. While paused, the VM retains the context of applications that were running in it at the time of the storage failure. When the connection between the VM and storage is restored, the VM returns to its running state. As a result, the tenant machine's session state is retained on recovery.

The new storage resiliency features also apply to guest clusters.

## Diagnostic improvements

To help diagnose issues with failover clusters, Windows Server 2016 includes:

- Several enhancements to cluster log files, such as Time Zone Information and DiagnosticVerbose log, make it easier to troubleshoot failover clustering issues. For more information, see Windows Server 2016 Failover Cluster Troubleshooting Enhancements - Cluster Log.
- A new type of active memory dump filters out most memory pages allocated to VMs, making the memory.dmp file smaller and easier to save or copy. For more information, see Windows Server 2016 Failover Cluster Troubleshooting Enhancements - Active Dump.

## Site-aware failover clusters

Windows Server 2016 includes site-aware failover clusters that enable group nodes in stretched clusters based on their physical location, or site. Cluster site-awareness enhances key operations during the cluster lifecycle, such as failover behavior, placement policies, heartbeat between the nodes, and quorum behavior. For more information, see Site-aware Failover Clusters in Windows Server 2016.

## Workgroup and Multi-domain clusters

In Windows Server 2012 R2 and earlier, a cluster can only be created between member nodes joined to the same domain. Windows Server 2016 breaks down these barriers and introduces the ability to create a Failover Cluster without Active Directory dependencies. You can now create failover clusters in the following configurations:

- Single-domain Clusters, which have all their nodes joined to the same domain.
- Multi-domain Clusters, which have nodes that are members of different domains.
- Workgroup Clusters, which have nodes that are member servers or workgroups that aren't domain-joined.

For more information, see Workgroup and Multi-domain clusters in Windows Server 2016

## Virtual Machine Load Balancing

Virtual Machine Load Balancing is a new feature in Failover Clustering that seamlessly load balances VMs across the nodes in a cluster. The feature identifies overcommitted nodes based on VM memory and CPU utilization on the node. It then live migrates the VMs from the overcommitted node to nodes with available bandwidth. You can adjust how aggressively the feature balances nodes to ensure optimal cluster performance and utilization. Load Balancing is enabled by default in Windows Server 2016 Technical Preview. However, Load Balancing is disabled when SCVMM Dynamic Optimization is enabled.

## Virtual Machine Start Order

Virtual machine Start Order is a new feature in Failover Clustering that introduces start order orchestration for VMs and other groups in a cluster. You can now group VMs into tiers, then create start order dependencies between different tiers. These dependencies ensure that the most important VMs, such as Domain Controllers or Utility VMs, start up first. VMs on lower-priority tiers don't start until after the VMs that they have a dependency on startup.

## Simplified SMB Multichannel and Multi-NIC Cluster Networks

Failover cluster networks are no longer limited to a single network interface card (NIC) per subnet or network. With Simplified Server Message Block (SMB) Multichannel and Multi-NIC Cluster Networks, network configuration is automatic and every NIC on the subnet can be used for cluster and workload traffic. This enhancement allows customers to maximize network throughput for Hyper-V, SQL Server Failover Cluster Instance, and other SMB workloads.

For more information, see Simplified SMB Multichannel and Multi-NIC Cluster Networks.

# Application development

## Internet Information Services (IIS) 10.0

New features provided by the IIS 10.0 web server in Windows Server 2016 include:

- Support for the HTTP/2 protocol in the Networking stack and integrated with IIS 10.0, allowing IIS 10.0 websites to automatically serve HTTP/2 requests for supported configurations. This allows numerous enhancements over HTTP/1.1 such as more efficient reuse of connections and decreased latency, improving load times for web pages.

- Ability to run and manage IIS 10.0 in Nano Server. See IIS on Nano Server.
- Support for Wildcard Host Headers, enabling administrators to set up a web server for a domain and then have the web server serve requests for any subdomain.
- A new PowerShell module (IISAdministration) for managing IIS.

For more information, see IIS.

## Distributed Transaction Coordinator (MSDTC)

Three new features are added in Microsoft Windows 10 and Windows Server 2016:

- A new interface for Resource Manager Rejoin can be used by a resource manager to determine the outcome of an in-doubt transaction after a database restarts due to an error. See IResourceManagerRejoinable::Rejoin for details.

- The DSN name limit is enlarged from 256 bytes to 3,072 bytes. See IDtcToXaHelperFactory::Create, IDtcToXaHelperSinglePipe::XARMCreate, or IDtcToXaMapper::RequestNewResourceManager for details.

- Improved tracing allowing you to set a registry key to include an image file path in the Tracelog file name so you can tell which Tracelog file to check. See How to enable diagnostic tracing for MS DTC on a Windows-based computer for details on configuring tracing for MSDTC.

# DNS Server

Windows Server 2016 contains the following updates for Domain Name System (DNS) Server.

## DNS policies

You can configure DNS policies to specify how a DNS server responds to DNS queries. You can configure DNS responses based on client IP address, time of day, and several other parameters. DNS policies can enable location-aware DNS, traffic management, load balancing, split-brain DNS, and other scenarios. For more information, see the DNS Policy Scenario Guide.

## RRL

You can enable Response Rate Limiting (RRL) on your DNS servers to prevent malicious systems from using your DNS servers to initiate a Distributed Denial of Service (DDoS) attack on a DNS client. RRL prevents your DNS server from responding to too many requests at once,

which protects it during scenarios when a botnet sends multiple requests at once to try to disrupt server operations.

## DANE support

You can use DNS-based Authentication of Named Entities (DANE) support (RFC 6394 and RFC 6698) to specify which certificate authority your DNS clients should expect certificates from for domain names hosted in your DNS server. This prevents a form of man-in-the-middle attack where a malicious actor corrupts a DNS cache and points a DNS name to their own IP address.

## Unknown record support

You can add records that the DNS server doesn't explicitly support by using the unknown record functionality. A record is unknown when the DNS server doesn't recognize its RDATA format. Windows Server 2016 supports unknown record types (RFC 3597), so you can add unknown records to Windows DNS server zones in binary on-wire format. The windows caching resolver can already process unknown record types. Windows DNS server doesn't perform record-specific processing for unknown records, but can send them in response to queries it receives.

## IPv6 root hints

Windows DNS server now includes IPv6 root hints published by the Internet Assigned Numbers Authority (IANA). Support for IPv6 root hints lets you make internet queries that use the IPv6 root servers to perform name resolutions.

## Windows PowerShell support

Windows Server 2016 includes new commands you can use to configure DNS in PowerShell. For more information, see Windows Server 2016 DnsServer module and Windows Server 2016 DnsClient module.

## Nano Server support for file-based DNS

You can deploy DNS servers in Windows Server 2016 on a Nano Server image. This deployment option is available if you're using file-based DNS. By running DNS server on a Nano Server image, you can run your DNS servers with reduced footprint, quick boot up, and minimal patching.

> ⓘ Note
>
> Active Directory integrated DNS isn't supported on Nano Server.

# DNS client

The DNS client service now offers enhanced support for computers with more than one network interface.

Multi-homed computers can also use DNS client service binding to improve server resolution:

- When you use a DNS server configured on a specific interface to resolve a DNS query, the DNS client binds to the interface before sending the query. This binding lets the DNS client specify the interface where name resolution should take place, optimizing communications between applications and DNS client over the network interface.
- If the DNS server you're using was designated by a Group Policy setting from the Name Resolution Policy Table (NRPT), the DNS client service doesn't bind to the specified interface.

> ⓘ Note
>
> Changes to the DNS Client service in Windows 10 are also present in computers running Windows Server 2016 and later.

# Remote Desktop Services

Remote Desktop Services (RDS) made the following changes for Windows Server 2016.

## App compatibility

RDS and Windows Server 2016 are compatible with many Windows 10 applications, creating a user experience that's almost identical to a physical desktop.

## Azure SQL Database

The Remote Desktop (RD) Connection Broker can now store all deployment information, such as connection states and user-host mappings, in a shared Azure Structured Query Language (SQL) Database. This feature lets you use a highly available environment without having to use

an SQL Server Always On Availability Group. For more information, see Use Azure SQL DB for your Remote Desktop Connection Broker high availability environment.

## Graphical improvements

Discrete Device Assignment for Hyper-V lets you map graphics processing units (GPUs) on a host machine directly to a virtual machine (VM). Any applications on the VM that need more GPUs than the VM can provide can use the mapped GPU instead. We also improved the RemoteFX vGPU, including support for OpenGL 4.4, OpenCL 1.1, 4K resolution, and Windows Server VMs. For more information, see Discrete Device Assignment.

## RD Connection Broker improvements

We improved how the RD Connection Broker handles connection during logon storms, which are periods of high sign in requests from users. The RD Connection Broker can now handle over 10,000 concurrent sign in requests! Maintenance improvements also make it easier for you to perform maintenance on your deployment by being able to quickly add servers back into the environment once they're ready to go back online. For more information, see Improved Remote Desktop Connection Broker Performance.

## RDP 10 protocol changes

Remote Desktop Protocol (RDP) 10 now uses the H.264/AVC 444 codec, which optimizes across both video and text. This release also includes pen remoting support. These new capabilities allow your remote session to feel more like a local session. For more information, see RDP 10 AVC/H.264 improvements in Windows 10 and Windows Server 2016.

## Personal session desktops

Personal session desktops is a new feature that lets you host your own personal desktop in the cloud. Administrative privileges and dedicated session hosts removes the complexity of hosting environments where users want to manage a remote desktop like a local desktop. For more information, see Personal Session Desktops.

# Kerberos authentication

Windows Server 2016 includes the following updates for Kerberos authentication.

## KDC support for Public Key Trust-based client authentication

Key Distribution Centers (KDCs) now support public key mapping. If you provision a public key for an account, the KDC supports Kerberos PKInit explicitly using that key. Because there's no certificate validation, Kerberos supports self-signed certificates but doesn't support authentication mechanism assurance.

Accounts you've configured to use Key Trust only uses Key Trust regardless of how you configured the UseSubjectAltName setting.

## Kerberos client and KDC support for RFC 8070 PKInit Freshness Extension

Starting with Windows 10, version 1607 and Windows Server 2016, Kerberos clients can use the RFC 8070 PKInit freshness extension for public key-based sign-ons. KDCs have the PKInit freshness extension disabled by default, so to enable it you must configure the KDC support for PKInit Freshness Extension KDC administrative template policy on all DCs in your domain.

The policy has the following settings available when your domain is in the Windows Server 2016 domain functional level (DFL):

- **Disabled**: The KDC never offers the PKInit Freshness Extension and accepts valid authentication requests without checking for freshness. Users don't receive the fresh public key identity SID.
- **Supported**: Kerberos supports PKInit Freshness Extension on request. Kerberos clients successfully authenticating with the PKInit Freshness Extension receive the fresh public key identity SID.
- **Required**: PKInit Freshness Extension is required for successful authentication. Kerberos clients that don't support the PKInit Freshness Extension will always fail when using public key credentials.

## Domain-joined device support for authentication using public key

If a domain-joined device can register its bound public key with a Windows Server 2016 domain controller (DC), then the device can authenticate with the public key using Kerberos PKInit authentication to a Windows Server 2016 DC.

Domain-joined devices with bound public keys registered with a Windows Server 2016 domain controller can now authenticate to a Windows Server 2016 domain controller using Kerberos Public Key Cryptography for Initial Authentication (PKInit) protocols. To learn more, see Domain-joined Device Public Key Authentication.

Key Distribution Centers (KDCs) now support authentication using Kerberos key trust.

For more information, see KDC support for Key Trust account mapping.

## Kerberos clients allow IPv4 and IPv6 address host names in Service Principal Names (SPNs)

Starting with Windows 10 version 1507 and Windows Server 2016, you can configure Kerberos clients to support IPv4 and IPv6 host names in SPNs. For more information, see Configuring Kerberos for IP Addresses.

To configure support for IP address host names in SPNs, create a TryIPSPN entry. This entry doesn't exist in the registry by default. You should place this entry on the following path:

```
text

HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System\Kerberos\Parameters
```

After creating the entry, change its DWORD value to 1. If this value isn't configured, Kerberos won't attempt IP address host names.

Kerberos authentication only succeeds if the SPN is registered in Active Directory.

## KDC support for Key Trust account mapping

Domain controllers now support Key Trust account mapping and fallback to existing AltSecID and User Principal Name (UPN) in the SAN behavior. You can configure the UseSubjectAltName variable to the following settings:

- Setting the variable to 0 makes explicit mapping required. Users must use either a Key Trust or set an ExplicitAltSecID variable.
- Setting the variable to 1, which is the default value, allows implicit mapping.
    - If you configure a Key Trust for an account in Windows Server 2016 or later, then KDC uses the KeyTrust for mapping.
    - If there's no UPN in the SAN, KDC attempts to use the AltSecID for mapping.
    - If there's a UPN in the SAN, KDC attempts to use the UPN for mapping.

# Active Directory Federation Services (AD FS)

AD FS for Windows Server 2016 contains the following updates.

## Sign in with Microsoft Entra multifactor authentication

AD FS 2016 builds upon the multifactor authentication (MFA) capabilities of AD FS in Windows Server 2012 R2. You can now allow sign-on that only requires a Microsoft Entra multifactor authentication code instead of a username or password.

- When you configure Microsoft Entra multifactor authentication as the primary authentication method, AD FS prompts the user for their username and the one-time password (OTP) code from the Azure Authenticator app.
- When you configure Microsoft Entra multifactor authentication as the secondary or extra authentication method, the user provides primary authentication credentials. Users can sign in by using Windows Integrated Authentication, which can request their username and password, smart card, or a user or device certificate. Next, the user sees a prompt for their secondary credentials, such as text, voice, or OTP-based Microsoft Entra multifactor authentication sign-in.
- The new built-in Microsoft Entra multifactor authentication adapter offers simpler setup and configuration for Microsoft Entra multifactor authentication with AD FS.
- Organizations can use Microsoft Entra multifactor authentication without needing an on-premises Microsoft Entra multifactor authentication server.
- You can configure Microsoft Entra multifactor authentication for intranet, extranet, or as part of any access control policy.

For more information about Microsoft Entra multifactor authentication with AD FS, see Configure AD FS 2016 and Microsoft Entra multifactor authentication.

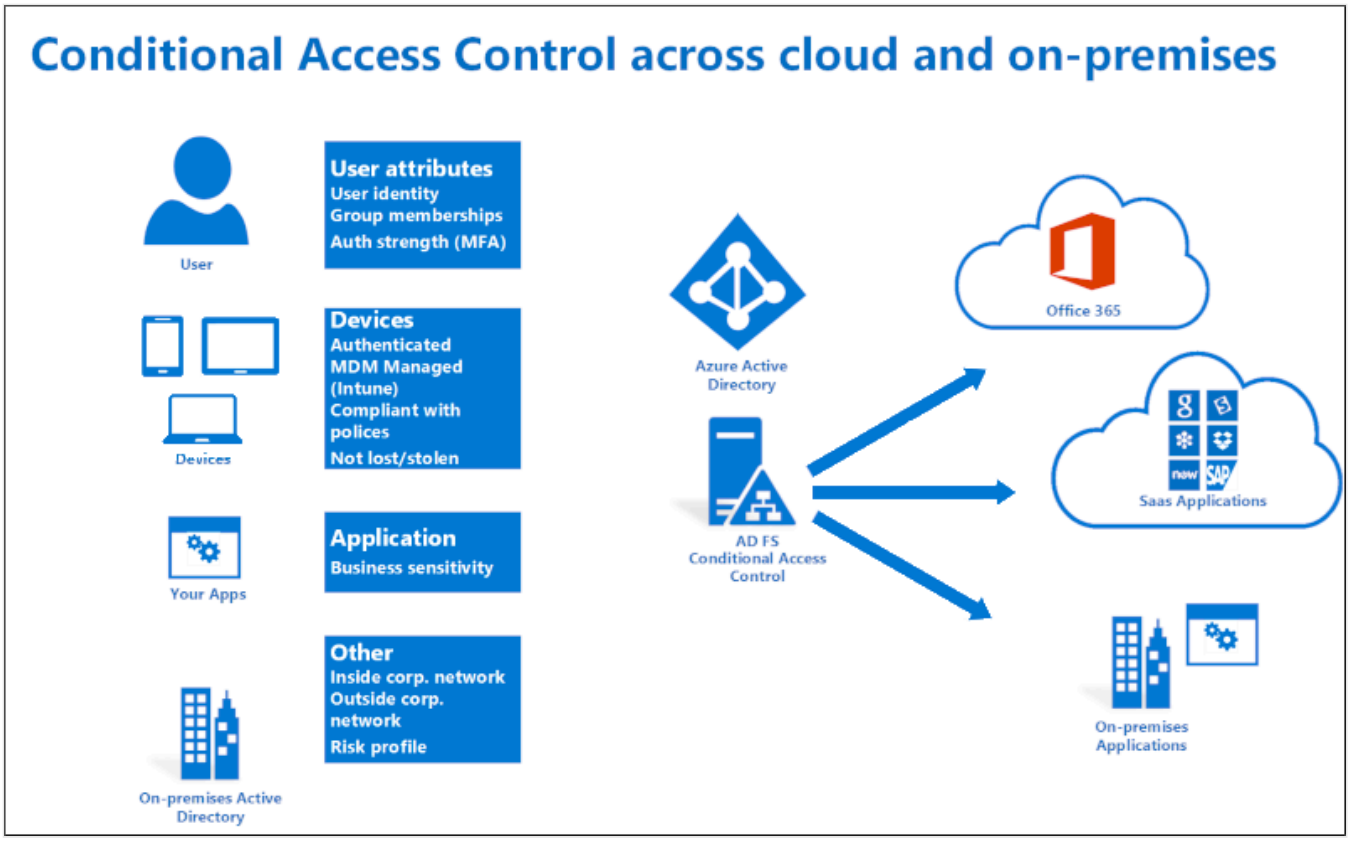
## Passwordless access from compliant devices

AD FS 2016 builds on previous device registration capabilities to enable sign-on and access control on devices based on their compliance status. Users can sign on using the device credential, and AD FS reevaluates compliance whenever device attributes change to ensure policies are being enforced. This feature enables the following policies:

- Enable Access only from devices that are managed and/or compliant.
- Enable Extranet Access only from devices that are managed and/or compliant.
- Require multifactor authentication for computers that aren't managed or compliant.

AD FS provides the on-premises component of conditional access policies in a hybrid scenario. When you register devices with Azure AD for conditional access to cloud resources, you can

also use the device identity for AD FS policies.




For more information about using device based conditional access in the cloud, see Azure Active Directory Conditional Access.

For more information about using device based conditional access with AD FS, see Planning for Device Based Conditional Access with AD FS and Access Control Policies in AD FS.

## Sign in with Windows Hello for Business

Windows 10 devices introduce Windows Hello and Windows Hello for Business, replacing user passwords with strong device-bound user credentials protected by a user's gesture, such as entering a PIN, a biometric gesture like a fingerprint, or facial recognition. With Windows Hello, users can sign in to AD FS applications from an intranet or extranet without requiring a password.

For more information about using Windows Hello for Business in your organization, see Enable Windows Hello for Business in your organization.

## Modern authentication

AD FS 2016 supports the latest modern protocols that provide a better user experience for Windows 10 and the latest iOS and Android devices and apps.

For more information, see AD FS Scenarios for Developers.

## Configure access control policies without having to know claim rules language

Previously, AD FS administrators had to configure policies by using the AD FS claim rule language, making it difficult to configure and maintain policies. With access control policies, administrators can use built-in templates to apply common policies. For example, you can use templates to apply the following policies:

- Permit intranet access only.
- Permit everyone and require MFA from extranet.
- Permit everyone and require MFA from a specific group.

The templates are easy to customize. You can apply extra exceptions or policy rules, and you can apply these changes to one or more applications for consistent policy enforcement.

For more information, see Access control policies in AD FS.

## Enable sign on with non-AD LDAP directories

Many organizations combine Active Directory with third-party directories. AD FS support for authenticating users stored in Lightweight Directory Access Protocol (LDAP) v3-compliant directories means you can now use AD FS in the following scenarios:

- Users in third party, LDAP v3-compliant directories.
- Users in Active Directory forests that don't have a configured Active Directory two-way trust.
- Users in Active Directory Lightweight Directory Services (AD LDS).

For more information, see Configure AD FS to authenticate users stored in LDAP directories.

## Customize sign in experience for AD FS applications

Previously, AD FS in Windows Server 2012 R2 provided a common sign-on experience for all relying party applications, with the ability to customize a subset of text-based content per application. With Windows Server 2016, you can customize not only the messages, but images, logo and web theme per application. Additionally, you can create new, custom web themes and apply these themes per relying party.

For more information, see AD FS user sign-in customization.

## Streamlined auditing for easier administrative management

In previous versions of AD FS, a single request could generate many audit events. Relevant information about sign-in or token issuance activities were often absent or spread across multiple audit events, making issues harder to diagnose. As a result, audit events were turned off by default. However, in AD FS 2016, the auditing process is more streamlined and relevant information easier to find. For more information, see Auditing enhancements to AD FS in Windows Server 2016.

## Improved interoperability with SAML 2.0 for participation in confederations

AD FS 2016 contains more SAML protocol support, including support for importing trusts based on metadata that contains multiple entities. This change enables you to configure AD FS to participate in confederations such as InCommon Federation and other implementations conforming to the eGov 2.0 standard.

For more information, see Improved interoperability with SAML 2.0.

## Simplified password management for federated Microsoft 365 users

You can configure AD FS to send password expiry claims to any relying party trusts or applications that it protects. How these claims appear varies between applications. For example, with Office 365 as your relying party, updates are implemented to Exchange and Outlook to notify federated users of their soon-to-be-expired passwords.

For more information, see Configure AD FS to send password expiry claims.

## Moving from AD FS in Windows Server 2012 R2 to AD FS in Windows Server 2016 is easier

Previously, migrating to a new version of AD FS required exporting the configuration settings from your Windows Server farm to a new, parallel server farm. AD FS on Windows Server 2016 makes the process easier by removing the requirement to have a parallel server farm. When you add a Windows Server 2016 server to a Windows Server 2012 R2 server farm, the new server behaves just like a Windows Server 2012 R2 server. When you're ready to upgrade and you removed the older servers, you can change the operational level to Windows Server 2016. For learn more, see Upgrading to AD FS in Windows Server 2016.

# Get started with Windows Server Insiders Preview

Article • 11/22/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

Join the Windows Insider Program for Windows Server and gain exclusive access to Windows Server Insider Previews and Remote Server Administration tools. By becoming a part of this community, you'll have the opportunity to help shape the future of Windows Server and be at the forefront of innovation!

## Where to get Windows Server Insiders Preview

If you're a registered Insider, you can access the Windows Insider Preview Downloads page directly to view available Window Server Preview builds. If you'd like to participate as an Insider, you can refer to the Getting started with the Windows Insider Program for Windows Server.

The following keys are valid only for preview builds:

Expand table

| Windows Server Version | Key |
|---|---|
| Standard | MFY9F-XBN2F-TYFMP-CCV49-RMYVH |
| Datacenter | 2KNJJ-33Y9H-2GXGX-KMQWH-G6H67 |
| Azure Edition | Keys aren't accepted |

> **Note**
>
> Downloads may be restricted in certain countries. To learn more, see Microsoft suspends new sales in Russia.

## Insiders Preview known issues

- During the OOBE install process, some users notice overlapping windows or graphics anomalies when using the mouse to proceed to the next step.

- First sign-in user privacy settings are limited where all features aren't available nor function as desired.
- Installing the **WinPE-Powershell** optional component via any method doesn't install properly and related cmdlets fail. Customers who are dependent on PowerShell in WinPE shouldn't use this build.
- The new **Feedback Hub** and **Terminal** apps aren't functioning properly in this release.
- We advise against using this build to validate upgrades from Windows Server 2019 or 2022 due to identified intermittent upgrade failures.
- Using the `wevtutil al` command to archive event logs causes the Windows Event Log service to crash and archive operation to fail. To resolve this issue, the service must be restarted by running the following in an elevated PowerShell prompt:

  ```PowerShell
  Start-Service EventLog
  ```

- If you have the Secure Launch or Dynamic Root of Trust for Measurement (DRTM) code path enabled, we recommend avoiding installing this build.

# How to provide feedback for Insiders Preview

Your feedback is valuable to us as it provides insight to what is currently working, capturing bugs, and suggestions of where improvements can be made. To learn how to provide feedback, see Deeper look at feedback.

Use your registered Windows 10 or Windows 11 device and open the Feedback Hub app. In the Feedback Hub app, provide us:

1. A title about the issue with the preview build number. Example, *Server Manager Issue in Windows Server Standard 25997*.
2. A detailed explanation of what is occurring.
3. For the **Category**, select **Windows Server**.
4. Attaching a screenshot of the issue is optional.
5. Complete submitting your feedback.

# See also

- What's new in Windows Server 2025
- Explore the Feedback Hub

---

# Feedback

Was this page helpful? Yes No

# Comparison of Windows Server editions

Article • 05/01/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

Use this article to compare the Standard, Datacenter, and Datacenter: Azure Edition of Windows Server to see which is most appropriate for you.

> **Tip**
>
> If you're looking for information about the locks and limits in Windows Server, see Comparison of locks and limits in Windows Server.

## Available roles and features

Expand table

| Feature | Subfeature | Standard edition | Datacenter edition | Datacenter: Azure Edition |
|---|---|---|---|---|
| .NET Framework 3.5 Features | | ✅ | ✅ | ✅ |
| .NET Framework 4.8 Features | | ✅ | ✅ | ✅ |
| Activation | | ✅ | ✅ | ✅ |
| | Automatic Virtual Machine Activation | ⚠[1] | ✅ | ✅[2] |
| | Key Management Service (KMS) | ✅ | ✅ | ❌[3] |
| Active Directory Certificate Services | | ✅ | ✅ | ✅ |
| | Certificate Enrollment Policy Web Service | ✅ | ✅ | ✅ |
| | Certificate Enrollment Web Service | ✅ | ✅ | ✅ |
| | Certification Authority | ✅ | ✅ | ✅ |

| Feature | Subfeature | Standard edition | Datacenter edition | Datacenter: Azure Edition |
|---|---|---|---|---|
| | Certification Authority Web Enrollment | ✅ | ✅ | ✅ |
| | Network Device Enrollment Service | ✅ | ✅ | ✅ |
| | Online Responder | ✅ | ✅ | ✅ |
| Active Directory Domain Services | | ✅ | ✅ | ✅ |
| Active Directory Federation Services | | ✅ | ✅ | ✅ |
| Active Directory Lightweight Directory Services | | ✅ | ✅ | ✅ |
| Active Directory Rights Management Services | | ✅ | ✅ | ✅ |
| Azure Extended Networking | | ❌ | ❌ | ✅ |
| Background Intelligent Transfer Service (BITS) | | ✅ | ✅ | ✅ |
| BitLocker Drive Encryption | | ✅ | ✅ | ✅ |
| BitLocker Network Unlock | | ✅[4] | ✅[4] | ✅[4] |
| BranchCache | | ✅ | ✅ | ✅ |
| Client for NFS | | ✅ | ✅ | ✅ |
| Data Center Bridging | | ✅ | ✅ | ✅ |
| Device Health Attestation | | ✅ | ✅ | ✅ |
| DHCP Server | | ✅ | ✅ | ✅ |
| Direct Play | | ✅[3] | ✅[4] | ✅[4] |
| DLNA codecs and web media streaming | | ✅[3] | ✅[4] | ✅[4] |

| Feature | Subfeature | Standard edition | Datacenter edition | Datacenter: Azure Edition |
|---|---|---|---|---|
| DNS Server | | ✅ | ✅ | ✅ |
| Enhanced Storage | | ✅ | ✅ | ✅ |
| Failover Clustering | | ✅ | ✅ | ✅ |
| Fax Server | | ✅ | ✅ | ✅ |
| File and Storage Services | | ✅ | ✅ | ✅ |
| | BranchCache for Network Files | ✅ | ✅ | ✅ |
| | Data Deduplication | ✅ | ✅ | ✅ |
| | DFS Namespaces | ✅ | ✅ | ✅ |
| | DFS Replication | ✅ | ✅ | ✅ |
| | File Server | ✅ | ✅ | ✅ |
| | File Server Resource Manager | ✅ | ✅ | ✅ |
| | File Server VSS Agent Service | ✅ | ✅ | ✅ |
| | iSCSI Target Server | ✅ | ✅ | ✅ |
| | iSCSI Target Storage Provider (VDS and VSS hardware providers) | ✅ | ✅ | ✅ |
| | Server for NFS | ✅ | ✅ | ✅ |
| | SMB 1.0/CIFS File Sharing Support | ✅ | ✅ | ✅ |
| | SMB Bandwidth Limit | ✅ | ✅ | ✅ |
| | SMB over QUIC | ✅ | ✅ | ✅ |
| | Work Folders | ✅ | ✅ | ✅ |
| | Storage Migration Service | ✅ | ✅ | ✅ |
| | Storage Migration Service Proxy | ✅ | ✅ | ✅ |
| | Storage Spaces | ✅ | ✅ | ✅ |

| Feature | Subfeature | Standard edition | Datacenter edition | Datacenter: Azure Edition |
|---|---|---|---|---|
| | Storage Spaces Direct | ✗ | ✓ | ✓ |
| | Storage Replica | ✓ | ✓ | ✓ |
| Group Policy Management | | ✓ | ✓ | ✓ |
| Host Guardian Hyper-V Support | | ✗ | ✓ | ✓ |
| Host Guardian Service | | ✓ | ✓ | ✓ |
| Hotpatching | | ✓[5] | ✓[5] | ✓ |
| I/O Quality of Service | | ✓ | ✓ | ✓ |
| IIS Hostable Web Core | | ✓ | ✓ | ✓ |
| IP Address Management (IPAM) Server | | ✓ | ✓ | ✓ |
| Management OData IIS Extension | | ✓ | ✓ | ✓ |
| Media Foundation | | ✓ | ✓ | ✓ |
| Message Queuing | | ✓ | ✓ | ✓ |
| | Message Queuing DCOM Proxy | ✓ | ✓ | ✓ |
| | Message Queuing Services | ✓ | ✓ | ✓ |
| Microsoft Defender Antivirus | | ✓ | ✓ | ✓ |
| Multipath I/O | | ✓ | ✓ | ✓ |
| MultiPoint Connector | | ✓ | ✓ | ✓ |
| Network ATC | | ✓ | ✓ | ✓ |
| Network Controller | | ✗ | ✓ | ✓ |
| Network Load Balancing | | ✓ | ✓ | ✓ |
| Network Policy and Access Services | | ✓[4] | ✓[4] | ✓[4] |
| Network Virtualization | | ✓ | ✓ | ✓ |

| Feature | Subfeature | Standard edition | Datacenter edition | Datacenter: Azure Edition |
|---|---|---|---|---|
| Print and Document Services | | ✅ | ✅ | ✅ |
| | Internet Printing | ✅[4] | ✅[4] | ✅[4] |
| | Line Printer Daemon (LPD) Service | ✅[4] | ✅[4] | ✅[4] |
| | Print Server | ✅[4] | ✅[4] | ✅[4] |
| Quality Windows Audio Video Experience | | ✅ | ✅ | ✅ |
| RAS Connection Manager Administration Kit (CMAK) | | ✅ | ✅ | ✅ |
| Remote Access | | ✅ | ✅ | ✅ |
| | DirectAccess and VPN (RAS) | ✅ | ✅ | ✅ |
| | Routing | ✅ | ✅ | ✅ |
| | Web Application Proxy | ✅ | ✅ | ✅ |
| Remote Assistance | | ✅[4] | ✅[4] | ✅[4] |
| Remote Desktop Services | | ✅[4] | ✅[4] | ✅[4] |
| Remote Differential Compression | | ✅ | ✅ | ✅ |
| Remote Server Administration Tools | | ✅ | ✅ | ✅ |
| RPC over HTTP Proxy | | ✅ | ✅ | ✅ |
| Setup and Boot Event Collection | | ✅ | ✅ | ✅ |
| Simple TCP/IP Services | | ✅[4] | ✅[4] | ✅[4] |
| SNMP Service | | ✅ | ✅ | ✅ |
| Software Load Balancer | | ✅ | ✅ | ✅ |
| System Data Archiver | | ✅ | ✅ | ✅ |

| Feature | Subfeature | Standard edition | Datacenter edition | Datacenter: Azure Edition |
|---|---|---|---|---|
| System Insights | | ✅ | ✅ | ✅ |
| Telnet Client | | ✅ | ✅ | ✅ |
| TFTP Client | | ✅[4] | ✅[4] | ✅[4] |
| Virtualization | | ✅ | ✅ | ✅ |
| | Containers | ✅ | ✅ | ❌ |
| | Hyper-V | ✅ | ✅ | ✅ |
| | GPU partitioning | ✅[6] | ✅ | ✅[6] |
| | VM Shielding Tools for Fabric Management | ✅ | ✅ | ✅ |
| Volume Activation Services | | ✅ | ✅ | ✅ |
| Web Server (IIS) | | ✅ | ✅ | ✅ |
| | FTP Server | ✅ | ✅ | ✅ |
| | Web Server | ✅ | ✅ | ✅ |
| WebDAV Redirector | | ✅ | ✅ | ✅ |
| Windows Biometric Framework | | ✅[4] | ✅[4] | ✅[4] |
| Windows Deployment Services | | ✅ | ✅ | ✅ |
| Windows Identity Foundation 3.5 | | ✅[4] | ✅[4] | ✅[4] |
| Windows Internal Database | | ✅ | ✅ | ✅ |
| Windows PowerShell | | ✅ | ✅ | ✅ |
| | Windows PowerShell 2.0 Engine | ✅ | ✅ | ✅ |
| | Windows PowerShell 5.1 | ✅ | ✅ | ✅ |
| | Windows PowerShell Desired State Configuration Service | ✅ | ✅ | ✅ |

| Feature | Subfeature | Standard edition | Datacenter edition | Datacenter: Azure Edition |
|---|---|---|---|---|
| | Windows PowerShell Web Access | ✅ | ✅ | ✅ |
| Windows Process Activation Service | | ✅ | ✅ | ✅ |
| Windows Server Management enabled by Azure Arc | | ✅[7] | ✅[7] | ✅[7] |
| | Azure Update Manager | ✅[7] | ✅[7] | ✅[7] |
| | Change Tracking and Inventory | ✅[7] | ✅[7] | ✅[7] |
| | Azure Machine Configuration | ✅[7] | ✅[7] | ✅[7] |
| | Windows Admin Center in Azure for Arc | ✅[7] | ✅[7] | ✅[7] |
| | Remote Support | ✅[7] | ✅[7] | ✅[7] |
| | Network HUD | ✅[7] | ✅[7] | ✅[7] |
| | Best Practices Assessment | ✅[7] | ✅[7] | ✅[7] |
| | Azure Site Recovery Configuration | ✅[7] | ✅[7] | ✅[7] |
| Windows Search Service | | ✅[4] | ✅[4] | ✅[4] |
| Windows Server Backup | | ✅ | ✅ | ✅ |
| Windows Server Migration Tools | | ✅ | ✅ | ✅ |
| Windows Server Update Services | | ✅ | ✅ | ✅ |
| Windows Standards-Based Storage Management | | ✅ | ✅ | ✅ |
| Windows Subsystem for Linux | | ✅ | ✅ | ✅ |
| Windows TIFF IFilter | | ✅[4] | ✅[4] | ✅[4] |

| Feature | Subfeature | Standard edition | Datacenter edition | Datacenter: Azure Edition |
|---|---|---|---|---|
| WinRM IIS Extension | | ✅ | ✅ | ✅ |
| WINS Server | | ✅ | ✅ | ✅ |
| Wireless LAN Service | | ✅ | ✅ | ✅ |
| WoW64 Support | | ✅ | ✅ | ✅ |
| XPS Viewer | | ✅[4] | ✅[4] | ✅[4] |

1. As a guest if hosted on a virtualization host activated with Datacenter edition
2. Datacenter: Azure Edition can be used as a nested host or guest
3. Activated by Azure and can't be configured as a KMS host
4. When installed as Server with Desktop Experience
5. Available as an Azure Arc-enabled service. To learn more about Azure Arc pricing, see the Azure Pricing Calculator
6. Available with Windows Server 2025 Standard, and is designed for standalone servers. Live migrate VMs between standalone nodes for planned downtime. If clustering is required for unplanned downtime, Windows Server 2025 Datacenter must be used instead.
7. Available for machines enrolled in Windows Server Management enabled by Azure Arc with Windows Server licenses that have active Software Assurances or Windows Server licenses that with active subscription licenses. To learn more about the available Azure benefits, billing, and requirements, see Windows Server Management enabled by Azure Arc.

# What is Azure Edition for Windows Server?

08/13/2025Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022

Windows Server Datacenter: Azure Edition is an edition of Windows Server focused on innovation and virtualization optimized to run on Azure. Azure Edition features a Long-Term Servicing Channel (LTSC) and yearly product updates, with two major product updates in the first 3 years. Azure Edition also brings new functionality to Windows Server users faster than the Standard and Datacenter editions of Windows Server.

The annual Azure Edition updates are delivered using Windows Update, rather than a full OS upgrade. As part of this annual update cadence, the Azure Edition Insider preview program gives the opportunity to access early builds - leading to general availability. To get started with Azure Edition Insider preview, visit the Azure Edition preview Azure Marketplace offer. Details regarding each preview is shared in release announcements posted to the Windows Server Insiders space on Microsoft Tech Community.

## Key differences

The following table summarizes the key differences:

Expand table

| Description | Windows Server Standard, Datacenter | Windows Server Datacenter: Azure Edition |
|---|---|---|
| New releases | Typically 2-3 years | Typically 2-3 years |
| Product updates | With new release | Yearly, with two major updates in the first 3 years |
| Support | 5 years of mainstream support, plus 5 years of extended support | 5 years of mainstream support, plus 5 years of extended support |
| Servicing channels | Long-Term Servicing Channel | Long-Term Servicing Channel |
| Who can use it? | All customers through all channels | Software Assurance, Windows Server subscription, and cloud customers only |
| Installation options | Server Core, Server with Desktop Experience, Nano Server container image | Server Core and Server with Desktop Experience only. Windows Server containers aren't supported. |
| Operating system environments (OSE) | Physical or virtual | Virtual only |

| Description | Windows Server Standard, Datacenter | Windows Server Datacenter: Azure Edition |
|---|---|---|
| Associated virtualization rights | 2 virtual OSE for Standard, Unlimited virtual OSEs for Datacenter | None |

Capabilities vary by image, see Getting started with Windows Server Datacenter: Azure Edition for more detail.

> **Tip**
>
> For more information, see the Microsoft Software Licensing Terms. The licensing terms might vary based on the distribution channel, for example, a Commercial Licensing program, Retail, Original Equipment Manufacturer (OEM), and so on.

# Key capabilities

## Hotpatch

Beginning with Windows Server 2022 Datacenter: Azure Edition, Hotpatch gives you the ability to apply security updates on your VM without rebooting. When used with Azure, Azure Guest Patching Service, along with Automanage for Window Server, automate the onboarding, configuration, and orchestration of hotpatching. To learn more, see Hotpatch for new virtual machines.

### Supported platforms

To learn more about what operating systems support Hotpatch for VMs running on Azure and Azure Stack HCI, see Supported platforms.

> **Note**
>
> Hotpatch isn't supported on Windows Server containers base images.

## SMB over QUIC

Beginning with Windows Server 2022 Datacenter: Azure Edition, SMB over QUIC offers an "SMB VPN" for telecommuters, mobile device users, and branch offices. SMB over QUIC provides

secure, reliable connectivity to edge file servers over untrusted networks like the Internet. QUIC is an IETF-standardized protocol used in HTTP/3, designed for maximum data protection with TLS 1.3 and requires encryption that can't be disabled. SMB behaves normally within the QUIC tunnel, meaning the user experience doesn't change. SMB features like multichannel, signing, compression, continuous availability, and directory leasing work normally.

SMB over QUIC is also integrated with Azure Automanage machine best practices for Windows Server to help make SMB over QUIC management easier. QUIC uses certificates to provide its encryption and organizations often struggle to maintain complex public key infrastructures. Azure Automanage machine best practices ensure that certificates don't expire without warning and that SMB over QUIC stays enabled for maximum continuity of service.

To learn more, see SMB over QUIC and SMB over QUIC management with Automanage machine best practices.

## Storage Replica compression for data transfer

Beginning with Update 1 for Windows Server 2022 Datacenter: Azure Edition, you can compress Storage Replica data between source and destination server. The compression results in fewer network packets to transfer the same amount of data, allowing for more throughputs, and less network utilization. Higher data throughput should also result in lowering synchronization time for when you need it most, for example in a disaster recovery scenario.

To learn more about Storage Replica features, see Storage Replica features

## Extended network for Azure

Beginning with Windows Server 2022 Datacenter: Azure Edition, Azure Extended Network enables you to stretch an on-premises subnet into Azure to let on-premises virtual machines keep their original on-premises private IP addresses when migrating to Azure. To learn more, see
Azure Extended Network.

# Get started with Windows Server Datacenter: Azure Edition

To get started using Azure Edition, use your preferred method to create an Azure or Azure Local VM, and select the *Windows Server Datacenter: Azure Edition* image that you would like to use.

> **Important**
>
> Some capabilities have specific configuration steps to perform during VM creation, and some capabilities that are in preview have specific opt-in and portal viewing requirements. See the individual capability articles to learn more about using that capability with your VM.

> **Caution**
>
> Once Windows Server Datacenter: Azure Edition is installed it isn't possible to switch the OS back to a non-Azure Edition OS. If you didn't mean to install Azure Edition, reinstalling the previous OS is required to switch back.

To learn more about creating virtual machine using Azure or Azure Local, see Create a Windows virtual machine in the Azure portal and Deploy Windows Server Azure Edition VMs in Azure Local.

# Next steps

- Comparison of Standard, Datacenter, and Datacenter: Azure Edition editions of Windows Server 2022
- Hotpatch for new virtual machines
- Enable Hotpatch for Azure Edition virtual machines built from ISO
- SMB over QUIC
- Extend your on-premises subnets into Azure using extended network for Azure

# Configure Windows Server Pay-as-you-go with Azure Arc

06/27/2025Applies to: ✅ Windows Server 2025

Azure Arc's Pay-as-you-go subscription licensing option is an alternative to the conventional perpetual licensing for Windows Server 2025. With Pay-as-you-go, you can deploy a Windows Server device, license it, and only pay for as much as you use. This feature is facilitated through Azure Arc and billed via your Azure subscription. You have the flexibility to disable Pay-as-you-go whenever necessary. Additionally, you can use Pay-as-you-go for free for the first seven days after enabling it as a trial. Windows Server Pay-as-you-go is an Azure service and the detailed usage rights are provided in the Microsoft Product Terms.

> **⊗ Caution**
>
> If you shut down or deprovision your device, without disabling Pay-as-you-go, billing continues. To avoid unexpected charges, disable Pay-as-you-go using the Azure portal, PowerShell, API, or by removing the device from Azure Arc.

Windows Server Pay-as-you-go shares the same pricing and model as Windows Server licensing on Microsoft Azure, but is designed for devices that are deployed outside of Microsoft Azure. Windows Server Pay-as-you-go with Azure Arc isn't supported on Microsoft Azure, as there are other ways to license Windows Server with Pay-as-you-go with that service.

Pay-as-you-go regulations differ from traditional perpetual licensing. For instance, in Pay-as-you-go, the cost is the same for both Standard and Datacenter editions, and there are no client access licenses (CALs) required for standard functionality. However, Remote Desktop Services (RDS) CALs are still required.

The Pay-as-you-go license only applies to the exact device where the feature is enabled on Microsoft Azure. Unlike traditional perpetual licenses, it doesn't provide additional rights for virtual machines (VM) running on the same server. Thus, the Automatic Virtual Machine Activation (AVMA) feature isn't available. Each VM requires its own separate license, regardless of the host server. The host and VMs can run different versions of the operating system (OS) and mix different license types.

## Prerequisites

Before you can set up Windows Server Pay-as-you-go, ensure that your device meets the following prerequisites:

- Your device must be running Windows Server 2025 Standard or Datacenter edition. If you're new to Windows Server, familiarize yourself with Feature update, clean install, or migrate to Windows Server for more details.
- Your device must be Azure Arc-enabled and be running Microsoft Azure connected machine agent version 1.47 or later. To learn more about how to onboard your device to Azure Arc, see Connect Windows Server machines to Azure through Azure Arc Setup.
- Your device isn't currently licensed (activated) with another license type, such as OEM, Retail, or Volume License (VL).
- An active internet connection is required.

# Set up Windows Server Pay-as-you-go

You can enable Windows Server Pay-as-you-go either while installing Windows Server or after the installation is complete. The following sections explain both options step by step.

## Choose a licensing method

During the installation of Windows Server, you can choose to set up Pay-as-you-go. This option is available on the **Choose a licensing method** screen where you would normally enter a product key. Once Pay-as-you-go is selected, providing a product key isn't required.

If you already have Windows Server installed and it's not activated, you can skip to the Manage Windows Server Pay-as-you-go section. If you're new to installing Windows Server, you can follow the steps in Install Windows Server from installation media and return to these steps when you reach the **Choose a licensing method** screen.

> **Note**
>
> Pay-as-you-go integration is only available during the operating system installation if you're running Windows Setup from a retail copy of the Windows Server media. If you're using Evaluation or Volume license (VL) media of Windows Server, then the Product key page is bypassed, and hence you can't opt in to Pay-as-you-go during the operating system installation. Nevertheless, you can enable Pay-as-you-go after installation as described in the Manage Windows Server Pay-as-you-go section.

There are two methods for setting up Pay-as-you-go, via the graphical wizard and through the command line. Follow these steps based on which edition of Windows you're installing:

GUI

Once you're at the **Choose a licensing method** screen, follow these steps:

1. Select **Pay-as-you-go** and then select **Next**.
2. Select your edition of Windows with **Desktop Experience** and then select **Next**.
3. Accept the license terms by selecting **Accept** to proceed.
4. Under **Select location to install Windows Server**, choose your disk and partition per your needs. Then select **Next**.
5. On the **Ready to install** screen, select **Install**

The install process automatically reboots your device as needed. You're prompted to create a password in order to proceed with the installation. Once complete and you're signed in, the **Azure Arc Setup** wizard is displayed. Follow these steps to continue setting up Pay-as-you-go:

1. On the Get **Started with Azure Arc** screen, select **Next**.
2. On the **Installing Azure Arc** screen, your device goes through several checks to install the Azure Connected Machine (AzCM) agent. Then select **Configure**.
3. On the **Configure Azure Arc** screen, select **Next**.
4. On the **Sign in to Azure** screen, select your Azure cloud, sign in, then select **Next**.
5. On the **Resource details** screen, continue filling in the required information, then select **Next**.
6. On the **Choose a licensing method for Windows Server** screen, select **Pay-as-you-go with Azure**, then select **Next**.
7. Once you reach the last screen during the configuration process, select **Finish**.

> ⓘ **Note**
>
> If you select **Use a product key** on the **Choose a licensing method for Windows Server** screen, and continue with the installation wizard, Pay-as-you-go won't be activated even if you had previously opted in during the OS installation.
>
> If you decide to opt out of using Pay-as-you-go, you need to manually enter a product key using any supported method. However, if you still wish to use Pay-as-you-go, even though it hasn't been enabled through the Azure Arc configuration wizard, you can activate it through the Azure portal.

## Manage Windows Server Pay-as-you-go

Managing Pay-as-you-go for Windows Server is possible using the Azure portal, PowerShell, or API. Even if you didn't choose to enable Pay-as-you-go during the initial installation of the OS

is also an option, as long as the necessary prerequisites are met.

To enable Windows Server Pay-as-you-go, follow these steps:

Azure portal

To use the Azure portal to enabled Windows Server Pay-as-you-go, follow these steps:

1. Navigate to your Azure portal, then search for and select **Machines - Azure Arc**.
2. Select the machine you want to enable Windows Server Pay-as-you-go.
3. Select the **Pay-as-you-go** tile.
4. Check the box next to **Pay-as-you-go with Azure**, and then select **Confirm**.

*Alternatively*, in the left pane menu by expanding **Licenses**, selecting **Windows Server**, and follow step 4.

To disable Windows Server Pay-as-you-go, follow these steps:

Azure portal

To use the Azure portal to disable Windows Server Pay-as-you-go, follow these steps:

1. Navigate to your Azure portal, then search for and select **Machines - Azure Arc**.
2. Select the machine you want to disable Windows Server Pay-as-you-go.
3. Select the **Pay-as-you-go** tile.
4. Uncheck the box next to **Pay-as-you-go with Azure**, and then select **Confirm**.
5. The **Deactivate Pay-as-you-go** notification appears asking if you'd like to deactivate. Select **Deactivate**.

*Alternatively*, in the left pane menu by expanding **Licenses**, selecting **Windows Server**, and follow step 4.

# See also

- Azure Arc-enabled servers
- Connect Windows Server machines to Azure through Azure Arc Setup

# What is Windows Server Annual Channel for Containers?

Article • 11/01/2024 • Applies to: ✅ Windows Server, version 23H2 and later

Windows Server Annual Channel for Containers is an operating system to host Windows Server containers. Annual Channel enables customers who are innovating quickly to take advantage of new operating system capabilities at a faster pace, focusing on containers, microservices, and portability.

Windows Server Annual Channel for Containers means new features and functionality being released on an annual basis. The more frequent releases of Annual Channel enable customers to take advantage of innovation more quickly, with a focus on containers and microservices. To learn more about the lifecycle, see the Windows Server Annual Channel lifecycle article. To learn more about difference between servicing channels, see Windows Server servicing channels.

## Supported platforms

A Windows Server, version 23H2 container host only supports the Windows Server 2022 Long Term Servicing Channel (LTSC) container image.

## Portability

Portability is an important feature introduced in Windows Server Annual Channel for Containers that lets users run workloads with different container image versions. Portability lets Windows Server 2022-based container images run on session hosts running later versions of Windows Server. This increased support helps container services like AKS to update the operating systems on container hosts on a more frequent basis without requiring you to update the containers themselves. Portability doesn't only streamline the upgrade process, but also helps developers take full advantage of the enhanced flexibility and compatibility that containers offer. For more information about portability, see Portability in Windows Server Annual Channel for containers.

## Get started with Windows Server Annual Channel for Containers

To get started using Annual Channel for Containers, use your preferred method to install Windows Server on your container host. Annual Channel is available to volume-licensed customers with Software Assurance and loyalty programs such as Visual Studio Subscriptions. You can get Annual Channel releases from:

- Volume Licensing Service Center (VLSC): Volume-licensed customers with Software Assurance can obtain this release by going to the Volume Licensing Service Center and select **Sign In**. Finally, select **Downloads and Keys**, search for Annual Channel, then download the media.

- Visual Studio Subscriptions: Visual Studio Subscribers can obtain Annual Channel releases by downloading them from the Visual Studio Subscriber download page. If you aren't already a subscriber, go to Visual Studio Subscriptions to sign up, and then visit the Visual Studio Subscriber downloads page. Releases obtained through Visual Studio Subscriptions are for development and testing only.

> ⚠ **Warning**
>
> Annual Channel can only be installed with the Server Core installation option. To move to an Annual Channel release from LTSC, or to upgrade an existing Annual Channel installation, you must perform a clean installation.

## Related content

- Portability in Windows Server Annual Channel for container

## Feedback

Was this page helpful? Yes No

# Hardware requirements for Windows Server

07/22/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

To install Windows Server correctly, your computer must meet the minimum hardware requirements outlined in this article. If your computer falls short of these requirements, the product might not install properly. Actual requirements vary based on your system configuration, applications, and features that are installed.

Unless otherwise specified, these minimum hardware requirements apply to all installation options (Server Core and Server with Desktop Experience) for both Windows Server Standard and Windows Server Datacenter editions.

> **ⓘ Important**
>
> The highly diverse scope of potential deployments makes it unrealistic to state recommended hardware requirements that would be applicable. Consult documentation for each of the server roles you intend to deploy to learn more about the resource needs of particular server roles. For the best results, conduct test deployments to determine appropriate hardware requirements for your particular deployment scenarios.

## Components

CPU

Processor performance depends not only on the clock frequency of the processor, but also on the number of processor cores and the size of the processor cache. The following are the processor requirements.

**Minimum**:

- 1.4-GHz 64-bit processor
- Compatible with x64 instruction set
- Support for NX and DEP
- Support for CMPXCHG16b, LAHF/SAHF, and PrefetchW instructions

- Support for Second Level Address Translation (EPT or NPT)
- Support for the SSE4.2 (Streaming SIMD Extensions 4.2) instruction set
- Support for the POPCNT instruction

You can utilize Coreinfo, which is a tool included in Windows Sysinternals, to verify the capabilities of your CPU.

## Secured-core server requirements

Secured-core is a set of integrated hardware, firmware, driver, and operating system (OS) security features that provide enhanced protection against advanced threats. Secured-core systems deliver security that starts before the OS boots and continues throughout system operation. Secured-core server features are available in Windows Server 2022 and later. To deploy a Secured-core server, your device must meet these additional requirements:

- **DMA Remapping (IOMMU)**: Support for *Intel VT-d* or *AMD-Vi* is required to ensure secure and efficient management of direct memory access (DMA) by devices, safeguarding against unauthorized memory access.
- **Kernel DMA Protection**: The system must have opt-in capability for Kernel DMA Protection, which helps prevent attacks that exploit external peripherals to gain unauthorized access.
- **DRTM (Dynamic Root of Trust for Measurement)**: This feature is needed to ensure a secure boot process by verifying the integrity of the system's startup environment, protecting against firmware-based attacks.

> ⓘ **Note**
>
> In addition to the specific requirements outlined for Secured-core server, the following features must be enabled on your hardware:
>
> - TPM 2.0
> - Secure Boot
> - Virtualization-based Security (VBS) support, including hardware virtualization extensions
>
> To learn more, see What is Secured-core server?.

# Other requirements

There are other hardware requirements to consider, depending on your scenario:

- DVD drive (if you intend to install the OS from DVD media)

The following items are required only for certain features:

- UEFI 2.3.1c-based system and firmware that supports secure boot.
- Trusted Platform Module (TPM).
- Integrated or dedicated graphics and monitor capable of Super VGA (1024 x 768) or higher-resolution.
- Keyboard and mouse (or other compatible pointing device).
- Internet access. (Fees might apply.)

> **Note**
>
> A TPM chip is required in order to use certain features, such as **BitLocker Drive Encryption**. If your computer has a TPM, it must meet these requirements:
>
> - Hardware-based TPMs must implement version 2.0 of the TPM specification.
> - TPMs that implement version 2.0 must have an EK certificate that's either pre-provisioned to the TPM by the hardware vendor or capable of being retrieved by the device during the first boot.
> - TPMs that implement version 2.0 must ship with SHA-256 PCR banks and implement PCRs 0 through 23 for SHA-256. It's acceptable to ship TPMs with a single switchable PCR bank that can be used for both SHA-1 and SHA-256 measurements.

# Comparison of locks and limits in Windows Server

06/27/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

Choosing the right edition of Windows Server is essential for meeting your organization's technical and business requirements. This article provides a side-by-side comparison of key operating system locks and limits across supported Windows Server editions. Use these tables to compare key features across Windows Server editions, such as:

- Maximum number of users
- Connection limits
- Virtualization rights
- Hardware support

This information can help you choose the version that best fits your needs.

Select a Windows Server release to view its locks and limits.

## Locks and limits

To view the locks and limits for Windows Server 2025, select the full comparison tab or the version differences tab.

Full Comparison

Expand table

| Locks and limits | Windows Server 2025 Standard | Windows Server 2025 Datacenter | Windows Server 2025 Datacenter: Azure Edition |
|---|---|---|---|
| Maximum number of users | Based on CALs | Based on CALs | Based on CALs |
| Maximum SMB connections | 16,777,216 | 16,777,216 | 16,777,216 |
| Maximum RRAS connections | Unlimited | Unlimited | Unlimited |

| Locks and limits | Windows Server 2025 Standard | Windows Server 2025 Datacenter | Windows Server 2025 Datacenter: Azure Edition |
|---|---|---|---|
| Maximum IAS connections | 2,147,483,647 | 2,147,483,647 | 2,147,483,647 |
| Maximum RDS connections | 65,535 | 65,535 | 65,535 |
| Maximum number of 64-bit sockets | 64 | 64 | 64 |
| Maximum number of cores | Unlimited | Unlimited | 2,048 logical processors |
| Maximum RAM | • 4 PB for hosts that support 5-level paging<br>• 256 TB for hosts that support 4-level paging | • 4 PB for hosts that support 5-level paging<br>• 256 TB for hosts that support 4-level paging | • 240 TB for a generation 2 virtual machine<br>• 1 TB for a generation 1 virtual machine |
| Can be used as virtualization guest | Yes; 2 virtual machines, plus one Hyper-V host per license | Yes; **unlimited virtual machines**, plus one Hyper-V host per license | Yes; **unlimited virtual machines**, plus one Hyper-V host per license |
| Windows Server Containers | Unlimited | Unlimited | Unlimited |
| Virtual OSE/Hyper-V isolated Containers | 2 | Unlimited | Unlimited |
| Windows Server Containers | Unlimited Windows containers and up to two Hyper-V containers | Unlimited Windows containers and Hyper-V containers | Unlimited Windows containers and Hyper-V containers |
| Storage Replica | 1 partnership and 1 resource group with a single 2TB volume | **Unlimited** | **Unlimited** |

# Features removed or no longer developed in Windows Server

09/24/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

Each release of Windows Server brings new features and improvements. Sometimes older features are removed to make room for better alternatives. This article details the features and functionalities removed or deprecated in Windows Server.

> **Tip**
>
> You can get early access to Windows Server builds by joining the Windows Insider Program for Business, which is a great way to test feature changes.

> **Note**
>
> Windows Server long-term servicing channel (LTSC) releases have a Fixed Lifecycle Policy that includes 5 years of mainstream support and 5 years of extended support for a total lifecycle of 10 years. To learn more, see:
>
> - Fixed Lifecycle Policy
> - Lifecycle Terms and Definitions

**This list is subject to change and might not include every affected feature or functionality.**

## Features no longer in development

We're no longer actively developing these features and might remove them from a future update. Some features were replaced with other features or functionality. Others are now available from different sources. *Deprecation* means that a feature, functionality, or service is no longer in active development. A deprecated feature might be removed in future releases. A deprecated component still ships in Windows Server, is supported for production deployments, and continues to receive security and quality updates per the product lifecycle.

Windows Server 2025

Expand table

| Feature | Explanation |
| --- | --- |
| Computer Browser | The Computer Browser driver and service are deprecated. The browser (browser protocol and service) is a dated and insecure device location protocol. This protocol, service, and driver were first disabled by default in Windows 10, with the removal of the SMB1 service. For more information on the CIFS Browser Protocol used by the Computer Browser drive and service, see MS-BRWS Common Internet File System. |
| Failover Clustering Cluster Sets | The Failover Clustering Cluster Sets feature is no longer in active feature development and is deprecated. |
| Network Load Balancing (NLB) | NLB is no longer in active feature development and is deprecated. Consider using a Software Load Balancer (SLB) as an alternative. To learn more about SLB, see What is Software Load Balancer (SLB) for SDN? |
| NTLM | NTLMv1 is removed. LANMAN and NTLMv2 are no longer under active feature development and are deprecated. NTLMv2 will continue to work but will be removed from Windows Server in a future release. Replace calls to NTLM with calls to Negotiate, which tries to authenticate with Kerberos and only fall back to NTLM when necessary. For more information, see The evolution of Windows authentication. |
| Remote Mailslots | Remote Mailslots are deprecated. The Remote Mailslot protocol, which was initially introduced in MS DOS, is a dated and simple IPC method that is both unreliable and insecure. This protocol was first disabled by default in Windows 11 Insider Preview Build. For more information on Remote Mailslots, see About Mailslots and [MS-MAIL]: Remote Mailslot Protocol. |
| TLS 1.0 TLS 1.1 | TLS versions 1.0 and 1.1 are deprecated per internet standards and regulatory bodies due to security concerns mentioned in RFC 8996. These versions are disabled by default in Windows Server 2025. For more information on TLS deprecation, see TLS 1.0 and TLS 1.1 deprecation in Windows. |
| WebDAV Redirector service | The WebDAV Redirector service is deprecated. The service isn't installed by default in Windows Server. For more information on the WebDAV Redirector service, see WebDAV - Win32 apps. |
| Windows Internal Database (WID) | WID is used by several roles, including AD FS, AD RMS, IPAM, RD Connection Broker, and WSUS. Consider using a free or full version of SQL Server for these roles. WID will be removed from Windows in a future release. To learn more, see SQL Server editions. |
| Windows Management Instrumentation | Beginning with Windows Server 2025, WMIC is available as a Feature on Demand (FOD). It can be added with the `DISM /Add-Capability` command. It will be removed from Windows in a future release. |

| Feature | Explanation |
| --- | --- |
| Command-line (WMIC) | PowerShell for WMI replaces the WMIC tool. Use PowerShell or programmatically query WMI as a replacement for WMIC. To learn more about WMIC depreciation, see WMI command line (WMIC) utility deprecation: Next steps. |
| VBScript | VBScript is available as an FOD and preinstalled in Windows Server 2025, before its removal from the operating system in a later release. As a replacement for VBScript, use PowerShell for automating tasks, custom actions, or scripts. To learn more about migrating to PowerShell, see The VBScript-to-Windows PowerShell Conversion Guide.<br><br>If you're using VBScript within your webpage, functionality is currently limited to browsers predating Internet Explorer 11. We recommend migrating your webpages to JavaScript, which provides cross-browser compatibility and modern browser support. |
| Windows Server Update Services (WSUS) | WSUS is no longer actively developed. All the existing capabilities and content continue to be available for your deployments. |

# Features removed

We're removing the following features and functionalities from the installed product image in Windows Server. Applications or code that depend on these features won't function in future releases unless you use an alternative method. Removal occurs when a feature or functionality is no longer available for use and is eliminated from a product or service. The removal of features and functionalities can take place in any release of Windows Server. Some features are deprecated but still ship in current versions of Windows Server. Components aren't removed from generally available supported versions that are within their product lifecycle.

Windows Server 2025

Expand table

| Feature | Explanation |
| --- | --- |
| Data Encryption Standard (DES) | DES, the symmetric-key block encryption cipher, isn't considered secure against modern cryptographic attacks. It's replaced by more robust encryption algorithms. DES was disabled starting with Windows Server 2008 R2 and is removed from Windows Server 2025 and later releases. |

| Feature | Explanation |
| --- | --- |
| IIS 6 Management Console (Web-Lgcy-Mgmt-Console) | The console is removed after being no longer developed in Windows Server 2019. You should also start migrating from IIS 6.0 or earlier versions to the latest version of IIS, which is always available in the most recent release of Windows Server. |
| NTLMv1 | Replace calls to NTLM with calls to Negotiate, which tries to authenticate with Kerberos and only falls back to NTLM when necessary. For more information, see The evolution of Windows authentication. |
| Windows PowerShell 2.0 | Starting with the September 2025 update, Windows Server 2025 no longer includes Windows PowerShell 2.0. If Windows PowerShell 2.0 was installed before the September update, you can no longer use it. Windows PowerShell 2.0 applications and components should be migrated to PowerShell 5.0 or later. To learn more about the deprecation, see Windows PowerShell 2.0 Deprecation. |
| SMTP Server | Because SMTP Server features were removed from Windows Server 2025, there's no replacement within the operating system. Consider using Exchange Server or a non-Microsoft SMTP server as an alternative. To learn more about enabling SMTP connection in Exchange Server, see Receive connectors in Exchange Server. |
| Wordpad | We recommend Microsoft Word for rich text documents like `.doc` and `.rtf`, and Windows Notepad for plain text documents like `.txt`. |

# Related content

- Deprecation: What it means in the Windows lifecycle

# Windows Server release information

Windows Server has two primary release channels: the Long-Term Servicing Channel (LTSC) and the Annual Channel (AC). The LTSC provides a longer-term option focusing on a traditional lifecycle of security and quality updates. The AC provides more frequent releases, focusing on containers and microservices, so you can take advantage of innovation more quickly.

Windows Server 2025 is the current LTSC release. Windows Server, version 23H2 is the latest AC release. The focus on virtualization, container, and microservice innovation continues with Azure Stack HCI, Windows containers, and AKS on Azure Stack HCI.

If you are an IT administrator and want to programmatically get information from this page, use the Windows Updates API in Microsoft Graph.

## Windows Server major versions by servicing option

(All dates are listed in ISO 8601 format: YYYY-MM-DD)

Expand table

| Windows Server version | Servicing option | Editions | Availability date | Mainstream support end date | Extended support end date | Latest update for ESU | Latest revision date | Latest |
|---|---|---|---|---|---|---|---|---|
| Windows Server 2025 | Long-Term Servicing Channel (LTSC) | Datacenter, Standard | 2024-11-01 | 2029-11-13 | 2034-11-14 | 2026-10 B | 2026-10-13 | 26100 |
| Windows Server 2022 | Long-Term Servicing Channel (LTSC) | Datacenter, Standard | 2021-08-18 | End of updates | 2031-10-14 | 2026-10 B | 2026-10-13 | 20348 |
| Windows Server 2019 (version 1809) | Long-Term Servicing Channel (LTSC) | Datacenter, Standard | 2018-11-13 | End of updates | 2029-01-09 | 2026-10 B | 2026-10-13 | 17763 |
| Windows Server 2016 (version 1607) | Long-Term Servicing Branch (LTSB) | Datacenter, Essentials, Standard | 2016-08-02 | End of updates | 2027-01-12 | 2026-10 B | 2026-10-13 | 14393 |

Note

Windows Server is governed by the Fixed Lifecycle Policy. See the Windows Lifecycle FAQ and Comparison of servicing channels for details regarding servicing requirements and other important information. To learn more about Windows Server's Lifecycle Policy, see a Windows Server Releases.

# Windows Server release history

The table below shows the history of all monthly security and non-security preview updates released for supported versions of Windows Server in the Long-Term Servicing Channel.

Note that the Update type column references each monthly update by the year and the month of its release, followed by a letter indicating the week of the month (B is the second week of the month; D is the fourth week of the month). Out-of-band updates are referenced as OOB. This shorthand notation aligns with what you see in Microsoft Intune. For more information, see Type of update releases.

Release notes for Windows Server are available on Windows Server 2025 update history, Windows Server 2022 update history, Windows Server 2019 update history, and Windows Server 2016 update history.

**Windows Server 2025 (OS build 26100)**

Expand table

| Servicing option | Update type | Availability date | Build | KB article |
|---|---|---|---|---|
| LTSC | 2026-10 B | 2026-10-13 | 26100.33650 | KB5126646 |
| LTSC | 2026-09 OOB | 2026-09-14 | 26100.33451 | KB5129235 |
| LTSC | 2026-09 B | 2026-09-08 | 26100.33438 | KB5122871 |
| LTSC | 2026-08 B | 2026-08-11 | 26100.33296 | KB5120233 |
| LTSC | 2026-07 B | 2026-07-14 | 26100.33158 | KB5099536 |
| LTSC | 2026-06 B | 2026-06-09 | 26100.32995 | KB5094125 |
| LTSC | 2026-05 B | 2026-05-12 | 26100.32860 | KB5087539 |
| LTSC | 2026-04 OOB | 2026-04-19 | 26100.32698 | KB5091157 |
| LTSC | 2026-04 B | 2026-04-14 | 26100.32690 | KB5082063 |
| LTSC | 2026-03 B | 2026-03-10 | 26100.32522 | KB5078740 |
| LTSC | 2026-02 B | 2026-02-10 | 26100.32370 | KB5075899 |
| LTSC | 2026-01 OOB | 2026-01-24 | 26100.32236 | KB5078135 |
| LTSC | 2026-01 OOB | 2026-01-17 | 26100.32234 | KB5077793 |

| | | | | |
|---|---|---|---|---|
| LTSC | 2026-01 B | 2026-01-13 | 26100.32230 | KB5073379 |
| LTSC | 2025-12 B | 2025-12-09 | 26100.7462 | KB5072033 |
| LTSC | 2025-11 OOB | 2025-11-18 | 26100.7178 | KB5072359 |
| LTSC | 2025-11 B | 2025-11-11 | 26100.7171 | KB5068861 |
| LTSC | 2025-10 OOB | 2025-10-23 | 26100.6905 | KB5070881 |
| LTSC | 2025-10 OOB | 2025-10-20 | 26100.6901 | KB5070773 |
| LTSC | 2025-10 B | 2025-10-14 | 26100.6899 | KB5066835 |
| LTSC | 2025-09 OOB | 2025-09-22 | 26100.6588 | KB5068221 |
| LTSC | 2025-09 B | 2025-09-09 | 26100.6584 | KB5065426 |
| LTSC | 2025-08 B | 2025-08-12 | 26100.4946 | KB5063878 |
| LTSC | 2025-07 OOB | 2025-07-13 | 26100.4656 | KB5064489 |
| LTSC | 2025-07 B | 2025-07-08 | 26100.4652 | KB5062553 |
| LTSC | 2025-06 B | 2025-06-10 | 26100.4349 | KB5060842 |
| LTSC | 2025-05 OOB | 2025-05-27 | 26100.4066 | KB5061977 |
| LTSC | 2025-05 B | 2025-05-13 | 26100.4061 | KB5058411 |
| LTSC | 2025-04 OOB | 2025-04-16 | 26100.3781 | KB5059087 |
| LTSC | 2025-04 B | 2025-04-08 | 26100.3775 | KB5055523 |
| LTSC | 2025-03 B | 2025-03-11 | 26100.3476 | KB5053598 |
| LTSC | 2025-02 B | 2025-02-11 | 26100.3194 | KB5051987 |
| LTSC | 2025-01 B | 2025-01-14 | 26100.2894 | KB5050009 |
| LTSC | 2024-12 B | 2024-12-10 | 26100.2605 | KB5048667 |
| LTSC | 2024-11 B | 2024-11-12 | 26100.2314 | KB5046617 |
| LTSC | 2024-10 A | 2024-11-01 | 26100.1742 | |

### ▼ Windows Server 2022 (OS build 20348)

Expand table

| Servicing option | Update type | Availability date | Build | KB article |
|---|---|---|---|---|
| LTSC | 2026-10 B | 2026-10-13 | 20348.5761 | KB5126655 |
| LTSC | 2026-09 OOB | 2026-09-14 | 20348.5631 | KB5129237 |
| LTSC | 2026-09 B | 2026-09-08 | 20348.5622 | KB5122882 |
| LTSC | 2026-08 B | 2026-08-11 | 20348.5499 | KB5120242 |

| | | | | |
|---|---|---|---|---|
| LTSC | 2026-07 B | 2026-07-14 | 20348.5386 | KB5099540 |
| LTSC | 2026-06 B | 2026-06-09 | 20348.5256 | KB5094128 |
| LTSC | 2026-05 B | 2026-05-12 | 20348.5139 | KB5087545 |
| LTSC | 2026-04 OOB | 2026-04-19 | 20348.5024 | KB5091575 |
| LTSC | 2026-04 B | 2026-04-14 | 20348.5020 | KB5082142 |
| LTSC | 2026-03 B | 2026-03-10 | 20348.4893 | KB5078766 |
| LTSC | 2026-02 OOB | 2026-03-02 | 20348.4776 | KB5082314 |
| LTSC | 2026-02 B | 2026-02-10 | 20348.4773 | KB5075906 |
| LTSC | 2026-01 OOB | 2026-01-24 | 20348.4651 | KB5078136 |
| LTSC | 2026-01 OOB | 2026-01-17 | 20348.4650 | KB5077800 |
| LTSC | 2026-01 B | 2026-01-13 | 20348.4648 | KB5073457 |
| LTSC | 2025-12 B | 2025-12-09 | 20348.4529 | KB5071547 |
| LTSC | 2025-11 B | 2025-11-11 | 20348.4405 | KB5068787 |
| LTSC | 2025-10 OOB | 2025-10-23 | 20348.4297 | KB5070884 |
| LTSC | 2025-10 B | 2025-10-14 | 20348.4294 | KB5066782 |
| LTSC | 2025-09 B | 2025-09-09 | 20348.4171 | KB5065432 |
| LTSC | 2025-08 B | 2025-08-12 | 20348.4052 | KB5063880 |
| LTSC | 2025-07 B | 2025-07-08 | 20348.3932 | KB5062572 |
| LTSC | 2025-06 B | 2025-06-10 | 20348.3807 | KB5060526 |
| LTSC | 2025-05 OOB | 2025-05-23 | 20348.3695 | KB5061906 |
| LTSC | 2025-05 B | 2025-05-13 | 20348.3692 | KB5058385 |
| LTSC | 2025-04 OOB | 2025-04-16 | 20348.3566 | KB5059092 |
| LTSC | 2025-04 OOB | 2025-04-11 | 20348.3561 | KB5058920 |
| LTSC | 2025-04 B | 2025-04-08 | 20348.3453 | KB5055526 |
| LTSC | 2025-03 B | 2025-03-11 | 20348.3328 | KB5053603 |
| LTSC | 2025-02 B | 2025-02-11 | 20348.3207 | KB5051979 |
| LTSC | 2025-01 OOB | 2025-01-18 | 20348.3095 | KB5052819 |
| LTSC | 2025-01 B | 2025-01-14 | 20348.3091 | KB5049983 |
| LTSC | 2024-12 B | 2024-12-10 | 20348.2966 | KB5048654 |
| LTSC | 2024-11 B | 2024-11-12 | 20348.2849 | KB5046616 |
| LTSC | 2024-10 B | 2024-10-08 | 20348.2762 | KB5044281 |

| | | | | |
|---|---|---|---|---|
| LTSC | 2024-09 B | 2024-09-10 | 20348.2700 | KB5042881 |
| LTSC | 2024-08 B | 2024-08-13 | 20348.2655 | KB5041160 |
| LTSC | 2024-07 B | 2024-07-09 | 20348.2582 | KB5040437 |
| LTSC | 2024-06 OOB | 2024-06-20 | 20348.2529 | KB5041054 |
| LTSC | 2024-06 B | 2024-06-11 | 20348.2527 | KB5039227 |
| LTSC | 2024-05 B | 2024-05-14 | 20348.2461 | KB5037782 |
| LTSC | 2024-04 B | 2024-04-09 | 20348.2402 | KB5036909 |
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| LTSC | 2023-12 B | 2023-12-12 | 20348.2159 | KB5033118 |
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| LTSC | 2023-08 B | 2023-08-08 | 20348.1906 | KB5029250 |
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| LTSC | 2023-03 B | 2023-03-14 | 20348.1607 | KB5023705 |
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| LTSC | 2023-01 B | 2023-01-10 | 20348.1487 | KB5022291 |
| LTSC | 2022-12 OOB | 2022-12-20 | 20348.1368 | KB5022553 |
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| LTSC | 2022-10 OOB | 2022-10-17 | 20348.1131 | KB5020436 |
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| LTSC | 2021-09 B | 2021-09-14 | 20348.230 | KB5005575 |
| LTSC | 2021-08 C | 2021-08-26 | 20348.202 | KB5005104 |

▼ **Windows Server 2019 (OS build 17763)**

Expand table

| Servicing option | Update type | Availability date | Build | KB article |
|---|---|---|---|---|
| LTSC | 2026-10 B | 2026-10-13 | 17763.9376 | KB5126651 |
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| LTSC | 2025-12 OOB | 2025-12-18 | 17763.8148 | KB5074975 |
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| LTSC | 2025-04 OOB | 2025-04-16 | 17763.7249 | KB5059091 |
| LTSC | 2025-04 OOB | 2025-04-11 | 17763.7240 | KB5058922 |
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| LTSC | 2022-05 B | 2022-05-10 | 17763.2928 | KB5013941 |
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| LTSC | 2022-02 B | 2022-02-08 | 17763.2565 | KB5010351 |
| LTSC | 2022-01 C | 2022-01-25 | 17763.2510 | KB5009616 |
| LTSC | 2022-01 OOB | 2022-01-18 | 17763.2458 | KB5010791 |
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| LTSC | 2022-01 OOB | 2022-01-04 | 17763.2369 | KB5010196 |
| LTSC | 2021-12 B | 2021-12-14 | 17763.2366 | KB5008218 |
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| LTSC | 2021-11 B | 2021-11-09 | 17763.2300 | KB5007206 |
| LTSC | 2021-10 C | 2021-10-19 | 17763.2268 | KB5006744 |
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| LTSC | 2021-07 OOB | 2021-07-06 | 17763.2029 | KB5004947 |
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| LTSC | 2021-06 B | 2021-06-08 | 17763.1999 | KB5003646 |
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| LTSC | 2021-03 C | 2021-03-25 | 17763.1852 | KB5000854 |
| LTSC | 2021-03 OOB | 2021-03-18 | 17763.1823 | KB5001638 |
| LTSC | 2021-03 OOB | 2021-03-15 | 17763.1821 | KB5001568 |
| LTSC | 2021-03 B | 2021-03-09 | 17763.1817 | KB5000822 |
| LTSC | 2021-02 C | 2021-02-16 | 17763.1790 | KB4601383 |
| LTSC | 2021-02 B | 2021-02-09 | 17763.1757 | KB4601345 |
| LTSC | 2021-01 C | 2021-01-21 | 17763.1728 | KB4598296 |
| LTSC | 2021-01 B | 2021-01-12 | 17763.1697 | KB4598230 |
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| LTSC | 2020-11 C | 2020-11-19 | 17763.1613 | KB4586839 |
| LTSC | 2020-11 OOB | 2020-11-17 | 17763.1579 | KB4594442 |
| LTSC | 2020-11 B | 2020-11-10 | 17763.1577 | KB4586793 |
| LTSC | 2020-10 C | 2020-10-20 | 17763.1554 | KB4580390 |
| LTSC | 2020-10 B | 2020-10-13 | 17763.1518 | KB4577668 |
| LTSC | 2020-09 C | 2020-09-16 | 17763.1490 | KB4577069 |
| LTSC | 2020-09 B | 2020-09-08 | 17763.1457 | KB4570333 |
| LTSC | 2020-08 C | 2020-08-20 | 17763.1432 | KB4571748 |

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|---|---|---|---|---|
| LTSC | 2020-08 B | 2020-08-11 | 17763.1397 | KB4565349 |
| LTSC | 2020-07 C | 2020-07-21 | 17763.1369 | KB4559003 |
| LTSC | 2020-07 B | 2020-07-14 | 17763.1339 | KB4558998 |
| LTSC | 2020-06 OOB | 2020-06-16 | 17763.1294 | KB4567513 |
| LTSC | 2020-06 B | 2020-06-09 | 17763.1282 | KB4561608 |
| LTSC | 2020-05 B | 2020-05-12 | 17763.1217 | KB4551853 |
| LTSC | 2020-04 C | 2020-04-21 | 17763.1192 | KB4550969 |
| LTSC | 2020-04 B | 2020-04-14 | 17763.1158 | KB4549949 |
| LTSC | 2020-03 OOB | 2020-03-30 | 17763.1132 | KB4554354 |
| LTSC | 2020-03 C | 2020-03-17 | 17763.1131 | KB4541331 |
| LTSC | 2020-03 B | 2020-03-10 | 17763.1098 | KB4538461 |
| LTSC | 2020-02 C | 2020-02-25 | 17763.1075 | KB4537818 |
| LTSC | 2020-02 B | 2020-02-11 | 17763.1039 | KB4532691 |
| LTSC | 2020-01 C | 2020-01-23 | 17763.1012 | KB4534321 |
| LTSC | 2020-01 B | 2020-01-14 | 17763.973 | KB4534273 |
| LTSC | 2019-12 B | 2019-12-10 | 17763.914 | KB4530715 |
| LTSC | 2019-11 B | 2019-11-12 | 17763.864 | KB4523205 |
| LTSC | 2019-10 C | 2019-10-15 | 17763.832 | KB4520062 |
| LTSC | 2019-10 B | 2019-10-08 | 17763.805 | KB4519338 |
| LTSC | 2019-09 OOB | 2019-10-03 | 17763.775 | KB4524148 |
| LTSC | 2019-09 C | 2019-09-24 | 17763.774 | KB4516077 |
| LTSC | 2019-09 OOB | 2019-09-23 | 17763.740 | KB4522015 |
| LTSC | 2019-09 B | 2019-09-10 | 17763.737 | KB4512578 |
| LTSC | 2019-08 C | 2019-08-17 | 17763.720 | KB4512534 |
| LTSC | 2019-08 B | 2019-08-13 | 17763.678 | KB4511553 |
| LTSC | 2019-07 C | 2019-07-22 | 17763.652 | KB4505658 |
| LTSC | 2019-07 B | 2019-07-09 | 17763.615 | KB4507469 |
| LTSC | 2019-06 OOB | 2019-06-26 | 17763.593 | KB4509479 |
| LTSC | 2019-06 C | 2019-06-18 | 17763.592 | KB4501371 |
| LTSC | 2019-06 B | 2019-06-11 | 17763.557 | KB4503327 |
| LTSC | 2019-05 C | 2019-05-21 | 17763.529 | KB4497934 |

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|---|---|---|---|---|
| LTSC | 2019-05 OOB | 2019-05-19 | 17763.504 | KB4505056 |
| LTSC | 2019-05 B | 2019-05-14 | 17763.503 | KB4494441 |
| LTSC | 2019-04 C | 2019-05-03 | 17763.475 | KB4495667 |
| LTSC | 2019-04 OOB | 2019-05-01 | 17763.439 | KB4501835 |
| LTSC | 2019-04 B | 2019-04-09 | 17763.437 | KB4493509 |
| LTSC | 2019-03 D | 2019-04-02 | 17763.404 | KB4490481 |
| LTSC | 2019-03 B | 2019-03-12 | 17763.379 | KB4489899 |
| LTSC | 2019-02 D | 2019-03-01 | 17763.348 | KB4482887 |
| LTSC | 2019-02 B | 2019-02-12 | 17763.316 | KB4487044 |
| LTSC | 2019-01 D | 2019-01-22 | 17763.292 | KB4476976 |
| LTSC | 2019-01 B | 2019-01-08 | 17763.253 | KB4480116 |
| LTSC | 2018-12 OOB | 2018-12-19 | 17763.195 | KB4483235 |
| LTSC | 2018-12 B | 2018-12-11 | 17763.194 | KB4471332 |
| LTSC | 2018-11 D | 2018-12-05 | 17763.168 | KB4469342 |
| LTSC | 2018-11 B | 2018-11-13 | 17763.134 | KB4467708 |
| LTSC | 2018-10 D | 2018-11-13 | 17763.107 | KB4464455 |
| LTSC | 2018-10 B | 2018-10-09 | 17763.55 | KB4464330 |
| LTSC | 2018-10 A | 2018-10-02 | 17763.1 | |

### ▼ Windows Server 2016 (OS build 14393)

Expand table

| Servicing option | Update type | Availability date | Build | KB article |
|---|---|---|---|---|
| LTSB | 2026-10 B | 2026-10-13 | 14393.9612 | KB5126868 |
| LTSB | 2026-09 OOB | 2026-09-14 | 14393.9514 | KB5129239 |
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| LTSB | 2026-07 B | 2026-07-14 | 14393.9339 | KB5099535 |
| LTSB | 2026-06 B | 2026-06-09 | 14393.9234 | KB5094122 |
| LTSB | 2026-05 B | 2026-05-12 | 14393.9140 | KB5087537 |
| LTSB | 2026-04 OOB | 2026-04-19 | 14393.9062 | KB5091572 |
| LTSB | 2026-04 B | 2026-04-14 | 14393.9060 | KB5082198 |

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| LTSB | 2026-03 B | 2026-03-10 | 14393.8957 | KB5078938 |
| LTSB | 2026-02 B | 2026-02-10 | 14393.8868 | KB5075999 |
| LTSB | 2026-01 B | 2026-01-13 | 14393.8783 | KB5073722 |
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| LTSB | 2025-10 B | 2025-10-14 | 14393.8519 | KB5066836 |
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| LTSB | 2025-01 B | 2025-01-14 | 14393.7699 | KB5049993 |
| LTSB | 2024-12 B | 2024-12-10 | 14393.7606 | KB5048671 |
| LTSB | 2024-11 B | 2024-11-12 | 14393.7515 | KB5046612 |
| LTSB | 2024-10 B | 2024-10-08 | 14393.7428 | KB5044293 |
| LTSB | 2024-09 B | 2024-09-10 | 14393.7336 | KB5043051 |
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| LTSB | 2024-07 B | 2024-07-09 | 14393.7159 | KB5040434 |
| LTSB | 2024-06 B | 2024-06-11 | 14393.7070 | KB5039214 |
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| LTSB | 2024-01 B | 2024-01-09 | 14393.6614 | KB5034119 |

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|---|---|---|---|---|
| LTSB | 2023-12 B | 2023-12-12 | 14393.6529 | KB5033373 |
| LTSB | 2023-11 B | 2023-11-14 | 14393.6452 | KB5032197 |
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| LTSB | 2023-09 B | 2023-09-12 | 14393.6252 | KB5030213 |
| LTSB | 2023-08 B | 2023-08-08 | 14393.6167 | KB5029242 |
| LTSB | 2023-07 B | 2023-07-11 | 14393.6085 | KB5028169 |
| LTSB | 2023-06 OOB | 2023-06-23 | 14393.5996 | KB5028623 |
| LTSB | 2023-06 B | 2023-06-13 | 14393.5989 | KB5027219 |
| LTSB | 2023-05 B | 2023-05-09 | 14393.5921 | KB5026363 |
| LTSB | 2023-04 B | 2023-04-11 | 14393.5850 | KB5025228 |
| LTSB | 2023-03 B | 2023-03-14 | 14393.5786 | KB5023697 |
| LTSB | 2023-02 B | 2023-02-14 | 14393.5717 | KB5022838 |
| LTSB | 2023-01 B | 2023-01-10 | 14393.5648 | KB5022289 |
| LTSB | 2022-12 B | 2022-12-13 | 14393.5582 | KB5021235 |
| LTSB | 2022-11 OOB | 2022-11-17 | 14393.5502 | KB5021654 |
| LTSB | 2022-11 B | 2022-11-08 | 14393.5501 | KB5019964 |
| LTSB | 2022-10 OOB | 2022-10-18 | 14393.5429 | KB5020439 |
| LTSB | 2022-10 B | 2022-10-11 | 14393.5427 | KB5018411 |
| LTSB | 2022-09 B | 2022-09-13 | 14393.5356 | KB5017305 |
| LTSB | 2022-08 B | 2022-08-09 | 14393.5291 | KB5016622 |
| LTSB | 2022-07 B | 2022-07-12 | 14393.5246 | KB5015808 |
| LTSB | 2022-06 B | 2022-06-14 | 14393.5192 | KB5014702 |
| LTSB | 2022-05 OOB | 2022-05-19 | 14393.5127 | KB5015019 |
| LTSB | 2022-05 B | 2022-05-10 | 14393.5125 | KB5013952 |
| LTSB | 2022-04 B | 2022-04-12 | 14393.5066 | KB5012596 |
| LTSB | 2022-03 B | 2022-03-08 | 14393.5006 | KB5011495 |
| LTSB | 2022-02 B | 2022-02-08 | 14393.4946 | KB5010359 |
| LTSB | 2022-01 OOB | 2022-01-17 | 14393.4889 | KB5010790 |
| LTSB | 2022-01 B | 2022-01-11 | 14393.4886 | KB5009546 |
| LTSB | 2022-01 OOB | 2022-01-05 | 14393.4827 | KB5010195 |
| LTSB | 2021-12 B | 2021-12-14 | 14393.4825 | KB5008207 |

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|---|---|---|---|---|
| LTSB | 2021-11 OOB | 2021-11-14 | 14393.4771 | KB5008601 |
| LTSB | 2021-11 B | 2021-11-09 | 14393.4770 | KB5007192 |
| LTSB | 2021-10 B | 2021-10-12 | 14393.4704 | KB5006669 |
| LTSB | 2021-09 B | 2021-09-14 | 14393.4651 | KB5005573 |
| LTSB | 2021-08 B | 2021-08-10 | 14393.4583 | KB5005043 |
| LTSB | 2021-07 OOB | 2021-07-29 | 14393.4532 | KB5005393 |
| LTSB | 2021-07 B | 2021-07-13 | 14393.4530 | KB5004238 |
| LTSB | 2021-07 OOB | 2021-07-07 | 14393.4470 | KB5004948 |
| LTSB | 2021-06 B | 2021-06-08 | 14393.4467 | KB5003638 |
| LTSB | 2021-05 B | 2021-05-11 | 14393.4402 | KB5003197 |
| LTSB | 2021-04 B | 2021-04-13 | 14393.4350 | KB5001347 |
| LTSB | 2021-03 OOB | 2021-03-18 | 14393.4288 | KB5001633 |
| LTSB | 2021-03 B | 2021-03-09 | 14393.4283 | KB5000803 |
| LTSB | 2021-02 B | 2021-02-09 | 14393.4225 | KB4601318 |
| LTSB | 2021-01 B | 2021-01-12 | 14393.4169 | KB4598243 |
| LTSB | 2020-12 B | 2020-12-08 | 14393.4104 | KB4593226 |
| LTSB | 2020-11 OOB | 2020-11-19 | 14393.4048 | KB4594441 |
| LTSB | 2020-11 B | 2020-11-10 | 14393.4046 | KB4586830 |
| LTSB | 2020-10 B | 2020-10-13 | 14393.3986 | KB4580346 |
| LTSB | 2020-09 B | 2020-09-08 | 14393.3930 | KB4577015 |
| LTSB | 2020-08 B | 2020-08-11 | 14393.3866 | KB4571694 |
| LTSB | 2020-07 B | 2020-07-14 | 14393.3808 | KB4565511 |
| LTSB | 2020-06 OOB | 2020-06-18 | 14393.3755 | KB4567517 |
| LTSB | 2020-06 B | 2020-06-09 | 14393.3750 | KB4561616 |
| LTSB | 2020-05 B | 2020-05-12 | 14393.3686 | KB4556813 |
| LTSB | 2020-04 C | 2020-04-21 | 14393.3659 | KB4550947 |
| LTSB | 2020-04 B | 2020-04-14 | 14393.3630 | KB4550929 |
| LTSB | 2020-03 C | 2020-03-17 | 14393.3595 | KB4541329 |
| LTSB | 2020-03 B | 2020-03-10 | 14393.3564 | KB4540670 |
| LTSB | 2020-02 C | 2020-02-25 | 14393.3542 | KB4537806 |
| LTSB | 2020-02 B | 2020-02-11 | 14393.3504 | KB4537764 |

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|---|---|---|---|---|
| LTSB | 2020-01 C | 2020-01-23 | 14393.3474 | KB4534307 |
| LTSB | 2020-01 B | 2020-01-14 | 14393.3443 | KB4534271 |
| LTSB | 2019-12 B | 2019-12-10 | 14393.3384 | KB4530689 |
| LTSB | 2019-11 B | 2019-11-12 | 14393.3326 | KB4525236 |
| LTSB | 2019-10 C | 2019-10-15 | 14393.3300 | KB4519979 |
| LTSB | 2019-10 B | 2019-10-08 | 14393.3274 | KB4519998 |
| LTSB | 2019-09 OOB | 2019-10-03 | 14393.3243 | KB4524152 |
| LTSB | 2019-09 C | 2019-09-24 | 14393.3242 | KB4516061 |
| LTSB | 2019-09 OOB | 2019-09-23 | 14393.3206 | KB4522010 |
| LTSB | 2019-09 B | 2019-09-10 | 14393.3204 | KB4516044 |
| LTSB | 2019-08 C | 2019-08-17 | 14393.3181 | KB4512495 |
| LTSB | 2019-08 B | 2019-08-13 | 14393.3144 | KB4512517 |
| LTSB | 2019-07 C | 2019-07-16 | 14393.3115 | KB4507459 |
| LTSB | 2019-07 B | 2019-07-09 | 14393.3085 | KB4507460 |
| LTSB | 2019-06 OOB | 2019-06-27 | 14393.3056 | KB4509475 |
| LTSB | 2019-06 C | 2019-06-18 | 14393.3053 | KB4503294 |
| LTSB | 2019-06 B | 2019-06-11 | 14393.3025 | KB4503267 |
| LTSB | 2019-05 C | 2019-05-23 | 14393.2999 | KB4499177 |
| LTSB | 2019-05 OOB | 2019-05-19 | 14393.2972 | KB4505052 |
| LTSB | 2019-05 B | 2019-05-14 | 14393.2969 | KB4494440 |
| LTSB | 2019-04 C | 2019-04-25 | 14393.2941 | KB4493473 |
| LTSB | 2019-04 OOB | 2019-04-25 | 14393.2908 | KB4499418 |
| LTSB | 2019-04 B | 2019-04-09 | 14393.2906 | KB4493470 |
| LTSB | 2019-03 C | 2019-03-19 | 14393.2879 | KB4489889 |
| LTSB | 2019-03 B | 2019-03-12 | 14393.2848 | KB4489882 |
| LTSB | 2019-02 C | 2019-02-19 | 14393.2828 | KB4487006 |
| LTSB | 2019-02 B | 2019-02-12 | 14393.2791 | KB4487026 |
| LTSB | 2019-01 C | 2019-01-17 | 14393.2759 | KB4480977 |
| LTSB | 2019-01 B | 2019-01-08 | 14393.2724 | KB4480961 |
| LTSB | 2018-12 OOB | 2018-12-19 | 14393.2670 | KB4483229 |
| LTSB | 2018-12 B | 2018-12-11 | 14393.2665 | KB4471321 |

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|---|---|---|---|---|
| LTSB | 2018-11 OOB | 2018-12-03 | 14393.2641 | KB4478877 |
| LTSB | 2018-11 C | 2018-11-27 | 14393.2639 | KB4467684 |
| LTSB | 2018-11 B | 2018-11-13 | 14393.2608 | KB4467691 |
| LTSB | 2018-10 C | 2018-10-18 | 14393.2580 | KB4462928 |
| LTSB | 2018-10 B | 2018-10-09 | 14393.2551 | KB4462917 |
| LTSB | 2018-09 C | 2018-09-20 | 14393.2515 | KB4457127 |
| LTSB | 2018-09 B | 2018-09-11 | 14393.2485 | KB4457131 |
| LTSB | 2018-08 C | 2018-08-30 | 14393.2457 | KB4343884 |
| LTSB | 2018-08 B | 2018-08-14 | 14393.2430 | KB4343887 |
| LTSB | 2018-07 OOB | 2018-07-30 | 14393.2396 | KB4346877 |
| LTSB | 2018-07 C | 2018-07-24 | 14393.2395 | KB4338822 |
| LTSB | 2018-07 OOB | 2018-07-16 | 14393.2368 | KB4345418 |
| LTSB | 2018-07 B | 2018-07-10 | 14393.2363 | KB4338814 |
| LTSB | 2018-06 C | 2018-06-21 | 14393.2339 | KB4284833 |
| LTSB | 2018-06 B | 2018-06-12 | 14393.2312 | KB4284880 |
| LTSB | 2018-05 C | 2018-05-17 | 14393.2273 | KB4103720 |
| LTSB | 2018-05 B | 2018-05-08 | 14393.2248 | KB4103723 |
| LTSB | 2018-04 C | 2018-04-17 | 14393.2214 | KB4093120 |
| LTSB | 2018-04 B | 2018-04-10 | 14393.2189 | KB4093119 |
| LTSB | 2018-03 OOB | 2018-03-29 | 14393.2156 | KB4096309 |
| LTSB | 2018-03 C | 2018-03-22 | 14393.2155 | KB4088889 |
| LTSB | 2018-03 B | 2018-03-13 | 14393.2125 | KB4088787 |
| LTSB | 2018-02 C | 2018-02-22 | 14393.2097 | KB4077525 |
| LTSB | 2018-02 B | 2018-02-13 | 14393.2068 | KB4074590 |
| LTSB | 2018-01 C | 2018-01-17 | 14393.2035 | KB4057142 |
| LTSB | 2018-01 B | 2018-01-03 | 14393.2007 | KB4056890 |
| LTSB | 2017-12 B | 2017-12-12 | 14393.1944 | KB4053579 |
| LTSB | 2017-11 C | 2017-11-27 | 14393.1914 | KB4051033 |
| LTSB | 2017-11 B | 2017-11-14 | 14393.1884 | KB4048953 |
| LTSB | 2017-10 OOB | 2017-11-02 | 14393.1797 | KB4052231 |
| LTSB | 2017-10 C | 2017-10-17 | 14393.1794 | KB4041688 |

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|---|---|---|---|---|
| LTSB | 2017-10 B | 2017-10-10 | 14393.1770 | KB4041691 |
| LTSB | 2017-09 C | 2017-09-28 | 14393.1737 | KB4038801 |
| LTSB | 2017-09 B | 2017-09-12 | 14393.1715 | KB4038782 |
| LTSB | 2017-08 OOB | 2017-08-28 | 14393.1670 | KB4039396 |
| LTSB | 2017-08 C | 2017-08-16 | 14393.1613 | KB4034661 |
| LTSB | 2017-08 B | 2017-08-08 | 14393.1593 | KB4034658 |
| LTSB | 2017-08 OOB | 2017-08-07 | 14393.1537 | KB4038220 |
| LTSB | 2017-07 C | 2017-07-18 | 14393.1532 | KB4025334 |
| LTSB | 2017-07 B | 2017-07-11 | 14393.1480 | KB4025339 |
| LTSB | 2017-06 C | 2017-06-27 | 14393.1378 | KB4022723 |
| LTSB | 2017-06 B | 2017-06-13 | 14393.1358 | KB4022715 |
| LTSB | 2017-05 B | 2017-05-09 | 14393.1198 | KB4019472 |
| LTSB | 2017-04 B | 2017-04-11 | 14393.1066 | KB4015217 |
| LTSB | 2017-03 OOB | 2017-03-20 | 14393.969 | KB4015438 |
| LTSB | 2017-03 B | 2017-03-14 | 14393.953 | KB4013429 |
| LTSB | 2017-01 B | 2017-01-10 | 14393.693 | KB3213986 |
| LTSB | 2016-12 B | 2016-12-13 | 14393.576 | KB3206632 |
| LTSB | 2016-11 C | 2016-12-09 | 14393.479 | KB3201845 |
| LTSB | 2016-11 B | 2016-11-08 | 14393.447 | KB3200970 |
| LTSB | 2016-10 D | 2016-10-27 | 14393.351 | KB3197954 |
| LTSB | 2016-10 B | 2016-10-11 | 14393.321 | KB3194798 |
| LTSB | 2016-09 D | 2016-09-29 | 14393.222 | KB3194496 |
| LTSB | 2016-09 B | 2016-09-20 | 14393.187 | KB3193494 |
| LTSB | 2016-09 B | 2016-09-13 | 14393.187 | KB3189866 |
| LTSB | 2016-08 E | 2016-08-31 | 14393.105 | KB3176938 |
| LTSB | 2016-08 D | 2016-08-23 | 14393.82 | KB3176934 |
| LTSB | 2016-08 B | 2016-08-09 | 14393.51 | KB3176495 |
| LTSB | 2016-08 A | 2016-08-02 | 14393.10 | KB3176929 |

## Windows Server hotpatch calendar

View a calendar of hotpatch updates for Windows Server 2025 and Windows Server 2022. With hotpatching, devices receive a baseline cumulative update the first month of each quarter in the calendar year. A restart is required to install that baseline update. During the next two months, devices receive a hotpatch update, which includes only security updates and can be installed without a restart.

### Windows Server 2025 (OS build 26100)

For details on what is included in each update, see Release notes for Hotpatch on Windows Server 2025 Datacenter Azure Edition.

#### Calendar year 2026

Expand table

| Month | Update type | Type | Availability date | Build | KB article |
|---|---|---|---|---|---|
| January | 2026.01 B | Baseline (Restart) | 2026-01-13 | 26100.32230 | KB5073379 |
| February | 2026.02 B | Hotpatch | 2026-02-10 | 26100.32313 | KB5075942 |
| March | 2026.03 B | Hotpatch | 2026-03-10 | 26100.32463 | KB5078736 |
| April | 2026.04 B | Baseline (Restart) | 2026-04-14 | 26100.32690 | KB5082063 |
| May | 2026.05 B | Hotpatch | 2026-05-12 | 26100.32772 | KB5087423 |
| June | 2026.06 B | Baseline (Restart) | 2026-06-09 | 26100.32995 | KB5094125 |
| July | 2026.07 B | Baseline (Restart) | 2026-07-14 | 26100.33158 | KB5099536 |
| August | 2026.08 B | Hotpatch | 2026-08-11 | 26100.33222 | KB5120228 |
| September | 2026.09 B | Baseline (Restart) | 2026-09-08 | 26100.33438 | KB5122871 |
| October | 2026.10 B | Baseline (Restart) | 2026-10-13 | 26100.33650 | KB5126646 |
| November | 2026.11 B | Hotpatch | | | |
| December | 2026.12 B | Hotpatch | | | |

#### ▼ Calendar year 2025

Expand table

| Month | Update type | Type | Availability date | Build | KB article |
|---|---|---|---|---|---|
| January | 2025.01 B | Baseline (Restart) | 2025-01-14 | 26100.2894 | KB5050009 |
| February | 2025.02 B | Hotpatch | 2025-02-11 | 26100.3107 | KB5052105 |
| March | 2025.03 B | Hotpatch | 2025-03-11 | 26100.3403 | KB5053636 |
| April | 2025.04 B | Baseline (Restart) | 2025-04-08 | 26100.3775 | KB5055523 |

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|---|---|---|---|---|---|
| May | 2025.05 B | Hotpatch | 2025-05-13 | 26100.3981 | KB5058497 |
| June | 2025.06 B | Hotpatch | 2025-06-10 | 26100.4270 | KB5060841 |
| July | 2025.07 B | Baseline (Restart) | 2025-07-08 | 26100.4652 | KB5062553 |
| August | 2025.08 B | Hotpatch | 2025-08-12 | 26100.4851 | KB5064010 |
| September | 2025.09 B | Hotpatch | 2025-09-09 | 26100.6508 | KB5065474 |
| October | 2025.10 B | Baseline (Restart) | 2025-10-14 | 26100.6899 | KB5066835 |
| November | 2025.11 B | Hotpatch | 2025-11-11 | 26100.7092 | KB5068966 |
| December | 2025.12 B | Hotpatch | 2025-12-09 | 26100.7392 | KB5072014 |

### ▼ Calendar year 2024

Expand table

| Month | Update type | Type | Availability date | Build | KB article |
|---|---|---|---|---|---|
| November | 2024.11 B | Hotpatch | 2024-11-12 | 26100.2240 | KB5046696 |
| December | 2024.12 B | Hotpatch | 2024-12-10 | 26100.2528 | KB5048794 |

### ▼ Windows Server 2022 (OS build 20348)

For details on what is included in each update, see Release notes for Hotpatch in Azure Automanage for Windows Server 2022.

### Calendar year 2026

Expand table

| Month | Update type | Type | Availability date | Build | KB article |
|---|---|---|---|---|---|
| January | 2026.01 B | Baseline (Restart) | 2026-01-13 | 20348.4648 | KB5073457 |
| February | 2026.02 B | Hotpatch | 2026-02-10 | 20348.4711 | KB5075943 |
| March | 2026.03 B | Hotpatch | 2026-03-10 | 20348.4830 | KB5078737 |
| April | 2026.04 B | Baseline (Restart) | 2026-04-14 | 20348.5020 | KB5082142 |
| May | 2026.05 B | Hotpatch | 2026-05-12 | 20348.5074 | KB5087424 |
| June | 2026.06 B | Baseline (Restart) | 2026-06-09 | 20348.5256 | KB5094128 |
| July | 2026.07 B | Baseline (Restart) | 2026-07-14 | 20348.5386 | KB5099540 |
| August | 2026.08 B | Hotpatch | 2026-08-11 | 20348.5440 | KB5120229 |
| September | 2026.09 B | Baseline (Restart) | 2026-09-08 | 20348.5622 | KB5122882 |

| | | | | | |
|---|---|---|---|---|---|
| October | 2026.10 B | Baseline (Restart) | 2026-10-13 | 20348.5761 | KB5126655 |
| November | 2026.11 B | Baseline (Restart) | | | |
| December | 2026.12 B | Baseline (Restart) | | | |

### ▼ Calendar year 2025

Expand table

| Month | Update type | Type | Availability date | Build | KB article |
|---|---|---|---|---|---|
| January | 2025.01 B | Baseline (Restart) | 2025-01-14 | 20348.3091 | KB5049983 |
| February | 2025.02 B | Hotpatch | 2025-02-11 | 20348.3148 | KB5052106 |
| March | 2025.03 B | Hotpatch | 2025-03-11 | 20348.3270 | KB5053638 |
| April | 2025.04 B | Baseline (Restart) | 2025-04-08 | 20348.3453 | KB5055526 |
| May | 2025.05 B | Hotpatch | 2025-05-13 | 20348.3630 | KB5058500 |
| June | 2025.06 B | Hotpatch | 2025-06-10 | 20348.3745 | KB5060525 |
| July | 2025.07 B | Baseline (Restart) | 2025-07-08 | 20348.3932 | KB5062572 |
| August | 2025.08 B | Hotpatch | 2025-08-12 | 20348.3989 | KB5063812 |
| September | 2025.09 B | Hotpatch | 2025-09-09 | 20348.4106 | KB5065306 |
| October | 2025.10 B | Baseline (Restart) | 2025-10-14 | 20348.4294 | KB5066782 |
| November | 2025.11 B | Hotpatch | 2025-11-11 | 20348.4346 | KB5068840 |
| December | 2025.12 B | Hotpatch | 2025-12-09 | 20348.4467 | KB5071413 |

### ▼ Calendar year 2024

Expand table

| Month | Update type | Type | Availability date | Build | KB article |
|---|---|---|---|---|---|
| January | 2024.01 B | Baseline (Restart) | 2024-01-09 | 20348.2227 | KB5034129 |
| February | 2024.02 B | Hotpatch | 2024-02-13 | 20348.2277 | KB5034860 |
| March | 2024.03 B | Hotpatch | 2024-03-12 | 20348.2333 | KB5035959 |
| April | 2024.04 B | Baseline (Restart) | 2024-04-09 | 20348.2402 | KB5036909 |
| May | 2024.05 B | Hotpatch | 2024-05-14 | 20348.2458 | KB5037848 |
| June | 2024.06 B | Hotpatch | 2024-06-11 | 20348.2522 | KB5039330 |
| July | 2024.07 B | Baseline (Restart) | 2024-07-09 | 20348.2582 | KB5040437 |
| August | 2024.08 B* | Baseline (Restart) | 2024-08-13 | 20348.2655 | KB5041160 |

| | | | | | |
|---|---|---|---|---|---|
| September | 2024.09 B | Hotpatch | 2024-09-10 | 20348.2695 | KB5042880 |
| October | 2024.10 B | Baseline (Restart) | 2024-10-08 | 20348.2762 | KB5044281 |
| November | 2024.11 B | Hotpatch | 2024-11-12 | 20348.2819 | KB5046698 |
| December | 2024.12 B | Hotpatch | 2024-12-10 | 20348.2908 | KB5048800 |

Last updated on 10/13/2026

# Windows Server 2025 known issues and notifications

Find information on known issues and the servicing status for Windows Server 2025. For immediate help with Windows update issues, click here if you are using a Windows device to open the Get Help app or go to support.microsoft.com. Follow @WindowsUpdate on X for Windows release health updates. If you are an IT administrator and want to programmatically get information from this page, use the Windows Updates API in Microsoft Graph.

**Current status as of November 6, 2024**

Windows Server 2025 is now generally available. It delivers security advancements and new hybrid cloud capabilities in a high performing, AI-capable platform. Windows Server 2025 is Microsoft's latest Long-Term Servicing Channel (LTSC) release for Windows Server. To download a free 180-day evaluation, visit the Microsoft Evaluation Center.

Windows Server 2025 is offered as an **Optional** update for Windows Server 2022 and Window Server 2019 devices, if organizations want to do an in-place upgrade. We recommend you use these methods to deploy Windows Server feature updates as Windows Server 2025 is not automatically installed.

To learn more about Windows Server's Lifecycle Policy, see the Windows Server 2025 lifecycle article.

### What is new in Windows Server 2025

Check its advanced features and innovations

### Get Windows Server known issues in Graph API

Data is available for supported Windows Server versions

See all messages >

## Known issues

See open issues, content updated in the last 30 days, and information on safeguard holds. To find a specific issue, use the search function on your browser (CTRL + F for Microsoft Edge).

Expand table

| Summary | Originating update | Status | Last updated |
|---|---|---|---|
| **Incorrect notifications that "Microsoft Defender Antivirus is turned off"**<br>Microsoft Defender Antivirus remains active and functioning correctly despite notifications following the latest update. | N/A | Resolved | 2026-09-30 16:42 PT |
| **USB audio devices might fail to start or produce no sound**<br>Some USB Audio Class 1.0 devices display Code 10 or fail in multichannel audio modes after installing the Sept. 8 update | OS Build 26100.33438<br>KB5122871<br>2026-09-08 | Mitigated | 2026-09-29 10:12 PT |
| **Remote Desktop Services might stop responding after Sept. 2026 security update**<br>Some Windows devices with Remote Desktop enabled might experience Remote Desktop Services (RDS) instability. | OS Build 26100.33438<br>KB5122871<br>2026-09-08 | Resolved<br>KB5129235 | 2026-09-17 19:22 PT |

# Issue details

## September 2026

### USB audio devices might fail to start or produce no sound

Expand table

| Status | Originating update | History |
|---|---|---|
| Mitigated | OS Build 26100.33438<br>KB5122871<br>2026-09-08 | Last updated: 2026-09-29, 10:12 PT<br>Opened: 2026-09-25, 18:40 PT |

After installing the September 8, 2026, Windows security update (KB5122871), some USB Audio Class 1.0 devices might fail to start or produce audio. Affected devices might experience one or more of the following symptoms:

- The device displays an error in Device Manager: "This device cannot start (Code 10)."
- No audio output.

- Volume controls are unresponsive or remain at zero.
- Sound settings are unresponsive or unavailable.
- Some devices might function in standard stereo configurations but fail when using multichannel audio features, including 8-channel or 3D audio modes. Some customers have reported that they're able to restore audio in these cases by switching to 2-channel mode. *(*Note: This behavior is resolved in the out-of-band update released on September 14, 2026. See **Resolution** section below.)*

This issue is limited to USB Audio Class 1.0 devices.

**Microsoft Support:** IT administrators who need an immediate workaround for the symptoms not addressed yet by the OOB update should contact Microsoft Support for Business for assistance.

**Resolution:** This issue is partially resolved in the out-of-band (OOB) update released on September 14, 2026, (KB5129235), and updates released after this date. This OOB update resolves the symptoms experienced on devices using 8-channel or 3D audio modes (last bullet point in the list of symptoms documented in this entry). We are working to address the other symptoms and will update this documentation when more information is available.

**Affected platforms:**

- Client: Windows 11, version 26H2; Windows 11, version 26H1; Windows 11, version 25H2; Windows 11, version 24H2; Windows 11, version 23H2; Windows 10, version 22H2; Windows 10, version 21H2; Windows 10 Enterprise LTSC 2019; Windows 10 Enterprise LTSC 2016
- Server: Windows Server 2025; Windows Server 2022; Windows Server 2019; Windows Server 2016; Windows Server 2012 R2; Windows Server 2012

Back to top

## Remote Desktop Services might stop responding after Sept. 2026 security update

Expand table

| Status | Originating update | History |
|---|---|---|
| Resolved KB5129235 | OS Build 26100.33438<br>KB5122871<br>2026-09-08 | Resolved: 2026-09-14, 10:00 PT<br>Opened: 2026-09-11, 11:19 PT |

After installing the September 2026 Windows security update (KB5122871), some organizations might experience issues with Remote Desktop Services (RDS).

In some environments, RDS might become unstable, resulting in RDP connections failing after several minutes, sign-in issues, or servers hanging at "Please wait for the Remote Desktop Configuration". Related tools,

including Microsoft Management Console (MMC), RDS Licensing Diagnoser, and File Explorer might also become unresponsive. Additionally, the Windows Update page might stop responding and continuously display a loading indicator.

**Note:** This issue does not affect Windows 365 or Azure Virtual Desktop.

**Resolution:** This issue is resolved by the out-of-band (OOB) update released on September 14, 2026 (KB5129235), and in updates released after this date.

IT administrators who deployed a temporary mitigation through Group Policy do not need to take any action before installing this OOB update.

This OOB update is cumulative and includes all improvements and security protections contained in previous Windows updates. As a best practice, we recommend installing the latest update available for your devices, as it contains important improvements and issue resolutions, including this one.

**Affected platforms:**

- Client: Windows 11, version 26H1; Windows 11, version 25H2; Windows 11, version 24H2; Windows 11, version 23H2; Windows 10, version 22H2; Windows 10, version 21H2; Windows 10 Enterprise LTSC 2019; Windows 10 Enterprise LTSC 2016
- Server: Windows Server 2025; Windows Server 2022; Windows Server 2019; Windows Server 2016; Windows Server 2012 R2; Windows Server 2012

Back to top

# August 2026

## Incorrect notifications that "Microsoft Defender Antivirus is turned off"

Expand table

| Status | Originating update | History |
|---|---|---|
| Resolved | N/A | Resolved: 2026-09-30, 16:42 PT<br>Opened: 2026-08-28, 15:34 PT |

After installing Microsoft Defender Antivirus update (version 4.18.26070.9), released on August 5, 2026, notifications might appear stating that "Microsoft Defender Antivirus is turned off," even though the antivirus is functioning correctly and all settings show it as active. These notifications can appear when Windows starts and intermittently afterward. They persist even if notification settings are turned off.

This issue can be observed in any version of Windows or Windows Server with Microsoft Defender Antivirus running with the latest Defender updates.

**Resolution:** This issue was resolved in Microsoft Defender Antivirus update (version 4.18.26080.4), released on September 17, 2026.

**Affected platforms:**

- Client: Windows 11, version 26H1; Windows 11, version 25H2; Windows 11, version 24H2; Windows 11, version 23H2; Windows 10, version 22H2; Windows 10, version 21H2; Windows 10 Enterprise LTSC 2019; Windows 10 Enterprise LTSC 2016
- Server: Windows Server 2025; Windows Server 2022; Windows Server 2019; Windows Server 2016; Windows Server 2012 R2; Windows Server 2012

Back to top

## Report a problem with Windows updates

To report an issue to Microsoft at any time, use the Feedback Hub app. To learn more, see Send feedback to Microsoft with the Feedback Hub app.

## Need help with Windows updates?

Search, browse, or ask a question on the Microsoft Support Community. If you are an IT pro supporting an organization, visit Windows release health on the Microsoft 365 admin center for additional details.

For direct help with your home PC, use the Get Help app in Windows or contact Microsoft Support. Organizations can request immediate support through Support for business.

## View this site in your language

This site is available in 11 languages: English, Chinese Traditional, Chinese Simplified, French (France), German, Italian, Japanese, Korean, Portuguese (Brazil), Russian, and Spanish (Spain). All text will appear in English if your browser default language is not one of the 11 supported languages. To manually change the display language, scroll down to the bottom of this page, click on the current language displayed on the bottom left of the page, and select one of the 11 supported languages from the list.

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Last updated on 10/15/2026

# Resolved issues in Windows Server 2025

Find information on recently resolved issues for Windows Server 2025. To find a specific issue, use the search function on your browser (CTRL + F for Microsoft Edge). For immediate help with Windows update issues, click here if you are using a Windows device to open the Get Help app or go to support.microsoft.com. Follow @WindowsUpdate on X for Windows release health updates. If you are an IT administrator and want to programmatically get information from this page, use the Windows Updates API in Microsoft Graph.

## Resolved issues

Expand table

| Summary | Originating update | Status | Date resolved |
|---|---|---|---|
| **Incorrect notifications that "Microsoft Defender Antivirus is turned off"**<br>Microsoft Defender Antivirus remains active and functioning correctly despite notifications following the latest update. | N/A | Resolved | 2026-09-30 16:42 PT |
| **Remote Desktop Services might stop responding after Sept. 2026 security update**<br>Some Windows devices with Remote Desktop enabled might experience Remote Desktop Services (RDS) instability. | OS Build 26100.33438<br>KB5122871<br>2026-09-08 | Resolved<br>KB5129235 | 2026-09-14 10:00 PT |
| **GIF functionality in the Windows Emoji Panel might become unavailable**<br>This issue occurs because the previous GIF provider (Tenor) retired its service on June 30, 2026. | N/A | Resolved<br>KB5120233 | 2026-08-11 10:00 PT |
| **Windows Server Update Services sync operations might have issues or time out**<br>Remediation efforts are underway.<br>Synchronization times are expected to return to normal as recovery progresses. | N/A | Resolved | 2026-07-20 13:22 PT |

| Summary | Originating update | Status | Date resolved |
| --- | --- | --- | --- |
| **Deleting a file from the Recycle Bin displays an internal filename in the dialog**<br>This issue only occurs within the confirmation dialog after installing the June Windows security update. | OS Build 26100.32995<br>KB5094125<br>2026-06-09 | Resolved<br>KB5099536 | 2026-07-14<br>10:00 PT |
| **Updates installed via WUSA might fail if installed from a shared folder**<br>This issue might occur when installing an update from a network folder containing multiple .msu files. | OS Build 26100.4349<br>KB5060842<br>2025-06-10 | Resolved<br>KB5094125 | 2026-06-09<br>10:00 PT |
| **Domain controllers may restart repeatedly after installing April security update**<br>Affected DCs will experience LSASS crashes. A mitigation is available through Microsoft Support. | OS Build 26100.32690<br>KB5082063<br>2026-04-14 | Resolved<br>KB5091157 | 2026-04-19<br>14:00 PT |
| **Windows Server 2022 and Server 2019 unexpectedly upgraded to Windows Server 2025**<br>This issue has been mitigated. It was observed when updates were managed through some third-party applications. | N/A | Resolved<br>KB5082063 | 2026-04-14<br>10:00 PT |
| **Apps might become unresponsive when saving files to cloud-backed storage**<br>Affected apps include Outlook, which can become unresponsive when accessing PST files stored on Microsoft OneDrive. | OS Build 26100.32230<br>KB5073379<br>2026-01-13 | Resolved<br>KB5078135 | 2026-01-23<br>14:00 PT |
| **Connection and authentication failures in Azure Virtual Desktop and Windows 365**<br>January 2026 Windows update causes Windows App credential prompt failures on Azure Virtual Desktop and Windows 365 | OS Build 26100.32230<br>KB5073379<br>2026-01-13 | Resolved<br>KB5077793 | 2026-01-17<br>14:00 PT |
| **IIS websites might fail to load**<br>Server-side applications that rely on HTTP.sys may experience issues with incoming connections. | OS Build 26100.6899<br>KB5066835<br>2025-10-14 | Resolved<br>KB5068861 | 2025-11-11<br>10:00 PT |
| **Directory synchronization fails for AD groups exceeding 10,000 members**<br>Issue affects Active Directory Domain Services (AD DS) synchronization, including Microsoft Entra Connect Sync | OS Build 26100.6584<br>KB5065426<br>2025-09-09 | Resolved<br>KB5068861 | 2025-11-11<br>10:00 PT |
| **Smartcard authentication issues might occur with the October 2025 Windows update**<br>This issue is related to a security change | OS Build 26100.6899<br>KB5066835<br>2025-10-14 | Resolved | 2025-10-22<br>17:31 PT |

| Summary | Originating update | Status | Date resolved |
|---|---|---|---|
| introduced for strengthening Windows Cryptographic Services. | | | |
| **USB mouse and keyboard not working in the Windows Recovery Environment (WinRE)** This issue affects USB devices only within WinRE after installing Windows updates released on October 14, 2025. | OS Build 26100.6899 KB5066835 2025-10-14 | Resolved KB5070773 | 2025-10-20 14:00 PT |

# Issue details

## September 2026

### Remote Desktop Services might stop responding after Sept. 2026 security update

Expand table

| Status | Originating update | History |
|---|---|---|
| Resolved KB5129235 | OS Build 26100.33438 KB5122871 2026-09-08 | Resolved: 2026-09-14, 10:00 PT Opened: 2026-09-11, 11:19 PT |

After installing the September 2026 Windows security update (KB5122871), some organizations might experience issues with Remote Desktop Services (RDS).

In some environments, RDS might become unstable, resulting in RDP connections failing after several minutes, sign-in issues, or servers hanging at "Please wait for the Remote Desktop Configuration". Related tools, including Microsoft Management Console (MMC), RDS Licensing Diagnoser, and File Explorer might also become unresponsive. Additionally, the Windows Update page might stop responding and continuously display a loading indicator.

**Note:** This issue does not affect Windows 365 or Azure Virtual Desktop.

**Resolution:** This issue is resolved by the out-of-band (OOB) update released on September 14, 2026 (KB5129235), and in updates released after this date.

IT administrators who deployed a temporary mitigation through Group Policy do not need to take any action before installing this OOB update.

This OOB update is cumulative and includes all improvements and security protections contained in previous Windows updates. As a best practice, we recommend installing the latest update available for your devices, as it contains important improvements and issue resolutions, including this one.

**Affected platforms:**

- Client: Windows 11, version 26H1; Windows 11, version 25H2; Windows 11, version 24H2; Windows 11, version 23H2; Windows 10, version 22H2; Windows 10, version 21H2; Windows 10 Enterprise LTSC 2019; Windows 10 Enterprise LTSC 2016
- Server: Windows Server 2025; Windows Server 2022; Windows Server 2019; Windows Server 2016; Windows Server 2012 R2; Windows Server 2012

Back to top

# August 2026

## Incorrect notifications that "Microsoft Defender Antivirus is turned off"

Expand table

| Status | Originating update | History |
|---|---|---|
| Resolved | N/A | Resolved: 2026-09-30, 16:42 PT<br>Opened: 2026-08-28, 15:34 PT |

After installing Microsoft Defender Antivirus update (version 4.18.26070.9), released on August 5, 2026, notifications might appear stating that "Microsoft Defender Antivirus is turned off," even though the antivirus is functioning correctly and all settings show it as active. These notifications can appear when Windows starts and intermittently afterward. They persist even if notification settings are turned off.

This issue can be observed in any version of Windows or Windows Server with Microsoft Defender Antivirus running with the latest Defender updates.

**Resolution:** This issue was resolved in Microsoft Defender Antivirus update (version 4.18.26080.4), released on September 17, 2026.

**Affected platforms:**

- Client: Windows 11, version 26H1; Windows 11, version 25H2; Windows 11, version 24H2; Windows 11, version 23H2; Windows 10, version 22H2; Windows 10, version 21H2; Windows 10 Enterprise LTSC 2019; Windows 10 Enterprise LTSC 2016
- Server: Windows Server 2025; Windows Server 2022; Windows Server 2019; Windows Server 2016; Windows Server 2012 R2; Windows Server 2012

Back to top

# July 2026

## Windows Server Update Services sync operations might have issues or time out

Expand table

| Status | Originating update | History |
| --- | --- | --- |
| Resolved | N/A | Resolved: 2026-07-20, 13:22 PT<br>Opened: 2026-07-17, 18:18 PT |

Microsoft has identified a service degradation affecting Windows Server Update Services (WSUS). Organizations might experience increased synchronization times or sync operation timeouts on WSUS servers. This issue began in recent days, with heightened impact observed starting July 13, 2026, and service-side mitigation on Saturday July 18, 2026. After this day, synchronization times and sync operations are expected to be restored to normal.

This issue is related to a buildup of publishing metadata which is present on existing WSUS server installations as well as from the service side.

**Resolution:** There are two parts to the resolution of this issue.

- Organizations with existing WSUS server installations that are experiencing long sync times can benefit from manual steps in order to clean up unneeded metadata. This metadata is present in existing WSUS installations but can be safely removed. Detailed guidance has been provided at the following KB article: https://support.microsoft.com/help/5121986 .
- On July 18, 2026, Microsoft deployed a service-side mitigation which returns synchronization times and sync operations back to normal for new WSUS installations and rebuilds. After this date, newly installed or rebuilt WSUS servers should not encounter this issue.

**Affected platforms:**

- Client: Windows 11, version 26H1; Windows 11, version 25H2; Windows 11, version 24H2; Windows 11, version 23H2; Windows 10, version 22H2; Windows 10, version 21H2; Windows 10, version 1809; Windows 10 Enterprise LTSC 2019; Windows 10 Enterprise LTSC 2016; Windows 10, version 1607
- Server: Windows Server 2025; Windows Server 2022; Windows Server, version 1809; Windows Server 2019; Windows Server 2016; Windows Server 2012 R2; Windows Server 2012

Back to top

# June 2026

## GIF functionality in the Windows Emoji Panel might become unavailable

Expand table

| Status | Originating update | History |
| --- | --- | --- |
| Resolved KB5120233 | N/A | Resolved: 2026-08-11, 10:00 PT<br>Opened: 2026-06-29, 20:45 PT |

Starting on June 30, 2026, you might not see GIF options in the Windows Emoji Panel on some devices and instead see a message stating "GIF service is not available." This issue occurs because the previous GIF provider (Tenor) retired its service on that date.

**Resolution:** This issue was resolved by Windows updates released August 11, 2026 KB5120233, and later. These updates include a new GIF provider (GIPHY). To restore the GIF functionality in the Windows Emoji Panel, install the latest available Windows update.

**Affected platforms:**

- Client: Windows 11, version 26H1; Windows 11, version 25H2; Windows 11, version 24H2; Windows 11, version 23H2
- Server: Windows Server 2025

Back to top

## Deleting a file from the Recycle Bin displays an internal filename in the dialog

Expand table

| Status | Originating update | History |
| --- | --- | --- |
| Resolved KB5099536 | OS Build 26100.32995<br>KB5094125<br>2026-06-09 | Resolved: 2026-07-14, 10:00 PT<br>Opened: 2026-06-18, 12:03 PT |

When permanently deleting a single item from the Recycle Bin, the confirmation dialog displays the internal Recycle Bin filename (for example, $Rxxxxx.ext) instead of the original filename. The Recycle Bin itself correctly displays the original filename, and restoring the item also restores it using the original filename.

This issue occurs after installing the Windows security update released on June 9, 2026 (KB5094125).

**Resolution:** This issue was resolved by Windows updates released July 14, 2026 (KB5099536), and later. We recommend you install the latest update for your device as it contains important improvements and issue resolutions, including this one. If your organization has installed an update released on July 14, 2026, or later, you do not need to use a Known Issue Rollback (KIR) or a special Group Policy to resolve this issue.

**Affected platforms:**

- Client: Windows 11, version 26H1; Windows 11, version 25H2; Windows 11, version 24H2; Windows 11, version 23H2; Windows 10, version 22H2; Windows 10, version 21H2; Windows 10 Enterprise LTSC 2019; Windows 10, version 1607
- Server: Windows Server 2025; Windows Server 2022; Windows Server 2019; Windows Server 2016; Windows Server 2012 R2; Windows Server 2012

Back to top

# April 2026

## Domain controllers may restart repeatedly after installing April security update

Expand table

| Status | Originating update | History |
|---|---|---|
| Resolved KB5091157 | OS Build 26100.32690<br>KB5082063<br>2026-04-14 | Resolved: 2026-04-19, 14:00 PT<br>Opened: 2026-04-16, 18:50 PT |

After installing the April 2026 Windows security update (KB5082063) and rebooting, domain controllers (DCs) in environments with multiple domains in the forest that use Privileged Access Management (PAM), might experience LSASS crashes during startup. As a result, affected DCs may restart repeatedly, preventing authentication and directory services from functioning, and potentially rendering the domain unavailable.

**Note:** This issue affects Windows Server only. It does not impact consumer PCs or personal devices. The scenario is unlikely to be observed on individual-use devices that are not managed by an IT department.

**Resolution:** This issue is resolved by the out-of-band (OOB) updates released on April 19, 2026.

- The standard OOB update is available as KB5091157. It can be downloaded from the Microsoft Update Catalog. For additional guidance, see How to download updates from the Microsoft Update Catalog.
- If your Windows Server 2025 device is enrolled in hotpatching, you should instead install the OOB hotpatch update KB5091470. This hotpatch OOB update is released through Windows Update and does not require the device to restart.

**Affected versions:**

- Client: None
- Server: Windows Server 2025; Windows Server 2022; Windows Server, version 23H2; Windows Server 2019; Windows Server 2016

Back to top

# January 2026

## Apps might become unresponsive when saving files to cloud-backed storage

Expand table

| Status | Originating update | History |
|---|---|---|
| Resolved KB5078135 | OS Build 26100.32230<br>KB5073379<br>2026-01-13 | Resolved: 2026-01-23, 14:00 PT<br>Opened: 2026-01-20, 22:10 PT |

After installing Windows updates released on or after January 13, 2026 (KB5073379), some applications might become unresponsive or experience unexpected errors when opening files from or saving files to cloud-backed storage, such as OneDrive or Dropbox.

For example, in some configurations of Outlook that store PST files on OneDrive, Outlook might become unresponsive and fail to reopen unless its process is terminated in Task Manager, or the system is restarted. In addition, sent emails might not appear in the Sent Items folder, and previously downloaded emails might be downloaded again. The affected Outlook configurations primarily involve classic Outlook, which is commonly tied to enterprise licensing and is not included with most home installations of Windows. To check your Outlook configuration, see Feature comparison between new Outlook and classic Outlook.

**Resolution**:

This issue is resolved in the out-of-band (OOB) update KB5078135, released on January 24, 2026, and available via the Microsoft Update Catalog, as well as in updates released after this date. We recommend installing the latest update for your device, as it contains important improvements and issue resolutions, including this one.

Note: To download updates from the Microsoft Update Catalog, follow the steps described on this article.

For devices that have not yet installed the OOB update, there is an optional workaround for Outlook-specific scenarios. Moving the Outlook PST files out of OneDrive should resolve this issue. For guidance, please see documentation at How to remove an Outlook .pst data file from OneDrive. In addition, email accounts can still be accessed via webmail, if supported by your email provider.

**Affected platforms:**

- Client: Windows 11, version 25H2; Windows 11, version 24H2; Windows 11, version 23H2; Windows 10, version 22H2; Windows 10 Enterprise LTSC 2021; Windows 10 Enterprise LTSC 2019

- Server: Windows Server 2025; Windows Server, version 23H2; Windows Server 2022; Windows Server 2019

Back to top

## Connection and authentication failures in Azure Virtual Desktop and Windows 365

Expand table

| Status | Originating update | History |
| --- | --- | --- |
| Resolved KB5077793 | OS Build 26100.32230<br>KB5073379<br>2026-01-13 | Resolved: 2026-01-17, 14:00 PT<br>Opened: 2026-01-14, 00:52 PT |

After installing the January 2026 Windows security update (KB5073379), credential prompt failures might occur in some remote connection applications. This includes remote desktop connections using the Windows App on Windows client devices, on Azure Virtual Desktop and Windows 365. The Windows App is affected by this issue on specific Windows builds and can experience sign-in failures.

Other remote connections and related applications may be similarly impacted.

**Resolution:** To address this issue, an out-of-band (OOB) update was released January 17, 2026, in the Microsoft Update Catalog. It can be found as KB5077793.

If you have not yet deployed the January 2026 Windows security update and your IT environment includes the affected applications and features, we recommend applying this OOB update instead. For additional guidance, see How to download updates from the Microsoft Update Catalog. As always, we recommend you install the latest update for your device as it contains important improvements and issue resolutions including this one.

If the OOB is not installed, one of the following connection options can be used as a temporary workaround:

- Use the Remote Desktop client for Windows to connect to Azure Virtual Desktop here (/previous-versions/remote-desktop-client/whats-new-windows?tabs=windows-msrdc-msi)
- Connect using the Windows App Web Client at windows.cloud.microsoft

**Affected platforms:**

- Client: Windows 11, version 25H2; Windows 11, version 24H2; Windows 11, version 23H2; Windows 10, version 22H2; Windows 10, version 21H2, Windows 10 Enterprise LTSC 2019
- Server: Windows Server 2025; Windows Server 2022; Windows Server 2019

Back to top

# October 2025

## IIS websites might fail to load

Expand table

| Status | Originating update | History |
| --- | --- | --- |
| Resolved KB5068861 | OS Build 26100.6899<br>KB5066835<br>2025-10-14 | Resolved: 2025-11-11, 10:00 PT<br>Opened: 2025-10-16, 16:06 PT |

**After further investigation, we concluded that this issue was not applicable to Windows Server 2025. It affects only Windows 11, version 25H2 and 24H2.**

**This issue can be disregarded by Windows Server users.**

To learn about the impact of this issue for Windows 11, select from the links below:

- Windows 11, version 25H2
- Windows 11, version 24H2

**The issue mentioned below was published prior to this finding and November 14, 2025 edits:**

Following installation of Windows updates releases on or after September 29, server-side applications that rely on HTTP.sys may experience issues with incoming connections. As a result, IIS websites might fail to load, displaying a message such as "Connection reset - error (ERR_CONNECTION_RESET)", or similar error. This includes websites hosted on http://localhost/, and other IIS connections.

**Affected platforms:**

- Client: Windows 11, version 25H2; Windows 11, version 24H2

Back to top

## Directory synchronization fails for AD groups exceeding 10,000 members

Expand table

| Status | Originating update | History |
| --- | --- | --- |
| Resolved KB5068861 | OS Build 26100.6584<br>KB5065426<br>2025-09-09 | Resolved: 2025-11-11, 10:00 PT<br>Opened: 2025-10-14, 17:49 PT |

Applications that use the Active Directory directory synchronization (DirSync) control for on-premises Active Directory Domain Services (AD DS), such as when using Microsoft Entra Connect Sync, can result in incomplete synchronization of large AD groups exceeding 10,000 members. This issue occurs only on Windows Server 2025 after installing the September 2025 Windows security update (KB5065426), or later updates.

**Resolution:** This issue was resolved by Windows updates released November 11, 2025, KB5068861, and updates released after that date. We recommend you install the latest update for your device as it contains important improvements and issue resolutions, including this one.

If you have installed updates released on or after November 11, 2025, (KB5068861), you do not need to use a Known Issue Rollback (KIR) or a special Group Policy to resolve this issue.

If you are using an update released before November 11, 2025, and have this issue, your IT administrator can resolve it by installing and configuring the special Group Policy listed below.

**Group Policy downloads with Group Policy name:**

- Download for Windows 11, Versions 24H2 and 25H2 and Windows Server 2025 -- Windows 11 24H2, Windows 11 25H2 and Windows Server 2025 KB5066835 251016_21401 Known Issue Rollback

The special Group Policy can be found in **Computer Configuration -> Administrative Templates -> Windows 11 24H2, Windows 11 25H2 and Windows Server 2025 KB5066835 251016_21401 Known Issue Rollback.** After installing the group policy, configure the **KB5066835 251016_21401 Known Issue Rollback** value to **Disabled** and restart Windows Server 2025 to apply the group policy setting. (Windows 11 is out of scope of this notification and guidance.) For information on deploying and configuring this special Group Policy, please see How to use Group Policy to deploy a Known Issue Rollback.

Alternatively, affected customers can apply the following registry key as a workaround to disable the feature change.

**Warning**: Serious problems might occur if you modify the registry incorrectly by using Registry Editor or by using another method. These problems might require that you reinstall the operating system. Microsoft cannot guarantee that these problems can be solved. Modify the registry at your own risk. For more information, see Windows registry for advanced users.

**Path:**
Computer\HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Policies\Microsoft\FeatureManagement\Overrides

**Name:** 2362988687

**Type:** REG_DWORD

**Value:** 0

**Affected platforms:**

- Client: None
- Server: Windows Server 2025

Back to top

## Smartcard authentication issues might occur with the October 2025 Windows update

Expand table

| Status | Originating update | History |
| --- | --- | --- |
| Resolved | OS Build 26100.6899<br>KB5066835<br>2025-10-14 | Resolved: 2025-10-22, 17:31 PT<br>Opened: 2025-10-17, 20:06 PT |

Smart card authentication and other certificate operations might intentionally fail after installing Windows Updates released on or after October 14, 2025 (KB5066835) that contain protections for the security vulnerability, CVE-2024-30098. As part of this cryptography improvement, RSA-based smart card certificates are required to use KSP (Key Storage Provider) instead of CSP (Cryptographic Service Provider).

Common symptoms for certificates that use CSP include:

- Smart cards not being recognized as CSP providers (Cryptographic Service Provider) in 32-bit applications
- Inability to sign documents
- Failures in applications relying on certificate-based authentication
- Users might observe error messages such as "invalid provider type specified" and "CryptAcquireCertificatePrivateKey error."

You can detect if your smart card will be affected by this security enforcement if, prior to installing the October 2025 Windows security update (KB5066835), the System log contains Smart Card Service or Microsoft-Windows-Smartcard-Server **Event ID: 624** with the message text: "Audit: This system is using CAPI for RSA cryptography operations. Please refer to the following link for more detail: https://go.microsoft.com/fwlink/?linkid=2300823."

**Resolution:**

For a permanent resolution, developers should update their authenticating app to perform Key Storage Retrieval using Key Storage API documented at Key Storage and Retrieval. Developers should complete this change before Windows updates released in April 2026, at which time the **DisableCapiOverrideForRSA** workaround listed below is planned to be removed.

**Workaround**:

If you encounter this issue, you can temporarily resolve it by setting the **DisableCapiOverrideForRSA** registry key value to 0. This is documented in CVE-2024-30098. Detailed steps to modify the registry key are listed below. **Note**: This option will be removed in Windows updates, planned for release in April 2026.

**Steps to Modify the Registry**

⚠ **Important**: Editing the registry incorrectly can cause system issues. Always back up the registry before making changes.

1. Open Registry Editor.

- Press Win + R, type regedit, and press Enter.
- If prompted by User Account Control, click Yes.

2. Navigate to the subkey.

- Go to: HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Cryptography\Calais

3. Edit the key and set the value.

- Inside Calais, check if key DisableCapiOverrideForRSA exists
- Double-click DisableCapiOverrideForRSA.
- In Value date, enter: 0

**Note**: The DisableCapiOverrideForRSA registry setting is NOT added by the default OS install or the installation of Windows Updates and must be manually added on each device.

4. Close and restart.

- Close Registry Editor.
- Restart the computer for changes to take effect.

**Affected platforms:**

- Client: Windows 11, version 25H2; Windows 11, version 24H2; Windows 11, version 23H2; Windows 11, version 22H2; Windows 10, version 22H2
- Server: Windows Server 2025; Windows Server 23H2; Windows Server 2022; Windows Server 2019; Windows Server 2016; Windows Server 2012 R2

Back to top

## USB mouse and keyboard not working in the Windows Recovery Environment (WinRE)

Expand table

| Status | Originating update | History |
| --- | --- | --- |
| Resolved KB5070773 | OS Build 26100.6899<br>KB5066835<br>2025-10-14 | Resolved: 2025-10-20, 14:00 PT<br>Opened: 2025-10-17, 22:18 PT |

After installing the Windows security update released on October 14, 2025 (KB5066835), USB devices, such as keyboards and mice, do not function in the Windows Recovery Environment (WinRE). This issue prevents navigation of any of the recovery options within WinRE. Note that the USB devices continue to work normally within the Windows operating system.

**Resolution:** This issue was resolved by the Windows out-of-band update, released October 20, 2025 (KB5070773), which is available via the Microsoft Update Catalog, and updates released after that date. We recommend you install the latest update for your device as it contains important improvements and issue resolutions, including this one.

**Workaround:** If your device is impacted by this issue and is unable to boot to Windows to install the latest Windows update, you can work around this issue using one of the following methods:

- If your PC has a touchscreen, you can use the touchscreen's touch keyboard to navigate within WinRE.
- If your PC has a PS/2 port, you can use a PS/2 keyboard or mouse to navigate within WinRE.
- If you had previously created a USB recovery drive, you can boot your computer from the recovery drive. This will take you directly to WinRE with restored USB functionality.
- OEMs and enterprises can use the Preboot Execution Environment (PXE) in Configuration Manager, or can deploy push-button reset features using the Windows Assessment and Deployment Kit (Windows ADK) and Windows Preinstallation Environment (WinPE) add-on to recover affected devices.

**Affected platforms:**

- Client: Windows 11, version 25H2; Windows 11, version 24H2
- Server: Windows Server 2025

Back to top

# August 2025

## Updates installed via WUSA might fail if installed from a shared folder

Expand table

| Status | Originating update | History |
| --- | --- | --- |
| Resolved KB5094125 | OS Build 26100.4349<br>KB5060842 | Resolved: 2026-06-09, 10:00 PT<br>Opened: 2025-08-15, 11:56 PT |

| | |
|---|---|
| | 2025-06-10 |

Windows updates installed using the Windows Update Standalone Installer (WUSA) might fail with error ERROR_BAD_PATHNAME, when the update is installed using WUSA or double-clicking a .msu file from a network share that contains multiple .msu files. These issues might occur on devices that installed updates released May 28, 2025 (KB5060842) and later.

WUSA is a method of installing updates using the Windows Update Agent API which is typically only employed in enterprise environments. It is not common in personal or home settings.

Please note that this issue does not occur when there is only one .msu file in the network share or when the .msu files are stored locally on the device. In addition, after installing an .msu file by double-clicking or using WUSA and restarting Windows, you might notice that the Update History page in Settings continues to indicate that a restart is required to complete the update. This is temporary and should be resolved on its own.

**Resolution**: This issue was resolved by Windows updates released June 9, 2026 (KB5094125), and later. We recommend you install the latest security update for your device as it contains important improvements and issue resolutions, including this one.

If you install an update released June 9, 2026 (KB5094125) or later, you do not need to use a workaround for this issue. If you are using an update released before this date, and are experiencing this issue, you have the option to work around it by saving the .msu files locally on the device and install the update from this location. Also, if you've restarted Windows after installing an .msu file via WUSA, please wait 15 minutes or more before checking the Update History page in Settings. After this short delay, the Settings app should properly indicate if the update installed successfully.

Additionally, this issue was addressed using Known Issue Rollback (KIR) beginning September 2025, and was resolved automatically for most home users and non-managed business devices. Restarting your Windows device might help the resolution apply to your device faster. IT admins can resolve this issue for managed devices which have installed the affected update and encountered this issue. It can be fixed by installing and configuring the Group policy listed below. For information on deploying and configuring these special Group Policy, please see How to use Group Policy to deploy a Known Issue Rollback. The special Group Policy can be found in **Computer Configuration** > **Administrative Templates** > **[Group Policy name].**

**Group Policy downloads with Group Policy name:**

- Download for Windows 11, version 24H2 and Windows Server, version 2025 - Windows 11 24H2 and Windows Server 2025 KB5062660 250806_17201 Known Issue Rollback.msi

**Affected platforms:**

- Client: Windows 11, version 25H2; Windows 11, version 24H2
- Server: Windows Server 2025

Back to top

# November 2024

## Windows Server 2022 and Server 2019 unexpectedly upgraded to Windows Server 2025

Expand table

| Status | Originating update | History |
|---|---|---|
| Resolved KB5082063 | N/A | Resolved: 2026-04-14, 10:00 PT<br>Opened: 2024-11-09, 12:16 PT |

Windows Server 2025 is intended to be offered as an **Optional** upgrade in **Windows Update** settings for devices running Windows Server 2019 and Windows Server 2022. Two scenarios were observed in certain environments:

- Some devices upgraded automatically to Windows Server 2025 (KB5044284). This was observed in environments that use third-party products to manage the update of clients and servers. Please verify whether third-party update software in your environment is configured not to deploy feature updates. This scenario has been mitigated.
- An upgrade to Windows Server 2025 was offered via a message in a banner displayed on the device's **Windows Update** page, under **Settings**. This message is intended for organizations that want to execute an in-place upgrade. This scenario has already been resolved.

The Windows Server 2025 feature update was released as an **Optional** update under the Upgrade Classification: "DeploymentAction=OptionalInstallation". Feature update metadata must be interpreted as **Optional** and not **Recommended** by patch management tools.

We advise organizations to use Microsoft-recommended methods to deploy Windows Server feature updates.

**Resolution**: This issue is resolved and Microsoft has re-enabled the upgrade offer via the **Windows Update** settings panel. For more information on Feature Update from the Windows Server Setting Dialog, see Perform an in-place upgrade of Windows Server at Microsoft Learn.

**Affected platforms:**

- Client: None
- Server: Windows Server 2025; Windows Server 2022; Windows Server 2019

Back to top

# Report a problem with Windows updates

To report an issue to Microsoft at any time, use the Feedback Hub app. To learn more, see Send feedback to Microsoft with the Feedback Hub app.

## Need help with Windows updates?

Search, browse, or ask a question on the Microsoft Support Community. If you are an IT pro supporting an organization, visit Windows release health on the Microsoft 365 admin center for additional details.

For direct help with your home PC, use the Get Help app in Windows or contact Microsoft Support. Organizations can request immediate support through Support for business.

## View this site in your language

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Last updated on 10/15/2026

# Windows Server 2022

known issues and notifications

Find information on known issues and the servicing status for Windows Server 2022. For immediate help with Windows update issues, click here if you are using a Windows device to open the Get Help app or go to support.microsoft.com. Follow @WindowsUpdate on X (formerly Twitter) for Windows release health updates. If you are an IT administrator and want to programmatically get information from this page, use the Windows Updates API in Microsoft Graph.

**Current status as of October 13, 2026**

As of October 13, 2026, Windows Server 2022 has reached end of mainstream support and transitioned to extended support. Windows Server 2022 will continue to receive monthly security updates at no additional cost through October 14, 2031. For more information, see the Fixed Lifecycle Policy.

Hotpatch update support for Windows Server 2022 Datacenter: Azure Edition has been extended through October 2027. Devices enrolled in hotpatch updates will continue to receive monthly security updates without requiring a restart. To learn more, see the Windows Server 2022 lifecycle page.

Windows Server 2025 is the latest Long-Term Servicing Channel (LTSC) release for Windows Server. To download a free 180-day evaluation, visit the Microsoft Evaluation Center.



**What is new in Windows Server 2025**

Check its advanced features and innovations



**Get Windows Server known issues in Graph API**

Data is available for supported Windows Server versions

See all messages >

## Known issues

See open issues, content updated in the last 30 days, and information on safeguard holds. To find a specific issue, use the search function on your browser (CTRL + F for Microsoft Edge).

Expand table

| Summary | Originating update | Status | Last updated |
|---|---|---|---|
| **USB audio devices might fail to start or produce no sound**<br>Some USB Audio Class 1.0 devices display Code 10 or fail in multichannel audio modes after installing the Sept. 8 update | OS Build 20348.5622<br>KB5122882<br>2026-09-08 | Resolved<br>KB5129237 | 2026-10-13<br>11:02 PT |
| **Incorrect notifications that "Microsoft Defender Antivirus is turned off"**<br>Microsoft Defender Antivirus remains active and functioning correctly despite notifications following the latest update. | N/A | Resolved | 2026-09-30<br>16:42 PT |
| **Remote Desktop Services might stop responding after Sept. 2026 security update**<br>Some Windows devices with Remote Desktop enabled might experience Remote Desktop Services (RDS) instability. | OS Build 20348.5622<br>KB5122882<br>2026-09-08 | Resolved<br>KB5129237 | 2026-09-17<br>19:22 PT |

# Issue details

## September 2026

### USB audio devices might fail to start or produce no sound

Expand table

| Status | Originating update | History |
|---|---|---|
| Resolved KB5129237 | OS Build 20348.5622<br>KB5122882<br>2026-09-08 | Resolved: 2026-09-14, 10:00 PT<br>Opened: 2026-09-11, 18:36 PT |

After installing the September 8, 2026, Windows security update (KB5122882), some USB Audio Class 1.0 devices might fail to start or produce audio. Affected devices might experience one or more of the following symptoms:

- The device displays an error in Device Manager: "This device cannot start (Code 10)."
- No audio output.

- Volume controls are unresponsive or remain at zero.
- Sound settings are unresponsive or unavailable.
- Some devices might function in standard stereo configurations but fail when using multichannel audio features, including 8-channel or 3D audio modes. Some customers have reported that they're able to restore audio in these cases by switching to 2-channel mode. *(*Note: This behavior is resolved in the out-of-band update released on September 14, 2026. See **Resolution** section below.)*

This issue is limited to USB Audio Class 1.0 devices.

**Resolution:** The symptoms experienced on devices using 8-channel or 3D audio modes (last bullet point in the list of symptoms documented in this entry) are resolved in the out-of-band (OOB) update released on September 14, 2026, (KB5129237 ), and updates released after this date. The remaining symptoms are resolved in the Windows security update released on October 13, 2026 (KB5126655 ) and later updates. We recommend you install the latest security update for your device as it contains important improvements and issue resolutions, including this one.

If you are an IT administrator that contacted Microsoft Support for Business to obtain an immediate workaround, you do not need to use the Known Issue Rollback (KIR) or special group policy to resolve this issue.

**Affected platforms:**

- Client: Windows 11, version 26H2; Windows 11, version 26H1; Windows 11, version 25H2; Windows 11, version 24H2; Windows 11, version 23H2; Windows 10, version 22H2; Windows 10, version 21H2; Windows 10 Enterprise LTSC 2019; Windows 10 Enterprise LTSC 2016
- Server: Windows Server 2025; Windows Server 2022; Windows Server 2019; Windows Server 2016; Windows Server 2012 R2; Windows Server 2012

Back to top

## Remote Desktop Services might stop responding after Sept. 2026 security update

Expand table

| Status | Originating update | History |
|---|---|---|
| Resolved KB5129237 | OS Build 20348.5622<br>KB5122882<br>2026-09-08 | Resolved: 2026-09-14, 10:00 PT<br>Opened: 2026-09-11, 11:19 PT |

After installing the September 2026 Windows security update (KB5122882 ), some organizations might experience issues with Remote Desktop Services (RDS).

In some environments, RDS might become unstable, resulting in RDP connections failing after several minutes, sign-in issues, or servers hanging at "Please wait for the Remote Desktop Configuration". Related tools, including Microsoft Management Console (MMC), RDS Licensing Diagnoser, and File Explorer might also become unresponsive. Additionally, the Windows Update page might stop responding and continuously display a loading indicator.

**Note:** This issue does not affect Windows 365 or Azure Virtual Desktop.

**Resolution:** This issue is resolved by the out-of-band (OOB) update released on September 14, 2026 (KB5129237), and in updates released after this date.

IT administrators who deployed a temporary mitigation through Group Policy do not need to take any action before installing this OOB update.

This OOB update is cumulative and includes all improvements and security protections contained in previous Windows updates. As a best practice, we recommend installing the latest update available for your devices, as it contains important improvements and issue resolutions, including this one.

**Affected platforms:**

- Client: Windows 11, version 26H1; Windows 11, version 25H2; Windows 11, version 24H2; Windows 11, version 23H2; Windows 10, version 22H2; Windows 10, version 21H2; Windows 10 Enterprise LTSC 2019; Windows 10 Enterprise LTSC 2016
- Server: Windows Server 2025; Windows Server 2022; Windows Server 2019; Windows Server 2016; Windows Server 2012 R2; Windows Server 2012

Back to top

# August 2026

## Incorrect notifications that "Microsoft Defender Antivirus is turned off"

Expand table

| Status | Originating update | History |
|---|---|---|
| Resolved | N/A | Resolved: 2026-09-30, 16:42 PT<br>Opened: 2026-08-28, 15:34 PT |

After installing Microsoft Defender Antivirus update (version 4.18.26070.9), released on August 5, 2026, notifications might appear stating that "Microsoft Defender Antivirus is turned off," even though the antivirus is functioning correctly and all settings show it as active. These notifications can appear when Windows starts and intermittently afterward. They persist even if notification settings are turned off.

This issue can be observed in any version of Windows or Windows Server with Microsoft Defender Antivirus running with the latest Defender updates.

**Resolution:** This issue was resolved in Microsoft Defender Antivirus update (version 4.18.26080.4), released on September 17, 2026.

**Affected platforms:**

- Client: Windows 11, version 26H1; Windows 11, version 25H2; Windows 11, version 24H2; Windows 11, version 23H2; Windows 10, version 22H2; Windows 10, version 21H2; Windows 10 Enterprise LTSC 2019; Windows 10 Enterprise LTSC 2016
- Server: Windows Server 2025; Windows Server 2022; Windows Server 2019; Windows Server 2016; Windows Server 2012 R2; Windows Server 2012

Back to top

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Last updated on 10/15/2026

# Resolved issues in Windows Server 2022

Find information on recently resolved issues for Windows Server 2022. To find a specific issue, use the search function on your browser (CTRL + F for Microsoft Edge). For immediate help with Windows update issues, click here if you are using a Windows device to open the Get Help app or go to support.microsoft.com. Follow @WindowsUpdate on X (formerly Twitter) for Windows release health updates. If you are an IT administrator and want to programmatically get information from this page, use the Windows Updates API in Microsoft Graph.

## Resolved issues

Expand table

| Summary | Originating update | Status | Date resolved |
|---|---|---|---|
| **USB audio devices might fail to start or produce no sound**<br>Some USB Audio Class 1.0 devices display Code 10 or fail in multichannel audio modes after installing the Sept. 8 update | OS Build 20348.5622<br>KB5122882<br>2026-09-08 | Resolved<br>KB5129237 | 2026-09-14<br>10:00 PT |
| **Incorrect notifications that "Microsoft Defender Antivirus is turned off"**<br>Microsoft Defender Antivirus remains active and functioning correctly despite notifications following the latest update. | N/A | Resolved | 2026-09-30<br>16:42 PT |
| **Remote Desktop Services might stop responding after Sept. 2026 security update**<br>Some Windows devices with Remote Desktop enabled might experience Remote Desktop Services (RDS) instability. | OS Build 20348.5622<br>KB5122882<br>2026-09-08 | Resolved<br>KB5129237 | 2026-09-14<br>10:00 PT |
| **Windows Server Update Services sync operations might have issues or time out**<br>Remediation efforts are underway. Synchronization times are expected to return to normal as recovery progresses. | N/A | Resolved | 2026-07-20<br>13:22 PT |

| Summary | Originating update | Status | Date resolved |
|---|---|---|---|
| **Deleting a file from the Recycle Bin displays an internal filename in the dialog**<br>This issue only occurs within the confirmation dialog after installing the June Windows security update. | OS Build 20348.5256<br>KB5094128<br>2026-06-09 | Resolved<br>KB5099540 | 2026-07-14<br>10:00 PT |
| **Domain controllers may restart repeatedly after installing April security update**<br>Affected DCs will experience LSASS crashes. A mitigation is available through Microsoft Support. | OS Build 20348.5020<br>KB5082142<br>2026-04-14 | Resolved<br>KB5091575 | 2026-04-19<br>14:00 PT |
| **Windows Server 2022 and Server 2019 unexpectedly upgraded to Windows Server 2025**<br>This issue has been mitigated. It was observed when updates were managed through some third-party applications. | N/A | Resolved<br>KB5082142 | 2026-04-14<br>10:00 PT |
| **Some devices might fail to shut down or hibernate**<br>This issue affects some Secure Launch-capable PCs with Virtual Secure Mode enabled after installing Jan '26 updates. | OS Build 20348.4648<br>KB5073457<br>2026-01-13 | Resolved<br>KB5075906 | 2026-02-10<br>10:00 PT |
| **Apps might become unresponsive when saving files to cloud-backed storage**<br>Affected apps include Outlook, which can become unresponsive when accessing PST files stored on Microsoft OneDrive. | OS Build 20348.4648<br>KB5073457<br>2026-01-13 | Resolved<br>KB5078136 | 2026-01-23<br>14:00 PT |
| **Connection and authentication failures in Azure Virtual Desktop and Windows 365**<br>January 2026 Windows update causes Windows App credential prompt failures on Azure Virtual Desktop and Windows 365 | OS Build 20348.4648<br>KB5073457<br>2026-01-13 | Resolved<br>KB5077800 | 2026-01-17<br>14:00 PT |
| **Smartcard authentication issues might occur with the October 2025 Windows update**<br>This issue is related to a security change introduced for strengthening Windows Cryptographic Services. | OS Build 20348.4294<br>KB5066782<br>2025-10-14 | Resolved | 2025-10-22<br>17:31 PT |

# Issue details

## September 2026

### USB audio devices might fail to start or produce no sound

Expand table

| Status | Originating update | History |
| --- | --- | --- |
| Resolved KB5129237 | OS Build 20348.5622<br>KB5122882<br>2026-09-08 | Resolved: 2026-09-14, 10:00 PT<br>Opened: 2026-09-11, 18:36 PT |

After installing the September 8, 2026, Windows security update (KB5122882), some USB Audio Class 1.0 devices might fail to start or produce audio. Affected devices might experience one or more of the following symptoms:

- The device displays an error in Device Manager: "This device cannot start (Code 10)."
- No audio output.
- Volume controls are unresponsive or remain at zero.
- Sound settings are unresponsive or unavailable.
- Some devices might function in standard stereo configurations but fail when using multichannel audio features, including 8-channel or 3D audio modes. Some customers have reported that they're able to restore audio in these cases by switching to 2-channel mode. *(*Note: This behavior is resolved in the out-of-band update released on September 14, 2026. See **Resolution** section below.)*

This issue is limited to USB Audio Class 1.0 devices.

**Resolution:** The symptoms experienced on devices using 8-channel or 3D audio modes (last bullet point in the list of symptoms documented in this entry) are resolved in the out-of-band (OOB) update released on September 14, 2026, (KB5129237), and updates released after this date. The remaining symptoms are resolved in the Windows security update released on October 13, 2026 (KB5126655) and later updates. We recommend you install the latest security update for your device as it contains important improvements and issue resolutions, including this one.

If you are an IT administrator that contacted Microsoft Support for Business to obtain an immediate workaround, you do not need to use the Known Issue Rollback (KIR) or special group policy to resolve this issue.

**Affected platforms:**

- Client: Windows 11, version 26H2; Windows 11, version 26H1; Windows 11, version 25H2; Windows 11, version 24H2; Windows 11, version 23H2; Windows 10, version 22H2; Windows 10, version 21H2; Windows 10 Enterprise LTSC 2019; Windows 10 Enterprise LTSC 2016
- Server: Windows Server 2025; Windows Server 2022; Windows Server 2019; Windows Server 2016; Windows Server 2012 R2; Windows Server 2012

Back to top

## Remote Desktop Services might stop responding after Sept. 2026 security update

Expand table

| Status | Originating update | History |
| --- | --- | --- |
| Resolved KB5129237 | OS Build 20348.5622<br>KB5122882<br>2026-09-08 | Resolved: 2026-09-14, 10:00 PT<br>Opened: 2026-09-11, 11:19 PT |

After installing the September 2026 Windows security update (KB5122882), some organizations might experience issues with Remote Desktop Services (RDS).

In some environments, RDS might become unstable, resulting in RDP connections failing after several minutes, sign-in issues, or servers hanging at "Please wait for the Remote Desktop Configuration". Related tools, including Microsoft Management Console (MMC), RDS Licensing Diagnoser, and File Explorer might also become unresponsive. Additionally, the Windows Update page might stop responding and continuously display a loading indicator.

**Note:** This issue does not affect Windows 365 or Azure Virtual Desktop.

**Resolution:** This issue is resolved by the out-of-band (OOB) update released on September 14, 2026 (KB5129237), and in updates released after this date.

IT administrators who deployed a temporary mitigation through Group Policy do not need to take any action before installing this OOB update.

This OOB update is cumulative and includes all improvements and security protections contained in previous Windows updates. As a best practice, we recommend installing the latest update available for your devices, as it contains important improvements and issue resolutions, including this one.

**Affected platforms:**

- Client: Windows 11, version 26H1; Windows 11, version 25H2; Windows 11, version 24H2; Windows 11, version 23H2; Windows 10, version 22H2; Windows 10, version 21H2; Windows 10 Enterprise LTSC 2019; Windows 10 Enterprise LTSC 2016
- Server: Windows Server 2025; Windows Server 2022; Windows Server 2019; Windows Server 2016; Windows Server 2012 R2; Windows Server 2012

Back to top

# August 2026

## Incorrect notifications that "Microsoft Defender Antivirus is turned off"

Expand table

| Status | Originating update | History |
| --- | --- | --- |
| Resolved | N/A | Resolved: 2026-09-30, 16:42 PT<br>Opened: 2026-08-28, 15:34 PT |

After installing Microsoft Defender Antivirus update (version 4.18.26070.9), released on August 5, 2026, notifications might appear stating that "Microsoft Defender Antivirus is turned off," even though the antivirus is functioning correctly and all settings show it as active. These notifications can appear when Windows starts and intermittently afterward. They persist even if notification settings are turned off.

This issue can be observed in any version of Windows or Windows Server with Microsoft Defender Antivirus running with the latest Defender updates.

**Resolution:** This issue was resolved in Microsoft Defender Antivirus update (version 4.18.26080.4), released on September 17, 2026.

**Affected platforms:**

- Client: Windows 11, version 26H1; Windows 11, version 25H2; Windows 11, version 24H2; Windows 11, version 23H2; Windows 10, version 22H2; Windows 10, version 21H2; Windows 10 Enterprise LTSC 2019; Windows 10 Enterprise LTSC 2016
- Server: Windows Server 2025; Windows Server 2022; Windows Server 2019; Windows Server 2016; Windows Server 2012 R2; Windows Server 2012

Back to top

# July 2026

## Windows Server Update Services sync operations might have issues or time out

Expand table

| Status | Originating update | History |
| --- | --- | --- |
| Resolved | N/A | Resolved: 2026-07-20, 13:22 PT<br>Opened: 2026-07-17, 18:18 PT |

Microsoft has identified a service degradation affecting Windows Server Update Services (WSUS). Organizations might experience increased synchronization times or sync operation timeouts on WSUS servers. This issue began in recent days, with heightened impact observed starting July 13, 2026, and service-side mitigation on

Saturday July 18, 2026. After this day, synchronization times and sync operations are expected to be restored to normal.

This issue is related to a buildup of publishing metadata which is present on existing WSUS server installations as well as from the service side.

**Resolution:** There are two parts to the resolution of this issue.

- Organizations with existing WSUS server installations that are experiencing long sync times can benefit from manual steps in order to clean up unneeded metadata. This metadata is present in existing WSUS installations but can be safely removed. Detailed guidance has been provided at the following KB article: https://support.microsoft.com/help/5121986 .
- On July 18, 2026, Microsoft deployed a service-side mitigation which returns synchronization times and sync operations back to normal for new WSUS installations and rebuilds. After this date, newly installed or rebuilt WSUS servers should not encounter this issue.

**Affected platforms:**

- Client: Windows 11, version 26H1; Windows 11, version 25H2; Windows 11, version 24H2; Windows 11, version 23H2; Windows 10, version 22H2; Windows 10, version 21H2; Windows 10, version 1809; Windows 10 Enterprise LTSC 2019; Windows 10 Enterprise LTSC 2016; Windows 10, version 1607
- Server: Windows Server 2025; Windows Server 2022; Windows Server, version 1809; Windows Server 2019; Windows Server 2016; Windows Server 2012 R2; Windows Server 2012

Back to top

# June 2026

## Deleting a file from the Recycle Bin displays an internal filename in the dialog

Expand table

| Status | Originating update | History |
|---|---|---|
| Resolved KB5099540 | OS Build 20348.5256<br>KB5094128<br>2026-06-09 | Resolved: 2026-07-14, 10:00 PT<br>Opened: 2026-06-18, 12:03 PT |

When permanently deleting a single item from the Recycle Bin, the confirmation dialog displays the internal Recycle Bin filename (for example, $Rxxxxx.ext) instead of the original filename. The Recycle Bin itself correctly displays the original filename, and restoring the item also restores it using the original filename.

This issue occurs after installing the Windows security update released on June 9, 2026 (KB5094128 ).

**Resolution:** This issue was resolved by Windows updates released July 14, 2026 (KB5099540), and later. We recommend you install the latest update for your device as it contains important improvements and issue resolutions, including this one. If your organization has installed an update released on July 14, 2026, or later, you do not need to use a Known Issue Rollback (KIR) or a special Group Policy to resolve this issue.

**Affected platforms:**

- Client: Windows 11, version 26H1; Windows 11, version 25H2; Windows 11, version 24H2; Windows 11, version 23H2; Windows 10, version 22H2; Windows 10, version 21H2; Windows 10 Enterprise LTSC 2019; Windows 10, version 1607
- Server: Windows Server 2025; Windows Server 2022; Windows Server 2019; Windows Server 2016; Windows Server 2012 R2; Windows Server 2012

Back to top

# April 2026

## Domain controllers may restart repeatedly after installing April security update

Expand table

| Status | Originating update | History |
|---|---|---|
| Resolved KB5091575 | OS Build 20348.5020<br>KB5082142<br>2026-04-14 | Resolved: 2026-04-19, 14:00 PT<br>Opened: 2026-04-16, 18:50 PT |

After installing the April 2026 Windows security update (KB5082142) and rebooting, domain controllers (DCs) in environments with multiple domains in the forest that use Privileged Access Management (PAM), might experience LSASS crashes during startup. As a result, affected DCs may restart repeatedly, preventing authentication and directory services from functioning, and potentially rendering the domain unavailable.

**Note:** This issue affects Windows Server only. It does not impact consumer PCs or personal devices. The scenario is unlikely to be observed on individual-use devices that are not managed by an IT department.

**Resolution:** This issue is resolved by the out-of-band (OOB) updates released on April 19, 2026.

- The standard OOB update is available as KB5091575. It can be downloaded from the Microsoft Update Catalog. For additional guidance, see How to download updates from the Microsoft Update Catalog.
- If your Windows Server 2022 device is enrolled in hotpatching, you should instead install the OOB hotpatch update KB5091576. This hotpatch OOB update is released through Windows Update and does not require the device to restart.

**Affected versions:**

- Client: None
- Server: Windows Server 2025; Windows Server 2022; Windows Server, version 23H2; Windows Server 2019; Windows Server 2016

Back to top

# January 2026

## Some devices might fail to shut down or hibernate

Expand table

| Status | Originating update | History |
|---|---|---|
| Resolved KB5075906 | OS Build 20348.4648<br>KB5073457<br>2026-01-13 | Resolved: 2026-02-10, 10:00 PT<br>Opened: 2026-01-15, 18:33 PT |

After installing Windows updates released on or after January 13, 2026 (KB5073457), some Secure Launch-capable PCs with Virtual Secure Mode (VSM) enabled are unable to shut down or enter hibernation. Instead, the device restarts. This issue does not affect devices with AMD or ARM64 processors and is limited to the affected platforms listed below.

**Resolution:** This issue was resolved by Windows updates released February 10, 2026 (KB5075906), and updates released after that date. We recommend you install the latest update for your device as it contains important improvements and issue resolutions, including this one.

**Affected platforms:**

- Client: Windows 11, version 23H2; Windows 10, version 22H2; Windows 10 Enterprise LTSC 2021; Windows 10 Enterprise LTSC 2019
- Server: Windows Server 2022; Windows Server 2019

Back to top

## Apps might become unresponsive when saving files to cloud-backed storage

Expand table

| Status | Originating update | History |
|---|---|---|

| | | |
|---|---|---|
| Resolved KB5078136 | OS Build 20348.4648<br>KB5073457<br>2026-01-13 | Resolved: 2026-01-23, 14:00 PT<br>Opened: 2026-01-20, 22:10 PT |

After installing Windows updates released on or after January 13, 2026 (KB5073457), some applications might become unresponsive or experience unexpected errors when opening files from or saving files to cloud-backed storage, such as OneDrive or Dropbox.

For example, in some configurations of Outlook that store PST files on OneDrive, Outlook might become unresponsive and fail to reopen unless its process is terminated in Task Manager, or the system is restarted. In addition, sent emails might not appear in the Sent Items folder, and previously downloaded emails might be downloaded again. The affected Outlook configurations primarily involve classic Outlook, which is commonly tied to enterprise licensing and is not included with most home installations of Windows. To check your Outlook configuration, see Feature comparison between new Outlook and classic Outlook.

**Resolution**:

This issue is resolved in the out-of-band (OOB) update KB5078136, released on January 24, 2026, and available via the Microsoft Update Catalog, as well as in updates released after this date. We recommend installing the latest update for your device, as it contains important improvements and issue resolutions, including this one.

Note: To download updates from the Microsoft Update Catalog, follow the steps described on this article.

For devices that have not yet installed the OOB update, there is an optional workaround for Outlook-specific scenarios. Moving the Outlook PST files out of OneDrive should resolve this issue. For guidance, please see documentation at How to remove an Outlook .pst data file from OneDrive. In addition, email accounts can still be accessed via webmail, if supported by your email provider.

**Affected platforms:**

- Client: Windows 11, version 25H2; Windows 11, version 24H2; Windows 11, version 23H2; Windows 10, version 22H2; Windows 10 Enterprise LTSC 2021; Windows 10 Enterprise LTSC 2019
- Server: Windows Server 2025; Windows Server, version 23H2; Windows Server 2022; Windows Server 2019

Back to top

## Connection and authentication failures in Azure Virtual Desktop and Windows 365

Expand table

| Status | Originating update | History |
|---|---|---|

| | | |
|---|---|---|
| Resolved KB5077800 | OS Build 20348.4648<br>KB5073457<br>2026-01-13 | Resolved: 2026-01-17, 14:00 PT<br>Opened: 2026-01-14, 00:52 PT |

After installing the January 2026 Windows security update (KB5073457), credential prompt failures might occur in some remote connection applications. This includes remote desktop connections using the Windows App on Windows client devices, on Azure Virtual Desktop and Windows 365. The Windows App is affected by this issue on specific Windows builds and can experience sign-in failures.

Other remote connections and related applications may be similarly impacted.

**Resolution:** To address this issue, an out-of-band (OOB) update was released January 17, 2026, in the Microsoft Update Catalog. It can be found as KB5077800.

If you have not yet deployed the January 2026 Windows security update and your IT environment includes the affected applications and features, we recommend applying this OOB update instead. For additional guidance, see How to download updates from the Microsoft Update Catalog. As always, we recommend you install the latest update for your device as it contains important improvements and issue resolutions including this one.

If the OOB is not installed, one of the following connection options can be used as a temporary workaround:

- Use the Remote Desktop client for Windows to connect to Azure Virtual Desktop here (/previous-versions/remote-desktop-client/whats-new-windows?tabs=windows-msrdc-msi)
- Connect using the Windows App Web Client at windows.cloud.microsoft

**Affected platforms:**

- Client: Windows 11, version 25H2; Windows 11, version 24H2; Windows 11, version 23H2; Windows 10, version 22H2; Windows 10, version 21H2, Windows 10 Enterprise LTSC 2019
- Server: Windows Server 2025; Windows Server 2022; Windows Server 2019

Back to top

# October 2025

## Smartcard authentication issues might occur with the October 2025 Windows update

Expand table

| Status | Originating update | History |
|---|---|---|
| Resolved | OS Build 20348.4294<br>KB5066782 | Resolved: 2025-10-22, 17:31 PT<br>Opened: 2025-10-17, 20:06 PT |

| 2025-10-14 |
| --- |

Smart card authentication and other certificate operations might intentionally fail after installing Windows Updates released on or after October 14, 2025 (KB5066782 ) that contain protections for the security vulnerability, CVE-2024-30098 . As part of this cryptography improvement, RSA-based smart card certificates are required to use KSP (Key Storage Provider) instead of CSP (Cryptographic Service Provider).

Common symptoms for certificates that use CSP include:

- Smart cards not being recognized as CSP providers (Cryptographic Service Provider) in 32-bit applications
- Inability to sign documents
- Failures in applications relying on certificate-based authentication
- Users might observe error messages such as "invalid provider type specified" and "CryptAcquireCertificatePrivateKey error."

You can detect if your smart card will be affected by this security enforcement if, prior to installing the October 2025 Windows security update (KB5066782 ), the System log contains Smart Card Service or Microsoft-Windows-Smartcard-Server **Event ID: 624** with the message text: "Audit: This system is using CAPI for RSA cryptography operations. Please refer to the following link for more detail: https://go.microsoft.com/fwlink/?linkid=2300823 ."

**Resolution:**

For a permanent resolution, developers should update their authenticating app to perform Key Storage Retrieval using Key Storage API documented at Key Storage and Retrieval. Developers should complete this change before Windows updates released in April 2026, at which time the **DisableCapiOverrideForRSA** workaround listed below is planned to be removed.

**Workaround**:

If you encounter this issue, you can temporarily resolve it by setting the **DisableCapiOverrideForRSA** registry key value to 0. This is documented in CVE-2024-30098 . Detailed steps to modify the registry key are listed below. **Note**: This option will be removed in Windows updates, planned for release in April 2026.

**Steps to Modify the Registry**

⚠ **Important**: Editing the registry incorrectly can cause system issues. Always back up the registry before making changes.

1. Open Registry Editor.

- Press Win + R, type regedit, and press Enter.
- If prompted by User Account Control, click Yes.

2. Navigate to the subkey.

- Go to: HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Cryptography\Calais

3. Edit the key and set the value.

- Inside Calais, check if key DisableCapiOverrideForRSA exists
- Double-click DisableCapiOverrideForRSA.
- In Value date, enter: 0

**Note**: The DisableCapiOverrideForRSA registry setting is NOT added by the default OS install or the installation of Windows Updates and must be manually added on each device.

4. Close and restart.

- Close Registry Editor.
- Restart the computer for changes to take effect.

**Affected platforms:**

- Client: Windows 11, version 25H2; Windows 11, version 24H2; Windows 11, version 23H2; Windows 11, version 22H2; Windows 10, version 22H2
- Server: Windows Server 2025; Windows Server 23H2; Windows Server 2022; Windows Server 2019; Windows Server 2016; Windows Server 2012 R2

Back to top

# November 2024

## Windows Server 2022 and Server 2019 unexpectedly upgraded to Windows Server 2025

Expand table

| Status | Originating update | History |
|---|---|---|
| Resolved KB5082142 | N/A | Resolved: 2026-04-14, 10:00 PT<br>Opened: 2024-11-09, 12:16 PT |

Windows Server 2025 is intended to be offered as an **Optional** upgrade in **Windows Update** settings for devices running Windows Server 2019 and Windows Server 2022. Two scenarios were observed in certain environments:

- Some devices upgraded automatically to Windows Server 2025 (KB5044284). This was observed in environments that use third-party products to manage the update of clients and servers. Please verify whether third-party update software in your environment is configured not to deploy feature updates. This scenario has been mitigated.

- An upgrade to Windows Server 2025 was offered via a message in a banner displayed on the device's **Windows Update** page, under **Settings**. This message is intended for organizations that want to execute an in-place upgrade. This scenario has already been resolved.

The Windows Server 2025 feature update was released as an **Optional** update under the Upgrade Classification: "DeploymentAction=OptionalInstallation". Feature update metadata must be interpreted as **Optional** and not **Recommended** by patch management tools.

We advise organizations to use Microsoft-recommended methods to deploy Windows Server feature updates.

**Resolution**: This issue is resolved and Microsoft has re-enabled the upgrade offer via the **Windows Update** settings panel. For more information on Feature Update from the Windows Server Setting Dialog, see Perform an in-place upgrade of Windows Server at Microsoft Learn.

**Affected platforms:**

- Client: None
- Server: Windows Server 2025; Windows Server 2022; Windows Server 2019

Back to top

# Report a problem with Windows updates

To report an issue to Microsoft at any time, use the Feedback Hub app. To learn more, see Send feedback to Microsoft with the Feedback Hub app.

# Need help with Windows updates?

Search, browse, or ask a question on the Microsoft Support Community. If you are an IT pro supporting an organization, visit Windows release health on the Microsoft 365 admin center for additional details.

For direct help with your home PC, use the Get Help app in Windows or contact Microsoft Support. Organizations can request immediate support through Support for business.

# View this site in your language

This site is available in 11 languages: English, Chinese Traditional, Chinese Simplified, French (France), German, Italian, Japanese, Korean, Portuguese (Brazil), Russian, and Spanish (Spain). All text will appear in English if your browser default language is not one of the 11 supported languages. To manually change the display language, scroll down to the bottom of this page,

click on the current language displayed on the bottom left of the page, and select one of the 11 supported languages from the list.

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Last updated on 10/15/2026

# Plan your Windows Server upgrade

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

Windows Server supports several methods for moving to a newer version, including in-place upgrade, clean install, migration, cluster rolling upgrade, and edition conversion. Each method has different trade-offs for downtime, complexity, and hardware requirements. Upgrading keeps your servers secure, supported, and able to use the latest features and performance improvements.

This article explains the available upgrade methods, the supported in-place upgrade paths by version, and the restrictions that apply.

## Factors for choosing an upgrade method

The best method depends on your environment and requirements. Consider the following factors when planning your upgrade:

- **Downtime tolerance** — An in-place upgrade requires a reboot but keeps you on the same hardware. A cluster rolling upgrade maintains workload availability throughout the process. A clean install or migration might require more downtime but gives you a fresh environment.
- **Hardware constraints** — If you're staying on the same hardware, an in-place upgrade or clean install is appropriate. If you're moving to new hardware, migration is the better choice.
- **Roles and features** — Not all roles support in-place upgrade. Check the role and feature migration matrix to verify support for your configuration.
- **Version gap** — Starting with Windows Server 2025, nonclustered systems can upgrade up to four versions at a time. For Windows Server 2022 and earlier, nonclustered systems can upgrade up to two versions at a time. Cluster rolling upgrades can only advance one version at a time. See the supported paths table for details.
- **Licensing** — Before you upgrade, ensure you have a valid product key and activation method for the target version. Unlike Windows client, each Windows Server upgrade requires a separate license.

## Windows Server upgrade methods

Use the following table to decide which method fits your scenario.

> **Important**
>
> Always back up your system and important files before performing an in-place upgrade, clean install, or migration to a later version of Windows Server.

Expand table

| Method | What it does | When to use it | More information |
|---|---|---|---|
| **In-place upgrade** | Installs a newer version of Windows Server over the existing one, keeping your settings, server roles, features, and data. You can perform an in-place upgrade by using installation media or receive it as a feature update through Windows Update. | You want the fastest path to a newer version on the same hardware, and your roles and features support in-place upgrade. | Perform an in-place upgrade |
| **Clean install** | Installs Windows Server on a new server or overwrites the existing OS. | You want a fresh start, your hardware is new, or an in-place upgrade isn't supported for your current configuration. | Install Windows Server |
| **Migration** | Moves roles or features from a source server to a different destination server running Windows Server. | You're moving to new hardware, or you need to migrate one role or feature at a time without upgrading the OS in-place. | Upgrade and migrate roles and features |
| **Cluster OS rolling upgrade** | Upgrades cluster node operating systems one at a time without stopping Hyper-V or Scale-Out File Server workloads. Clusters can only upgrade one version at a time. You can perform a rolling upgrade by using installation media or receive it as a feature update through Windows Update. For failover clusters running on Azure Local, use the Lifecycle Manager (LCM) instead. | You're running a failover cluster and need to maintain availability during the upgrade. | Cluster OS rolling upgrade |
| **License conversion** | Converts one edition of Windows Server to another edition of the same release by using a command and a product key (for example, Standard to Datacenter). | You need to change your Windows Server edition or switch between retail, volume- | Convert Windows Server editions and license types |

| Method | What it does | When to use it | More information |
| --- | --- | --- | --- |
| | | licensed, and OEM licenses. | |

# Supported in-place upgrade paths by version

Upgrade to the latest version of Windows Server to get the latest features, security updates, and best performance.

Starting with Windows Server 2025, nonclustered systems can upgrade up to four versions at a time. You can upgrade directly to Windows Server 2025 from Windows Server 2012 R2 and later. For Windows Server 2022 and earlier, nonclustered systems can upgrade up to two versions at a time. If you're using a Cluster OS rolling upgrade, you can only upgrade one version at a time.

For step-by-step instructions for how to perform an in-place upgrade, see Perform an in-place upgrade of Windows Server.

Select the tab for your upgrade method to see the supported paths:

Installation media

Expand table

| Upgrade from / to | Windows Server 2012 R2 | Windows Server 2016 | Windows Server 2019 | Windows Server 2022 | Windows Server 2025 |
| --- | --- | --- | --- | --- | --- |
| **Windows Server 2012** | Yes | Yes | No | No | No |
| **Windows Server 2012 R2** | No | Yes | Yes | No | Yes |
| **Windows Server 2016** | No | No | Yes | Yes | Yes |
| **Windows Server 2019** | No | No | No | Yes | Yes |
| **Windows Server 2022** | No | No | No | No | Yes |

| Upgrade from / to | Windows Server 2012 R2 | Windows Server 2016 | Windows Server 2019 | Windows Server 2022 | Windows Server 2025 |
|---|---|---|---|---|---|
| **Windows Server 2025** | No | No | No | No | Yes |

## Upgrade restrictions for licensed versions of Windows Server

The following restrictions apply to in-place upgrades where Windows Server is already licensed (not an evaluation version):

- Upgrades from 32-bit to 64-bit architectures aren't supported. All releases since Windows Server 2008 R2 are 64-bit only.
- Upgrades from one language to another aren't supported.
- If the server is an Active Directory domain controller, you can't convert it to a retail version. See Upgrade domain controllers to Windows Server for important information.
- Upgrades from prerelease versions (previews) of Windows Server aren't supported. Perform a clean install instead.
- Switching from a Server Core installation to a Server with Desktop Experience installation (or vice versa) during an in-place upgrade isn't supported.
- Upgrades from a previous Windows Server installation to an evaluation copy aren't supported. Install evaluation versions as clean installs.
- By default, an upgrade retains the existing edition. For example, Standard upgrades to Standard and Datacenter upgrades to Datacenter.
- You can change from Standard to Datacenter or Datacenter: Azure Edition, or from Datacenter to Datacenter: Azure Edition, during an upgrade. You can't downgrade from Datacenter to Standard, or from Datacenter: Azure Edition to Standard or Datacenter.
- If your server uses NIC Teaming, disable NIC Teaming before the in-place upgrade. You can re-enable it when the in-place upgrade is complete. For more information, see NIC Teaming overview.
- In-place upgrades on Windows Server configured to boot from VHD aren't supported.
- In-place upgrades from Windows Storage Server editions aren't supported.
- Some public and private cloud providers support in-place upgrades. Check with your cloud provider for details.

## End-of-support Windows Server versions

Support for Windows Server 2012 and Windows Server 2012 R2 ended October 10, 2023. Extended Security Updates (ESU) are available, with one option to migrate your on-premises servers to Azure, where you can continue to run them on virtual machines. For more information, see Extended Security Updates overview.

## Related content

- Perform an in-place upgrade of Windows Server
- Convert Windows Server editions and license types
- Upgrade and migrate roles and features in Windows Server
- Cluster OS rolling upgrade
- Cluster-Aware Updating overview

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Last updated on 04/14/2026

# Server Core vs Server with Desktop Experience install options

Article • 04/15/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

When you install Windows Server using the setup wizard, you can choose between Server Core or Server with Desktop Experience install options. With Server Core, the standard graphical user interface (the Desktop Experience) isn't installed; you manage the server from the command line using PowerShell, the Server Configuration tool (SConfig), or by remote methods. Server with Desktop Experience installs the standard graphical user interface and all tools, including client experience features.

We recommend that you choose the Server Core install option unless you have a particular need for the extra user interface elements and graphical management tools that are included in the Server with Desktop Experience install option.

The setup wizard lists the install options. In this list, editions without **Desktop Experience** are the Server Core install options:

- Windows Server Standard
- Windows Server Standard with Desktop Experience
- Windows Server Datacenter
- Windows Server Datacenter with Desktop Experience

> **Note**
>
> Unlike some previous releases of Windows Server, you can't convert between Server Core and Server with Desktop Experience after installation. You'll need to do a clean installation if you install later decide to use a different option.

## Differences

There are some key differences between Server Core and Server with Desktop Experience:

Expand table

| Component | Server Core | Server with Desktop Experience |
|---|---|---|
| User interface | Minimal, command line driven (PowerShell, SConfig, cmd) | Standard Windows graphical user interface |

| Component | Server Core | Server with Desktop Experience |
| --- | --- | --- |
| Disk space | Smaller requirement | Larger requirement |
| Install, configure, uninstall server roles locally | PowerShell | Server Manager or PowerShell |
| Roles and Features | Some roles and features aren't available. For more information, see Roles, Role Services, and Features not in Windows Server - Server Core.<br><br>Some of the features from Server with Desktop Experience for application compatibility can be installed with the App Compatibility Feature on Demand (FOD). | All roles and features are available, including those for application compatibility. |
| Remote management | Yes, can be managed remotely using GUI tools, such as Windows Admin Center, Remote Server Administration Tools (RSAT), or Server Manager, or by PowerShell. | Yes, can be managed remotely using GUI tools, such as Windows Admin Center, Remote Server Administration Tools (RSAT), or Server Manager, or by PowerShell. |
| Potential attack surface | Greatly reduced attack surface | No reduction |
| Microsoft Management Console | Not installed - can be installed with the App Compatibility Feature on Demand (FOD). | Installed |

> **Note**
>
> For RSAT, you must use the version included with Windows 10 or later.

# Windows Server servicing channels

06/25/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

Beginning in September 2023 Windows Server has two primary release channels available, the Long-Term Servicing Channel and the Annual Channel. The Long-Term Servicing Channel (LTSC) provides a longer term option focuses on providing a traditional lifecycle of quality and security updates, whereas the Annual Channel (AC) provides more frequent releases. The more frequent releases of the AC deliver faster innovation, focusing on containers and microservices.

## Long-Term Servicing Channel (LTSC)

With the Long-Term Servicing Channel, a new major version of Windows Server is typically released every 2-3 years. Users are entitled to five years of mainstream support and five years of extended support. This channel provides systems with a long servicing option and consistency, and can be installed with Server Core or Server with Desktop Experience installation options.

## Annual Channel (AC)

Windows Server Annual Channel for Containers is an operating system to host Windows Server containers. The Annual Channel enables customers who are innovating quickly to take advantage of new operating system capabilities at a faster pace, focused on containers and microservices. To learn more about Windows Server Annual Channel for Containers, see our TechCommunity announcement.

Each release in this channel is supported for 24 months from the initial release. This channel can only be installed with the Server Core installation option. The Annual Channel is available to volume-licensed customers with Software Assurance and loyalty programs such as Visual Studio Subscriptions.

An Annual Channel release isn't an update, it's the next Windows Server release in the Annual Channel. To move to an Annual Channel release, you must perform a clean installation.

Releases of Windows Server in the Annual Channel typically occur every 12 months. The 24 month support lifecycle for each release is 18 months of mainstream support, plus 6 months of extended support. To learn more about the lifecycle, see Windows Server 2022 lifecycle. Each release is named based on the release cycle; for example, **version 23H2** is a release in the second half of the year 2023.

# Key differences

The following table summarizes the key differences between the channels:

Expand table

| Description | Long-Term Servicing Channel | Annual Channel |
|---|---|---|
| Recommended scenarios | General purpose file servers, Microsoft and non-Microsoft workloads, traditional apps, infrastructure roles, software-defined Datacenter, and hyper-converged infrastructure | Containerized applications running on container hosts benefiting from faster innovation |
| New releases | Typically 2–3 years | Typically 12 months |
| Support | 5 years of mainstream support, plus 5 years of extended support | 18 months of mainstream support, plus 6 months of extended support |
| Activation | All Windows Server activation keys | Windows Server Datacenter activation keys |
| Licensing | All licensing programs | Software Assurance customers only |
| Get media | All distribution channels | Volume Licensing Service Center (VLSC) and Visual Studio Subscriptions only |
| Installation options | Server Core and Server with Desktop Experience | Server Core for a container host only |

# Device compatibility

The minimum hardware requirements to run the Annual Channel releases are the same as the most recent Long-Term Servicing Channel release of Windows Server. Most hardware drivers continue to function in these releases.

# Servicing

Both the Long-Term Servicing Channel and the Annual Channel releases are supported with security updates and nonsecurity updates up to the dates listed in the Microsoft Lifecycle pages. The difference is the length of time that the release is supported, as described in the Annual Channel (AC) section of this article.

## Servicing tools

There are many tools with which you can service Windows Server. Each option has its pros and cons, ranging from capabilities and control to simplicity and low administrative requirements. The following are examples of the servicing tools available to manage updates:

- **Windows Update (stand-alone)**: This option is only available for servers that are connected to the Internet and have Windows Update enabled.
- **Windows Server Update Services (WSUS)** provides extensive control over Windows Server and Windows client updates and is natively available in the Windows Server operating system. You can defer updates, add an approval layer, and choose to deploy them to specific computers or groups of computers whenever ready.
- **Microsoft Endpoint Configuration Manager** provides the greatest control over servicing. You can defer updates, approve them, and have multiple options for targeting deployments and managing bandwidth usage and deployment times.

You can continue using the same process for Annual Channel Releases; for example, if you already use Configuration Manager to manage updates, you can continue to use it. Similarly, if you're using WSUS, you can continue to use that.

# Where to get Annual Channel

You can obtain Annual Channel releases from the following places:

- Volume Licensing Service Center (VLSC): Volume-licensed customers with Software Assurance can obtain this release by going to the Volume Licensing Service Center and select **Sign In**. Finally, select **Downloads and Keys**, search for Annual Channel, then download the media.

- Visual Studio Subscriptions: Visual Studio Subscribers can obtain Annual Channel releases by downloading them from the Visual Studio Subscriber download page. If you aren't already a subscriber, go to Visual Studio Subscriptions to sign up, and then visit the Visual Studio Subscriber downloads page. Releases obtained through Visual Studio Subscriptions are for development and testing only.

# Activating Annual Channel releases

You need to activate your installation using your activation keys obtained from the VLSC. If you're using KMS, Annual Channel releases use the same CSVLK of the last LTSC release before their release. For example, an Annual Channel released with or after Windows Server 2022 would use the Windows Server 2022 CSVLK. For more information, see KMS client setup keys.

# How to tell whether a server is running an LTSC or AC release

Long-Term Servicing Channel releases could be released at the same time as a new version of the Annual Channel. To determine whether a server is running Annual Channel release, you must look at the operating system version. The product name doesn't reflect the servicing channel. To determine whether a server is running an LTSC or AC release, you can run the Get-ComputerInfo PowerShell command. The following example is a computer running Windows Server 2022 Datacenter Edition (LTSC).

To determine the operating system version, run the following command:

PowerShell

```
Get-ComputerInfo  | fl WindowsProductName,OSDisplayVersion
```

Here's an example output from a computer running Windows Server LTSC.

Output

```
WindowsProductName : Windows Server 2022 Datacenter
OSDisplayVersion   : 21H2
```

Here's an example output from a computer running Windows Server Annual Channel for Containers.

Output

```
WindowsProductName : Windows Server 2022 Datacenter
OSDisplayVersion   : 23H2
```

> **Tip**
>
> `OSDisplayVersion` only applies to Windows Server 2022 and later. Annual Channel releases don't apply to Windows Server 2019 and earlier. If you're running Windows Server 2019 or earlier, you're running an LTSC release.

The following table lists the Windows Server LTSC and AC releases and their corresponding operating system versions.

Expand table

| Channel | Operating system display version |
| --- | --- |
| LTSC | 21H2 |
| Annual Channel | 23H2 |

The guidance is intended to help identify and differentiate between LTSC and AC for lifecycle and general inventory purposes only. It isn't intended for application compatibility or to represent a specific API surface. App developers should refer to other guidance to ensure compatibility. Components, APIs, and functionality might be added during a system's lifecycle or might not yet be available. To learn more about using programmatically determining the version, see Operating System Version.

# Microsoft server applications compatibility for Windows Server

Article • 01/13/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

We understand that application compatibility is one of the top reasons that customers choose Windows Server. Throughout the Windows Server development cycle, Microsoft regularly tests a large suite of Microsoft and non-Microsoft applications to ensure our products are compatible with as many applications as possible. These tests cover a wide range of non-Microsoft enterprise applications, including development tools, graphics tools, security tools, business office suites, storage management, antivirus, backup or recovery, and utilities.

The following table lists Microsoft server applications that support installation and functionality on Window Server. This information is for quick reference and isn't intended to replace the individual product specifications, requirements, announcements, or general communications of each individual server application. Refer to official documentation for each product to fully understand compatibility and options.

> **Tip**
>
> If you're a software vendor partner looking for more information on Windows Server compatibility with non-Microsoft applications, visit the Commercial App Certification portal.

Expand table

| Product | Supported on Server Core | Supported on Server with Desktop Experience | Product Web Link |
|---|---|---|---|
| Configuration Manager (version 2409) | ✅[2] | ✅ | Support for Windows Server 2025 |
| Exchange Server | ✅ | ✅ | Exchange Server supportability matrix |
| Microsoft 365 Apps | ❌ | ✅ | Windows and Office configuration support |

| Product | Supported on Server Core | Supported on Server with Desktop Experience | Product Web Link |
|---|---|---|---|
| | | | matrix |
| Project Server Subscription Edition | ✅ | ✅ | Software requirements for Project Server Subscription Edition |
| SharePoint Server Subscription Edition | ✅ | ✅ | System requirements for SharePoint Server Subscription edition |
| SQL Server 2019 | ✅[1] | ✅ | Hardware and Software Requirements for Installing SQL Server 2019 |
| SQL Server 2022 | ✅[1] | ✅ | Hardware and Software Requirements for Installing SQL Server 2022 |
| System Center Data Protection Manager | ✅[1] | ✅ | Preparing your environment for System Center Data Protection Manager |
| System Center Operations Manager | ✅[1] | ✅ | System requirements for System Center Operations Manager |
| System Center Virtual Machine Manager | ✅[1] | ✅ | System requirements for System Center Virtual Machine Manager |

1. Might have limitations or require the Server Core App Compatibility Feature on Demand (FOD). For more information, see specific product or Feature on Demand documentation.

2. Yes, as a managed client and distribution point but not as a site server.

# Feedback

Was this page helpful? Yes No

# Azure Hybrid Benefit for Windows Server

09/03/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

Azure Hybrid Benefit enables commercial customers to use their qualifying on-premises licenses to get Windows virtual machines (VMs) on Azure at a reduced cost. This article focuses on the benefits of using qualifying Windows Server licenses to get cost savings for Windows Server VMs in Azure, Azure Local, and Azure Kubernetes Service (AKS) hybrid deployments.

For other Azure hybrid benefits (for example, Microsoft SQL Server), see Azure Hybrid Benefit.

## What qualifies you for Azure Hybrid Benefit?

To qualify for Azure Hybrid Benefit for Windows Server, you need on-premises core licenses for Windows Server from an applicable program with active Software Assurance or qualifying subscription licenses. Software Assurance and qualifying subscription licenses are only available as part of certain commercial licensing agreements. To learn more about commercial licensing, see Microsoft Licensing Resources. To learn more about Windows Server core licenses, see Windows Server product licensing.

> **Important**
>
> - Workloads using Azure Hybrid Benefit can run only during the Software Assurance or subscription license term. When the Software Assurance or subscription license term approaches expiration, you must either renew your agreement with either Software Assurance or a subscription license, disable the hybrid benefit functionality, or deprovision those workloads that are using Azure Hybrid Benefit.
> - The Microsoft Product Terms for your program take precedent over this article. For more information, see **Microsoft Azure Product Terms** and select your program to show the terms.

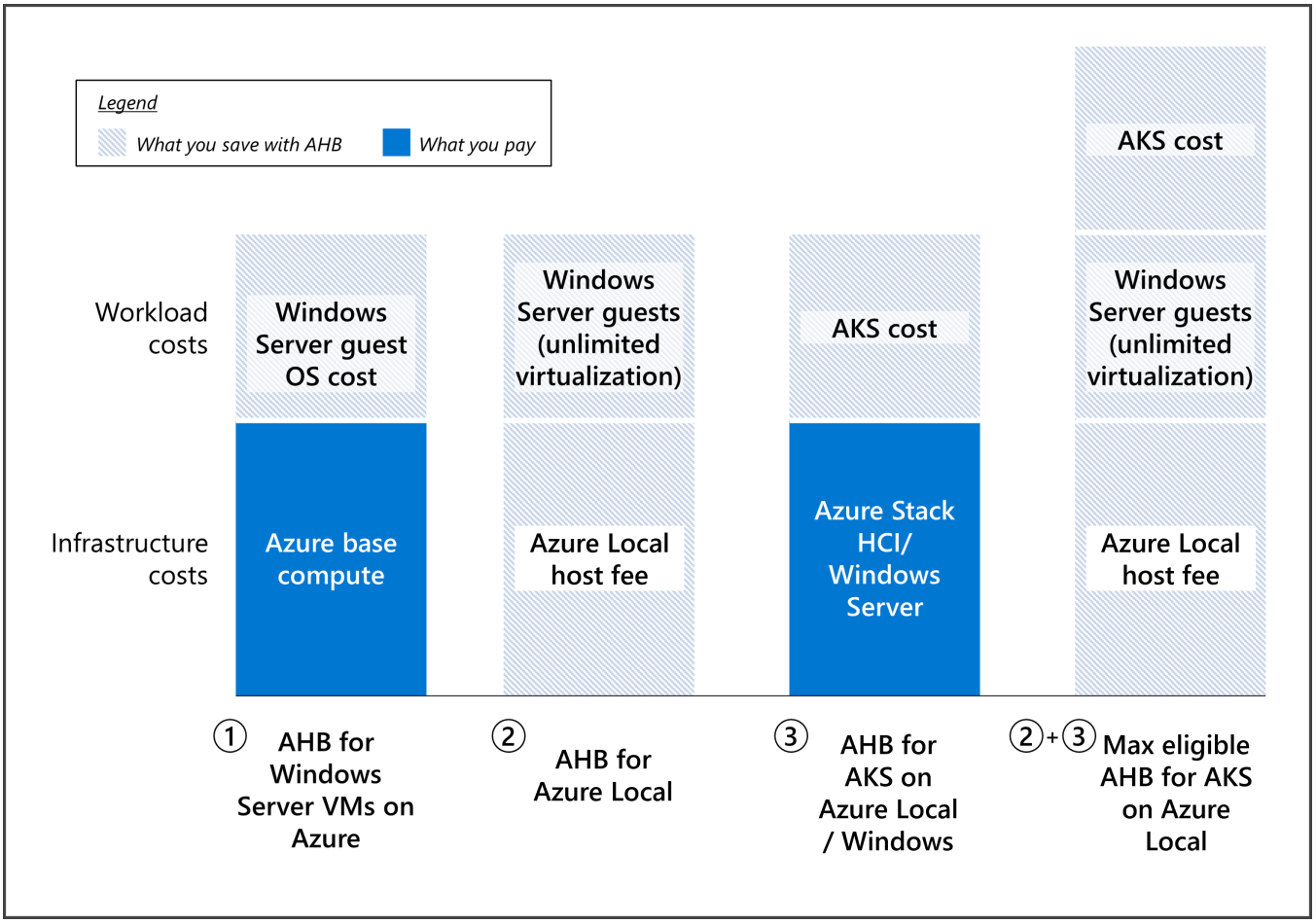
## What's included in Azure Hybrid Benefit?

Customers with Windows Server licensed by an applicable program with active Software Assurance or qualifying subscription licenses can use Azure Hybrid Benefit to further reduce

costs in the cloud and in datacenter and edge locations.

Azure Hybrid Benefit includes the following cost savings:

- **Windows Server VMs on Azure:** The license for Windows Server is covered by Azure Hybrid Benefit, so you only need to pay for the base compute rate of the VM. The base compute rate is equal to the Linux rate for VMs.
- **Azure Local:** The Azure Local host fee and Windows Server subscription fee are waived with Azure Hybrid Benefit. That is, unlimited virtualization rights are provided at no extra cost. You still pay other costs associated with Azure Local (for example, customer-managed hardware, Azure services, and workloads).
- **AKS:** Run AKS on Windows Server and Azure Local at no extra cost. You still pay for the underlying host infrastructure and any licenses for Windows containers unless you're also eligible for Azure Hybrid Benefit for Azure Local. With Azure Hybrid Benefit for Azure Local, you can waive fees for the Azure Local host and Windows Server subscription.




# Pricing for Azure Hybrid Benefit

To evaluate your potential cost savings, you can use these resources:

- **Windows VMs on Azure:** Windows Virtual Machine Pricing. Use the Azure Pricing Calculator to estimate cost savings, or compare Windows VM pricing with and without Azure Hybrid Benefit.
- **Azure Local:** Azure Local pricing.
- **Azure Kubernetes Service (AKS):** AKS on Azure Local pricing.

# Getting Azure Hybrid Benefit

Select the tab for your scenario.

Azure

Follow the guidance in this section to get and maintain Azure Hybrid Benefit for your Windows VMs in Azure.

## Licensing prerequisites

To qualify for Azure Hybrid Benefit for Windows VMs in Azure, you must meet the following licensing prerequisites.

### Types of license

- Windows Server Standard with active Software Assurance or subscription.
- Windows Server Datacenter with active Software Assurance or subscription.

### Number of licenses

You need a minimum of 8 core licenses (Datacenter or Standard edition) per VM. For example, 8 core licenses are still required if you run a 4-core instance. You might also run instances larger than 8 cores by allocating licenses equal to the core size of the instance. For example, 12 core licenses are required for a 12-core instance. For customers with processor licenses, each processor license is equivalent to 16 core licenses.

### Azure Migration Allowance

- **Windows Server Standard edition:** Licenses must be used either on-premises or in Azure, but not at the same time. The only exception is on a one-time basis, for up to 180 days, to allow you to migrate the same workloads to Azure.

- **Windows Server Datacenter edition:** For VM Licensing, when migrating workloads to Azure, licenses allow simultaneous usage on-premises and in Azure indefinitely. For Dedicated Host Licensing, when migrating workloads to Azure, licenses allow simultaneous usage on-premises and in Azure for a period of 180 days from when the licenses are allocated to Azure.

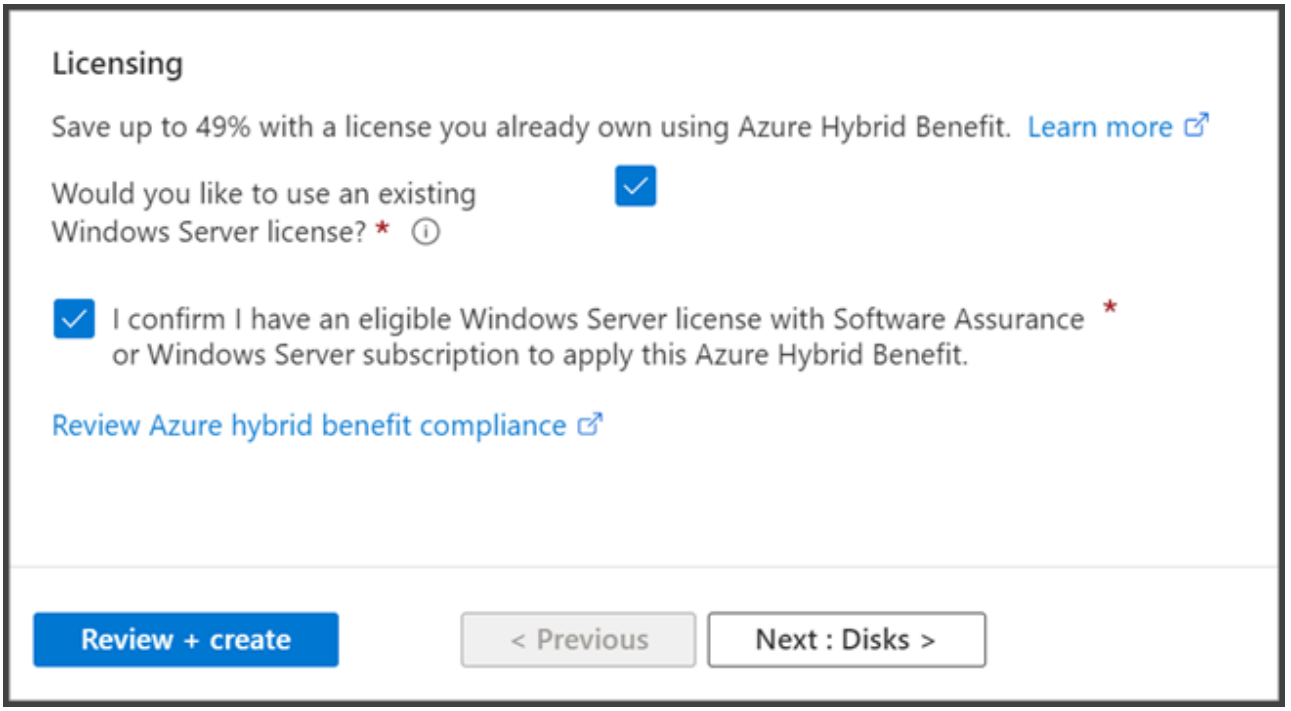
### Unlimited virtualization

Unlimited Virtualization Rights refers to the right to use any number of Windows Server VMs on a host.

- **Windows Server Datacenter edition:** You can use any number of Windows Server VMs on an Azure dedicated host if you allocate Windows Server Datacenter licenses with active Software Assurance or subscription for all the available physical cores on that Azure server.
- **Windows Server Standard edition:** Unlimited Virtualization Rights aren't available.

## How to apply Azure Hybrid Benefit for Windows VMs in Azure

To learn how to deploy Windows Server VMs in Azure with Azure Hybrid Benefit, follow the steps in Explore Azure Hybrid Benefit for Windows VMs. One way to activate Azure Hybrid Benefit for a Windows Server VM is to check the box under **Licensing** during VM creation, as shown in the following screenshot.



## How to maintain compliance

If you apply Azure Hybrid Benefit to your Windows Server VMs, verify the number of eligible licenses and the Software Assurance (or subscription) coverage period before you activate this benefit. Use the preceding guidelines to make sure you deploy the correct number of Windows Server VMs with this benefit.

If you already have Windows Server VMs running with Azure Hybrid Benefit, perform an inventory to see how many units you're running, and check this number against your Software Assurance or subscription licenses. You can contact your Microsoft licensing specialist to validate your Software Assurance licensing position.

To see and count all VMs that are deployed with Azure Hybrid Benefit in an Azure subscription, list all VMs and virtual machine scale sets.

You can also look at your Microsoft Azure bill to determine how many VMs with Azure Hybrid Benefit for Windows Server you're running. You can find information about the number of instances with the benefit under **Additional Info**:

JSON

```
"
{"ImageType":"WindowsServerBYOL","ServiceType":"Standard_A1","VMName":"","Usag
eType":"ComputeHR"}"
```

Billing isn't applied in real time. Expect a delay of several hours after you activate a Windows Server VM with Azure Hybrid Benefit before the VM shows on your bill.

To get a comprehensive view of your licensing position, perform an inventory in each of your Azure subscriptions. Confirm that you're fully licensed for the Windows Server VMs running with Azure Hybrid Benefit. You don't need to take any further action.

Perform an inventory regularly to make sure you're using any license benefits that you're entitled to. Regular inventories can help you reduce costs and make sure that you always have enough licenses to cover the Windows Server VMs you've deployed with Azure Hybrid Benefit.

If you don't have enough eligible Windows Server licenses for your deployed VMs, you have three choices:

- Purchase extra Windows Server licenses covered by Software Assurance or subscription through a commercial licensing agreement.
- Disable Azure Hybrid Benefit for some of your VMs and purchase them at regular Azure hourly rates.

- Deallocate some VMs.

> **Note**
>
> Microsoft reserves the right to audit customers at any time to verify eligibility for Azure Hybrid Benefit utilization.

# FAQ: Azure Hybrid Benefit

## Which regions are eligible for Azure Hybrid Benefit?

Azure Hybrid Benefit is available across all Azure regions and sovereign clouds.

## What happens to my benefits if my Software Assurance or subscription expires?

To use these benefits, your Software Assurance or qualifying subscription must be active. If you choose not to renew your Software Assurance or subscription when it expires, you need to remove your benefits from your resources in the Azure portal.

## What is Software Assurance?

Software Assurance is a comprehensive Volume Licensing program. Software Assurance is only available through Volume Licensing and is purchased when you buy or renew a Volume Licensing agreement. It's included with some agreements and is an optional purchase with others. Software Assurance benefits include new product version rights, support, license mobility rights, and a unique set of technologies and services to maximize your IT investments.

For information about Volume Licensing, see Microsoft Licensing. To learn more about Software Assurance benefits, and how each benefit can help meet your business needs, see Software Assurance benefits.

## What is a subscription license?

Subscription licenses are licenses to run the software only during the term of the subscription. Subscription licenses don't include perpetual rights to run the software.

## How can customers get Software Assurance?

You can purchase Software Assurance through Volume Licensing. Your Software Assurance benefits are activated in the Volume Licensing Service Center (VLSC). If your organization has a Microsoft Products and Services Agreement (MPSA), the Business Center is your destination for easy management of your Software Assurance benefits.

## See also

- Azure Hybrid Benefit product page
- Explore Azure Hybrid Benefit for Windows VMs
- Azure Hybrid Benefit for Azure Stack HCI

# Extended Security Updates for Windows Server overview

06/05/2025Applies to: ✅ Windows Server 2012, ✅ Windows Server 2012 R2

The Extended Security Update (ESU) program is a last resort option for customers who need to run certain legacy Microsoft products past the end of support. Windows Server Long Term Servicing Channel (LTSC) has a minimum of 10 years of support: five years for mainstream support and five years for extended support, which includes regular security updates.

However, once products reach the end of support, it also means the end of security updates and bulletins. This scenario can cause security or compliance issues and put business applications at risk. Microsoft recommends that you upgrade to the current version of Windows Server for the most advanced security, performance, and innovation.

Extended support for Windows Server 2012 and Windows Server 2012 R2 ended on October 10, 2023.

> **Tip**
>
> You can find information on support dates on **Microsoft Lifecycle**.

## What are Extended Security Updates?

Extended Security Updates for Windows Server include security updates and bulletins rated *critical* and *important* for a maximum period of time from the end of extended support, depending on the version. They're available free of charge for servers hosted in Azure, and available to purchase for servers not hosted in Azure. Extended Security Updates don't include new features, customer-requested nonsecurity hotfixes, or design change requests. For more information, see Lifecycle FAQ - Extended Security Updates.

With Extended Security Updates, the different phases for these versions of Windows Server are as follows:



If you haven't already upgraded your servers, you can do the following things to protect your applications and data during the transition:

- Migrate the affected existing Windows Server workloads as-is to Azure Virtual Machines (VM). Migrating to Azure automatically provides Extended Security Updates for the defined period. There's no extra charge for Extended Security Updates on top of an Azure VM's cost, and you don't need to do any other configuration.
- Purchase an Extended Security Update subscription for your servers and remain protected until you're ready to upgrade to a newer Windows Server version. When you have an Extended Security Update subscription, Microsoft provides updates for the defined period. Once you purchase a subscription, you must get a product key and install it on each applicable server. For more information, see How to get Extended Security Updates.

The following table lists the Extended Security Update duration for each version of Windows Server.

Expand table

| Product version | Hosted | ESU duration | ESU end date |
|---|---|---|---|
| Windows Server 2012<br>Windows Server 2012 R2 | Azure* | Three years | October 13, 2026 |
| Windows Server 2012<br>Windows Server 2012 R2 | Not in Azure | Three years | October 13, 2026 |

* Includes the Azure Stack portfolio of products that extend Azure services and capabilities to your environment of choice.

> ⚠ **Warning**
>
> After the period of Extended Security Updates ends, we'll stop providing updates. We recommend you update your version of Windows Server to a more recent version as soon as possible.

# Migrate to Azure

If your on-premises servers are running a version of Windows Server that has reached or is nearing the end of extended support, you can move them to Azure. In Azure, you can continue to run these servers as virtual machines. When you migrate to Azure, you not only stay

compliant with security updates, but also add cloud innovation to your work. The benefits of migrating to Azure include:

- Security updates in Azure.
- Get Windows Server critical and important security updates for a certain period of time, included at no extra charge.
- Upgrades in Azure free of charge.
- Adopt more cloud services whenever you're ready.
- By migrating SQL Server to Azure VMs, you get three more years of Windows Server critical security updates, included at no extra charge. You can also modernize your SQL Server to Azure SQL Managed Instance.
- Benefit from Azure Hybrid Benefit, which lets you use existing Windows Server licenses and SQL Server licenses for cloud savings unique to Azure.

To get started migrating, learn how to upload a generalized VHD and use it to create new VMs in Azure, or use Shared Image Galleries in Azure.

You can also read the Migration Guide for Windows Server for help with the following things:

- Analyze your existing IT resources.
- Assess the current state of your deployment.
- Decide whether to move certain services and applications to the cloud. Or, keep them on-premises and upgrade to the latest version of Windows Server. Choose the option that works best for your organization.

# Upgrade on-premises

If you need to keep your servers on-premises instead of migrating to Azure and the cloud, you have two choices for how to proceed:

- Build new servers with a supported version of Windows Server and migrate your applications and data.
- Upgrade in-place to a supported version of Windows Server.

In-place upgrades can typically upgrade Windows Server through at least one version, sometimes even two versions. For example, Windows Server 2012 R2 can upgrade in-place to Windows Server 2025. As you upgrade, you can also migrate to Azure at any time. For more information about your on-premises upgrade options, see supported upgrade paths for Windows Server.

# Upgrade SQL Server in parallel with your Windows Servers

If you're running a version of SQL Server that reached or is reaching the end of extended support, you can also benefit from Extended Security Updates for SQL Server. For more information, see Extended Security Updates for SQL Server and Windows Server .

## Related content

- Learn how to get Extended Security Updates (ESU) for Windows Server.
- Overview of Windows Server upgrades

# What is Secured-core server?

Article • 12/05/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Azure Local, versions 23H2 and 22H2

Secured-core is a collection of capabilities that offers built-in hardware, firmware, driver and operating system security features. The protection provided by Secured-core systems begins before the operating system boots and continues whilst running. Secured-core server is designed to deliver a secure platform for critical data and applications.

Secured-core server is built on three key security pillars:

- Creating a hardware backed root of trust.
- Defense against firmware level attacks.
- Protecting the OS from the execution of unverified code.

## What makes a Secured-core server

The Secured-core initiative started with Windows PCs through a deep collaboration between Microsoft and PC manufacturing partners to provide the most elevated Windows security ever. Microsoft has expanded the partnership further with server manufacturing partners to help ensure Windows Server delivers a secure operating system environment.

Windows Server integrates closely with hardware to provide increasing levels of security:

- Recommended baseline: The recommended minimum for all systems to provide foundational system integrity using TPM 2.0 for a hardware root of trust and Secure Boot. TPM2.0 and Secure boot are required for Windows Server hardware certification. To learn more, see Microsoft raises the security standard for next major Windows Server release
- Secured-core server: Recommended for systems and industries requiring higher levels of assurance. Secured-core server builds on the previous features and uses advanced processor capabilities to provide protection from firmware attacks.

The following table shows how each security concept and feature are used to create a Secured-core server.

Expand table

| Concept | Feature | Requirement | Recommended baseline | Secured-Core server |
|---|---|---|---|---|
| **Create a hardware backed root of trust** | | | | |
| | Secure Boot | Secure Boot is enabled in the Unified Extensible Firmware Interface (UEFI) BIOS by default. | ✓ | ✓ |
| | Trusted Platform Module (TPM) 2.0 | Meet the latest Microsoft requirements for the Trusted Computing Group (TCG) specification. | ✓ | ✓ |
| | Certified for Windows Server | Demonstrates that a server system meets Microsoft's highest technical bar for security, reliability and manageability. | ✓ | ✓ |
| | Boot DMA protection | Support on devices that have the Input/Output Memory Management Unit (IOMMU). For example, Intel VT-D or AMD-Vi. | | ✓ |
| **Defend against firmware level attacks** | | | | |
| | System Guard Secure Launch | Enabled in the operating system with Dynamic Root of Trust for Measurement (DRTM) compatible Intel and AMD hardware. | | ✓ |
| **Protect the OS from execution of unverified code** | | | | |

| Concept | Feature | Requirement | Recommended baseline | Secured-Core server |
|---|---|---|---|---|
| | Virtualization-based Security (VBS) | Requires the Windows hypervisor, which is only supported on 64-bit processors with virtualization extensions, including Intel VT-X and AMD-v. | ✓ | ✓ |
| | Hypervisor Enhanced Code Integrity (HVCI) | Hypervisor Code Integrity (HVCI)-compatible drivers plus VBS requirements. | ✓ | ✓ |

## Create a hardware backed root of trust

UEFI Secure boot is a security standard that protects your servers from malicious rootkits by verifying your systems boot components. Secure boot verifies a trusted author has digitally signed the UEFI firmware drivers and applications. When the server is started, the firmware checks the signature of each boot component including firmware drivers and the OS. If the signatures are valid, the server boots and the firmware gives control to the OS.

To learn more about the boot process, see Secure the Windows boot process.

TPM 2.0 provides a secure, hardware-backed storage for sensitive keys and data. Every component loaded during the boot process is measured and the measurements stored in the TPM. By verifying the hardware root-of-trust it elevates the protection provided by capabilities like BitLocker, which uses TPM 2.0 and facilitates the creation of attestation-based workflows. These attestation-based workflows can be incorporated into zero-trust security strategies.

Learn more about Trusted Platform Modules and how Windows uses the TPM.

Along with Secure Boot and TPM 2.0, Windows Server Secured-core uses Boot DMA protection on compatible processors that have the Input/Output Memory Management Unit (IOMMU). For example, Intel VT-D or AMD-Vi. With boot DMA protection, systems are protected from Direct Memory Access (DMA) attacks during boot and during the operating system runtime.

## Defend against firmware level attacks

Endpoint protection and detection solutions usually have limited visibility of firmware, given that firmware runs underneath of the operating system. Firmware has a higher level of access and privilege than operating system and hypervisor kernel, making it an attractive target for attackers. Attacks targeting firmware undermine other security measures implemented by the operating system, making it more difficult to identify when a system or user has been compromised.

Beginning with Windows Server 2022, System Guard Secure Launch protects the boot process from firmware attacks by using hardware capabilities from AMD and Intel. With processor support for Dynamic Root of Trust for Measurement (DRTM) technology, Secured-core servers put firmware in a hardware-backed sandbox helping to limit the effects of vulnerabilities in highly privileged firmware code. System Guard uses the DRTM capabilities that are built into compatible processors to launch the operating system, ensuring the system launches into a trusted stated using verified code.

## Protect the OS from execution of unverified code

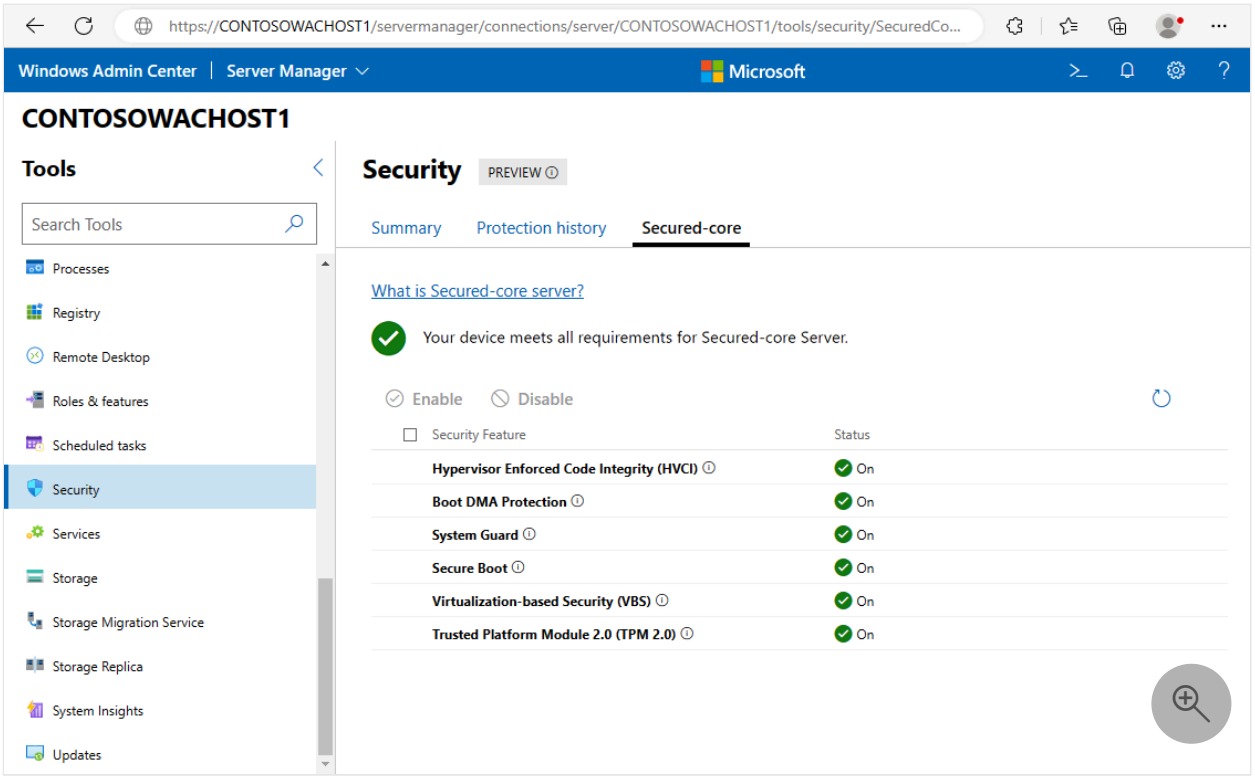
Secured-core server uses Virtualization Based Security (VBS) and hypervisor-protected code integrity (HVCI) to create and isolate a secure region of memory from the normal operating system. VBS uses the Windows hypervisor to create a Virtual Secure Mode (VSM) to offer security boundaries within the operating system, which can be used for other security solutions.

HVCI, commonly referred to as Memory integrity protection, is a security solution that helps ensure that only signed and trusted code is allowed to execute in the kernel. Using only signed and trusted code prevents attacks that attempt to modify the kernel mode code. For example, attacks that modify drivers, or exploits such as WannaCry that attempt to inject malicious code into the kernel.

To learn more about VBS and hardware requirements, see Virtualization-based Security.

# Simplified management

You can view and configure the OS security features of Secured-core systems using Windows PowerShell or the security extension in Windows Admin Center. With Azure Local integrated systems, manufacturing partners have further simplified the configuration experience for customers so that Microsoft's best server security is available right out of the box.




Learn more about Windows Admin Center.

# Preventative defense

You can proactively defend against and disrupt many of the paths attackers use to exploit systems by enabling Secured-core functionality. Secured-core server enables advanced security features at the bottom layers of the technology stack, protecting the most privileged areas of the system before many security tools are aware of exploits. It also occurs without the need for extra tasks or monitoring by IT and SecOps teams.

# Begin your Secured-core journey

You can find hardware certified for Secured-core server from the Windows Server Catalog, and Azure Local servers in the Azure Local Catalog. These certified servers come fully equipped with industry-leading security mitigations built into the hardware, firmware, and the operating system to help thwart some of the most advanced attack vectors.

# Next steps

Now you understand what Secured-core server is, here are some resources to get you started. Learn about how:

- Configure Secured-core server.

- Microsoft brings advanced hardware security to Server and Edge with Secured-core in the Microsoft Security Blog.
- New Secured-core servers are now available from the Microsoft ecosystem to help secure your infrastructure in the Microsoft Security Blog.
- Building Windows-compatible devices, systems, and filter drivers across all Windows Platforms in Windows Hardware Compatibility Program Specifications and Policies.

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## Feedback

Was this page helpful? Yes No

# Configure Secured-core server

Article • 12/05/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

Secured-core is a collection of capabilities that offers built-in hardware, firmware, driver and operating system security features. This article shows you how to configure Secured-core server by using Windows Admin Center, the Windows Server Desktop Experience, and Group Policy.

Secured-core server is designed to deliver a secure platform for critical data and applications. For more information, see What is Secured-core server?

## Prerequisites

Before you can configure Secured-core server, you must have the following security components installed and enabled in the BIOS:

- Secure Boot.
- Trusted Platform Module (TPM) 2.0.
- System firmware must meet preboot DMA protection requirements and set appropriate flags in ACPI tables to opt into and enable Kernel DMA Protection. To learn more about Kernel DMA Protection, see Kernel DMA Protection (Memory Access Protection) for OEMs.
- A processor with support enabled in the BIOS for:
  - Virtualization extensions.
  - Input/Output Memory Management Unit (IOMMU).
  - Dynamic Root of Trust for Measurement (DRTM).
  - Transparent Secure Memory Encryption is also required for AMD based systems.

> **ⓘ Important**
>
> Enabling each of the security features in the BIOS can vary based on your hardware vendor. Make sure to check your hardware manufacturer's Secured-core server enablement guide.

You can find hardware certified for Secured-core server from the Windows Server Catalog, and Azure Local servers in the Azure Local Catalog.

# Enable security features

To configure Secured-core server you need to enable specific Windows Server security features, select the relevant method and follow the steps.

GUI

Here's how to enable Secured-core server using the user interface.

1. From the Windows desktop, open the **Start** menu, select **Windows Administrative Tools**, open **Computer Management**.
2. In Computer management, select **Device Manager**, resolve any device error if necessary.
   a. For AMD based systems, confirm the DRTM Boot Driver device is present before continuing
3. From Windows desktop, open the **Start** menu, select **Windows Security**.
4. Select **Device security** > **Core isolation details**, then enable **Memory Integrity** and **Firmware Protection**. You might not be able to enable Memory Integrity until you've enabled Firmware Protection first and restarted your server.
5. Restart your server when prompted.

Once your server has restarted, your server is enabled for Secured-core server.

# Verify Secured-core server configuration

Now that you've configured Secured-core server, select the relevant method to verify your configuration.

GUI

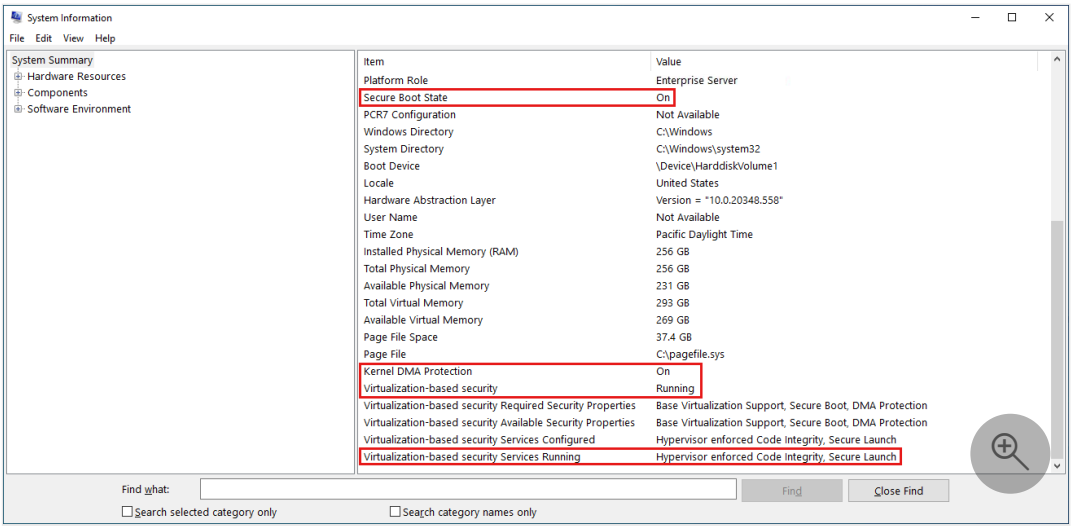
Here's how to verify your Secured-core server is configured using the user interface.

1. From the Windows desktop, open the **Start** menu, type `msinfo32.exe` to open System Information. From the System Summary page, confirm:

   a. **Secure Boot State** and **Kernel DMA Protection** is On.

   b. **Virtualization-based security** is Running.

   c. **Virtualization-based security Services** Running shows **Hypervisor enforced Code Integrity** and **Secure Launch**.




# Next steps

Now that you've configured Secured-core server, here are some resources to learn more about:

- Virtualization-based Security (VBS)
- Memory integrity and VBS enablement
- System Guard Secure Launch

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# Feedback

**Was this page helpful?** Yes No

# Hotpatch for Windows Server

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022

Hotpatching is a way to install OS security updates on Windows Server without having to restart your machine. Hotpatching patches the in-memory code of running processes without the need to restart the process. Hotpatching also provides the following benefits:

- Fewer binaries mean updates install faster and consume less disk and CPU resources.
- Lower workload impact with less need to restart your machine.
- Better protection, as the Hotpatch update packages are scoped to Windows security updates that install faster without requiring you to restart your machine.
- Reduces the time exposed to security risks and change windows, and easier patch orchestration with Azure Update Manager.

This article describes Hotpatch for Windows Server, which uses the Windows Server support and management model on Azure, Azure Local, and Azure Arc-connected machines.

> **Note**
>
> This article doesn't cover Hotpatch for Windows client, which is separate from the Windows Server support and management model described here. For hotpatch guidance for Windows 11 Enterprise, version 24H2 or later, see Hotpatch updates. On Windows client, Windows Autopatch manages hotpatch updates through Microsoft Intune, and only eligible devices that meet the prerequisites on that page receive them.

## Supported platforms

### Azure and Azure Local virtual machines

The following table lists the exact combinations of publisher, OS offer, and SKU that support Hotpatching for Windows Server 2022 and Windows Server 2025 on Azure. Virtual machines (VMs) you create on Azure Local using these combinations also support Hotpatching.

> **Note**

Windows Server container base images, custom images, or any other combination of publisher, offer, and SKU aren't supported.

Expand table

| Publisher | OS Offer | SKU |
|---|---|---|
| MicrosoftWindowsServer | WindowsServer | 2022-Datacenter-Azure-Edition-Core |
| MicrosoftWindowsServer | WindowsServer | 2022-Datacenter-Azure-Edition-Core-smalldisk |
| MicrosoftWindowsServer | WindowsServer | 2022-Datacenter-Azure-Edition-Hotpatch |
| MicrosoftWindowsServer | WindowsServer | 2022-Datacenter-Azure-Edition-Hotpatch-smalldisk |
| MicrosoftWindowsServer | WindowsServer | 2025-Datacenter-Azure-Edition |
| MicrosoftWindowsServer | WindowsServer | 2025-Datacenter-Azure-Edition-smalldisk |
| MicrosoftWindowsServer | WindowsServer | 2025-Datacenter-Azure-Edition-Core |
| MicrosoftWindowsServer | WindowsServer | 2025-Datacenter-Azure-Edition-Core-smalldisk |

For more information about the available images, see Windows Server on Azure Marketplace.

## Azure Arc-connected machines

**Important**

Azure Arc-enabled Hotpatch for Windows Server 2025 is now available at no extra cost. To learn more, see Simplified access to Hotpatching enabled by Azure Arc for Windows Server 2025.

Azure Arc-connected Windows Server 2025 machines can receive Hotpatches if you enable the feature on Azure Arc Portal. To start using Azure Arc-enabled Hotpatch, connect Azure Arc to machines using one of the following editions:

- Windows Server 2025 Datacenter Edition
- Windows Server 2025 Standard Edition

# How Hotpatch works

Hotpatch first establishes a baseline with the current Cumulative Update for Windows Server. Every three months, the baseline periodically refreshes with the latest Cumulative Update. You then receive Hotpatch releases for the next two months after the Cumulative Update. For example, if January is a Cumulative Update, February and March would have Hotpatch releases. For more information about the Hotpatch release schedule, see the Windows Server Hotpatch Calendar.

There are two types of baselines: **Planned baselines** and **Unplanned baselines**.

- **Planned baselines** are released on a regular cadence, with Hotpatch releases in between. Planned baselines include all the updates in a comparable Latest Cumulative Update for that month and require you to restart your machine.
  - For example, a planned one year release period might include four planned baseline releases in a calendar year and eight Hotpatch releases.

- **Unplanned baselines** are released during an unplanned important update, such as a zero-day fix, when that particular update can't be released as a Hotpatch. When unplanned baselines release, a Hotpatch release is replaced with an unplanned baseline for that month. Unplanned baselines also include all the updates in a comparable Latest Cumulative Update for that month, and therefore require you to restart your machine.
  - Because these events are unplanned, developers can't predict unplanned baselines in advance.

Hotpatch updates don't require you to restart your machine. Because Hotpatches patch the in-memory code of running processes without needing to restart them, your applications aren't affected. This lack of restarting doesn't affect the performance or functionality implications of the patch itself.

## Supported updates

Hotpatch covers Windows Security updates and maintains parity with the content of security updates issued to in the regular non-Hotpatch Windows update channel.

There are a few important things you need to consider when enabling Hotpatch on a supported version of Windows Server. You still need to restart your machine to install updates that aren't included in the Hotpatch program. You also need to periodically restart after installing a new baseline. Restarting keeps your VM in sync with nonsecurity patches included in the latest cumulative updates.

The following patches currently aren't included in the Hotpatch program and require you to update your machine during Hotpatch release months:

- Nonsecurity updates for Windows
- .NET updates
- Non-Windows updates, such as drivers, firmware updates, and so on.

# Patch orchestration process

Hotpatch is an extension of Windows Update and typical management processes. However, the tools for patch management vary depending on which platform you're using.

## Azure

- VMs you create in Azure using a supported Windows Server image have Automatic VM Guest Patching enabled by default.
- Hotpatch automatically downloads and applies patches classified as Critical or Security to your VM.
- Hotpatch applies patches during off-peak hours in the VM time zone.
- Azure manages patches for you, applying patches according to availability-first principles.
- Azure monitors VM health through platform health signals to detect patching failures.

> **Note**
>
> You can't create Azure Virtual Machine Scale Sets with Uniform orchestration on Azure Edition images with Hotpatch. To learn more about which features Uniform orchestration for scale sets supports, see A comparison of Flexible, Uniform, and availability sets.

## Azure Local

Azure Local can orchestrate Hotpatch updates for VMs using the following tools:

- Group Policy configures Windows Update client settings.
- SConfig configures Windows Update client settings for Server Core.

- Non-Microsoft patch management solutions.

## Azure Arc-connected machines

Azure Arc-connected machines can install and manage Hotpatch updates using the following tools:

- Azure Update Manager
- Group Policy configures Windows Update client settings.
- SConfig configures Windows Update client settings for Server Core.
- Non-Microsoft patch management solutions.

For more information about which tools Hotpatch uses, check out our Azure Update Manager documentation.

## Understand the patch status for your VM in Azure

To view the patch status for your VM, open the Overview page for your VM in the Azure portal. From there, under **Operations**, select **Updates**. You should see the patch status and most recently installed patches under **Recommended updates**.

In the **Recommended updates** page, you can see the Hotpatch status of your VM and if there are any available patches for your VM. As we stated in How Hotpatch works, Automatic VM Guest Patching automatically installs all Critical and Security patches on your VM.

Patches outside of those two categories aren't automatically installed, and are instead displayed in the **Update compliance** tab as a list of available patches. You can also check the **Update history** tab to view patch installation details for update deployments on your VM from the past 30 days.

Automatic VM Guest Patching regularly runs assessments of available patches, which you can view in the **Updates** tab. You can manually start an assessment by selecting the **Assess Now** button. You can also install patches on-demand by selecting the **Install updates now** button. This option lets you choose whether to install all updates under specific patch classifications. You can also select individual updates to include or exclude by providing a list of knowledge base articles. However, keep in mind that patches you install manually don't follow availability-first principles and might require you to restart your VM.

You can also view installed patches by running the Get-HotFix cmdlet in PowerShell or by viewing the **Settings** menu in Desktop Experience.

## Rollback support for Hotpatching

Hotpatch updates don't support automatic rollback. If you experience an issue during or after an update, you must uninstall the latest update and install the last functional baseline update. This process requires that you restart the VM.

## Next steps

- Enable Hotpatch for Azure Arc-enabled servers
- Automatic VM Guest Patching
- Enable Hotpatch for Azure Edition virtual machines built from ISO
- Azure Update Management

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Last updated on 09/08/2026

# Install Windows Server from installation media

10/13/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

Learn how to create a bootable USB drive or DVD that can be used to install Windows Server on your devices. Creating a Windows Server installation media is a crucial step in setting up a new server or upgrade from an existing one. This article aims to guide you on how to create installation media and install Windows Server using either the Server Core or Desktop Experience install options.

## Prerequisites

Setup media for the target version of Windows Server can be obtained from Original Equipment Manufacturer (OEM), Retail, Visual Studio subscription, and the Volume Licensing Service Center (VLSC) channels. Your device must also not be running in Azure. If you want to install Windows Server in Azure, see Quickstart: Create a Windows virtual machine in the Azure portal. The following are required before installing Windows Server:

- **Hardware**
  - Your device must meet or exceed the minimum hardware requirements. To learn more, see Hardware requirements for Windows Server.
  - A minimum of an 8 GB USB flash drive.
  - An optical drive capable of burning a DVD+/-RW dual-layer disk.
  - A dual-layer 8.5 GB DVD.

- **Software**
  - Determine which version of Windows Server is appropriate for your environment. To learn more, see Comparison of Windows Server editions.
  - Make sure you have a valid product key or subscription license for your product. Product keys and activation methods might vary depending on the distribution channel you received Windows Server media from, such as a Commercial Licensing program, Retail, or OEM.

> ⓘ **Note**

For users who desire to install Windows Server in a virtual environment through Windows Hyper-V, the minimum RAM requirements differ. To learn more, see the Hardware requirements for Windows Server components tab.

# Create a bootable USB flash drive

Generally, creating a bootable USB drive can be performed using the diskpart.exe utility. An alternative to manually preparing your USB drive, users can run the following PowerShell script to make the USB drive bootable. The Windows Server installation ISO file must be mounted before running this script. To mount the ISO file, perform the following steps:

1. Locate your Windows Server installation ISO file.
2. Right-click on your ISO file and select **Mount**.

Mounting the ISO file creates a virtual optical drive, with its own drive letter, where users can migrate the ISO contents to their USB drive.

Once you plug in your USB drive, it should provide you with a drive letter. Once you know the drive letter of both the USB drive and mounted ISO, run one of the following scripts. Each script accounts for three retries before skipping over to the next set of files.

Format as NTFS

This script makes your USB drive bootable in NTFS format.

PowerShell

```
# Select USB drive letter
$usbDriveLetter = Read-Host "Enter USB drive letter (Ex: E)"

# Format USB drive
Format-Volume -DriveLetter $usbDriveLetter -FileSystem NTFS -
NewFileSystemLabel "WinServerUSB" -Confirm:$false | Out-Null

# Select ISO file mount point
$isoMountPointDriveLetter = Read-Host "Enter ISO mount point drive letter (Ex:
F)"

# Copy ISO files to USB drive
$source = "$($isoMountPointDriveLetter):\"
$destination = "$($usbDriveLetter):\"
robocopy $source $destination /COPYALL /Z /E /SEC /R:3 /W:3

# Make USB drive bootable
$usbDriveNumber = (Get-WmiObject -Class Win32_DiskDrive | Where-Object
{$_.InterfaceType -eq "USB" -and $_.DeviceID -like "*$usbDriveLetter"}).Index
```

```
bootsect /nt60 $usbDriveLetter`:

# Task completion notification
Write-Host "USB Creation Complete!"
Start-Sleep -Seconds 2
```

# Create a bootable DVD

1. Insert a dual-layer 8.5 GB DVD into your optical drive.
2. Locate your Windows Server installation ISO file.
3. Right-click on your ISO file and select **Burn disc image**.
4. The **Windows Disc Image Burner** wizard prompts you to select which **Disc burner** drive you'd like to use.
5. Select the appropriate drive and then select the **Burn** button.
6. This step is optional, but recommended to ensure that data integrity is verified against any errors or discrepancies between the burned disk and the original ISO image. Select the **Verify disc after burning** checkbox to detect any differences.

# Configure BIOS settings

Before you can install Windows Server via USB or DVD, the computer's boot order in the BIOS must be changed to ensure that the system boots from either the USB or DVD. Accessing the BIOS can vary depending on your hardware. In most cases, you can access the BIOS by pressing a specific key while your device boots. Typically, the key is either **F2**, **F10**, **F12**, or **Delete**. Refer to the user manual for your device. On certain hardware, once POST completes, there's a short window allowed for accessing the BIOS. You might need to repeatedly press one of the keys as soon as POST completes.

Once you're in the BIOS settings, navigate through the menus until you see the **Boot Order** or **Boot Sequence** option. Refer to the user manual for your device to locate this setting. Once you select the boot order setting, you can edit the boot order priority based in top-down order. Move booting from the USB or optical drive to the top of the list depending on which media installation method you're using. Once appropriate changes are made, save and exit from the BIOS. Your device reboots automatically in applying these changes.

> ⓘ **Note**
>
> If your system has two disks and you attempt to install Windows Server 2025 on the second disk, the installation might fail. This is because a Legacy BIOS system requires the OS to be installed on the primary disk enumerated by the BIOS, typically BIOS disk 0. If

your BIOS allows explicit configuration of the primary BIOS enumerated disk, ensure you select the designated disk during installation. You can run `SELECT DISK=SYSTEM` in diskpart to identify the first BIOS enumerated disk.

# Install Windows Server

Once modifications are made to the boot order and you select to boot from either the USB or DVD drive, follow these steps to install Windows Server.

Desktop Experience

1. Plug in your USB drive or DVD into your optical drive, then reboot your device.
2. Once your device boots, you're prompted to press any key to boot from your installation media.
3. Under **Select language settings**, select your language, time and currency format, then select **Next**.
4. Under **Select keyboard settings**, select your keyboard language, then select **Next**.
5. Under **Select setup option**, select **Install Windows Server**, select **I agree everything will be deleted including files, apps, and settings**, then select **Next**.
6. Under **Choose a licensing method**, select the option that best serves your environment, then select **Next**:
    a. **Use a product key** - This option is for users who have an OEM, Retail, or Volume License (VL) key. If this license type is selected, proceed with the next steps.
    b. **Pay-as-you-go** - This option is for users who desire to use their Azure subscription license. This option is only available for Windows Server 2025 and has its own set of prerequisites. If this license type is selected, see Set up Windows Server Pay-as-you-go to continue the installation process.
7. Under **Select image**, select your Windows Server version, then select **Next**.
8. Under **Applicable notices and license terms**, review the software terms, then select **Accept**.
9. Under **Select location to install Windows Server**, select the disk that you want to install Windows Server, then select **Next**.

10. Under **Ready to install**, select **Install**.
11. Once your device reboots a few times, the **License terms** appear. Select **Accept** to proceed.
12. Under **Customize settings**, provide a complex password for the Administrator account, then select **Finish**.
13. Once you log into the Administrator account, review the **Send diagnostic data to Microsoft** information, then select **Accept**.

> **Note**
>
> To manage and configure your Windows Server Core environment, you have the option of using the Server Configuration tool (**SConfig**). To learn more, see Administer a Server Core server and Configure a Server Core installation of Windows Server and Azure Local with the Server Configuration tool (SConfig).

## See also

- Get started with Windows Server
- What's new in Windows Server 2025
- Install or Uninstall Roles, Role Services, or Features
- Hyper-V Technology Overview
- Overview of file sharing using the SMB 3 protocol in Windows Server

# Install the Application Compatibility Feature on Demand on Server Core

07/18/2025Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019

The Application Compatibility Feature on Demand (FOD) is an optional feature package designed to enhance the compatibility of Server Core installations in Windows Server. Starting with Windows Server 2019, you can install this feature at any time to improve the compatibility of applications with Server Core installations of Windows Server and provide extra tools for daily tasks. This article explains the benefits of the Application Compatibility Feature on Demand, outlines its installation process, and provides instructions for adding it to a server or a custom Windows image.

For more information on other Features on Demand, see Features On Demand.

## Why install the Application Compatibility Feature on Demand

The Application Compatibility Feature on Demand for Server Core includes a subset of binaries and packages from the Server with Desktop Experience installation option. This optional package is available from Windows Update or on a separate ISO, but it can only be added to Server Core installations and images.

The two primary benefits the Application Compatibility Feature on Demand provides are:

- Increased compatibility of Server Core for server applications.
- Adding OS components not normally included with Server Core that can help with administration tasks and compatibility for software tools used in acute troubleshooting and debugging scenarios.

The operating system components that are available as part of the Application Compatibility Feature on Demand include:

Expand table

| Component | Filename | First available |
| --- | --- | --- |
| Device Manager | `devmgmt.msc` | Windows Server 2019 |
| Disk Management | `diskmgmt.msc` | Windows Server 2019 |

| Component | Filename | First available |
| --- | --- | --- |
| Event Viewer | `eventvwr.msc` | Windows Server 2019 |
| Failover Cluster Manager | `cluadmin.msc` | Windows Server 2019 |
| File Explorer | `explorer.exe` | Windows Server 2019 |
| Hyper-V Manager | `virtmgmt.msc` | Windows Server 2022 |
| Microsoft Management Console | `mmc.exe` | Windows Server 2019 |
| Performance Monitor | `perfmon.exe` | Windows Server 2019 |
| Resource Monitor | `resmon.exe` | Windows Server 2019 |
| Task Scheduler | `taskschd.msc` | Windows Server 2022 |
| Windows PowerShell Integrated Scripting Environment (ISE) | `powershell_ise.exe` | Windows Server 2019 |

# Prerequisites

Before you begin, ensure that you meet the following prerequisites:

- The Application Compatibility Feature on Demand can only be installed on Server Core installations of Windows Server. Don't attempt to add the Application Compatibility Feature on Demand to servers with the Desktop Experience installation option.
- You need to be signed in with an administrator account on the Server Core computer that you want to add the Application Compatibility Feature on Demand to.
- The following Windows Server features require extra configuration:
  - Failover Cluster Manager (`cluadmin.msc`) requires the Failover Clustering Windows Server feature to be installed first.
  - IIS Management Console (`Web-Mgmt-Console`) relies on the Application Compatibility Feature on Demand to be installed, as it requires the Microsoft Management Console (`mmc.exe`) to run.

- If you want to add the Application Compatibility Feature on Demand to a custom Windows image (WIM), you need the ISO image file for the version of Windows Server you want to create a custom image for.

# Install the Application Compatibility Feature on Demand

Installing the Application Compatibility Feature on Demand involves adding a specialized package to your Server Core installation that provides extra tools and compatibility features normally found in Server with Desktop Experience.

The installation process depends on whether you want to install the Application Compatibility Feature on Demand from Windows Update or an ISO image. You can install the feature directly from Windows Update by running a PowerShell command. For an ISO image, you need to download the relevant Windows Server Languages and Optional Features ISO, mount it locally, and install the feature from that source.

After you install the Application Compatibility Feature on Demand and restart of the server, the command console window frame color changes to a different shade of blue.

Select the relevant tab for your preferred installation method.

Windows Update

To install the Application Compatibility Feature on Demand on a Server Core installation of Windows Server from Windows Update:

1. Sign in to the server with an administrator account.
2. In `SConfig`, use option **15** to exit `SConfig` to PowerShell.
3. Install the Application Compatibility Feature on Demand by running the following command. The command takes several minutes to complete.

   PowerShell

   ```
   Add-WindowsCapability -Online -Name
   "ServerCore.AppCompatibility~~~~0.0.1.0"
   ```

   The output is similar to the following example:

   Output

```
Path          :
Online        : True
RestartNeeded : True
```

4. Once the command completes, restart the server to apply the changes, then install the latest operating system updates.

> **Important**
>
> If you do an in-place upgrade of Windows Server to a newer version, the Application Compatibility Feature on Demand doesn't remain in place. You need to install it again after the upgrade. Alternatively, you can add the Application Compatibility Feature on Demand to a custom Windows image (WIM) that you use to install Windows Server. Adding the Application Compatibility Feature on Demand to a custom image ensures that it's present after the upgrade completes. For more information, see Add the Application Compatibility Feature on Demand to a custom WIM image section.

## Add the Application Compatibility Feature on Demand to a custom WIM image

If you add the Application Compatibility Feature on Demand to a custom Windows image (WIM) and use that image to install Windows Server, it's automatically installed during the installation process. It remains in place after an in-place upgrade of Windows Server to a newer version.

To add the Application Compatibility Feature on Demand to a custom WIM image, follow these steps. Be sure to change `<values>` with your own.

1. Download the ISO image file that contains Features on Demand for the version of Windows Server you want to create a custom image. Save the ISO image to a folder to the same folder where you have the Windows Server ISO image file. The ISO image is available for the following versions of Windows Server:

   - Windows Server 2025
   - Windows Server 2022
   - Windows Server 2019

2. Mount both the Languages and Optional Features ISO and the Windows Server ISO by running the following commands in an elevated PowerShell session:

```
PowerShell

$isoFolder = "<ISO folder path>"
$fodIsoFilename = "<FOD_ISO_filename.iso>"
$wsIsoFilename = "<Windows_Server_ISO_filename.iso>"

$fodIso = Mount-DiskImage -ImagePath "$isoFolder\$fodIsoFilename"
$wsIso = Mount-DiskImage -ImagePath "$isoFolder\$wsIsoFilename"
```

3. Run the following command to get the drive letters that the FOD ISO and Windows Server ISO are mounted to:

```
PowerShell

$fodDriveLetter = ($fodIso | Get-Volume).DriveLetter
$wsDriveLetter = ($wsIso | Get-Volume).DriveLetter
```

4. Copy the contents of the Windows Server ISO image to a local folder, for example, **C:\SetupFiles\WindowsServer\Files**, by running the following commands. The copy operation might take some time.

```
PowerShell

$wsFiles = "<Windows Server files path>"
New-Item -ItemType Directory -Path $wsFiles

Copy-Item -Path ${wsDriveLetter}:\* -Destination $wsFiles -Recurse
```

5. Get the image name you want to modify within the `install.wim` file by running the following command. The `install.wim` file is located inside the **sources** folder of the Windows Server ISO image. The names of the images available in this `install.wim` file are in the output.

```
PowerShell

$installWimPath = "<Windows Server Files Path>\sources\install.wim"

Get-WindowsImage -ImagePath $installWimPath
```

6. Mount the `install.wim` file in a new folder by running the following command:

   - `$wimImageName` - Enter the name of the image you want to mount from the output of the previous command. The example here uses **Windows Server 2022 Datacenter**.

- `$wimMountFolder` - Specify an empty folder to use when accessing the contents of the install.wim file.

```
PowerShell

$wimImageName = "<Image name, for example Windows Server 2022 Datacenter>"
$wimMountFolder = "<WIM folder path>"

New-Item -ItemType Directory -Path $wimMountFolder
Set-ItemProperty -Path $installWimPath -Name IsReadOnly -Value $false
Mount-WindowsImage -ImagePath $installWimPath -Name $wimImageName -Path
$wimMountFolder
```

7. Add the capabilities and packages you want to the mounted `install.wim` image by running the following commands (depending on the version), replacing the sample variable values with your own.

   - For Windows Server 2022 and later:

   ```
   PowerShell

   $capabilityName = "ServerCore.AppCompatibility~~~~0.0.1.0"

   Add-WindowsCapability -Path $wimMountFolder -Name $capabilityName -
   Source "${fodDriveLetter}:\LanguagesAndOptionalFeatures" -LimitAccess
   ```

   - For previous versions of Windows Server:

   ```
   PowerShell

   $capabilityName = "ServerCore.AppCompatibility~~~~0.0.1.0"

   Add-WindowsCapability -Path $wimMountFolder -Name $capabilityName -
   Source "${fodDriveLetter}:\" -LimitAccess
   ```

8. Dismount and commit changes to the `install.wim` file by running the following command:

   ```
   PowerShell

   Dismount-WindowsImage -Path $wimMountFolder -Save
   ```

You can now install Windows Server using the custom WIM image that has the Application Compatibility Feature on Demand included and it remains in place after an in-place upgrade of Windows Server to a newer version.

# Install Internet Explorer 11 on Server Core

You can install Internet Explorer 11 on a Server Core installation of Windows Server 2022 and previous versions. Internet Explorer requires the Application Compatibility Feature on Demand to be installed first. If you need to install it, see the section Install the Application Compatibility Feature on Demand. You don't need to install Internet Explorer to add the Application Compatibility Feature on Demand.

> **Tip**
>
> In Windows Server 2022, although you can add Internet Explorer 11 to Server Core installations of Windows Server, **Microsoft Edge** should be used instead. Microsoft Edge has **Internet Explorer mode** (IE mode) built-in, so you can access legacy Internet Explorer-based websites and applications straight from Microsoft Edge. For more information on the product lifecycle for Internet Explorer, see **Lifecycle FAQ - Internet Explorer and Microsoft Edge**.

Select the relevant tab for your preferred installation method.

Windows Update

To install Internet Explorer 11 on a Server Core installation of Windows Server from Windows Update:

1. Make sure you installed the Application Compatibility Feature on Demand on the Server Core installation of Windows Server.
2. Once again follow the steps in the section Install the Application Compatibility Feature on Demand, but for step 3, run the following command instead:

   PowerShell

   ```
   Add-WindowsCapability -Online -Name
   "Browser.InternetExplorer~~~~0.0.11.0"
   ```

   The output is similar to the following example:

   Output

   ```
   Path          :
   Online        : True
   RestartNeeded : True
   ```

3. Once the command completes, restart the server to apply the changes, then install the latest operating system updates.

4. After the server restarts, you can access Internet Explorer 11 by going back to a PowerShell prompt from `SConfig`, then running the following command:

```PowerShell
& "$env:ProgramFiles\Internet Explorer\iexplore.exe"
```

> **Important**
>
> Double-clicking to open locally saved `.htm` files isn't supported. However, you can **right-click** and choose **Open with Internet Explorer**, or you can open it directly from Internet Explorer by selecting **File**, then **Open** and browsing to the file.

**Note:** The author created this article with assistance from AI. Learn more

# Perform an in-place upgrade of Windows Server

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

An in-place upgrade moves your server from an older version of Windows Server to a newer version while keeping your settings, server roles, and data intact. For example, you can upgrade from Windows Server 2012 R2 to Windows Server 2025. When you upgrade in place, your environment stays running on a supported version with the latest security and performance features, all without rebuilding.

This article shows you how to upgrade by using Windows Server Setup from installation media or the feature update in Windows Update.

## Prerequisites

- To determine the Windows Server versions and supported upgrade paths, see Supported in-place upgrade paths by version.
- Administrative rights on the target server.
- A full backup of your server, including the operating system, apps, data, and any VMs. Perform a restore test to confirm the backup is valid and recoverable. You can use Windows Server Backup or a partner backup solution.
- A scheduled maintenance window, as downtime is required during the upgrade.
- Knowledge of:
  - The roles and features that support an in-place upgrade. See Upgrade and migrate roles and features in Windows Server.
  - The Microsoft server applications that are supported on Windows Server. See Microsoft server applications compatibility for Windows Server.
  - Any non-Microsoft application vendor support requirements.
- A server that:
  - Meets or exceeds the hardware requirements for Windows Server.

    - Isn't clustered. If you're running a cluster, use the Cluster-Aware Updating feature or a cluster operating system rolling upgrade instead.

- A valid product key and activation method. Keys and methods can vary depending on the distribution channel that you received the Windows Server media from, such as a Commercial Licensing program, a retail channel, or an OEM.
- A location to store files away from your server, such as a USB flash drive or network location.
- If Configuration Manager is installed on a Windows Server 2012 or Windows Server 2012 R2 server, follow the preupgrade and post-upgrade instructions at Upgrade on-premises infrastructure that supports Configuration Manager.

> **ⓘ Important**
>
> While most Windows Server roles support an in-place upgrade, Active Directory Domain Controllers are an exception. Although an in-place upgrade might work, don't upgrade servers that run the Active Directory Domain Services role. For more information, see **Upgrade domain controllers to a newer version of Windows Server**.

Depending on whether you upgrade by using installation media or Windows Update, you need to meet additional prerequisites. Select the tab for your upgrade method:

Installation media

The installation media upgrade method applies to non-Azure servers only. To upgrade Windows Server in an Azure virtual machine (VM), see In-place upgrade for VMs running Windows Server in Azure or use the Windows Update method.

- Installation media (ISO image, USB drive, or DVD) for the version of Windows Server that you want to upgrade to.
- You can get installation media for your target version of Windows Server from an original equipment manufacturer (OEM), a retail distribution channel, a Visual Studio subscription, or the Microsoft 365 admin center.

# Collect pre-upgrade diagnostic information

Collect diagnostic information from your server before upgrading in case the upgrade fails. Store the diagnostic files where you can access them if your server goes down.

To collect your information:

1. Open an elevated PowerShell prompt, make a note of your current directory, and run the following commands:

   **PowerShell**

   ```
   Get-ComputerInfo -Property WindowsBuildLabEx,WindowsEditionID | Out-File -
   FilePath .\computerinfo.txt
   systeminfo.exe | Out-File -FilePath systeminfo.txt
   ipconfig /all | Out-File -FilePath ipconfig.txt
   ```

   The `Get-ComputerInfo` command requires PowerShell 5.1 or later. If your Windows Server version doesn't include PowerShell 5.1, open Registry Editor and find the `BuildLabEx` and `EditionID` values under `HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion`.

2. Open File Explorer, go to the directory you noted, and copy the files to a USB flash drive or network location off your computer.

Back up your server operating system, apps, and VMs after collecting your system information. Shut down or migrate any VMs currently running on the server. Make sure no VMs are running during the upgrade.

# Perform the in-place upgrade

You can perform the in-place upgrade by using installation media or Windows Update. Select the tab for your upgrade method:

Installation media

> **Tip**
>
> Looking for a way to upgrade without installation media? Use the Windows Update tab to upgrade without needing installation media.

Run Windows Server Setup from your installation media to do the in-place upgrade. This procedure applies to non-Azure, nonclustered servers running Windows Server 2012 R2 or

later. During the upgrade, your server restarts several times.

To do the in-place upgrade by using installation media:

1. Mount or insert the installation media. Open File Explorer, go to the root of the installation media, and then open `setup.exe`. For example, if you mounted an ISO image or inserted a DVD as drive D, the file path is `D:\setup.exe`. If **User Account Control** prompts you to allow Setup to make changes, select **Yes**.

2. By default, Setup automatically downloads updates for the installation. If you're okay with the default settings, select **Next** to continue.

   If you don't want Setup to automatically download updates, select **Change how Setup downloads updates**, select the appropriate option for your environment, and then select **Next**.

3. If prompted, enter your product key, and then select **Next**.

4. Select the edition of Windows Server you want to install, and then select **Next**.

5. Review the applicable notices and license terms. If you agree to the terms, select **Accept**.

6. Select **Keep files, settings, and apps** to do an in-place upgrade, and then select **Next**.

7. After Setup finishes analyzing your device, it displays the **Ready to install** screen. To start the in-place upgrade, select **Install**.

The in-place upgrade starts, and the screen displays the progress. After the in-place upgrade finishes, your server restarts.

# Verify the in-place upgrade

After the server restarts, verify the upgrade succeeded by checking the Windows Server version and testing your applications.

1. Open an elevated PowerShell prompt. Run the following command to verify that the version and edition match what you expected.

   **PowerShell**

```
Get-ComputerInfo -Property WindowsProductName
```

2. Ensure all applications are running and client connections are successful.

If your server isn't working as expected after the in-place upgrade, analyze the Setup log files in the `C:\Windows\Panther` directory. Also download the SetupDiag tool to analyze the Setup log files.

If you need technical assistance, contact Microsoft Support.

## Related content

- Plan your Windows Server upgrade
- Add or remove roles and features
- Windows Server management overview
- Get started with Windows Admin Center
- Key Management Services (KMS) activation planning

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Last updated on 04/14/2026

# Upgrade and migrate roles and features in Windows Server

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

Upgrading or migrating roles and features in Windows Server is essential for maintaining security, performance, and supportability. This article provides an overview which roles and features can be upgraded in-place or migrated to a new server. It also provides links to detailed migration guides for each role and feature, as well as information about the Windows Server Migration Tools and Storage Migration Service. Use this information to plan a smooth transition to newer versions of Windows Server, minimize downtime, and ensure continued reliability for your workloads.

You can update roles and features to later versions of Windows Server in two ways:

- **Migration:** Move roles and features to a new server running the newer version of Windows Server.
- **In-place upgrade:** Install the new version of Windows Server over the existing one, keeping roles and features in place.

Some roles and features support both methods, while others might only support migration.

You can migrate many roles and features by using Windows Server Migration Tools. This feature is built into Windows Server and helps you move roles and features to a new server. For file servers and storage, use the Storage Migration Service.

The migration guides support migrations of specified roles and features from one server to another (not in-place upgrades). Unless the guides say otherwise, you can migrate between physical and virtual computers. You can also migrate between Windows Server installations that use either Server with Desktop Experience or Server Core.

> **ⓘ Important**
>
> Before you begin migrating roles and features, verify that both source and destination servers are running the most current updates that are available for their operating systems.

Before migrating or upgrading to a new version of Windows Server, review the support lifecycle policy for that version. Make sure you understand how long it's supported. You can search for lifecycle information for the specific Windows Server release you plan to use.

## Windows Server Migration Tools

Windows Server Migration Tools enables you to migrate server roles, features, operating system settings, and other data and shares to servers, including later versions of Windows Server. It's a feature of Windows Server and so it's easily installed using the Add Roles and Features wizard, or PowerShell. Learn more about how to install, use, and remove Windows Server Migration Tools.

> **Note**
>
> Cross-subnet migrations using Windows Server Migration Tools is available with Windows Server 2012 and later releases. Previous versions of Windows Server Migration Tools only support migrations in the same subnet.

## Upgrade and migration matrix

Expand table

| Server Role | Upgradeable in-place? | Migration Supported? | Can migration be completed without downtime? |
|---|---|---|---|
| Active Directory Certificate Services | Yes | Yes | No |
| Active Directory Domain Services | Not recommended. Use clean OS install to get AD performance and feature improvements. | Yes | Yes |
| Active Directory Federation Services | No | Yes | No (new nodes need to be added to the farm) |
| Active Directory Lightweight Directory Services | Yes | Yes | Yes |
| Active Directory Rights Management Services | Yes | Yes | No |

| Server Role | Upgradeable in-place? | Migration Supported? | Can migration be completed without downtime? |
|---|---|---|---|
| DHCP Server | Yes | Yes | Yes |
| DNS Server | Yes | Yes | No |
| Failover Clustering | Yes with Cluster OS Rolling Upgrade or when the server is removed by the cluster for upgrade and then added to a different cluster. | Yes | Yes for Failover Clusters with Hyper-V VMs or Failover Clusters running the Scale-out File Server role. See Cluster OS Rolling Upgrade. |
| File and Storage Services | Yes | Varies by subfeature | No |
| Hyper-V | Yes with Cluster OS Rolling Upgrade process | Yes | Yes for Failover Clusters with Hyper-V VMs or Failover Clusters running the Scale-out File Server role. See Cluster OS Rolling Upgrade. |
| Print and Fax Services | No | Yes (using `Printbrm.exe`) | No |
| Remote Desktop Services | Yes, for all subroles, but mixed mode farm isn't supported | Yes | No |
| Web Server (IIS) | Yes | Yes | No |
| Windows Server Essentials Experience | Yes | Yes | No |
| Windows Server Update Services | Yes | Yes | No |
| Work Folders | Yes | Yes | Yes with Cluster OS Rolling Upgrade process. |

# Migration guides

Below you can find links to migration guides for specific Windows Roles and Features.

## Active Directory

- Migrate the Certification Authority
- Migrate Active Directory Federation Services Role Service

- Upgrading AD RMS to Windows Server 2016
- Upgrade domain controllers to a newer version of Windows Server
- Active Directory Domain Services and Domain Name System (DNS) Server Migration Guide

## BranchCache

- BranchCache Migration Guide

## DHCP

- Migrate DHCP Server
- Migrate existing DHCP failover deployment

## Failover Clustering

- Upgrade a Windows Server failover cluster with a cluster OS rolling upgrade
- Upgrading Failover Clusters on the same hardware
- Migrate Cluster Roles
- Migrate Clustered Services and Applications to Windows Server 2012

## File and Storage Services

- Storage Migration Service
- Migrate File and Storage Services
- Upgrade a Storage Spaces Direct cluster

## Hyper-V

- Migrate Hyper-V
- Upgrade virtual machine version in Hyper-V on Windows or Windows Server

## Network Policy Server

- Migrate Network Policy Server
- Migrate Health Registration Authority

## Print and Document Services

- Migrate Print and Document Services

## Remote Desktop Services

- Migrate Remote Desktop Services
- Migrate your Remote Desktop Services Client Access Licenses (RDS CALs)

## Routing and Remote Access

- Overview of the DirectAccess to Always On VPN migration
- RRAS Migration Guide

## Web Server (IIS)

- Web Server (IIS)

## Windows Server Update Services

- Migrate Windows Server Update Services to Windows Server 2012 R2

# Other Windows migration guides

- Local User and Group Migration Guide
- IP Configuration Migration Guide

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Last updated on 04/27/2026

# Convert Windows Server editions and license types

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

After you install Windows Server, you can convert between editions (for example, evaluation to retail, or Standard to Datacenter) and switch between licensing types (retail, volume-licensed, and OEM). This article provides step-by-step procedures for each conversion scenario.

For information about supported in-place upgrade paths and restrictions, see Plan your Windows Server upgrade.

## Prerequisites

Before you convert your Windows Server edition or license type, make sure you have:

- The product key for the target edition or license type.
- An elevated command prompt or PowerShell session on the server you want to convert.
- Administrative access to the server.

## Convert an evaluation version to a retail version

You can convert evaluation versions and editions of Windows Server to retail versions and editions. For example, if you install the evaluation version of Standard (Desktop Experience) edition, you can convert it to the retail version of either the Standard (Desktop Experience) edition or the Datacenter (Desktop Experience) edition.

However, you can't convert all Windows Server evaluation versions and editions to all retail versions or editions. For example, if you install the evaluation Datacenter edition, you can only convert it to the retail Datacenter edition, not to the retail Standard edition.

Since Windows Server 2016, you can't convert Desktop Experience evaluation versions to Core retail versions. If you install the Standard Core evaluation version, you can convert it only to retail Datacenter Core, not to retail Standard Core.

It's important to run the `DISM /online /Get-TargetEditions` command as instructed in the following procedure to determine which retail versions you can convert to. If the retail version

you want isn't listed as a target version, you need to do a fresh install of the retail version you want.

> ⓘ **Note**
>
> To verify that your server is running an evaluation version, you can run either of the following commands at an elevated command prompt:
>
> - Run `DISM /online /Get-CurrentEdition` and make sure the current edition name includes `Eval`.
> - Run `slmgr.vbs /dlv` and make sure the output includes `EVAL`.

If Windows isn't activated, the bottom right-hand corner of the desktop shows the time remaining in the evaluation period.

## Windows Server Standard or Datacenter

If your server is running an evaluation version of Windows Server Standard or Datacenter edition, you can convert it to an available retail version. Run the following commands in an elevated command prompt or PowerShell session.

1. Determine the current edition name by running the following command. The output is an abbreviated form of the edition name. For example, Windows Server Datacenter (Desktop Experience) Evaluation edition is `ServerDatacenterEval`.

   **Windows Command Prompt**

   ```
   DISM /online /Get-CurrentEdition
   ```

2. Verify which editions the current installation can be converted to by running the following command. From the output, make a note of the edition name you want to convert to.

   **Windows Command Prompt**

   ```
   DISM /online /Get-TargetEditions
   ```

3. Run the following command to save the Microsoft Software License Terms for Windows Server, which you can then review. Replace the `<target edition>` placeholder with the edition name you noted from the previous step.

**Windows Command Prompt**

```
DISM /online /Set-Edition:<target edition> /GetEula:C:\license.rtf
```

4. Enter the new edition name and corresponding retail product key in the following command. The set edition process requires you to accept the Microsoft Software License Terms for Windows Server that you saved previously.

   **Windows Command Prompt**

   ```
   DISM /online /Set-Edition:<target edition> /ProductKey:<product key>
   /AcceptEula
   ```

   For example:

   **Windows Command Prompt**

   ```
   DISM /online /Set-Edition:ServerDatacenter /ProductKey:ABCDE-12345-ABCDE-
   12345-ABCDE /AcceptEula
   ```

> **Tip**
>
> For more information about Dism.exe, see DISM Command-line options.

> **Important**
>
> You can't convert an Active Directory domain controller from an evaluation to a retail version. In this case, install another domain controller on a server that runs a retail version. Then migrate any FSMO roles that are held by the evaluation domain controller. Finally, remove Active Directory Domain Services (AD DS) from the domain controller that runs on the evaluation version. For more information, see Upgrade Domain Controllers to Windows Server.

## Windows Server Essentials

If the server is running Windows Server Essentials, you can convert it to the full retail version by entering a retail, volume license, or OEM key in the following command at an elevated command prompt:

**Windows Command Prompt**

```
slmgr.vbs /ipk <license key>
```

# Convert Windows Server Standard edition to Datacenter edition

At any time after installing Windows Server, you can convert Windows Server Standard edition to Datacenter edition. You can also run `setup.exe` from the installation media to upgrade or repair the installation, sometimes called in-place repair. If you run `setup.exe` to upgrade or repair in-place on any edition of Windows Server, the result is the same edition you started with.

You can convert the Standard edition of Windows Server to the Datacenter edition as follows:

1. Determine that Windows Server Standard is the current edition name by running the following command. The output is an abbreviated form of the edition name, for example Windows Server Standard (Desktop Experience) edition is `ServerStandard`.

   **Windows Command Prompt**

   ```
   DISM /online /Get-CurrentEdition
   ```

2. Verify that Windows Server Datacenter is a valid option to convert to by running the following command:

   **Windows Command Prompt**

   ```
   DISM /online /Get-TargetEditions
   ```

3. Enter `ServerDatacenter` and your retail product key in the following command:

   **Windows Command Prompt**

   ```
   DISM /online /Set-Edition:ServerDatacenter /ProductKey:<product key>
   /AcceptEula
   ```

# Convert between retail, volume-licensed, and OEM licenses

At any time after installing Windows Server, you can freely convert between a retail license, a volume-licensed license, or an OEM license. The edition (Standard or Datacenter) remains the

same during this conversion. If you're starting with an evaluation version, convert it to the retail version first and then convert between the versions by running the following command from an elevated command prompt. Provide your volume-license, retail, or OEM product key.

**Windows Command Prompt**

```
slmgr.vbs /ipk <product key>
```

## Related content

- Plan your Windows Server upgrade
- Perform an in-place upgrade of Windows Server
- Server Core vs Server with Desktop Experience install options
- In-place upgrade for VMs running Windows Server in Azure

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Last updated on 04/03/2026

# Automatic Virtual Machine Activation in Windows Server

07/14/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

Automatic Virtual Machine Activation (AVMA) acts as a proof-of-purchase mechanism, helping to ensure that Windows products are used in accordance with the Product Use Rights and Microsoft Software License Terms. AVMA lets you activate Windows Server virtual machines (VM) on a Windows Server Hyper-V host that is properly activated, even in disconnected environments. AVMA binds the VM activation to the licensed virtualization host and activates the VM when it starts up. When you use AVMA, you can get real-time reporting on usage and historical data on the license state of the VM. Reporting and tracking data is available on the virtualization host.

## Practical applications

On virtualization hosts, AVMA offers several benefits. Server data center managers can use AVMA to do the following tasks:

- Activate VMs in remote locations.
- Activate VMs with or without an internet connection.
- Track VM usage and licenses from the virtualization host, without requiring any access rights on the virtualized systems.

Service Provider License Agreement (SPLA) partners and other hosting providers don't have to share product keys with tenants or access a tenant's VM to activate it. VM activation is transparent to the tenant when AVMA is used. Hosting providers can use the server logs to verify license compliance and to track client usage history.

## System requirements

To run guest VMs on a virtualization server host, you must activate the host. You can obtain host activation keys from the Microsoft 365 admin center or your OEM provider.

> **Note**
>
> In a failover cluster, each virtualization server host in the cluster must be activated for guest VMs to stay activated, regardless of which server they run on.

AVMA requires a Windows Server Datacenter edition with the Hyper-V server host role installed. The Windows Server version of the host determines which versions it can activate in a guest VM. The following table lists the guest VM versions that each host version is able to activate. A host version can access all the editions (Datacenter, Standard, or Essentials) of its eligible guest VM versions.

Expand table

| Server host version | Windows Server 2025 guest VM | Windows Server 2022 guest VM | Windows Server 2019 guest VM | Windows Server 2016 guest VM | Windows Server 2012 R2 guest VM |
|---|---|---|---|---|---|
| Windows Server 2025 | X | X | X | X | X |
| Windows Server 2022 | | X | X | X | X |
| Windows Server 2019 | | | X | X | X |
| Windows Server 2016 | | | | X | X |
| Windows Server 2012 R2 | | | | | X |

> **Note**
>
> AVMA doesn't work with other server virtualization technologies.

# How to implement AVMA

To activate VMs with AVMA, you use a generic AVMA key (detailed in AVMA keys) that corresponds to the version of Windows Server that you want to activate. To create a VM and activate it with an AVMA key, follow these steps:

1. On the server that hosts the VMs, install, and configure the Microsoft Hyper-V Server role. Ensure that the server is successfully activated. For more information, see Install Hyper-V Server.

2. Create a virtual machine and install a supported Windows Server operating system on it.

> **Important**
>
> The Data Exchange integration service (also known as Key-Value Pair Exchange) must be enabled in the VM settings for AVMA to work. It's enabled by default for new VMs.

3. After installing Windows Server on the VM, install the AVMA key on the VM. From PowerShell or an elevated command prompt, run the following command:

```
slmgr /ipk <AVMA_key>
```

The VM automatically activates, providing that the virtualization host itself is activated.

> **Tip**
>
> You can also add the AVMA keys in any Unattend setup file.

# AVMA keys

Windows Server 2025

Expand table

| Edition | Key |
|---|---|
| Datacenter | YQB4H-NKHHJ-Q6K4R-4VMY6-VCH67 |
| Datacenter: Azure Edition | 6NMQ9-T38WF-6MFGM-QYGYM-88J4F |
| Standard | WWVGQ-PNHV9-B89P4-8GGM9-9HPQ4 |

# Reporting and tracking

The Key-Value Pair (KVP) exchange between the virtualization host and the VM provides real-time tracking data for the guest operating systems, including activation information. This

activation information is stored in the Windows registry of the VM. Historical data about AVMA requests is logged in Event Viewer on the virtualization host.

For more information about KVP, see Data Exchange: Using key-value pairs to share information between the host and guest on Hyper-V.

> **Important**
>
> KVP data isn't secured. It can be modified and isn't monitored for changes. KVP data should be removed if the AVMA key is replaced with another product key (retail, OEM, or volume licensing key).

Since the AVMA activation process is transparent, error messages aren't displayed. However, AVMA requests are also logged on the virtualization host in Event Viewer in the Application log with Event ID **12310**, and on the VM with Event ID **12309**. The following events are captured on the VMs:

Expand table

| Notification | Description |
|---|---|
| AVMA Success | The VM was activated. |
| Invalid Host | The virtualization host is unresponsive. This event can happen when the server isn't running a supported version of Windows. |
| Invalid Data | This event usually results from a failure in communication between the virtualization host and the VM, often caused by corruption, encryption, or data mismatch. |
| Activation Denied | The virtualization host couldn't activate the guest operating system because the AVMA ID didn't match. |

# Key Management Services (KMS) activation planning

Article • 03/17/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

The following information outlines initial planning considerations that you need to review for Key Management Services (KMS) activation.

KMS uses a client-server model to active clients and is used for volume activation. KMS clients connect to a KMS server, called the KMS host, for activation. The KMS host must reside on your local network.

KMS hosts don't need to be dedicated servers, and KMS can be cohosted with other services. You can run a KMS host on any physical or virtual system that is running a supported Windows Server or Windows client operating system. A KMS host running on a Windows Server operating system can activate computers running both server and client operating systems. However, a KMS host running on a Windows client operating system can only activate computers also running client operating systems.

To use KMS, a KMS host needs a key that activates, or authenticates, the KMS host with Microsoft. This key is sometimes referred to as the KMS host key, but is formally known as a Microsoft Customer Specific Volume License Key (CSVLK). You can get this key from the Product Keys section of the Volume Licensing Service Center for the following agreements: Open, Open Value, Select, Enterprise, and Services Provider License. You can also get assistance by contacting your local Microsoft Activation Center.

## Operational requirements

KMS can activate physical and virtual computers, but to qualify for KMS activation, a network must have a minimum number of computers (called the activation threshold). KMS clients activate only after this threshold is met. To ensure that the activation threshold is met, a KMS host counts the number of computers that are requesting activation on the network.

KMS hosts count the most recent connections. When a client or server contacts the KMS host, the host adds the machine ID to its count and then returns the current count value in its response. The client or server activates if the count is high enough. Clients activate if the count is 25 or higher. Servers and volume editions of Microsoft Office products

activate if the count is five or greater. The KMS only counts unique connections from the past 30 days, and only stores the 50 most recent contacts.

KMS activations are valid for 180 days, a period known as the activation validity interval. KMS clients must renew their activation by connecting to the KMS host at least once every 180 days to stay activated. By default, KMS client computers attempt to renew their activation every seven days. After a client's activation is renewed, the activation validity interval begins again.

A single KMS host can support an unlimited number of KMS clients. If you have more than 50 clients, we recommend that you have at least two KMS hosts in case one of your KMS hosts becomes unavailable. Most organizations can operate with as few as two KMS hosts for their entire infrastructure.

After the first KMS host is activated, the CSVLK that is used on the first host can be used to activate up to five more KMS hosts on your network for a total of six. After a KMS host is activated, administrators can reactivate the same host up to nine times with the same key.

If your organization needs more than six KMS hosts, you can request more activations for your organization's CSVLK. For example, if you have 10 physical locations under one volume licensing agreement and you want each location to have a local KMS host. To request this exception, contact your local Microsoft Activation Center.

Computers that are running volume licensing editions of Windows Server and Windows client are, by default, KMS clients with no extra configuration needed.

If you're converting a computer from a KMS host, MAK, or retail edition of Windows to a KMS client, installing the applicable KMS client setup key is necessary. For more information, see KMS client setup keys.

# Network requirements

KMS activation requires TCP/IP connectivity. KMS hosts and clients are configured by default to use Domain Name System (DNS). KMS hosts use DNS dynamic updates to automatically publish the information that KMS clients need to find and connect to them. You can accept these default settings, or if you have special network and security configuration requirements, you can manually configure KMS hosts and clients.

By default, a KMS host is configured to use TCP on port 1688.

# Activation versions

The following table summarizes KMS host and client versions for networks that include Windows Server and Windows client devices.

> **Important**
>
> Windows Updates might be required on the KMS server to support activation of newer clients. If you receive activation errors, check that you have the appropriate updates listed below this table.

Windows Server 2025

Expand table

| CSVLK group | CSVLK can be hosted on | Windows editions activated by this KMS host |
| --- | --- | --- |
| Volume License for Windows Server 2025 | • Windows Server 2025[1]<br>• Windows Server 2022[1]<br>• Windows Server 2019 | • Windows Server 2025 (all editions)<br>• Windows Server 2022 (all editions)<br>• Windows Server Semi-Annual Channel<br>• Windows Server 2019 (all editions)<br>• Windows Server 2016 (all editions)<br>• Windows Server 2012 R2 (all editions)<br>• Windows Server 2012 (all editions)<br>• Windows Server 2008 R2 (all editions)<br>• Windows Server 2008 (all editions)<br><br>• Windows 11 Enterprise LTSC 2024<br>• Windows 11 Enterprise/Enterprise N<br>• Windows 11 Professional/Professional N<br>• Windows 11 Professional for Workstations/Professional N for Workstations<br>• Windows 11 for Education/Education N<br>• Windows 10 Enterprise LTSC/LTSC N/LTSB<br>• Windows 10 Enterprise/Enterprise N<br>• Windows 10 Professional/Professional N<br>• Windows 10 Professional for Workstations/Professional N for Workstations<br>• Windows 10 for Education/Education N<br>• Windows 8.1 Enterprise<br>• Windows 8.1 Professional<br>• Windows 7 Enterprise<br>• Windows 7 Professional |

1. Azure activates Windows Server Datacenter: Azure Edition, meaning it can't be configured as a KMS host

# KMS host required updates

Depending on which operating system your KMS host is running and which operating systems you want to activate, you might need to install one or more of the following updates. Updates are required when you want to activate a version of Windows that is newer than the version your KMS host is running.

> **Note**

The updates listed are the minimum required. Where later cumulative updates or monthly rollups are listed as an option, install the latest available version for your operating system to benefit from security and other fixes.

Expand table

| KMS host OS version | KMS client OS version(s) to activate | Required update |
|---|---|---|
| Windows Server 2022 | - Windows Server 2025 | February 13, 2024 - KB5034765 or later cumulative update |
| Windows Server 2019 | - Windows Server 2025<br>- Windows Server 2022 | February 13, 2024 - KB5034768 or later cumulative update<br>June 8, 2021 - KB5003646 or later cumulative update |
| Windows Server 2016 | - Windows Server 2022<br>- Windows Server 2019 | June 8, 2021 - KB5003638 or later cumulative update |
| Windows Server 2016 | - Windows Server 2019 | December 3, 2018 - KB4478877 or later cumulative update |
| Windows Server 2012 R2 | - Windows Server 2019<br>- Windows Server 2016<br>- Windows 10 | November 27, 2018 - KB4467695 (Preview of Monthly Rollup) or later monthly rollup |
| Windows Server 2012 R2 | - Windows Server 2016<br>- Windows 10 | July 2016 update rollup for Windows 8.1 and Windows Server 2012 R2 or later monthly rollup |
| Windows Server 2012 | - Windows Server 2016<br>- Windows Server 2012 R2<br>- Windows 10 | July 2016 update rollup for Windows Server 2012 or later monthly rollup |
| Windows Server 2008 R2 | - Windows Server 2012 R2<br>- Windows Server 2012<br>- Windows 10 | Update that enables Windows 7 and Windows Server 2008 R2 KMS hosts to activate Windows 10 |
| Windows 8.1 | - Windows 10 | July 2016 update rollup for Windows 8.1 and Windows Server 2012 R2 or later monthly rollup |
| Windows 7 | - Windows 10 | Update that enables Windows 7 and Windows Server 2008 R2 KMS hosts to activate Windows 10 |

## Feedback

Was this page helpful? Yes No

# Strengthen Key Management Services (KMS) activation with hardware-based trust

Applies to: ✅ Windows Server 2025

KMS Hardware-Secured activation is a strengthened form of Key Management Services (KMS) volume activation that uses Trusted Platform Module (TPM) 2.0 attestation to confirm that activation requests come from trusted, hardware-backed environments. It binds activation to the integrity of the device, which mitigates spoofing and helps ensure that only verified, validated hosts participate in activation.

A KMS host that isn't hardware-secured relies on software alone to validate activation, which leaves your environment more exposed to unverified or emulated KMS hosts.

Beginning with the 2026-08 Cumulative Update for Windows Server 2025, KMS host readiness messaging tells you whether a host meets the hardware-based security requirements, so you can check and prepare your environment. The activation changes that build on hardware-based trust apply to future versions of Windows Server. To learn more, see Strengthening Key Management Service (KMS) with hardware-based trust.

After you read this article, you can identify whether a KMS host is Hardware-Secured, understand how it differs from a host that isn't Hardware-Secured, and know which requirements your environment must meet.

## How KMS Hardware-Secured activation works

KMS Hardware-Secured activation uses TPM 2.0 attestation to establish trust in the KMS host before it participates in activation. The TPM provides a hardware-backed measurement of the host, and attestation validates that measurement so that activation requests originate from a trusted, verified environment.

TPM attestation works in three parts:

- **Hardware identity**: The KMS host uses TPM-backed attestation to prove its hardware identity, which Microsoft verifies before the host activates devices.
- **Platform integrity**: The TPM confirms the KMS host's integrity.
- **Activation flow**: After verification, the KMS host securely serves activation requests for Windows devices.

This approach ties activation to a trusted host, not just to a software configuration that someone can copy or spoof.

Because the device hardware anchors trust rather than software configuration alone, KMS Hardware-Secured activation:

- Binds activation to the device's integrity.
- Resists tampering by binding activation secrets to hardware.
- Mitigates spoofing and the use of emulated or unverified KMS hosts.
- Provides perpetual activation without periodic renewal.

# KMS hardware-secured and KMS not hardware-secured hosts

A KMS host operates in one of two states. A KMS hardware-secured host adds a hardware root of trust, whereas a KMS not hardware-secured host relies on software alone. Understanding the difference helps you plan the right path for your environment.

Expand table

| Aspect | KMS hardware-secured host | KMS not hardware-secured host |
|---|---|---|
| Basis of trust | Hardware root of trust established through TPM 2.0 attestation. | Software only, which is vulnerable to spoofing. |
| Hardware requirement | TPM 2.0 installed and enabled. | None. |
| Activation renewal | Perpetual activation with no renewal required. | Standard KMS activation on Windows Server 2025. A future version of Windows Server might require periodic renewal. |
| When to use | Recommended state, for the strongest activation trust and the least ongoing maintenance. | Transition state, while you prepare hardware and complete the move to a KMS hardware-secured host. |

## When to choose a KMS hardware-secured host

A *KMS hardware-secured* host is the recommended state. This host uses TPM 2.0 attestation to establish hardware-based trust and provides perpetual activation without renewal. Choose this state when you want the strongest activation trust and the least ongoing maintenance.

## What to expect on a KMS not hardware-secured host

A non-hardware-secured KMS host uses standard KMS activation on Windows Server 2025 and continues to activate clients as it does today. In a future version of Windows Server, a non-hardware-secured host might require periodic renewal to keep clients activated. Prepare your hardware now to adopt KMS hardware-secured activation ahead of that change. To learn more, see Strengthening Key Management Service (KMS) with hardware-based trust.

# Requirements for a KMS hardware-secured host

To operate as a KMS hardware-secured host, a server must meet all the following requirements:

- The host has a TPM 2.0 chip installed and enabled.
- The server hardware is certified in the Windows Server Catalog.

# Environment readiness for KMS hardware-secured activation

To prepare for KMS hardware-secured activation, assess what you have today, watch for readiness signals, and plan any hardware changes you need.

## Assess your current KMS setup

Start by understanding your existing KMS infrastructure so you know which hosts already qualify and which need changes:

- Inventory your KMS hosts, including both physical and virtual hosts.
- Verify that each host has a TPM 2.0 chip that's installed and enabled.
- Confirm that each host's server hardware is certified in the Windows Server Catalog.

To confirm that a host supports the TPM attestation capability, run the following command in an elevated PowerShell session:

**PowerShell**

```powershell
Get-TpmSupportedFeature -FeatureList "Key Attestation"
```

A response that shows `Key Attestation` confirms the server supports the TPM attestation capability for KMS Hardware-Secured activation.

## Monitor readiness signals

Beginning with the 2026-08 Cumulative Update for Windows Server 2025, KMS host readiness messaging helps you plan ahead. This messaging is informational and doesn't affect activation. Clients continue to activate as they do today. The messaging shows which hosts already meet the hardware-based security requirements, so you can prepare for KMS Hardware-Secured activation in a future version of Windows Server. To learn more about the change, see Strengthening Key Management Service (KMS) with hardware-based trust.

Readiness signals appear on two surfaces:

- **Command line**: Running `slmgr /dlv` shows either "This device is eligible to serve as a KMS host with hardware-based security." or "This device doesn't meet the requirements for using a KMS host with hardware-based security."
- **Event logs**: Warning entries appear under **Applications and Services Logs** > **Key Management Service**.

## Plan your hardware transition

For hosts that don't yet qualify, plan the changes that move them to a KMS Hardware-Secured state:

- Install and enable TPM 2.0 on hosts that don't have it.
- Move to Windows Server–certified hardware where the current hardware isn't certified.
- Validate readiness after you make hardware changes to confirm the host qualifies as a KMS Hardware-Secured host.

# Related content

- Key Management Services (KMS) activation planning
- Create a Key Management Services (KMS) activation host
- Strengthening Key Management Service (KMS) with hardware-based trust (Windows IT Pro Blog)

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 08/11/2026

# Create a Key Management Services (KMS) activation host

Article • 04/14/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

Key Management Services (KMS) uses a client-server model to activate Windows clients. KMS is used for volume activation on your local network. KMS clients connect to a KMS server, called the KMS host, for activation. The KMS clients that a KMS host can activate are dependent on the host key used to activate the KMS host.

This article walks you through the steps you need to create a KMS host. For more information about KMS and initial planning considerations, see Key Management Services (KMS) activation planning.

## Prerequisites

A single KMS host can support an unlimited number of KMS clients. If you have more than 50 clients, we recommend that you have at least two KMS hosts in case one of your KMS hosts becomes unavailable. Most organizations can operate with as few as two KMS hosts for their entire infrastructure.

KMS hosts don't need to be dedicated servers. You can host other services on a KMS host. You can run a KMS host on any physical or virtual system that runs a supported Windows Server or Windows client operating system.

The version of Windows you use for your KMS host determines the version of Windows you can activate for your KMS clients. For help with deciding which version is appropriate for your environment, see the table of activation versions.

By default, KMS hosts publish Service (SRV) resource records in your Domain Name System (DNS). As a result, KMS clients can automatically discover the KMS host and activate without the need for any configuration on the KMS client. You can disable automatic publishing and create the records manually. These steps are needed for automatic activation if the DNS service doesn't support dynamic updates.

To create a KMS host, you need the following prerequisites:

- A computer running Windows Server or Windows. A KMS host running on Windows Server can activate computers running both server and client operating systems. However, a KMS host running on a Windows client operating system can only activate computers also running client operating systems.

- A user account that's a member of the Administrators group on the KMS host.
- KMS and client versions that are compatible with each other, and a Windows version that can host KMS. For more information, see Key Management Services (KMS) activation planning.
- A KMS host key for your organization. You can get this key from the Microsoft 365 admin center. For more information, see Find and use product keys for volume licensing.
- Access to the internet for activating the KMS host or performing a phone activation.

# Install and configure a KMS host

To install and configure a KMS host, take the steps in the following sections.

## Install the Volume Activation Services role

To install the Volume Activation Services role, run the following command in an elevated PowerShell session:

PowerShell

```
Install-WindowsFeature -Name VolumeActivation -IncludeManagementTools
```

## Configure Windows Firewall

Configure Windows Firewall to allow KMS to receive network traffic. You can allow this traffic for any network profiles, which is the default setting, or for any combination of Domain, Private, and Public network profiles. By default, a KMS host is configured to use Transmission Control Protocol (TCP) on port 1688.

To configure the firewall rule to allow network traffic for only the Domain and Private network profiles, run the following command:

PowerShell

```
Set-NetFirewallRule -Name SPPSVC-In-TCP -Profile Domain,Private -Enabled True
```

## Use the Volume Activation Tools wizard to configure the host

1. Open the Volume Activation Tools wizard by running the following command:

   PowerShell

```
vmw.exe
```

2. On the introduction page, select **Next**.
3. For the activation type, select **Key Management Service (KMS)**. For the server, enter **localhost** to configure the local server. If you want to configure a different server, enter its host name. Select **Next**.
4. Select **Install your KMS host key**, enter the product key for your organization, and then select **Commit**.
5. After the product key is installed, select **Next** to activate the product.
6. Under **Select product**, select the product you want to activate, and then select an activation method. To activate the key online, select **Activate online**, and then select **Commit**. If you're prompted to confirm activation of the KMS host, select **Yes**.

## Complete the configuration

After the activation finishes successfully, the KMS host configuration appears.

- If the configuration settings meet your requirements:
    1. Select **Close** to exit the wizard. The system creates DNS records, and you can start activating KMS clients.
    2. If you need to create SRV records manually to publish the KMS host, see Manually create DNS records later in this article.
- If you want to change the configuration settings:
    1. Select **Next**.
    2. Change the configuration values based on your requirements, and then select **Commit**.

> **Note**
>
> You can now start activating KMS clients. However, a network must first have a minimum number of computers (called the activation threshold). KMS hosts keep track of the number of recent connections. When a client or server contacts the KMS host, the host notes the machine ID, increments the count of contacts, and then returns the current count in its response. The client or server is activated if the count is high enough:
>
> - Windows clients are activated if the count is 25 or greater.

- Windows Server and volume editions of Microsoft Office products are activated if the count is five or greater.

In the count, KMS includes only unique connections from the past 30 days, and it stores only the 50 most recent contacts.

# Manually create DNS records

If your DNS service doesn't support dynamic updates, you must manually create the resource records in order to publish the KMS host. Create DNS resource records for KMS manually with your DNS service by using the following information. If you change the default port number in the KMS host configuration, also adjust the port number that you use for the resource records.

Expand table

| Property | Value |
|---|---|
| Type | SRV |
| Service/Name | _vlmcs |
| Protocol | _tcp |
| Priority | 0 |
| Weight | 0 |
| Port number | 1688 |
| Hostname | <FQDN-of-KMS-host> |

You should also disable publishing on all KMS hosts if your DNS service doesn't support dynamic updates. For instructions, see Disable publishing of DNS records later in this article. By disabling publishing, you prevent event logs from collecting failed DNS publishing events.

**Tip**

Manually created resource records can also coexist with resource records that KMS hosts automatically publish in other domains as long as all records are maintained to prevent conflicts.

## Disable publishing of DNS records

To disable publishing of DNS records by the KMS host:

1. Open the Volume Activation Tools wizard by running the following command:

   ```
   PowerShell

   vmw.exe
   ```

2. On the introduction page, select **Next**.

3. For the activation type, select **Key Management Service (KMS)**. For the server, enter **localhost** to configure the local server. If you want to configure a different server, enter its host name. Select **Next**.

4. Select **Skip to Configuration**, and then select **Next**.

5. Next to **DNS Records**, clear the **Publish** checkbox, and then select **Commit**.

# Replace a KMS host

07/30/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

This article outlines the steps and best practices for migrating the Key Management Services (KMS) host role to a new server. Migrating the KMS host is often necessary when the existing host's operating system (OS) is approaching its end-of-support. It might also be required due to operational or organizational changes that mandate moving the role to a different server. The instructions in this guide provide a seamless migration process while maintaining uninterrupted activation services for Microsoft Windows and Microsoft Office clients. No data migration is required during this process as the KMS Host doesn't rely on any database or backup.

Before proceeding with your installation and licensing tasks, review the following details regarding your new KMS host:

- Ensure you have access to your new Customer Specific Volume License Key (CSVLK). Your CSVLK is also known as the **KMS host key** for the Microsoft Windows OS and Microsoft Office, which are obtained through the Microsoft 365 Admin Center. The CSVLK has a predefined activation limit. If you encounter an error stating that you exceeded the activation limit, you can reset the key per request. To learn more, see Find and use product keys for volume licensing.
- If you're unable to find your CSVLK, contact support for licensing assistance.

## Prerequisites

- The **Volume Activation Services** role must be installed on the device that acts as the new KMS host. To learn more, see Install or Uninstall Roles, Role Services, or Features.

  *Alternatively*, you can run the following PowerShell command:

  ```PowerShell
  Install-WindowsFeature -Name VolumeActivation -IncludeManagementTools
  ```

- You must be a member of the following groups:
  - Administrators
  - Domain Admins
  - Enterprise Admins

- The CSVLK must be valid and accessible from your organization's licensing portal or the Microsoft 365 Admin Center. Ensure you have the appropriate CSVLK for the products you're activating, such as:
  - Microsoft Windows OS CSVLK
  - Microsoft Office CSVLK

- Your OS must have the latest Windows updates installed before configuring your device as a KMS host. To learn more, see KMS host required updates.

# Verify existing KMS hosts

Before you migrate to a new KMS host, it's recommended to inventory your environment to identify all existing KMS hosts. This helps ensure there aren't unauthorized or unnecessary KMS hosts present. Unapproved KMS hosts might appear if a CSVLK was used to activate a device that shouldn't serve as a KMS host. Only authorized servers should be activated with a CSVLK and configured as KMS hosts. Performing these actions can be done in an elevated command prompt or PowerShell window.

# Retrieve a KMS host

Command Prompt

To retrieve KMS hosts by the Domain Name System (DNS), run the following command:

Windows Command Prompt

```
nslookup -type=srv _vlmcs._tcp
```

To retrieve KMS hosts by the Fully Qualified Domain Name (FQDN), run the following command:

Windows Command Prompt

```
nslookup -type=SRV _vlmcs._tcp.mydomain.com
```

To retrieve KMS hosts by a specific DNS server, such as 8.8.8.8, run the following command:

Windows Command Prompt

```
nslookup -type=SRV _vlmcs._tcp.mydomain.com 8.8.8.8
```

If you discover unauthorized KMS hosts, you can revert them to KMS clients by running the following command in an elevated window and then restart the device. Replace the Generic Volume License Key (GVLK) with your GVLK:

```
slmgr.vbs /ipk <GVLK>
```

# Verify KMS host product activation

To verify which products the current KMS host is activating and ensure the new KMS host activates the same Windows OS and Microsoft Office clients, run the following command:

Windows Command Prompt

```
cscript %windir%\system32\slmgr.vbs /dlv All
```

PowerShell

```
cscript $env:windir\system32\slmgr.vbs /dlv All
```

Check the output to determine whether the KMS host is processing activation requests for the Windows OS, Microsoft Office, or both. The partial product keys displayed can help you match these KMS host keys (CSVLK) with your records. Additionally, review the KMS Host Event log to identify which clients are sending activation requests to this KMS host.

# Prepare the KMS host

Before configuring your environment as a KMS host, start with a clean installation of the target OS on the new server. To learn how to install your Windows Server OS, see Install Windows Server from installation media. Ensure that all available OS updates and security patches are applied and reboot as necessary. There are two options available for setting up your KMS host.

Windows OS KMS host

After you prepare the host OS, the next step is to configure it to serve as a KMS host. See

Create a Key Management Services (KMS) activation host.

# Verify firewall settings

Before proceeding on how to manage a KMS host, ensure that the firewall exception is configured for port **1688** to accept activation requests from KMS clients. Additionally, port **135** (Anonymous RPC) also needs to be configured.

GUI

1. Select **Start**, type **wf.msc**, then select it to open the **Windows Defender Firewall with Advanced Security**.
2. In the left pane, select **Inbound Rules**.
3. In the right pane, select **New Rule** to open the **New Inbound Rule Wizard**.
4. Under **Rule Type**, select **Port**, then select **Next**.
5. Under **Protocol and Ports**, select **TCP**, enter **1688** in the **Specific local ports** field, then select **Next**.
6. Under **Action**, ensure **Allow the connection** is selected, then select **Next**.
7. Under **Profile**, **Domain**, **Private**, and **Public** are selected by default. Select **Next**.
8. Under **Name**, provide any name desired for your rule, such as "KMS Host", then select **Finish**.

Repeat these steps to configure port **135**.

To verify that traffic is allowed through port **1688** or **135**, run the following command where values for **ComputerName** are the device name or IP address:

PowerShell

```
Test-NetConnection -ComputerName "MyDevice" -Port 1688
```

When the connection succeeds, the entry **TcpTestSucceeded** equals **True**, while if the connection to port 1688 can't be made (for instance, if no service is listening or because of a firewall or network issue) then **TcpTestSucceeded** equals **False**.

# Register a KMS host

After you configure the KMS host for Windows OS and Microsoft Office, it might automatically register with DNS if your domain permissions allow. If manual registration is required, follow the steps in Manually create DNS records.

After registering the new KMS host in DNS, you can remove the old KMS host from DNS. Client devices begin sending activation requests to the new KMS host, though it might take some time for the activation count to reach the required minimum thresholds.

> ⓘ Note
>
> - **KMS Count client activation threshold**: KMS requires at least **25** unique activation requests from the client or server OS to begin activating the client OS.
> - **KMS Count server activation threshold**: KMS requires at least **5** unique activation requests from the server or client OS to begin activating the server OS.

To confirm successful migration, review the *Event Log* on the new KMS host (**Event Viewer** > **Application and Services Log** > **Key Management Service**). You can also run the following command and review the output:

```
slmgr.vbs /dlv
```

Once client devices are sending activation requests to the new host, you can safely remove the old KMS host from DNS.

> ⓘ Note
>
> - To remove the KMS host functionality from old KMS host, install your GVLK and restart your device.
> - Best practice is to completely shut down the old KMS host to ensure that client devices transition to the new KMS host.
> - If a device was configured for a specific KMS host using `slmgr.vbs /skms`, executing `slmgr.vbs /ckms` clears that configuration and allows the devices to automatically detect the new KMS host.

# Troubleshoot KMS activation

Performing these actions must be done in an elevated command prompt or PowerShell window. To further troubleshoot other issues, see Guidelines for troubleshooting the Key Management Service (KMS).

# See also

- Key Management Services (KMS) activation planning
- Slmgr.vbs options for obtaining volume activation information

# KMS client activation and product keys

06/25/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

To use Key Management Services (KMS), you need to have a KMS host available on your local network. Computers that activate with a KMS host need to have a specific product key. This key is sometimes referred to as the KMS client key, but it's formally known as a Microsoft Generic Volume License Key (GVLK). Computers that are running volume licensing editions of Windows Server and Windows client are, by default, KMS clients with no extra configuration needed as the relevant GVLK is already there.

There are some scenarios where you'll need to add the GVLK to the computer you wish to activate against a KMS host, such as:

- Converting a computer from using a Multiple Activation Key (MAK)
- Converting a retail license of Windows to a KMS client
- If the computer was previously a KMS host

To use the GVLKs listed, you need a KMS host on your local network. If you don't have one, you can learn how to create a KMS host.

The KMS host requires its own key, known as the KMS host key, to activate or authenticate with Microsoft. This key is available in the Microsoft 365 admin center for Open, Open Value, Select, Enterprise, and Services Provider License agreements. You can get help from the local Microsoft Activation Center.

> ⊗ **Caution**
>
> When you activate Windows or Windows Server with a KMS host, the keys provided here are intended for volume licensing scenarios and **can't be used** with retail editions. KMS client keys **won't activate** or serve as a retail license key. If you installed a retail copy of Windows, you must use another activation method, like a MAK or by purchasing a genuine retail license. To learn more, see:
>
> - Activate Windows.
> - Windows Server pricing and licensing.

## Install a product key

If you're converting a computer from a KMS host, MAK, or retail edition of Windows to a KMS client, install the applicable product key (GVLK) from the list in this article. To install a client product key, open an administrative command prompt on the client, and run the following command and then press **Enter**:

Windows Command Prompt

```
slmgr /ipk <product key>
```

For example, to install the product key for Windows Server 2022 Datacenter Edition, run the following command and then press **Enter**:

Windows Command Prompt

```
slmgr /ipk WX4NM-KYWYW-QJJR4-XV3QB-6VM33
```

# Generic Volume License Keys

In the tables that follow, you'll find the GVLKs for each version and edition of Windows. LTSC is *Long-Term Servicing Channel*, while LTSB is *Long-Term Servicing Branch*.

## Windows 11 and Windows 10 Semi-Annual Channel

See the Windows lifecycle fact sheet for information about supported versions and end of service dates.

Expand table

| Operating system edition | KMS Client Product Key |
|---|---|
| Windows 11 Pro<br>Windows 10 Pro | W269N-WFGWX-YVC9B-4J6C9-T83GX |
| Windows 11 Pro N<br>Windows 10 Pro N | MH37W-N47XK-V7XM9-C7227-GCQG9 |
| Windows 11 Pro for Workstations<br>Windows 10 Pro for Workstations | NRG8B-VKK3Q-CXVCJ-9G2XF-6Q84J |
| Windows 11 Pro for Workstations N<br>Windows 10 Pro for Workstations N | 9FNHH-K3HBT-3W4TD-6383H-6XYWF |
| Windows 11 Pro Education<br>Windows 10 Pro Education | 6TP4R-GNPTD-KYYHQ-7B7DP-J447Y |

| Operating system edition | KMS Client Product Key |
| --- | --- |
| Windows 11 Pro Education N<br>Windows 10 Pro Education N | YVWGF-BXNMC-HTQYQ-CPQ99-66QFC |
| Windows 11 Education<br>Windows 10 Education | NW6C2-QMPVW-D7KKK-3GKT6-VCFB2 |
| Windows 11 Education N<br>Windows 10 Education N | 2WH4N-8QGBV-H22JP-CT43Q-MDWWJ |
| Windows 11 Enterprise<br>Windows 10 Enterprise | NPPR9-FWDCX-D2C8J-H872K-2YT43 |
| Windows 11 Enterprise N<br>Windows 10 Enterprise N | DPH2V-TTNVB-4X9Q3-TJR4H-KHJW4 |
| Windows 11 Enterprise G<br>Windows 10 Enterprise G | YYVX9-NTFWV-6MDM3-9PT4T-4M68B |
| Windows 11 Enterprise G N<br>Windows 10 Enterprise G N | 44RPN-FTY23-9VTTB-MP9BX-T84FV |

## Windows Enterprise LTSC and LTSB

Windows 11 LTSC 2024

Windows 10 LTSC 2021, 2019

Expand table

| Operating system edition | KMS Client Product Key |
| --- | --- |
| Windows 11 Enterprise LTSC 2024<br>Windows 10 Enterprise LTSC 2021<br>Windows 10 Enterprise LTSC 2019 | M7XTQ-FN8P6-TTKYV-9D4CC-J462D |
| Windows 11 Enterprise N LTSC 2024<br>Windows 10 Enterprise N LTSC 2021<br>Windows 10 Enterprise N LTSC 2019 | 92NFX-8DJQP-P6BBQ-THF9C-7CG2H |

## Earlier versions of Windows Client

Windows 8.1

Expand table

| Operating system edition | KMS Client Product Key |
|---|---|
| Windows 8.1 Pro | GCRJD-8NW9H-F2CDX-CCM8D-9D6T9 |
| Windows 8.1 Pro N | HMCNV-VVBFX-7HMBH-CTY9B-B4FXY |
| Windows 8.1 Enterprise | MHF9N-XY6XB-WVXMC-BTDCT-MKKG7 |
| Windows 8.1 Enterprise N | TT4HM-HN7YT-62K67-RGRQJ-JFFXW |

# Windows Server LTSC

Windows Server 2025

Expand table

| Operating system edition | KMS Client Product Key |
|---|---|
| Windows Server 2025 Standard | TVRH6-WHNXV-R9WG3-9XRFY-MY832 |
| Windows Server 2025 Datacenter | D764K-2NDRG-47T6Q-P8T8W-YP6DF |
| Windows Server 2025 Datacenter: Azure Edition | XGN3F-F394H-FD2MY-PP6FD-8MCRC |

# Windows Server Semi-Annual Channel

## Windows Server, versions 20H2, 2004, 1909, 1903, and 1809

Expand table

| Operating system edition | KMS Client Product Key |
|---|---|
| Windows Server Standard | N2KJX-J94YW-TQVFB-DG9YT-724CC |
| Windows Server Datacenter | 6NMRW-2C8FM-D24W7-TQWMY-CWH2D |

**Important**

Windows Server, version 20H2 reached end of service on August 9, 2022 and is no longer receiving security updates. This includes the retirement of Windows Server Semi-Annual Channel (SAC) with no future releases.

Customers using Windows Server SAC should move to Azure Stack HCI. Alternatively, customers might use the Long-Term Servicing Channel of Windows Server.

## Earlier versions of Windows Server

Windows Server, version 1803

Expand table

| Operating system edition | KMS Client Product Key |
|---|---|
| Windows Server Standard | PTXN8-JFHJM-4WC78-MPCBR-9W4KR |
| Windows Server Datacenter | 2HXDN-KRXHB-GPYC7-YCKFJ-7FVDG |

# Troubleshooting Windows volume activation

Article • 05/19/2022 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

Product activation is the process of validating software after it's installed on a specific computer. Activation confirms that the product is genuine (not a fraudulent copy) and that the product key or serial number is valid and has not been compromised or revoked. Activation also establishes a link or relationship between the product key and the installation.

Volume activation is the process of activating volume-licensed products. To become a volume licensing customer, an organization must set up a volume licensing agreement with Microsoft. Microsoft offers customized volume licensing programs that accommodate the organization's size and purchasing preference. For more information, see the Microsoft Volume Licensing Service Center.

The Windows Server 2016 Activation Guide focuses on the Key Management Service (KMS) activation technology. This section addresses common issues and provides troubleshooting guidelines for KMS and several other volume activation technologies.

## Best practices for volume activation

The following articles provide technical information and best practices for Microsoft's volume activation technologies.

### Key Management Service (KMS)

- Plan for volume activation
- Understanding KMS
- Deploying KMS Activation
- Configuring KMS Hosts
- Configuring DNS
- Activate using Key Management Service

### Active Directory-based activation (ADBA)

- Deploy Active-Directory-based Activation

- Activate using Active Directory-based activation
- Active Directory-Based Activation overview

### Multiple Activation Key (MAK) activation

- Using MAK Activation
- Understanding MAK Activation
- Activating MAK Clients

### Subscription activation

- Windows 10 Subscription Activation
- Deploy Windows 10 Enterprise licenses
- Windows 10 Enterprise E3 in CSP

## Resources for troubleshooting activation issues

The following articles provide guidelines and information about tools for troubleshooting volume activation issues:

- Guidelines for troubleshooting the Key Management Service (KMS)
- Slmgr.vbs options for obtaining volume activation information
- Example: Troubleshooting ADBA clients that do not activate

The following articles provide guidance for addressing more specific activation issues:

- Resolving common activation error codes
- KMS activation: known issues
- MAK activation: known issues
- Guidelines for troubleshooting DNS-related activation issues
- How to rebuild the Tokens.dat file

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## Feedback

**Was this page helpful?** Yes No

# Guidelines for troubleshooting the Key Management Service (KMS)

09/15/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

Enterprise customers set up Key Management Service (KMS) as part of their deployment process because it lets them use a simple, straightforward process to activate Windows in their environments. Usually, once you set up the KMS host, the KMS clients connect to the host automatically and activate on their own. However, sometimes the process doesn't work as expected. This article walks you through how to troubleshoot any issues you might encounter.

For more information about event log entries and the `slmgr.vbs` script, see Volume Activation Technical Reference.

## Where to begin troubleshooting KMS

Let's start with a quick refresher on how KMS activation works. KMS is a client-server model that has some similarities to Dynamic Host Configuration Protocol (DHCP). However, instead of handing out IP addresses to clients on their request, KMS enables product activation. KMS is also a renewal model, in which the clients try to reactivate on a regular interval. There are two roles: the *KMS host* and the *KMS client*.

- The KMS host runs the activation service and enables activation in the environment. To configure a KMS host, you must install KMS key from the Volume License Service Center (VLSC) and then activate the service.
- The KMS client is the Windows operating system (OS) that you deploy in the environment and need to activate. KMS clients can run any edition of Windows that uses volume activation. The KMS clients come with a preinstalled key, called the *Generic Volume License Key (GVLK)* or *KMS Client Setup Key*. The presence of the GVLK is what makes a system a KMS client. The KMS clients use Domain Name System (DNS) SRV records (`_vlmcs._tcp`) to identify the KMS host. Next, the clients automatically try to discover and use this service to activate themselves. During the 30-day out-of-the-box grace period, they try to activate every two hours. After you activate the KMS clients, they try to renew their activation every seven days.

From a troubleshooting perspective, you might have to look at both the host and client sides to figure out why an issue is happening.

# Troubleshooting on the KMS host

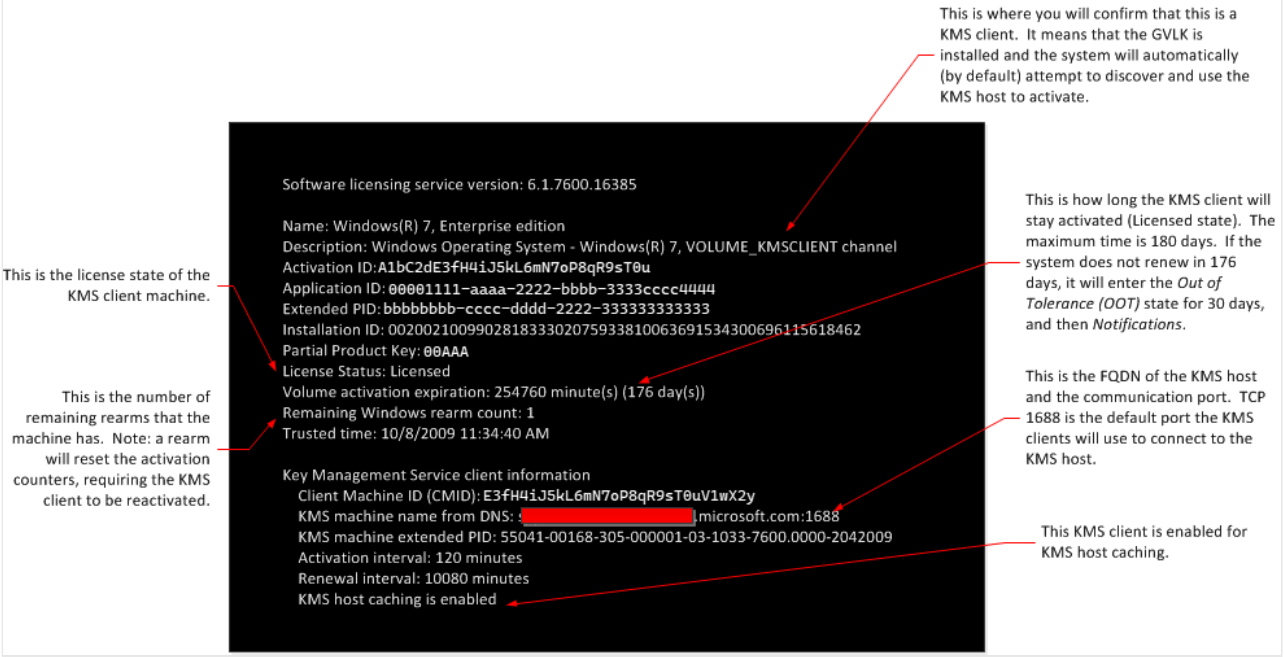
When you're examining the KMS host during troubleshooting, there are two areas you should look at:

- Check the status of the host software license service using the `slmgr.vbs` command in a command-line prompt.
- Check the Event Viewer for events related to licensing or activation.

# Slmgr.vbs and the Software Licensing service

You can use the `slmgr.vbs` command-line tool and the Event Viewer to troubleshoot activation issues on KMS clients. To see verbose output from the Software Licensing service, open an elevated Command Prompt window, or PowerShell window, and run `slmgr.vbs /dlv`. The following screenshot shows the results of this command for the KMS client and KMS host respectively:

KMS client




Here are some variables you should pay attention to in the output while troubleshooting:

- The *Version Information* is at the top of the `slmgr.vbs /dlv` output. The version information is useful for determining whether the service is up-to-date. Making sure everything's up to date is important because the KMS service supports different KMS host keys. You can use this data to evaluate whether or not the version you're currently using supports the KMS host key you're trying to install. For more information about updates, see An update is available for Windows Vista and for

Windows Server 2008 to extend KMS activation support for Windows 7 and for Windows Server 2008 R2.

- The *Name* indicates which edition of Windows is running on the KMS host system. You can use this information to troubleshoot issues that involve adding or changing the KMS host key. For example, you can use this information to verify if the OS edition supports the key you're trying to use.
- The *Description* shows you which key is currently installed. Use this field to verify whether the key that first activated the service was the correct one for the KMS clients you've deployed.
- The *License Status* shows the status of the KMS host system. The value should be **Licensed**. Any other value means you should reactivate the host.
- The *Current Count* displays a count between **0** and **50**. The count is cumulative between OSs and indicates the number of valid systems that have tried to activate within a **30-day** period.

  If the count is **0**, either the service was recently activated or no valid clients are connected to the KMS host.

  The count doesn't increase above **50**, no matter how many valid systems exist in the environment. The count is set to cache only twice the maximum license policy returned by a KMS client. The maximum policy set by the Windows client OS requires a count of **25** or higher from the KMS host to activate itself. Therefore, the highest count the KMS host can have are either **50** or 2 x 25. In environments that contain only Windows Server KMS clients, the maximum count on the KMS host is **10**. This limit is because the threshold for Windows Server editions is **5** (2 x 5 or 10).

  A common issue related to the count happens when the environment has an activated KMS host and enough clients, but the count doesn't increase beyond one. When this issue happens, it means the deployed client image wasn't configured correctly, so the systems don't have unique Client Machine IDs (CMIDs). For more information, see the following articles:

  - The KMS current count doesn't increase when you add new Windows Vista or Windows 7-based client computers to the network.
  - KMS Host Client Count not Increasing Due to Duplicate CMIDs.

  Another reason why the count might not be increasing is that there are too many KMS hosts in the environment and the count is distributed over all of them.

- **Listening on Port**. Communication with KMS uses anonymous RPC. By default, the clients use the 1688 TCP port to connect to the KMS host. Make sure that this port is open between your KMS clients and the KMS host. You can change or configure the port on the KMS host. During their communication, the KMS host sends the port designation to the KMS clients. If you change the port on a KMS client, the port designation is overwritten when that client contacts the host.

We often get asked about the *cumulative requests* section of the `slmgr.vbs /dlv` output. Generally, this data isn't helpful for troubleshooting. The KMS host keeps an ongoing record of the state of each KMS client that tries to activate or reactivate. Failed requests indicate the KMS host doesn't support certain KMS clients. For example, if a Windows 7 KMS client tries to activate against a KMS host that was activated by using a Windows Vista KMS key, the activation fails.

The *Requests with License Status* lines describe all possible license states, both past and present. From a troubleshooting perspective, this data is relevant only if the count isn't increasing as expected. In that case, you should see the number of failed requests increasing. To resolve this issue, you should check the product key that was used to first activate the KMS host system. Also, notice that the cumulative request values reset only if you reinstall the KMS host system.

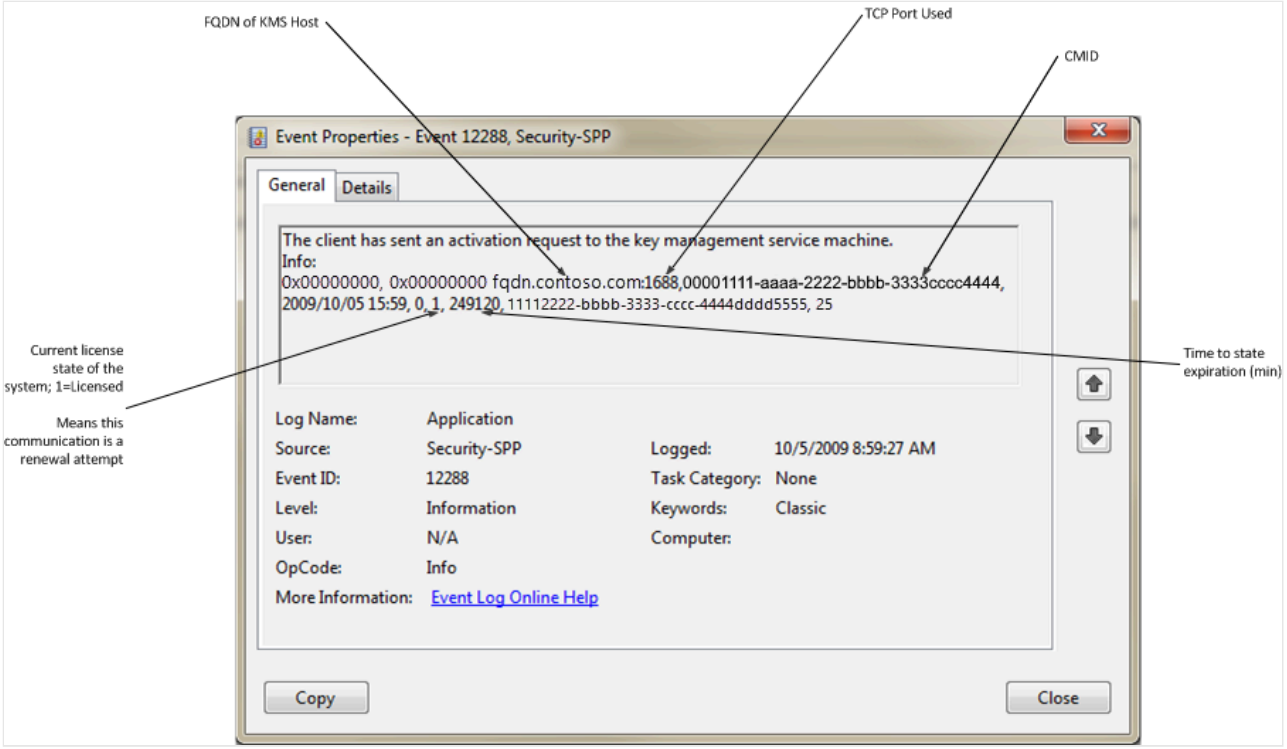
# KMS event IDs

The following sections describe client events that you should be familiar with to help you troubleshoot potential issues more efficiently. When a KMS client successfully activates or reactivates, the client logs event ID 12288 and event ID 12289.

KMS client

**Event ID 12288**:

The following screenshot showing a segment of an event ID 12288 entry from the KMS event log:



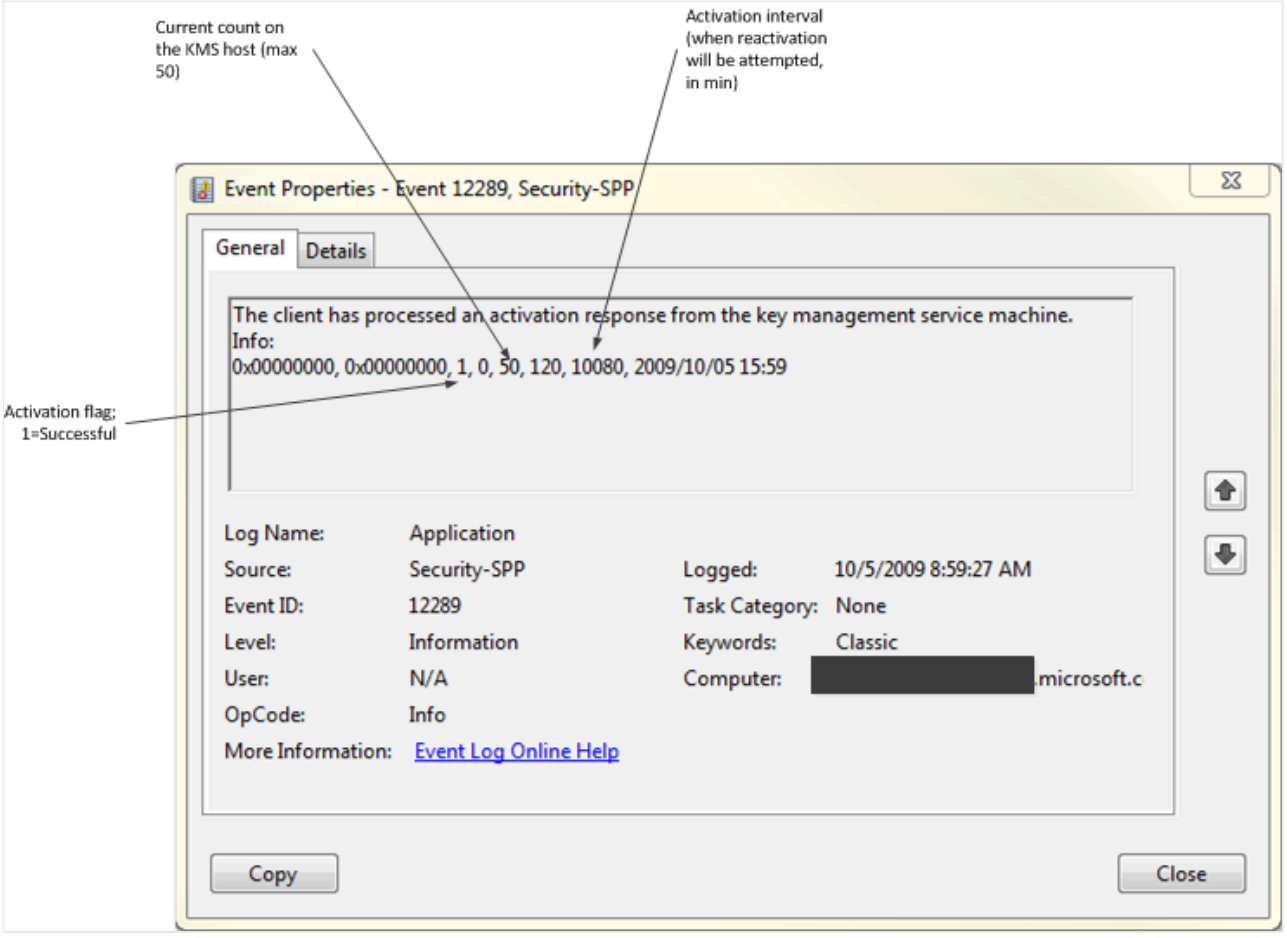

If you see only event ID 12288 without a corresponding event ID 12289, either the KMS client couldn't reach the KMS host, the KMS host didn't respond, or the client didn't receive the host's response. In these cases, you must verify that the KMS host is discoverable and that the KMS clients can contact it.

The most relevant information in event ID 12288 is the data in the *Info* field. For example, the Info field shows the current state of the client and which FQDN and TCP port the client used when it tried to activate. You can use the FQDN to troubleshoot scenarios where the count on a KMS host doesn't increase. For example, if there are too many KMS hosts available to the clients (either legitimate or unsupported systems), then the count might be distributed over all of them.

An unsuccessful activation doesn't always mean that the client has event ID 12288 and not 12289. A failed activation or reactivation might also have both events. In this case, you have to examine the second event to verify the reason for the failure.

**Event ID 12289**:

The following screenshot showing a segment of an event ID 12289 entry from the KMS event log:



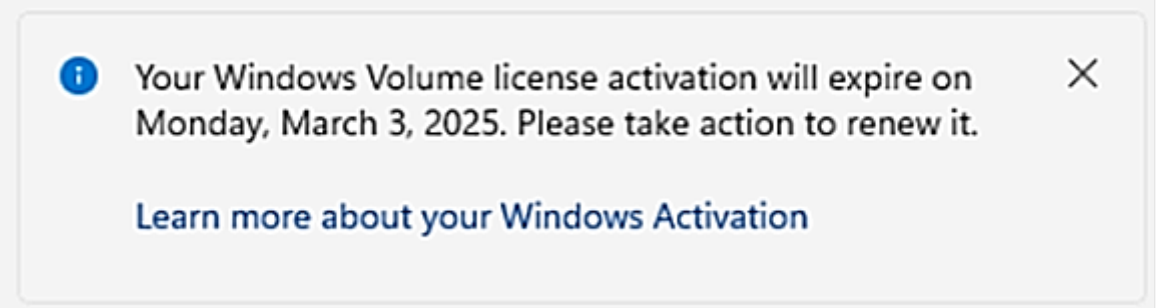

The *Info* section of event ID 12289 provides the following information:

- *Activation Flag*, which indicates whether the activation succeeded (**1**) or failed (**0**).
- *Current Count on the KMS Host*, which shows the count value on the KMS host when the client tries to activate. If activation fails, it might be because the count is insufficient for this client OS or that there aren't enough systems in the environment to build the count.

# KMS client renewal requirements

KMS activation operates on a renewal model, where each client device must connect to a KMS host server at least once every 180 days to maintain activation. By default, KMS clients attempt to renew their activation every seven days. When renewal succeeds, the activation validity period resets for another 180 days.

For any reason, if the renewal requirement isn't met, you're prompted with a toast notification 30 days before the 180 days for renewal. The renewal toast notification is also displayed in the activation pane:




## Troubleshoot KMS client renewal requirements

- You receive this notification *only* if you're on a corporate-owned or managed device. If this is incorrect, obtain a Retail key from the Microsoft Store.
- If you're using a corporate-owned or managed device, engage with your IT administrator. Here are some possible reasons why you might receive this message:
  - **The KMS host is decommissioned**: The IT administrator must configure the KMS client version with the KMS Server. Refer to Activate using Key Management Service.
  - **Listening on a different port**: Communication with the KMS uses anonymous RPC. By default, clients use TCP port 1688 to connect to the KMS host. Ensure this port is open between your KMS clients and the KMS host. You can configure the port on the KMS host through the Windows Firewall with Advanced Security.
  - **Verify DNS Configuration**: By default, the KMS clients use the automatic discovery process to query DNS for a list of servers. To learn more, see Guidelines for troubleshooting DNS-related activation issues.

# What does support ask for?

If your activations aren't working as expected after troubleshooting, you can contact Microsoft Support for technical assistance. The Support Engineer typically asks for the following information:

- `slmgr.vbs /dlv` output from the KMS host and KMS client systems.
- Event logs from both the KMS host (Key Management Service log) and KMS client systems (Application log).

# Slmgr.vbs options for obtaining volume activation information

07/07/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

Slmgr.vbs is a Visual Basic script included in Windows that serves as a command-line tool for managing the operating system's licensing and activation. It allows you to install and change product keys, activate Windows, and check its current activation or licensing status. It also supports tasks such as extending the activation grace period (rearming) and troubleshooting activation-related issues.

The syntax of the Slmgr.vbs script and tables in this article describe each command-line option.

Windows Command Prompt

```
slmgr.vbs [<ComputerName> [<User> <Password>]] [<Options>]
```

> **Note**
>
> In this article, square brackets ([ ]) enclose optional arguments, and angle brackets (< >) enclose placeholders. When you type these statements, omit the brackets and replace the placeholders by using corresponding values.
>
> For information about other software products that use volume activation, see the documents written for those applications.

## Using Slmgr on remote computers

To manage remote clients, use the Volume Activation Management Tool (VAMT) version 1.2 or later, or create custom WMI scripts that are aware of the differences between platforms. For more information, see WMI Properties and Methods for Volume Activation.

> **Important**
>
> Because of WMI changes in Windows 7 and Windows Server 2008 R2, the Slmgr.vbs script isn't intended to work across platforms. Using Slmgr.vbs to manage a Windows 7 or Windows Server 2008 R2 system from the Windows Vista operating system isn't supported. Trying to manage an older system from Windows 7 or Windows

Server 2008 R2 generates a specific version mismatch error. For example, running **cscript slmgr.vbs <vista_machine_name> /dlv** produces the following output:

```
Microsoft (R) Windows Script Host Version 5.8
Copyright (C) Microsoft Corporation. All rights reserved.

The remote machine does not support this version of SLMgr.vbs
```

# General Slmgr.vbs options

Expand table

| Option | Description |
|---|---|
| [<ComputerName>] | Name of a remote computer. (The default is the local computer.) |
| [<User>] | Account that has the required privilege on the remote computer. |
| [<Password>] | Password for the account that has the required privileges on the remote computer. |

# Global options

Expand table

| Option | Description |
|---|---|
| /ipk <ProductKey> | Tries to install a 5×5 product key. The product key provided by the parameter is confirmed valid and applicable to the installed operating system.<br><br>If it's not, an error is returned.<br><br>If the key is valid and applicable, the key is installed. If a key is already installed, it's silently replaced.<br><br>To prevent instability in the license service, the system should be restarted or the Software Protection Service should be restarted.<br><br>This operation must be run from an elevated Command Prompt window, or the Standard User Operations registry value must be set to allow unprivileged users extra access to the Software Protection Service. |

| Option | Description |
| --- | --- |
| /ato [<Activation ID>] | For retail editions and volume systems that have a KMS host key or a Multiple Activation Key (MAK) installed, **/ato** prompts Windows to try online activation.<br><br>For systems that have a Generic Volume License Key (GVLK) installed, **/ato** prompts a KMS activation attempt. Systems that are set to suspend automatic KMS activation attempts (**/stao**) still try KMS activation when **/ato** is run.<br><br>**Note:** Starting in Windows 8 (and Windows Server 2012), the **/stao** option is deprecated. Use the **/act-type** option instead.<br><br>The parameter <**Activation ID**> expands **/ato** support to identify a Windows edition installed on the computer. Specifying the <**Activation ID**> parameter isolates the effects of the option to the edition associated with that Activation ID. Run **slmgr.vbs /dlv all** to get the activation IDs for the installed version of Windows. If you need to support other applications, see the guidance provided by the application for further instruction.<br><br>KMS activation doesn't require elevated privileges. However, online activation does require elevation, or the Standard User Operations registry value must be set to allow unprivileged users extra access to the Software Protection Service. |
| /dli [<Activation ID> \| All] | Display license information.<br><br>By default, **/dli** displays the license information for the installed active Windows edition. Specifying the <**Activation ID**> parameter displays the license information for the specified edition that's associated with that activation ID. Specifying **All** as the parameter displays license information for all applicable installed products.<br><br>This operation doesn't require elevated privileges. |
| /dlv [<Activation ID> \| All] | Display detailed license information.<br><br>By default, **/dlv** displays the license information for the installed operating system. Specifying the <**Activation ID**> parameter displays the license information for the specified edition associated with that activation ID. Specifying the **All** parameter displays license information for all applicable installed products.<br><br>This operation doesn't require elevated privileges. |
| /xpr [<Activation ID>] | Display the activation expiration date for the product. By default, the date refers to the current Windows edition and is primarily useful for KMS clients, because MAK and retail activation is perpetual. |

| Option | Description |
| --- | --- |
| | Specifying the <**Activation ID**> parameter displays the activation expiration date of the specified edition that's associated with that activation ID. This operation doesn't require elevated privileges. |

# Advanced options

Expand table

| Option | Description |
| --- | --- |
| /cpky | Some servicing operations require the product key to be available in the registry during Out of Box Experience (OOBE) operations. The **/cpky** option removes the product key from the registry to prevent this key from being stolen by malicious code.<br><br>For retail installations that deploy keys, the best practice is to run this option. This option isn't required for MAK and KMS host keys, because this is the default behavior for those keys. This option is required only for other types of keys whose default behavior isn't to clear the key from the registry.<br><br>This operation must be run in an elevated Command Prompt window. |
| /ilc <license_file> | This option installs the license file specified by the required parameter. These licenses might be installed as a troubleshooting measure, to support token-based activation, or as part of a manual installation of an on-boarded application.<br><br>Licenses aren't validated during this process: License validation is out of scope for Slmgr.vbs. Instead, validation is handled by the Software Protection Service at runtime.<br><br>This operation must be run from an elevated Command Prompt window, or the Standard User Operations registry value must be set to allow unprivileged users extra access to the Software Protection Service. |
| /rilc | This option reinstalls all licenses stored in %SystemRoot%\system32\oem and %SystemRoot%\System32\spp\tokens. These are "known-good" copies that were stored during installation.<br><br>Any matching licenses in the Trusted Store are replaced. Any additional licenses—for example, Trusted Authority (TA) Issuance Licenses (ILs), licenses for applications—aren't affected.<br><br>This operation must be run in an elevated Command Prompt window, or the Standard User Operations registry value must be set to allow unprivileged users extra access to the Software Protection Service. |

| Option | Description |
| --- | --- |
| /rearm | This option resets the activation timers. The **/rearm** process is also called by **sysprep /generalize**.<br><br>This operation does nothing if the **HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\SoftwareProtectionPlatform\SkipRearm** registry entry is set to **1**. See Registry Settings for Volume Activation for details about this registry entry.<br><br>This operation must be run in an elevated Command Prompt window, or the Standard User Operations registry value must be set to allow unprivileged users extra access to the Software Protection Service. |
| /rearm-app <Application ID> | Resets the licensing status of the specified app. |
| /rearm-sku <Application ID> | Resets the licensing status of the specified SKU. |
| /upk [<Application ID>] | This option uninstalls the product key of the current Windows edition. After a restart, the system will be in an Unlicensed state unless a new product key is installed.<br><br>Optionally, you can use the <**Activation ID**> parameter to specify a different installed product.<br><br>This operation must be run from an elevated Command Prompt window. |
| /dti [<Activation ID>] | Displays installation ID for offline activation. |
| /atp <Confirmation ID> | Activate product by using user-provided confirmation ID. |

# KMS client options

Expand table

| Option | Description |
| --- | --- |
| /skms <Name[:Port] \| : Port> [<Activation ID>] | This option specifies the name and, optionally, the port of the KMS host computer to contact. Setting this value disables auto-detection of the KMS host.<br><br>If the KMS host uses Internet Protocol version 6 (IPv6) only, the address must be specified in the format <hostname>:<port>. IPv6 addresses contain colons (:), which the Slmgr.vbs script doesn't parse correctly. |

| Option | Description |
| --- | --- |
| | This operation must be run in an elevated Command Prompt window. |
| /skms-domain <FQDN> [<Activation ID>] | Sets the specific DNS domain in which all KMS SRV records can be found. This setting has no effect if the specific single KMS host is set by using the **/skms** option. Use this option, especially in disjoint namespace environments, to force KMS to ignore the DNS suffix search list and look for KMS host records in the specified DNS domain instead. |
| /ckms [<Activation ID>] | This option removes the specified KMS host name, address, and port information from the registry and restores KMS autodiscovery behavior.<br><br>This operation must be run in an elevated Command Prompt window. |
| /skhc | This option enables KMS host caching (default). After the client discovers a working KMS host, this setting prevents the Domain Name System (DNS) priority and weight from affecting further communication with the host. If the system can no longer contact the working KMS host, the client tries to discover a new host.<br><br>This operation must be run in an elevated Command Prompt window. |
| /ckhc | This option disables KMS host caching. This setting instructs the client to use DNS autodiscovery each time it tries KMS activation (recommended when using priority and weight).<br><br>This operation must be run in an elevated Command Prompt window. |

# KMS host configuration options

Expand table

| Option | Description |
| --- | --- |
| /sai <Interval> | This option sets the interval, in minutes, for unactivated clients to try to connect to KMS. The activation interval must be between 15 minutes and 30 days, although the default value (two hours) is recommended.<br><br>The KMS client initially picks up this interval from the registry but switches to the KMS setting after it receives the first KMS response.<br><br>This operation must be run in an elevated Command Prompt window. |
| /sri <Interval> | This option sets the renewal interval, in minutes, for activated clients to try to connect to KMS. The renewal interval must be between 15 minutes and 30 days. This option is set initially on both the KMS server and client sides. The default value is 10,080 minutes (7 days). |

| Option | Description |
| --- | --- |
| | The KMS client initially picks up this interval from the registry but switches to the KMS setting after it receives the first KMS response.<br><br>This operation must be run in an elevated Command Prompt window. |
| /sprt <Port> | This option sets the port on which the KMS host listens for client activation requests. The default TCP port is 1688.<br><br>This operation must be run from an elevated Command Prompt window. |
| /sdns | Enable DNS publishing by the KMS host (default).<br><br>This operation must be run in an elevated Command Prompt window. |
| /cdns | Disable DNS publishing by the KMS host.<br><br>This operation must be run in an elevated Command Prompt window. |
| /spri | Set the KMS priority to normal (default).<br><br>This operation must be run in an elevated Command Prompt window. |
| /cpri | Set the KMS priority to low.<br><br>Use this option to minimize contention from KMS in a co-hosted environment, which can cause KMS starvation depending on what other applications or server roles are active. Use this option with care.<br><br>This operation must be run in an elevated Command Prompt window. |
| /act-type [<Activation-Type>] [<Activation ID>] | This option sets a value in the registry that limits volume activation to a single type. Activation type **1** limits activation to Active Directory only. **2** limits it to KMS activation. **3** limits it to token-based activation. The **0** option allows any activation type and is the default value. |

# Token-based activation configuration options

Expand table

| Option | Description |
| --- | --- |
| /lil | List the installed token-based activation issuance licenses. |
| /ril <ILID> <ILvID> | Remove an installed token-based activation issuance license.<br><br>This operation must be run from an elevated Command Prompt window. |

| Option | Description |
| --- | --- |
| /stao | Set the **Token-based Activation Only** flag, disabling automatic KMS activation.<br><br>This operation must be run in an elevated Command Prompt window.<br><br>This option was removed in Windows Server 2012 R2 and Windows 8.1. Use the **/act–type** option instead. |
| /ctao | Clear the **Token-based Activation Only** flag (default), enabling automatic KMS activation.<br><br>This operation must be run in an elevated Command Prompt window.<br><br>This option was removed in Windows Server 2012 R2 and Windows 8.1. Use the **/act–type** option instead. |
| /ltc | List valid token-based activation certificates that can activate installed software. |
| /fta <Certificate Thumbprint> [<PIN>] | Force token-based activation by using the identified certificate. The optional personal identification number (PIN) is provided to unlock the private key without a PIN prompt if you use certificates that are protected by hardware (for example, smart cards). |

# Active Directory-based activation configuration options

Expand table

| Option | Description |
| --- | --- |
| /ad-activation-online <Product Key> [<Activation Object name>] | Collects Active Directory data and starts Active Directory forest activation using the credentials that the command prompt is running. Local administrator access isn't required. However, Read/Write access to the activation object container in the root domain of the forest is required. |
| /ad-activation-get-IID <Product Key> | This option starts Active Directory forest activation in phone mode. The output is the installation ID (IID) that can be used to activate the forest over the telephone if internet connectivity isn't available. When the IID is provided in the activation phone call, a CID is returned that's used to complete activation. |
| /ad-activation-apply-cid <Product Key> <Confirmation ID> | When you use this option, enter the CID that was provided in the activation telephone call to complete activation |

| Option | Description |
| --- | --- |
| [<Activation Object name>] | |
| [/name: <AO_Name>] | Optionally, you can append the **/name** option to any of these commands to specify a name for the activation object stored in Active Directory. The name must not exceed 40 Unicode characters. Use double quotation marks to explicitly define the name string.<br><br>In Windows Server 2012 R2 and Windows 8.1, you can append the name directly after **/ad-activation-online <Product Key>** and **/ad-activation-apply-cid** without having to use the **/name** option. |
| /ao-list | Displays all of the activation objects that are available to the local computer. |
| /del-ao <AO_DN><br><br>/del-ao <AO_RDN> | Deletes the specified activation object from the forest. |

# Related content

- Volume Activation Technical Reference
- Volume Activation Overview

# KMS activation: known issues

Article • 08/22/2022 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

**Try our Virtual Agent** - It can help you quickly identify and fix common issues related to KMS and MAK activation

This article describes common questions and issues that can arisse during Key Management Service (KMS) activations, and provides guidance for addressing the issues.

> ⓘ **Note**
>
> If you suspect that your issue is related to DNS, see Common troubleshooting procedures for KMS and DNS issues.

## Should I back up KMS host information?

Backup is not required for KMS hosts. However, if you use a tool to routinely clean up event logs, the activation history stored in the logs can be lost. If you use the event log to track or document KMS activations, periodically export the Key Management Service event log from the Applications and Services Logs folder of Event Viewer.

If you use System Center Operations Manager, the System Center Data Warehouse database stores event log data for reporting, therefore you do not have to back up the event logs separately.

## Is the KMS client computer activated?

On the KMS client computer, open the **System** control panel, and look for the **Windows is activated** message. Alternatively, run Slmgr.vbs and use the the **/dli** command-line option.

## The KMS client computer does not activate

Verify that the KMS activation threshold is met. On the KMS host computer, run Slmgr.vbs and use the **/dli** command-line option to determine the host's current count. Until the KMS host has a count of 25, Windows 7 client computers cannot be activated. Windows Server 2008 R2 KMS clients require a KMS count of 5 for activation. For more information about KMS requirements, see the Volume Activation Planning Guide.

On the KMS client computer, look in the Application event log for event ID 12289. Check this event for the following information:

- Is the result code **0**? Anything else is an error.
- Is the KMS host name in the event correct?
- Is the KMS port correct?
- Is the KMS host accessible?
- If the client is running a non-Microsoft firewall, does the outbound port have to be configured?

On the KMS host computer, look in the KMS event log for event ID 12290. Check this event for the following information:

- Did the KMS host log a request from the client computer? Verify that the name of the KMS client computer is listed. Verify that the client and KMS host can communicate. Did the client receive the response?
- If no event is logged from the KMS client, the request did not reach the KMS host or the KMS host was unable to process it. Make sure that routers do not block traffic using TCP port 1688 (if the default port is used) and that stateful traffic to the KMS client is allowed.

# What does this error code mean?

Except for KMS events that have event ID 12290, Windows logs all activation events to the Application event log under the event provider name Microsoft-Windows-Security-SPP. Windows logs KMS events to the Key Management Service log in the Applications and Services folder. IT pros can run Slui.exe to display a description of most activation-related error codes. The general syntax for this command is as follows:

Windows Command Prompt

```
slui.exe 0x2a ErrorCode
```

For example, if event ID 12293 contains error code 0x8007267C, you can display a description of that error by running the following command:

```
```

Windows Command Prompt

```
slui.exe 0x2a 0x8007267C
```

For more information about specific error codes and how to address them, see Resolving common activation error codes.

## Clients are not adding to the KMS count

To reset the client computer ID (CMID) and other product-activation information, run **sysprep /generalize** or **slmgr /rearm**. Otherwise, each client computer looks identical, and the KMS host does not count them as separate KMS clients.

## KMS hosts are unable to create SRV records

Domain Name System (DNS) may restrict Write access or may not support dynamic DNS (DDNS). In this case, give the KMS host Write access to the DNS database, or create the service (SRV) resource record (RR) manually. For more information about KMS and DNS issues, see Common troubleshooting procedures for KMS and DNS issues.

## Only the first KMS host is able to create SRV records

If the organization has more than one KMS host, the other hosts might not able to update the SRV RR unless the SRV default permissions are changed. For more information about KMS and DNS issues, see Common troubleshooting procedures for KMS and DNS issues.

## I installed a KMS key on the KMS client

KMS keys should be installed only on KMS hosts, not on KMS clients. Run **slmgr.vbs -ipk <SetupKey>**. For tables of keys that you can use to configure the computer as a KMS client, see KMS client setup keys. These keys are publicly known and are edition-specific. Remember to delete any unnecessary SRV RRs from DNS, and then restart the computers.

## A KMS host failed

If a KMS host fails, you must install a KMS host key on a new host and then activate the host. Make sure that the new KMS host has an SRV RR in the DNS database. If you install the new KMS host using the same computer name and IP address as the failed KMS host, the new KMS host can use the DNS SRV record of the failed host. If the new host has a different computer name, you can manually remove the DNS SRV RR of the failed host or (if scavenging is enabled in DNS) let DNS automatically remove it. If the network is using DDNS, the new KMS host automatically creates a new SRV RR on the DNS server. The new KMS host then starts collecting client renewal requests and begins activating clients as soon as the KMS activation threshold is met.

If your KMS clients use auto-discovery, they automatically select another KMS host if the original KMS host does not respond to renewal requests. If the clients do not use auto-discovery, you must manually update the KMS client computers that were assigned to the failed KMS host by running **slmgr.vbs /skms**. To avoid this scenario, configure the KMS clients to use auto-discovery. For more information, see the Volume Activation Deployment Guide.

---

# Feedback

**Was this page helpful?** Yes No

# MAK activation: known issues

Article • 08/22/2022 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

**Try our Virtual Agent** - It can help you quickly identify and fix common issues related to KMS and MAK activation

This article describes common issues that can occur during Multiple Activation Key (MAK) activations, and provides guidance for addressing those issues.

## How can I tell whether my computer is activated?

On the computer, open the **System** control panel and look for **Windows is activated**. Alternatively, run Slmgr.vbs and use the **/dli** command-line option.

## The computer does not activate over the internet

Make sure that the required ports are open in the firewall. For a list of ports, see the Volume Activation Deployment Guide.

## Internet and telephone activation fail

Contact a local Microsoft Activation Center. For the telephone numbers of Microsoft Activation Centers worldwide, go to Microsoft Licensing Activation Centers worldwide telephone numbers. Make sure to provide the Volume License agreement information and proof of purchase when you call.

## Slmgr.vbs /ato returns an error code

If Slmgr.vbs returns a hexadecimal error code, determine the corresponding error message by running the following script:

```
Windows Command Prompt
```

```
slui.exe 0x2a 0x <ErrorCode>
```

For more information about specific error codes and how to address them, see Resolving common activation error codes.

## Feedback

Was this page helpful? Yes No

# Guidelines for troubleshooting DNS-related activation issues

Article • 05/19/2022 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

You may have to use some of these methods if one or more of the following conditions are true:

- You use volume-licensed media and a Volume License generic product key to install one of the following operating systems:
  - Windows Server 2019
  - Windows Server 2016
  - Windows Server 2012 R2
  - Windows Server 2012
  - Windows Server 2008 R2
  - Windows Server 2008
  - Windows 10
  - Windows 8.1
  - Windows 8
- The activation wizard cannot connect to a KMS host computer.

When you try to activate a client system, the activation wizard uses DNS to locate a corresponding computer that's running the KMS software. If the wizard queries DNS and does not find the DNS entry for the KMS host computer, the wizard reports an error.

Review the following list to find an approach that fits your circumstances:

- If you cannot install a KMS host or if you cannot use KMS activation, try the Change the product key to an MAK procedure.
- If you have to install and configure a KMS host, use the Configure a KMS host for the clients to activate against procedure.
- If the client cannot locate your existing KMS host, use the following procedures to troubleshoot your routing configurations. These procedures are arranged from the simplest to the most complex.
  - Verify basic IP connectivity to the DNS server
  - Verify the KMS host configuration
  - Determine the type of routing issue
  - Verify the DNS configuration
  - Manually create a KMS SRV record

- Manually assign a KMS host to a KMS client
- Configure the KMS host to publish in multiple DNS domains

# Change the product key to an MAK

If you cannot install a KMS host or, for some other reason, you cannot use KMS activation, change the product key to an MAK. If you downloaded Windows images from the Microsoft Developer Network (MSDN), or from TechNet, the stock-keeping units (SKUs) that are listed below the media are generally volume licensed-media, and the product key that's provided is an MAK key.

To change the product key to an MAK, follow these steps:

1. Open an elevated Command Prompt window. To do this, press the Windows logo key+X, right-click **Command Prompt**, and then select **Run as administrator**. If you are prompted for an administrator password or for confirmation, type the password or provide confirmation.
2. At the command prompt, run the following command:

```
Windows Command Prompt

slmgr -ipk xxxxx-xxxxx-xxxxx-xxxxx-xxxxx
```

> **Note**
>
> The **xxxxx-xxxxx-xxxxx-xxxxx-xxxxx** placeholder represents your MAK product key.

Return to the procedure list.

# Configure a KMS host for the clients to activate against

KMS activation requires that a KMS host be configured for the clients to activate against. If there are no KMS hosts configured in your environment, install and activate one by using an appropriate KMS host key. After you configure a computer on the network to host the KMS software, publish the Domain Name System (DNS) settings.

For information about the KMS host configuration process, see Activate using Key Management Service and Install and Configure VAMT.

Return to the procedure list.

# Verify basic IP connectivity to the DNS server

Verify basic IP connectivity to the DNS server by using the ping command. To do this, follow these steps on both the KMS client that is experiencing the error and the KMS host computer:

1. Open an elevated Command Prompt window.
2. At the command prompt, run the following command:

Windows Command Prompt

```
ping <DNS_Server_IP_address>
```

> **Note**
>
> If the output from this command does not include the phrase "Reply from," there is a network problem or DNS issue that you must resolve before you can use the other procedures in this article. For more information about how to troubleshoot TCP/IP issues if you cannot ping the DNS server, see Advanced troubleshooting for TCP/IP issues.

Return to the procedure list.

# Verify the configuration of the KMS host

Check the registry of the KMS host server to determine whether it is registering with DNS. By default, a KMS host server dynamically registers a DNS SRV record one time every 24 hours.

> **Important**
>
> Follow the steps in this section carefully. Serious problems might occur if you modify the registry incorrectly. Before you modify it, back up the registry for restoration in case problems occur.

To check this setting, follow these steps:

1. Start Registry Editor. To do this, right-click **Start**, select **Run**, type **regedit**, and then press Enter.
2. Locate the **HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\SoftwareProtectionPlatform** subkey (previously **SL** instead of **SoftwareProtectionPlatform** in Windows Server 2008 and Windows Vista), and check the value of the **DisableDnsPublishing** entry. This entry has the following possible values:

   - **0** or undefined (default): The KMS host server registers a SRV record once every 24 hours.
   - **1**: The KMS host server does not automatically register SRV records. If your implementation does not support dynamic updates, see Manually create a KMS SRV record.

3. If the **DisableDnsPublishing** entry is missing, create it (the type is DWORD). If dynamic registration is acceptable, leave the value undefined or set it to **0**.

Return to the procedure list.

# Determine the type of routing issue

You can use the following commands to determine whether this is a name resolution issue or an SRV record issue.

1. On a KMS client, open an elevated Command Prompt window.

2. At the command prompt, run the following commands:

   Windows Command Prompt

   ```
   cscript \windows\system32\slmgr.vbs -skms <KMS_FQDN>:<port>
   cscript \windows\system32\slmgr.vbs -ato
   ```

   > **Note**
   >
   > In this command, <KMS_FQDN> represents the fully qualified domain name (FQDN) of the KMS host computer and <port> represents the TCP port that KMS uses.

   If these commands resolve the problem, this is an SRV record issue. You can you can troubleshoot it by using one of the commands that are documented in the Manually assign a KMS host to a KMS client procedure.

3. If the problem persists, run the following commands:

```
Windows Command Prompt

cscript \windows\system32\slmgr.vbs -skms <IP Address>:<port>
cscript \windows\system32\slmgr.vbs -ato
```

> **Note**
>
> In this command, <IP Address> represents the IP address of the KMS host computer and <port> represents the TCP port that KMS uses.

   If these commands resolve the problem, this is most likely a name resolution issue. For additional troubleshooting information, see the Verify the DNS configuration procedure.

4. If none of these commands resolves the problem, check the computer's firewall configuration. Any activation communications that occur between KMS clients and the KMS host use the 1688 TCP port. The firewalls on both the KMS client and the KMS host must allow communication over port 1688.

Return to the procedure list.

# Verify the DNS configuration

> **Note**
>
> Unless otherwise stated, follow these steps on a KMS client that has experienced the applicable error.

1. Open an elevated Command Prompt window
2. At the command prompt, run the following command:

```
Windows Command Prompt

IPCONFIG /all
```

3. From the command results, note the following information:

   - The assigned IP address of the KMS client computer
   - The IP address of the Primary DNS server that the KMS client computer uses

- The IP address of the default gateway that the KMS client computer uses
- The DNS suffix search list that the KMS client computer uses

4. Verify that the KMS host SRV records are registered in DNS. To do this, follow these steps:
   a. Open an elevated Command Prompt window.
   b. At the command prompt, run the following command:

```
Windows Command Prompt

nslookup -type=all _vlmcs._tcp>kms.txt
```

   c. Open the KMS.txt file that the command generates. This file should contain one or more entries that resemble the following entry:

```
_vlmcs._tcp.contoso.com SRV service location:
priority = 0
weight = 0
port = 1688 svr hostname = kms-server.contoso.com
```

> **Note**
>
> In this entry, contoso.com represents the domain of the KMS host.

   i. Verify the IP address, host name, port, and domain of the KMS host.
   ii. If these **_vlmcs** entries exist, and if they contain the expected KMS host names, go to Manually assign a KMS host to a KMS client.

> **Note**
>
> If the nslookup command finds the KMS host, it does not mean that the DNS client can find the KMS host. If the **nslookup** command finds the KMS host, but you still cannot activate by using the KMS host, check the other DNS settings, such as the primary DNS suffix and the search list of the DNS suffix.

5. Verify that the search list of the primary DNS suffix contains the DNS domain suffix that is associated with the KMS host. If the search list does not include this information, go to the Configure the KMS host to publish in multiple DNS domains procedure.

Return to the procedure list.

# Manually create a KMS SRV record

To manually create an SRV record for a KMS host that uses a Microsoft DNS server, follow these steps:

1. On the DNS server, open DNS Manager. To open DNS Manager, select **Start**, select **Administrative Tools**, and then select **DNS**.
2. Select the DNS server on which you have to create the SRV resource record.
3. In the console tree, expand **Forward Lookup Zones**, right-click the domain, and then select **Other New Records**.
4. Scroll down the list, select **Service Location (SRV)**, and then select **Create Record**.
5. Type the following information:

    - Service: **_VLMCS**
    - Protocol: **_TCP**
    - Port number: **1688**
    - Host offering the service: <***FQDN of the KMS host***>

6. When you are finished, select **OK**, and then select **Done**.

To manually create an SRV record for a KMS host that uses a BIND 9.x-compliant DNS server, follow the instructions for that DNS server, and provide the following information for the SRV record:

- Name: **_vlmcs._TCP**
- Type: **SRV**
- Priority: **0**
- Weight: **0**
- Port: **1688**
- Hostname: <***FQDN or A-Name of the KMS host***>

To configure a BIND 9.x-compatible DNS server to support KMS auto-publishing, configure the DNS server to enable resource record updates from KMS hosts. For example, add the following line to the zone definition in Named.conf or in Named.conf.local:

Windows Command Prompt

```
allow-update { any; };
```

# Manually assign a KMS host to a KMS client

By default, the KMS clients use the automatic discovery process. According to this process, a KMS client queries DNS for a list of servers that have published `_vlmcs` SRV records within the membership zone of the client. DNS returns the list of KMS hosts in a random order. The client picks a KMS host and tries to establish a session on it. If this attempt works, the client caches the name of the KMS host and tries to use it for the next renewal attempt. If the session setup fails, the client randomly picks another KMS host. We highly recommend that you use the automatic discovery process.

However, you can manually assign a KMS host to a particular KMS client. To do this, follow these steps.

1. On a KMS client, open an elevated Command Prompt window.
2. Depending on your implementation, follow one of these steps:

    - To assign a KMS host by using the FQDN of the host, run the following command:

      Windows Command Prompt

      ```
      cscript \windows\system32\slmgr.vbs -skms <KMS_FQDN>:<port>
      ```

    - To assign a KMS host by using the version 4 IP address of the host, run the following command:

      Windows Command Prompt

      ```
      cscript \windows\system32\slmgr.vbs -skms <IPv4Address>:<port>
      ```

    - To assign a KMS host by using the version 6 IP address of the host, run the following command:

      Windows Command Prompt

      ```
      cscript \windows\system32\slmgr.vbs -skms <IPv6Address>:<port>
      ```

    - To assign a KMS host by using the NETBIOS name of the host, run the following command:

      Windows Command Prompt

      ```
      cscript \windows\system32\slmgr.vbs -skms <NETBIOSName>:<port>
      ```

    - To revert to automatic discovery on a KMS client, run the following command:

Windows Command Prompt

```
cscript \windows\system32\slmgr.vbs -ckms
```

> **Note**
>
> These commands use the following placeholders:
> - **<KMS_FQDN>** represents the fully qualified domain name (FQDN) of the KMS host computer
> - **<IPv4Address>** represents the IP version 4 address of the KMS host computer
> - **<IPv6Address>** represents the IP version 6 address of the KMS host computer
> - **<NETBIOSName>** represents the NETBIOS name of the KMS host computer
> - **<port>** represents the TCP port that KMS uses.

# Configure the KMS host to publish in multiple DNS domains

> **Important**
>
> Follow the steps in this section carefully. Serious problems might occur if you modify the registry incorrectly. Before you modify it, back up the registry for restoration in case problems occur.

As described in Manually assign a KMS host to a KMS client, KMS clients typically use the automatic discovery process to identify KMS hosts. This process requires that the `_vlmcs` SRV records must be available in the DNS zone of the KMS client computer. The DNS zone corresponds to either the primary DNS suffix of the computer or to one of the following:

- For domain-joined computers, the computer's domain as assigned by the DNS system (such as Active Directory Domain Services (AD DS) DNS).
- For workgroup computers, the computer's domain as assigned by the Dynamic Host Configuration Protocol (DHCP). This domain name is defined by the option that has the code value of 15 as defined in Request for Comments (RFC) 2132.

By default, a KMS host registers its SRV records in the DNS zone that corresponds to the domain of the KMS host computer. For example, assume that a KMS host joins the contoso.com domain. In this scenario, the KMS host registers its `_vlmcs` SRV record under the contoso.com DNS zone. Therefore, the record identifies the service as `_VLMCS._TCP.CONTOSO.COM`.

If the KMS host and KMS clients use different DNS zones, you must configure the KMS host to automatically publish its SRV records in multiple DNS domains. To do this, follow these steps:

1. On the KMS host, start Registry Editor.
2. Locate and then select the **HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\SoftwareProtectionPlatform** subkey (previously **SL** instead of **SoftwareProtectionPlatform** in Windows Server 2008 and Windows Vista).
3. In the **Details** pane, right-click a blank area, select **New**, and then select **Multi-String Value**.
4. For the name of the new entry, enter **DnsDomainPublishList**.
5. Right-click the new **DnsDomainPublishList** entry, and then select **Modify**.
6. In the **Edit Multi-String** dialog box, type each DNS domain suffix that KMS publishes on a separate line, and then select **OK**.

   > ⓘ **Note**
   >
   > For Windows Server 2008 R2, the format for **DnsDomainPublishList** differs. For more information, see the Volume Activation Technical Reference Guide.

7. Use the Services administrative tool to restart the Software Protection service (previously the Software Licensing service in Windows Server 2008 and Windows Vista). This operation creates the SRV records.
8. Verify that by using a typical method, the KMS client can contact the KMS host that you configured. Verify that the KMS client correctly identifies the KMS host both by name and by IP address. If either of these verifications fails, investigate this DNS client resolver issue.
9. To clear any previously cached KMS host names on the KMS client, open an elevated Command Prompt window on the KMS client, and then run the following command:

   Windows Command Prompt

   ```
   cscript C:\Windows\System32\slmgr.vbs -ckms
   ```

# Feedback

Was this page helpful? Yes No

# Rebuild the Tokens.dat file

Article • 05/19/2022 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

When you troubleshoot Windows activation issues, you may have to rebuild the Tokens.dat file. This article describes in detail how to do this.

## Resolution

To rebuild the Tokens.dat file, follow these steps:

1. Open an elevated Command Prompt window: **For Windows 10**
   a. Open the **Start** menu, and enter **cmd**.
   b. In the search results, right-click **Command Prompt**, and the select **Run as administrator**.

   **For Windows 8.1**
   a. Swipe in from the right edge of the screen, and then tap **Search**. Or, if you are using a mouse, point to the lower-right corner of the screen, and then select **Search**.
   b. In the search box, enter **cmd**.
   c. Swipe across or right-click the displayed **Command Prompt** icon.
   d. Tap or click **Run as administrator**.

   **For Windows 7**
   a. Open the **Start** menu, and enter **cmd**.
   b. In the search results, right-click **cmd.exe**, and the select **Run as administrator**.

2. Enter the list of commands that is appropriate for your operating system.

   For Windows 10, Windows Server 2016 and later versions of Windows, enter the following commands in sequence:

   Windows Command Prompt

```
net stop sppsvc
cd %Systemdrive%\Windows\System32\spp\store\2.0\
ren tokens.dat tokens.bar
net start sppsvc
cscript.exe %windir%\system32\slmgr.vbs /rilc
```

For Windows 8.1, Windows Server 2012 and Windows Server 2012 R2, enter the following commands in sequence:

Windows Command Prompt

```
net stop sppsvc
cd %Systemdrive%\Windows\System32\spp\store\
ren tokens.dat tokens.bar
net start sppsvc
cscript.exe %windir%\system32\slmgr.vbs /rilc
```

For Windows 7, Windows Server 2008 and Windows Server 2008 R2, enter the following commands in sequence:

Windows Command Prompt

```
net stop sppsvc
cd
%Systemdrive%\Windows\ServiceProfiles\NetworkService\AppData\Roaming\Mi
crosoft\SoftwareProtectionPlatform
ren tokens.dat tokens.bar
net start sppsvc
cscript.exe %windir%\system32\slmgr.vbs /rilc
```

3. Restart the computer.

# More information

After you rebuild the Tokens.dat file, you must reinstall your product key by using one of the following methods:

- At the same elevated prompt command, type the following command, and then press Enter:

  Windows Command Prompt

  ```
  cscript.exe %windir%\system32\slmgr.vbs /ipk <Product key>
  ```

  > **Important**
  >
  > Do not use the **/upk** switch to uninstall a product key. To install a product key over an existing product key, use the **/ipk** switch.

- Right-click **My Computer**, select **Properties**, and then select **Change product key**.

For more information about KMS client setup keys, see KMS client setup keys.

## Feedback

Was this page helpful? Yes No

# Azure Connected Machine agent deployment options

You can connect machines in your hybrid environment directly with Azure by using different methods, depending on your requirements and the tools you prefer to use.

## Onboarding methods

The following table highlights each method so that you can determine which one works best for your deployment. For detailed information, follow the links to view the steps for each method.

Expand table

| Supported OS Type | Method | Description |
|---|---|---|
| Linux and Windows | Interactively | Manually install the agent on a single or small number of machines to connect machines by using a deployment script.<br>From the Azure portal, you can generate a script and run it on the machine to automate the installation and configuration steps of the agent. |
| Linux and Windows | Interactively or at scale | Connect machines by using PowerShell. |
| Linux and Windows | At scale | Connect machines at scale with Ansible using the Azure Arc onboarding role in Ansible Core or Ansible Automation Platform. |
| Linux and Windows | At scale | Connect machines by using a service principal to install the agent at scale noninteractively. |
| Linux and Windows | At scale | Install the Azure Arc agent on VMware virtual machines (VMs) at scale by using Azure Arc-enabled VMware vSphere. With Azure Arc-enabled VMware vSphere, you can connect your VMware vCenter server to Azure, automatically discover your VMware VMs, and install the Azure Arc agent on them. Requires VMware tools on VMs. |
| Linux and Windows | At scale | Connect your AWS cloud through the multicloud connector enabled by Azure Arc and enable the Arc onboarding solution to autodiscover and onboard EC2 VMs. |
| Windows | Interactively | Connect machines from Windows Admin Center. |
| Windows | Interactively | Connect Windows Server machines to Azure through Azure Arc setup. |

| Supported OS Type | Method | Description |
|---|---|---|
| Windows | At scale | Connect machines by running PowerShell scripts with Configuration Manager. |
| Windows | At scale | Connect machines with a Configuration Manager custom task sequence. |
| Windows | At scale | Connect Windows machines by using Group Policy. |

> **Important**
>
> You can't install the Connected Machine agent on an Azure VM. The installation script warns you and rolls back if it detects that the server is running in Azure.

Be sure to review the basic prerequisites and network configuration requirements before you deploy the agent and any specific requirements listed in the steps for the onboarding method that you choose. To learn more about what changes the agent makes to your system, see Overview of the Azure Connected Machine agent.

## Automatic connection for SQL Server

When you connect a Windows or Linux server to Azure Arc that also has Microsoft SQL Server installed, the SQL Server instances are automatically connected to Azure Arc as well. SQL Server enabled by Azure Arc provides a detailed inventory and additional management capabilities for your SQL Server instances and databases. As part of the connection process, an extension is deployed to your Azure Arc-enabled server, and new roles are applied to your SQL Server and databases. If you don't want to automatically connect your SQL Servers to Azure Arc, you can opt out by adding a tag to the Windows or Linux server with the name `ArcSQLServerExtensionDeployment` and value `Disabled` when connecting it to Azure Arc.

For more information, see Manage automatic connection for SQL Server enabled by Azure Arc.

## Related content

- Learn about the Azure Connected Machine agent prerequisites and network requirements.
- Review the Planning and deployment guide for Azure Arc-enabled servers.
- Learn about reconfiguring, upgrading, and removing the Connected Machine agent.
- Try out Azure Arc-enabled servers by using Azure Arc Jumpstart.

Last updated on 08/12/2026

# Connect Windows Server machines to Azure through Azure Arc Setup

You can onboard Windows Server machines directly to Azure Arc through a graphical wizard included in Windows Server. The wizard automates the onboarding process by checking the necessary prerequisites for successful Azure Arc onboarding, then fetching and installing the latest version of the Azure Connected Machine (AzCM) agent. When the wizard process completes, you're directed to your Window Server machine in the Azure portal, where you can view and manage it like any other Azure Arc-enabled resource.

If the Windows Server machine is already running in Azure, there's no need to onboard to Azure Arc.

For Windows Server 2022, Azure Arc Setup is an optional component that you can remove by using the **Remove Roles and Features Wizard**. For Windows Server 2025 and later, Azure Arc Setup is a Feature On Demand. The procedures for removal and enablement differ between OS versions.

> **Note**
>
> The Azure Arc Setup feature only applies to Windows Server 2022 and later. It was released in the Cumulative Update of 10/10/2023.

## Automatic connection for SQL Server

When you connect a Windows or Linux server to Azure Arc that also has Microsoft SQL Server installed, the SQL Server instances are automatically connected to Azure Arc as well. SQL Server enabled by Azure Arc provides a detailed inventory and additional management capabilities for your SQL Server instances and databases. As part of the connection process, an extension is deployed to your Azure Arc-enabled server, and new roles are applied to your SQL Server and databases. If you don't want to automatically connect your SQL Servers to Azure Arc, you can opt out by adding a tag to the Windows or Linux server with the name `ArcSQLServerExtensionDeployment` and value `Disabled` when connecting it to Azure Arc.

For more information, see Manage automatic connection for SQL Server enabled by Azure Arc.

## Prerequisites

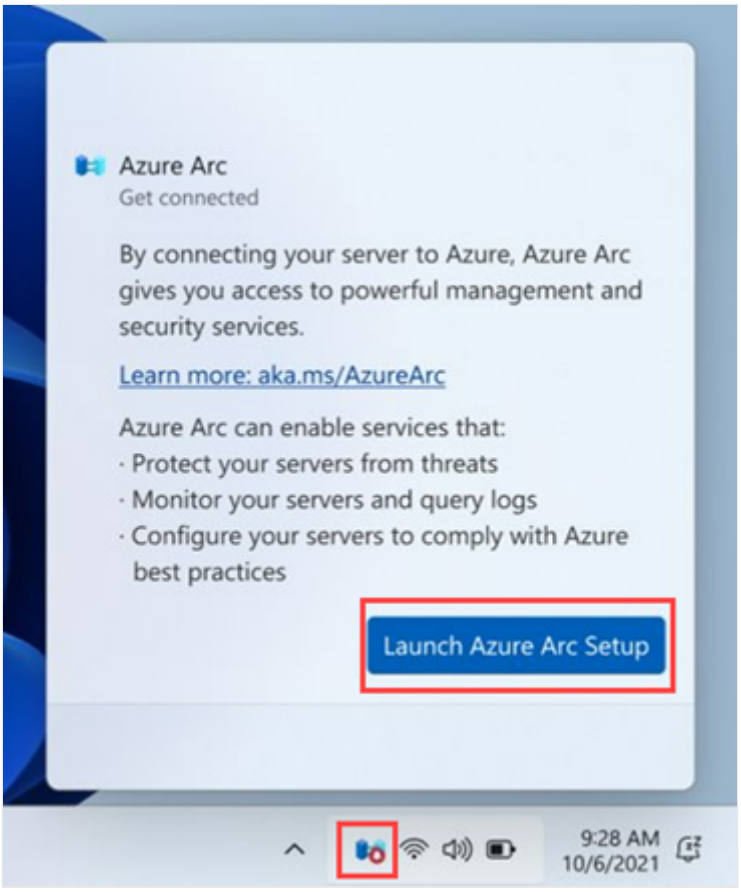
- Review the Azure Arc-enabled servers prerequisites and verify that your subscription, your Azure account, and resources meet the requirements.

- An Azure subscription. If you don't have one, create a free account before you begin.
- Modern browser (Microsoft Edge) for authentication to Microsoft Azure. Configuration of the Azure Connected Machine agent requires authentication to your Azure account, either through interactive authentication on a modern browser or device code authentication on a separate device (if the machine doesn't have a modern browser).

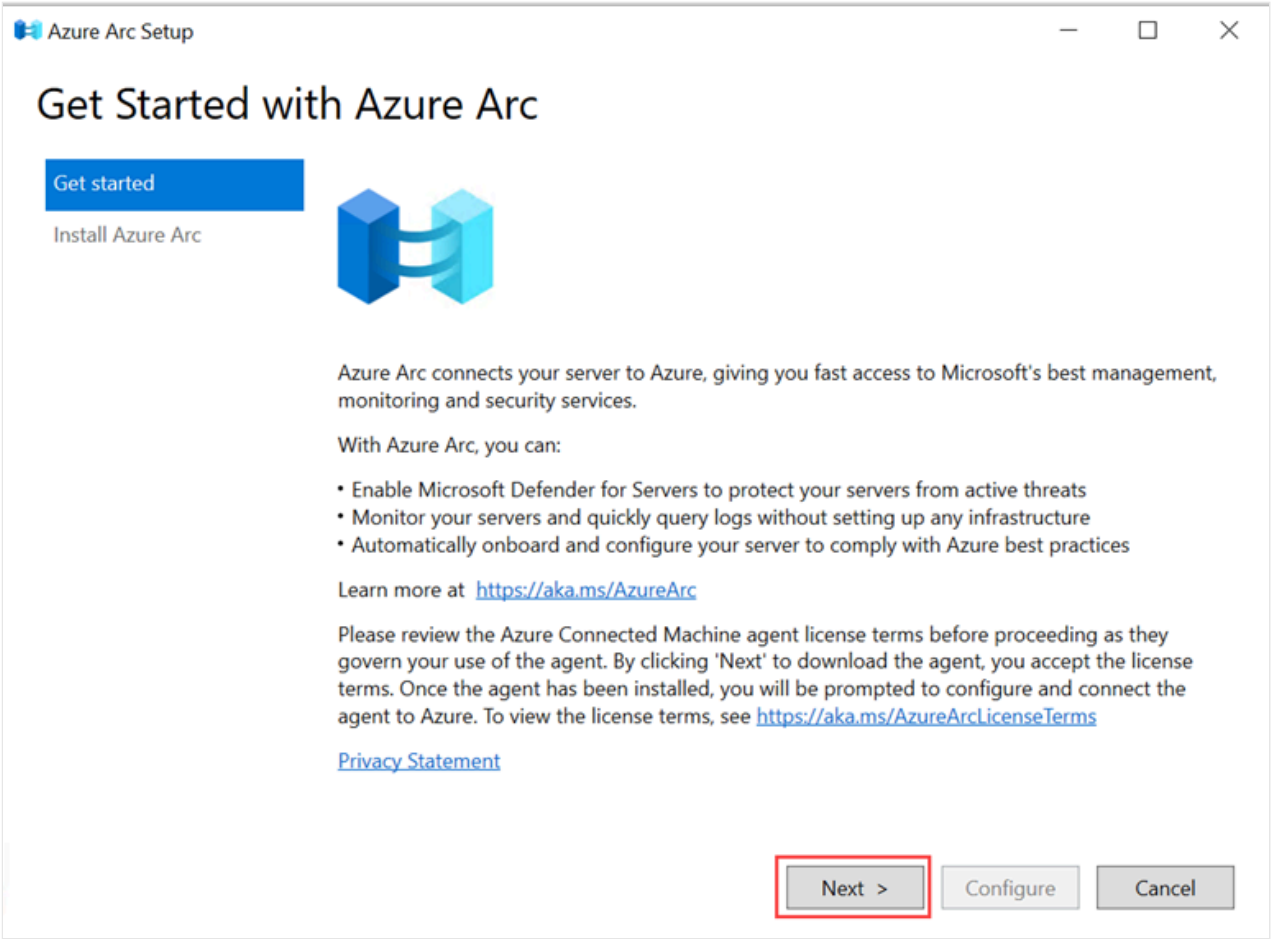
# Launch Azure Arc Setup and connect to Azure Arc

The Azure Arc Setup wizard is launched from a system tray icon at the bottom of the Windows Server machine when the Azure Arc Setup feature is enabled. This feature is enabled by default. Alternatively, you can launch the wizard from a pop-up window in the Server Manager or from the Windows Server Start menu.

1. Select the Azure Arc system tray icon, and then select **Launch Azure Arc Setup**.




2. The introduction window of the Azure Arc Setup wizard explains the benefits of onboarding your machine to Azure Arc. When you're ready to proceed, select **Next**.




3. The wizard automatically checks for the prerequisites necessary to install the Azure Connected Machine agent on your Windows Server machine. Once this process completes and the agent is installed, select **Configure**.

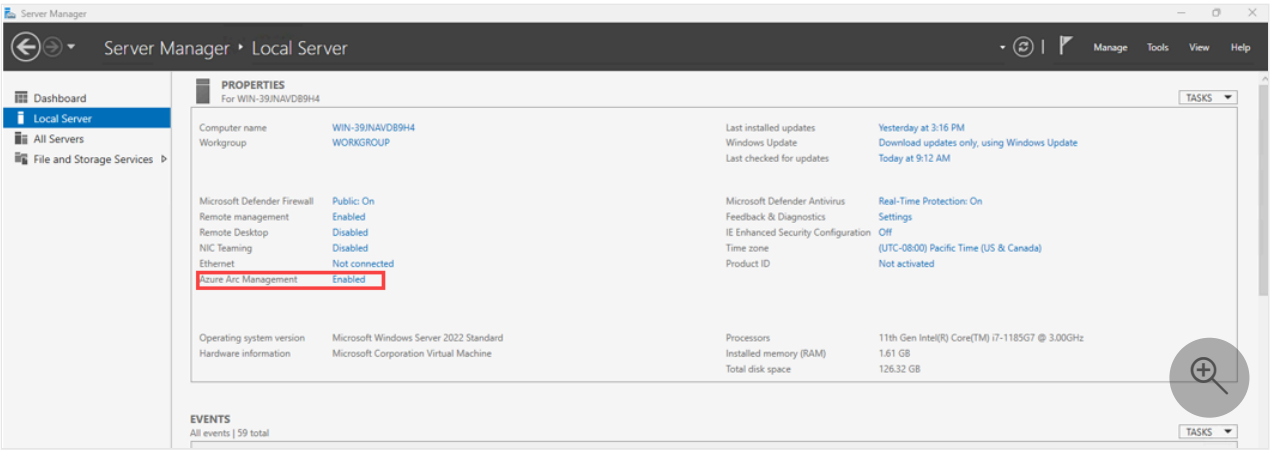
4. The configuration window explains the steps required to configure the Azure Connected Machine agent. When you're ready to begin configuration, select **Next**.

5. Sign in to Azure by selecting the applicable Azure cloud and then selecting **Sign in to Azure**. Provide your sign-in credentials.

6. In **Resource details**, specify details for how your machine should be represented in Azure, including the tenant, subscription, and resource group. Select **Next**.

7. Once the configuration completes and your machine is onboarded to Azure Arc, select **Finish**.

8. Go to the Server Manager and select **Local Server** to view the status of the machine in the **Azure Arc Management** field. A successfully onboarded machine has a status of **Enabled**.




# Server Manager functions

You can select the **Enabled/Disabled** link in the **Azure Arc Management** field of the Server Manager to launch different functions based on the status of the machine:

- If Azure Arc Setup isn't installed, selecting **Enabled/Disabled** launches the **Add Roles and Features Wizard**.
- If Azure Arc Setup is installed and the Azure Connected Machine agent isn't installed, selecting **Disabled** launches `AzureArcSetup.exe`, the executable file for the Azure Arc Setup wizard.
- If Azure Arc Setup is installed and the Azure Connected Machine agent is already installed, selecting **Enabled/Disabled** launches `AzureArcConfiguration.exe`, the executable file for configuring the Azure Connected Machine agent to work with your machine.

# View the connected machine

The Azure Arc system tray icon at the bottom of your Windows Server machine indicates if the machine is connected to Azure Arc. A red symbol means the machine doesn't have the Azure Connected Machine agent installed.

To view a connected machine in Azure Arc, select the icon and then select **View machine in Azure**. You can then view the machine in the Azure portal, just as you would other Azure Arc-enabled resources.

# Uninstall Azure Arc Setup

> ⓘ **Note**

Uninstalling Azure Arc Setup doesn't uninstall the Azure Connected Machine agent from the machine. For instructions on uninstalling the agent, see Managing and maintaining the Connected Machine agent.

To uninstall Azure Arc Setup from a Windows Server 2022 machine:

1. In the Server Manager, go to the **Remove Roles and Features Wizard**.
2. On the **Features** page, uncheck the box for **Azure Arc Setup**.
3. On the confirmation page, select **Restart the destination server automatically if required**, and then select **Remove**.

To uninstall Azure Arc Setup through PowerShell, run the following command:

PowerShell

```
Disable-WindowsOptionalFeature -Online -FeatureName AzureArcSetup
```

To uninstall Azure Arc Setup from a Windows Server 2025 machine:

1. Open the Settings app on the machine and select **System**, and then select **Optional features**.
2. Select **AzureArcSetup**, and then select **Remove**.

To uninstall Azure Arc Setup from a Windows Server 2025 machine from the command line, run the following line of code:

```
DISM /online /Remove-Capability /CapabilityName:AzureArcSetup~~~~
```

# Related content

- Learn how to troubleshoot the Azure Connected Machine agent.
- Review the Planning and deployment guide to plan for deploying Azure Arc-enabled servers at any scale and implement centralized management and monitoring.
- For more information about Windows Server licensing on a pay-per-usage basis for your Arc-enabled servers, see Windows Server Pay-as-you-go.

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Last updated on 02/05/2026

# How to get Extended Security Updates (ESU) for Windows Server

Article • 05/23/2025 • Applies to: ✅ Windows Server 2012, ✅ Windows Server 2012 R2

Extended Security Updates (ESU) for Windows Server include security updates and bulletins rated *critical* and *important*. Before using ESU, you should read Extended Security Updates for Windows Server Overview to understand what ESUs are, how long they're available for, and what your options are.

How you get ESUs depends on where your server is hosted. You can get access to ESUs through the following options.

- **Azure virtual machines** - Applicable virtual machines (VMs) hosted in Azure are automatically enabled for ESUs and these updates are provided free of charge, there's no need to deploy a MAK key or take any other action. To learn more, see Extended Security Updates on Azure.
- **Azure Arc-enabled servers** - If your servers are on-premises or in a hosted environment, you can enroll your Windows Server 2012 and 2012 R2 or SQL Server 2012 machines for Extended Security Updates via the Azure portal through Azure Arc. Once enrolled you're billed monthly via your Azure subscription. To learn more, see Extended Security Updates enabled by Azure Arc . [1]
- **Non-Azure physical and virtual machines** - If you can't connect using Azure Arc, use Extended Security Updates on non-Azure VMs, by using a Multiple Activation Key (MAK) and applying it to the relevant servers. This MAK key lets the Windows Update servers know that you can continue to receive security updates. To learn more, see Access your Multiple Activation Key from the Microsoft 365 Admin Center . [1]

[1] When using Azure Arc-enabled servers and non-Azure machines you must purchase ESUs. To purchase ESUs, you must have Software Assurance through Volume Licensing Programs such as an Enterprise Agreement (EA), Enterprise Agreement Subscription (EAS), Enrollment for Education Solutions (EES), or Server and Cloud Enrollment (SCE).

> **Note**
>
> It might take 3-5 business days for your Multiple Activation Key to become available after purchasing ESUs for on-premises VMs or physical servers. Your organization might also require time to plan and deploy the new keys. Before purchasing ESUs, you should keep these timelines in mind.

# Extended Security Updates on Azure

Applicable virtual machines (VMs) hosted in Azure are automatically enabled for ESU and these updates are provided free of charge. You don't need to configure anything, and there's no extra charge for using ESUs with Azure VMs. ESUs are automatically delivered to Azure VMs if they're configured to receive updates.

> **Note**
>
> Extended Security Updates are also free in some Azure products. These products include Azure Dedicated Host, Azure VMware Solution, Azure Nutanix Solution, Azure Local, and Azure Stack Hub and Edge. Some of these products might require extra configuration. Contact Microsoft Support for more help.
>
> Azure Classic VMs (Microsoft.ClassicCompute) also require extra configuration to receive Extended Security Updates since they don't have access to the Azure Instance Metadata Service that determines ESUs eligibility.

# Extended Security Updates enabled by Azure Arc

ESUs are automatically delivered to Azure Arc-enabled servers if they're connected and enrolled for ESUs through Azure Arc. This can also apply to non-Azure servers connected to Azure Arc.

You can enroll in ESUs at scale by using Azure Policy or Azure portal, there's no upfront charge and you're billed monthly via your Azure subscription. You also don't need to activate product keys.

Azure Arc-enabled servers also enable to you to use other Azure services, such as:

- Azure Update Manager.
- Microsoft Defender for Cloud.
- Azure Policy (Machine Configuration).
- Azure Monitor (VM Insights).

From September 2023, you're able to activate Windows Server 2012 and 2012 R2 ESUs through Azure Arc. You can connect Windows Server 2012 and 2012 R2 servers to Azure Arc today, Connect hybrid machines with Azure Arc-enabled servers.

To prepare for activating Windows Server 2012 and 2012R2 ESUs on your Arc-enabled servers, follow these steps:

1. Sign in to the Azure portal.
2. In the search bar, enter *Servers - Azure Arc* and select the matching service entry.
3. Add your existing Windows Server 2012 or 2012 R2 machine to Azure Arc. To learn about getting started with Azure Arc-enabled servers, see Connect hybrid machines with Azure Arc-enabled servers.

To learn more about ESUs with Azure Arc, see Prepare to deliver Extended Security Updates for Windows Server 2012 and Deliver Extended Security Updates for Windows 2012 and 2012 R2.

## Access your Multiple Activation Key from the Microsoft 365 Admin Center

Customers who can't connect to Azure Arc to apply ESUs can use Multiple Activation Keys (MAK) through Microsoft 365 Admin Center:

1. Sign in to the Microsoft 365 Admin Center.
2. Select **Your products** > **Volume licensing** > **View contracts**
3. Select your agreement number used to purchase ESUs, the three dots beside it (More Actions icon), then select **View product keys**. All the product keys available to the agreement shown on this page.
4. Once you have your MAK, install the new key on your eligible servers. To learn more about installing and activating your MAK, see our Tech Community blog post Obtaining Extended Security Updates for eligible Windows devices.

## Download and installation of Extended Security Updates

Delivery, download, and application of ESUs for Windows Server is no different than other Windows Updates. The updates provided through ESUs are only *Security* updates.

Before you can download and install ESUs, you must have installed the latest Servicing Stack Update (SSU) and the Licensing Preparation Package. To learn more about the steps required to install the latest SSU and Licensing Preparation Package, see KB5031043: Procedure to continue receiving security updates after extended support has ended on October 10, 2023.

You can install the updates using whatever tools and processes you already have in place. The only difference is that the system must be registered with the key that was generated in the

previous section. Once the system is registered, the updates will download and install.

For VMs hosted in Azure, the process of enabling the server for ESUs is automatically completed for you. Updates should download and install without extra configuration.

# Deliver Extended Security Updates for Windows Server

This article provides steps to enable delivery of Extended Security Updates (ESUs) to Windows Server machines onboarded to Arc-enabled servers. You can enable ESUs to these machines individually or at scale. The steps apply to Windows Server 2012/2012 R2 and Windows Server 2016. Use the selector at the top of the article to choose the operating system version you're licensing.

## Before you begin

Plan and prepare to onboard your machines to Azure Arc-enabled servers. To learn more, see Prepare to deliver Extended Security Updates for Windows Server.

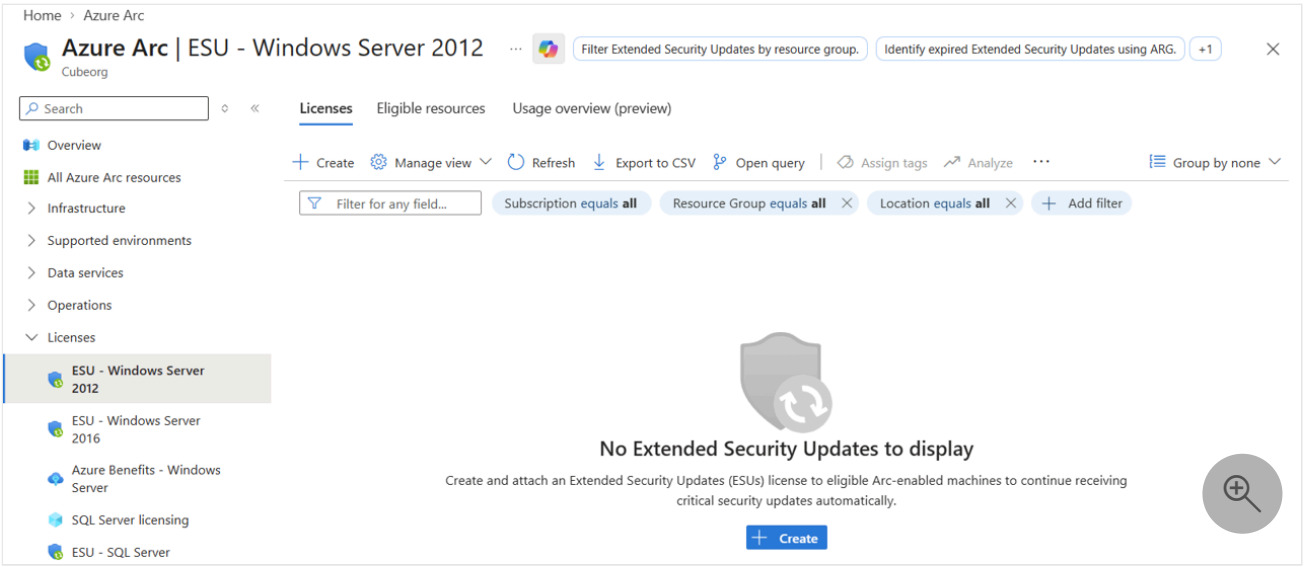
You also need the Contributor role in Azure RBAC to create and assign ESUs to Arc-enabled servers.

## Manage ESU licenses

> **Important**
>
> The Windows Server 2012 and Windows Server 2012 R2 ESU period ends on October 13, 2026. The October 13, 2026 security update is the final update provided through ESUs. At midnight Coordinated Universal Time (UTC) on October 14, 2026, ESU licenses enabled by Azure Arc deactivate. The license resources remain available to view in Azure, but servers linked to them aren't eligible for security updates released after October 13, 2026. Migrate your workloads or upgrade to a supported version of Windows Server before the ESU period ends.

1. From your browser, sign in to the Azure portal.

2. In the service menu, under **Licenses**, select **Windows Server ESU licenses**.




From here, you can view and create ESU **Licenses** and view **Eligible resources** for ESUs.

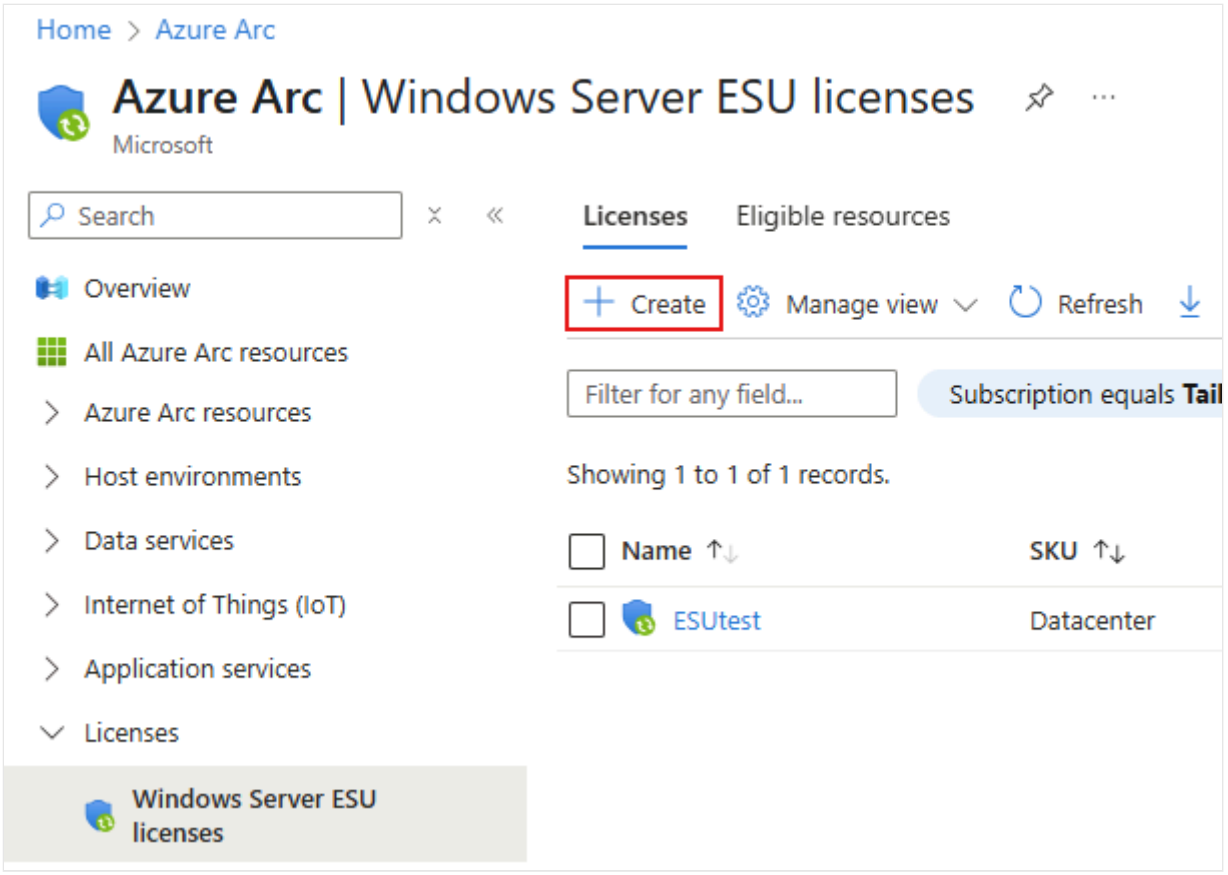
# Create Azure Arc Windows Server licenses

First, provision Extended Security Update licenses from Azure Arc. Link these licenses to one or more Arc-enabled servers that you select in the next section.

> **Note**
>
> To provision ESU licenses, you must attest to their SA or SPLA coverage.

After you provision an ESU license, specify the SKU (Standard or Datacenter), type of cores (Physical or vCore), and number of cores. You can also provision an Extended Security Update license in a deactivated state so that it doesn't initiate billing or be functional on creation. You can modify the cores associated with the license after provisioning.

The **Licenses** tab displays Azure Arc Windows Server licenses that are available. From here, you can select an existing license to apply or create a new license.




1. To create a new Windows Server license, select **Create**, and then provide the information required to configure the license on the page.

   For details on how to complete this step, see License provisioning guidelines for Extended Security Updates for Windows Server.

2. Review the information provided, and then select **Create**.

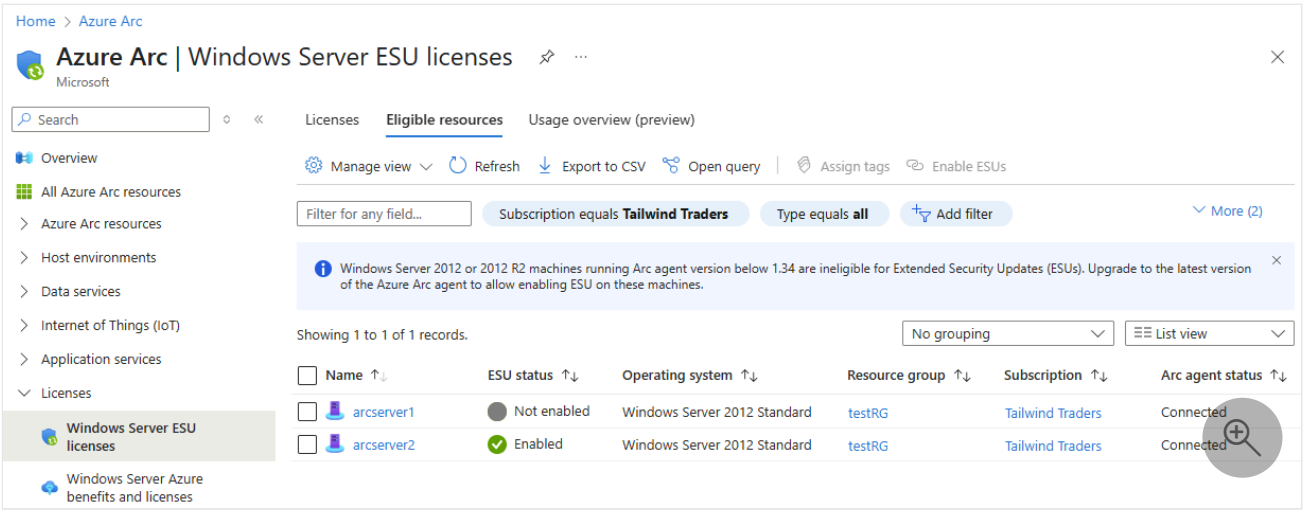
   The license you created appears in the list. You can link it to one or more Arc-enabled servers by following the steps in the next section.

## Link ESU licenses to Arc-enabled servers

Select one or more Arc-enabled servers to link to an Extended Security Update license. After you link a server to an activated ESU license, the server can receive Windows Server ESUs.

> **Note**
>
> You have the flexibility to configure your patching solution of choice to receive these updates – whether that's Update Manager, Windows Server Update Services, Microsoft Updates, Microsoft Endpoint Configuration Manager, or a third-party patch management solution.

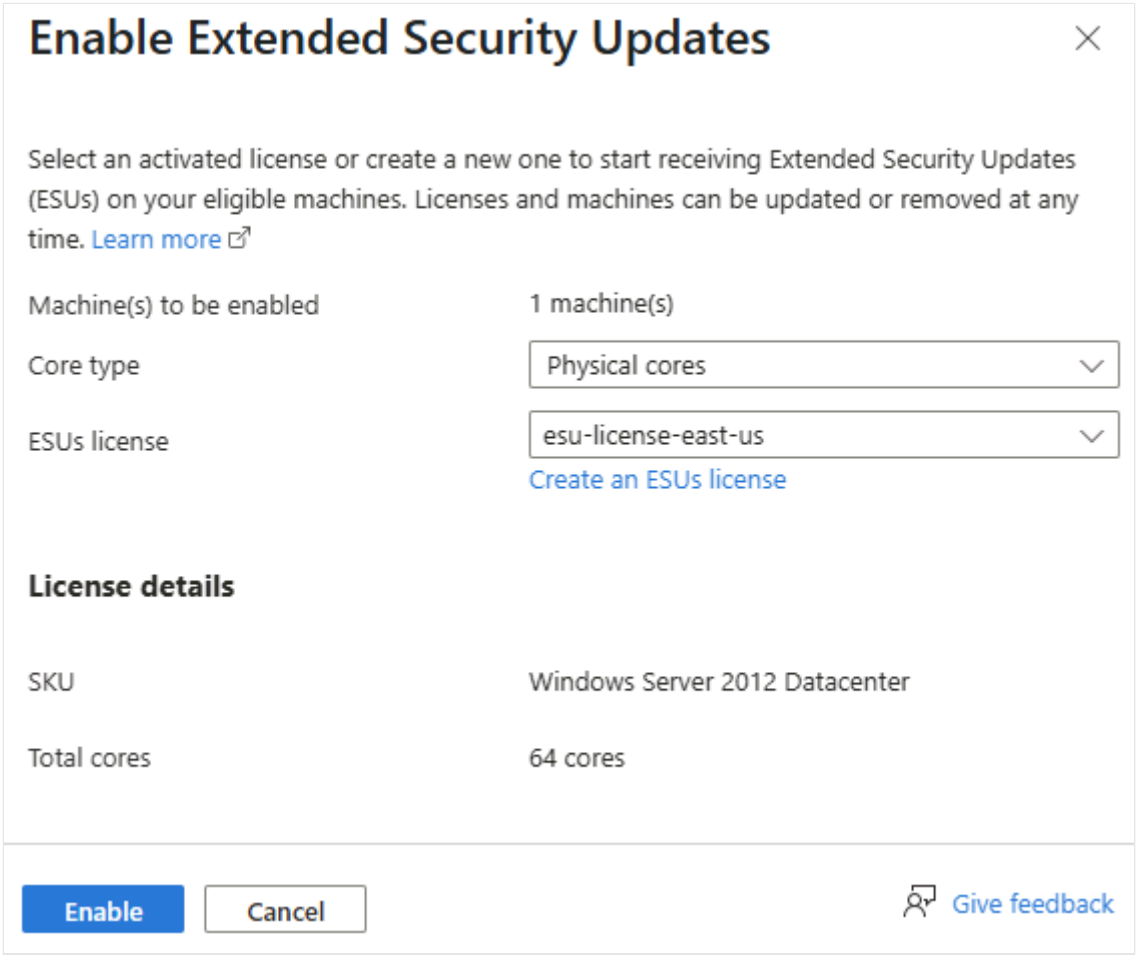
1. Select the **Eligible resources** tab to view a list of all your Arc-enabled servers running Windows Server 2012 and 2012 R2.




The **ESUs status** column indicates whether the machine is enabled for ESUs.

2. To enable ESUs for one or more machines, select them in the list, and then select **Enable ESUs**.

On the **Enable Extended Security Updates** pane, it shows the number of machines selected to enable ESU and the Windows Server licenses available to apply. Select a license to link to the selected machines and then select **Enable**.



> **Note**
>
> You can create a license from this page, rather than choosing an existing one, by selecting **Create an ESUs license**.

When you return to the **Eligible resources** tab, you see the status of the selected machines shows **Enabled**.

If problems occur during the enablement process, see Troubleshoot delivery of Extended Security Updates for Windows Server for assistance.

## Enable ESUs at scale by using Azure Policy

To link servers at scale to an Azure Arc Extended Security Update license and lock down license modification or creation, consider using the following built-in Azure policies:

- Enable Extended Security Updates (ESUs) license to keep Windows 2012 machines protected after their support lifecycle has ended (preview)
- Deny Extended Security Updates (ESUs) license creation or modification (preview)

Specify Azure policies to a targeted subscription or resource group for both auditing and management scenarios.

## Additional scenarios

You might be eligible to receive Extended Security Updates patches at no extra cost in some scenarios. Two of these scenarios supported by Azure Arc are Dev/Test (Visual Studio) and Disaster Recovery (Entitled benefit DR instances from Software Assurance or subscription only. Both scenarios require that you're already using Windows Server 2012/R2 ESUs enabled by Azure Arc for billable, production machines.

> **⚠ Warning**
>
> Don't create a Windows Server 2012/R2 ESU License for only Dev/Test or Disaster Recovery workloads. Don't provision an ESU License only for nonbillable workloads. You pay for all of the cores provisioned with an ESU license. You don't pay for dev/test cores on the license if you tag them according to the following qualifications.

To qualify for these scenarios, you must already have:

- **Billable ESU License.** You must already have provisioned and activated a WS2012 Arc ESU License to link to regular Azure Arc-enabled servers running in production environments (such as normally billed ESU scenarios). Provision this license only for billable cores, not for cores that are eligible for free Extended Security Updates, such as dev/test cores.
- **Arc-enabled servers.** Onboard your Windows Server 2012 and Windows Server 2012 R2 machines to Azure Arc-enabled servers for Dev/Test with Visual Studio subscriptions or Disaster Recovery.

To enroll Azure Arc-enabled servers eligible for ESUs at no extra cost, follow these steps to tag and link:

1. Tag both the WS2012 Arc ESU License (created for the production environment with cores for only the production environment servers) and the nonproduction Azure Arc-enabled servers with one of the following name-value pairs, corresponding to the appropriate exception:

   a. Name: "ESU Usage"; Value: "WS2012 VISUAL STUDIO DEV TEST"

   b. Name: "ESU Usage"; Value: "WS2012 DISASTER RECOVERY"

   If you're using the ESU License for multiple exception scenarios, mark the license with the tag: Name: "ESU Usage"; Value: "WS2012 MULTIPURPOSE"

2. Link the tagged license (created for the production environment with cores only for the production environment servers) to your tagged nonproduction Azure Arc-enabled Windows Server 2012 and Windows Server 2012 R2 machines. **Don't license cores for these servers or create a new ESU license for only these servers.**

This linking doesn't trigger a compliance violation or enforcement block, so you can extend the application of a license beyond its provisioned cores. The expectation is that the license only includes cores for production and billed servers. Any additional cores result in overbilling.

> **ⓘ Important**
>
> Adding these tags to your license doesn't make the license free or reduce the number of license cores that are chargeable. By using these tags, you can link your Azure machines to existing licenses that are already configured with payable cores without needing to create any new licenses or add more cores to your free machines.

**Example:**

- You have eight Windows Server 2012 R2 Standard instances, each with eight physical cores. Six of these Windows Server 2012 R2 Standard machines are for production, and two of these Windows Server 2012 R2 Standard machines are eligible for free ESUs because the operating system was licensed through a Visual Studio Dev Test subscription.
  - First, provision and activate a regular ESU License for Windows Server 2012/R2 that's Standard edition and has 48 physical cores to cover the six production machines. Link this regular, production ESU license to your six production servers.
  - Next, reuse this existing license, don't add any more cores or provision a separate license, and link this license to your two nonproduction Windows Server 2012 R2 standard machines. Tag the ESU license and the two nonproduction Windows Server 2012 R2 Standard machines with Name: "ESU Usage" and Value: "WS2012 VISUAL STUDIO DEV TEST".
  - This results in an ESU license for 48 cores, and you pay for those 48 cores. You don't pay for the extra 16 cores of the dev test servers that you added to this license, as long as you tag the ESU license and the dev test server resources appropriately.

> **Note**
>
> You need a regular production license to start with, and you pay only for the production cores.

## Upgrading from Windows Server 2012/2012 R2

When upgrading a Windows Server 2012/2012R machine to Windows Server 2016 or above, it's not necessary to remove the Connected Machine agent from the machine. The new operating system will be visible for the machine in Azure within a few minutes of upgrade completion. Upgraded machines no longer require ESUs and are no longer eligible for them. Any ESU license associated with the machine isn't automatically unlinked from the machine. See Unlink a license for instructions on doing so manually.

## Assess WS2012 ESU patch Status

To detect whether your Azure Arc-enabled servers are patched with the most recent Windows Server 2012/R2 Extended Security Updates, you can use the Azure Policy Extended Security Updates should be installed on Windows Server 2012 Arc machines. This policy definition, powered by Machine Configuration, identifies if the server has received the most recent ESU

Patches. This is observable from the Guest Assignment and Azure Policy Compliance views built into the Azure portal.

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Last updated on 10/05/2026

# Enable Hotpatch for Azure Edition virtual machines on Azure Local

Article • 12/05/2024 •

Applies to: ✅ Windows Server 2022 Datacenter: Azure Edition hosted on Azure Local

Hotpatch for Windows Server 2022 Datacenter: Azure Edition virtual machines (VMs) hosted on Azure Local allows you to install security updates on an ISO-deployed machine on Azure Local without requiring a reboot after installation. You can use Hotpatch with both Desktop Experience and Server Core. This article will teach you how to configure Hotpatch after installing or upgrading the operating system using an ISO.

> **Note**
>
> If you're using the Azure marketplace, don't follow the steps in this article. Instead, use the following images from Azure Marketplace that are ready for Hotpatching:
>
> - Windows Server 2022 Datacenter: Azure Edition Hotpatch - Gen2
> - Windows Server 2022 Datacenter: Azure Edition Core - Gen2

When using Hotpatch for your ISO deployed machine on Azure Local, there are a few important differences with the Hotpatch experience compared with using Hotpatch as part of Azure Automanage for Azure VMs.

The differences include:

- Hotpatch configuration isn't available via Azure Update Manager.
- Hotpatch can't be disabled.
- Automatic Patching orchestration isn't available.
- Orchestration must be performed manually (for example, using Windows Update via SConfig).

## Prerequisites

To enable Hotpatch, you must have the following prerequisites ready before you start:

- Windows Server 2022 Datacenter: Azure Edition hosted on a supported platform, such as Azure or Azure Local with Azure benefits enabled.
  - Azure Local must be version 21H2 or later.
- Review the How hotpatch works section of the Hotpatch for new virtual machines article.

- Outbound network access or an outbound port rule allowing HTTPS (TCP/443) traffic to the following endpoints:
  - `go.microsoft.com`
  - `software-static.download.prss.microsoft.com`

# Prepare your computer

Before you can enable Hotpatch for your VM, you must prepare your computer using the following steps:

1. Sign-in to your machine. If you're on Server core, from the SConfig menu, enter option **15**, then press `Enter` to open a PowerShell session. If you're on the desktop experience, remote desktop into your VM and launch PowerShell.

2. Enable virtualization-based security by running the following PowerShell command to configure the correct registry settings:

   PowerShell

   ```powershell
   $registryPath = "HKLM:\SYSTEM\CurrentControlSet\Control\DeviceGuard"
   $parameters = $parameters = @{
       Path = $registryPath
       Name = "EnableVirtualizationBasedSecurity"
       Value = "0x1"
       Force = $True
       PropertyType = "DWORD"
   }
   New-ItemProperty @parameters
   ```

3. Restart your computer.

4. Configure the Hotpatch table size in the registry by running the following PowerShell command:

   PowerShell

   ```powershell
   $registryPath = "HKLM:\SYSTEM\CurrentControlSet\Control\Session
   Manager\Memory Management"
   $parameters = $parameters = @{
       Path = $registryPath
       Name = "HotPatchTableSize"
       Value = "0x1000"
       Force = $True
       PropertyType = "DWORD"
   }
   New-ItemProperty @parameters
   ```

5. Configure the Windows Update endpoint for Hotpatch in the registry by running the following PowerShell command:

```PowerShell
$registryPath = "HKLM:\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Update\TargetingInfo\DynamicInstalled\Hotpatch.amd64"
$nameParameters = $parameters = @{
    Path = $registryPath
    Name = "Name"
    Value = "Hotpatch Enrollment Package"
    Force = $True
}
$versionParameters = $parameters = @{
    Path = $registryPath
    Name = "Version"
    Value = "10.0.20348.1129"
    Force = $True
}
New-Item $registryPath -Force
New-ItemProperty @nameParameters
New-ItemProperty @versionParameters
```

Now you've prepared your computer, you can install the Hotpatch servicing package.

# Install Hotpatch servicing package

> **Note**
>
> The Hotpatch Prerequisite KB is currently not published in the Microsoft Update catalog.

To be able to receive Hotpatch updates, you'll need to download and install the Hotpatch servicing package. In your PowerShell session, complete the following steps:

1. Download the (KB5003508) Microsoft Update Standalone Package from the Microsoft Update Catalog and copy it to your computer using the following PowerShell command:

```PowerShell
$parameters = @{
     Source = "https://go.microsoft.com/fwlink/?linkid=2211714"
     Destination = ".\KB5003508.msu"
}
Start-BitsTransfer @parameters
```

2. To install the Standalone Package, run the following command:

```
PowerShell

wusa.exe .\KB5003508.msu
```

3. Follow the prompts. Once it's completed, select Finish.

4. To verify the installation, run the following command:

```
PowerShell

Get-HotFix | Where-Object {$_.HotFixID -eq "KB5003508"}
```

> **Note**
>
> When using Server Core, updates are set to be manually installed by default. You can change this setting using the SConfig utility.

## Next steps

Now you've set up your computer for Hotpatch, here are some articles that might help you with updating your computer:

- Patch a Server Core installation.
- Learn more about Windows Server Update Services (WSUS).

---

## Feedback

**Was this page helpful?** Yes No

# Enable Hotpatch for Azure Arc-enabled servers

Applies to: ✅ Windows Server 2025

> **Important**
>
> Azure Arc-enabled Hotpatch for Windows Server 2025 is now available for a monthly subscription fee. To learn more about pricing, see Tired of all the restarts? Get hotpatching for Windows Server.

Hotpatch allows you to update your Windows Server installation without requiring your users to restart after installation. This feature minimizes downtime spent on updates and keeps your users running their workloads uninterrupted. For more information about how Hotpatch works, see Hotpatch for Windows Server.

Windows Server 2025 features the ability to enable Hotpatch for Azure Arc-enabled servers. In order to use Hotpatch on Azure Arc-enabled servers, all you have to do is deploy the Connected Machine agent and enable Windows Server Hotpatch. This article describes how to enable Hotpatch.

## Prerequisites

Before you can enable Hotpatch on Arc-enabled servers for Windows Server 2025, you need to satisfy the following requirements.

- A server must be running Windows Server 2025 (build 26100.1742 or later). Preview versions or Windows Server Insiders builds aren't supported because hotpatches aren't created for prerelease operating systems.
- The machine should be running one of the following editions of Windows Server.
  - Windows Server 2025 Standard
  - Windows Server 2025 Datacenter
  - Windows Server 2025 Datacenter: Azure Edition. This edition does **not** need to be Azure Arc-enabled, Hotpatch is already enabled by default. The remaining technical prerequisites still apply.
- Both **Server with Desktop Experience** and **Server Core** installation options are supported.
- The physical or virtual machine you intend to enable Hotpatch on needs to satisfy the requirements for Virtualization-based security (VBS), also known as Virtual Secure Mode

(VSM). At bare minimum, the machine has to use Unified extensible firmware interface (UEFI) with Secure boot enabled. Therefore, for a virtual machine (VM) on Hyper-V, it needs to be a Generation 2 virtual machine.

- An Azure subscription. If you don't already have one, create a free account before you begin.
- Your server and infrastructure should satisfy the Connected Machine agent prerequisites for enabling Azure Arc on a server.
- The machine should be connected to Azure Arc (Arc-enabled). To learn more about onboarding your machine to Azure Arc, see Azure Connected Machine agent deployment options.

## Check and enable Virtual Secure Mode if necessary

When you enable Hotpatch using the Azure portal, it checks whether Virtual Secure Mode (VSM) is running on the machine. If VSM isn't running, enabling hotpatch fails, and you'll have to enable VSM.

Alternatively, you can check the VSM status manually before enabling Hotpatch. VSM might be already enabled if you previously configured other features that (like Hotpatch) depend on VSM. Common examples of such features include Credential guard or Virtualization-based protection of code integrity, also known as Hypervisor-protected code integrity (HVCI).

> **Tip**
>
> You can use Group policy or another centralized management tool to enable one or more of the following features.
>
> - Credential guard
> - Credential Guard protected machine accounts
> - Virtualization-based protection of code integrity
> - System Guard Secure Launch and SMM protection
> - Kernel Mode Hardware-enforced Stack Protection
> - Secured-core server
>
> Configuring any of these features also enables VSM.

To verify VSM is configured and running, select your preferred method and examine the output.

PowerShell

PowerShell

```
Get-CimInstance -Namespace 'root/Microsoft/Windows/DeviceGuard' -ClassName
'win32_deviceGuard' | Select-Object -ExpandProperty
'VirtualizationBasedSecurityStatus'
```

If the command output is `2`, VSM is configured and running. In this case, proceed directly to Enable Hotpatch on Windows Server 2025.

If the output isn't `2`, you need to enable VSM.

▼ To enable VSM, expand this section.

Enable VSM using one of the following commands.

PowerShell

PowerShell

```
New-ItemProperty -Path 'HKLM:\System\CurrentControlSet\Control\DeviceGuard' -
Name 'EnableVirtualizationBasedSecurity' -PropertyType 'Dword' -Value 1 -Force
```

**Tip**

After you enable VSM, you need to restart your server.

After you reboot, run one the following commands again verify that the output is now `2` to make sure VSM is now running.

PowerShell

PowerShell

```
Get-CimInstance -Namespace 'root/Microsoft/Windows/DeviceGuard' -ClassName
'win32_deviceGuard' | Select-Object -ExpandProperty
'VirtualizationBasedSecurityStatus'
```

If the output still isn't `2`, VSM on your machine needs troubleshooting. The most likely reason is that the physical or virtual hardware requirements aren't met. Refer to documentation from the vendor of your hardware or virtualization platform. For example, here's documentation for VMware vSphere, Activate Virtualization-based Security on an Existing Virtual Machine.

Once you successfully enabled VSM and made sure it's running, proceed to the next section.

# Enable Hotpatch on Windows Server 2025

1. Connect the machine to Azure Arc, if it wasn't Arc-enabled previously.
2. After you connected the machine to Azure Arc, sign in to the Azure Arc portal and go to **Azure Arc** → **Machines**.
3. Select the name of your machine.
4. Select **Hotpatch**, then select **Confirm**.
5. Wait about 10 minutes for the changes to apply. If the update stays stuck on the *Pending* status, proceed to troubleshooting Azure Arc agent.

# Using Hotpatch on Windows Server 2025

Whenever a Hotpatch is available from Windows Update, you should receive a prompt to install it. Since these updates aren't released every month, you might need to wait until the next Hotpatch is published.

You can optionally automate hotpatch installation using update management tools such as Azure Update Manager (AUM).

# Known issues

## Multiple updates released in October 2025

In October 2025, Microsoft released several updates that were offered to some Windows Server customers. If you enrolled in hotpatch or plan to enroll, and plan to install hotpatch updates in November and December 2025, ensure your Windows Server machines are running on **exactly** one of the following update levels.

- October 14, 2025 - **KB5066835** (OS Build 26100.6899)
- October 24, 2025 - **KB5070893** (OS Build 26100.6905) Security Update for Windows Server Update Services

If you have one of the other October updates installed, this will result in regular non-hotpatch updates until, and including, the next baseline month that is currently scheduled for January 2026. These updates require a reboot each month. In particular, the following update, if installed, makes the machine incompatible with upcoming hotpatches: October 23, 2025 - **KB5070881** (OS Build 26100.6905) Out-of-band, and so does any other update not explicitly listed in this section.

## Feature licensing issue in October 2025 updates

An issue was identified with the October 2025 security updates for Windows Server 2025. This may impact customers running October 14, 2025 - KB5066835 (OS Build 26100.6899) update or later. Due to this issue, the following unexpected behavior can be observed.

- Enabling Windows Server hotpatching via Azure Arc on new machines may fail or not complete as expected. Instead, feature enablement remains in the "In Progress" state until the issue is resolved.
- On machines previously enabled for Windows Server hotpatching, the feature license may expire, and this will prevent the next Hotpatch from being installed. Instead, the next update will cause a reboot if no action is taken.

Hotpatching on Windows Server 2025 Datacenter: Azure Edition isn't affected by this issue.

To resolve this issue, a series of manual steps is recommended. Failure to apply either workaround will result in regular non-hotpatch updates until, and including, the next baseline month that is currently scheduled for January 2026. These updates require a reboot each month.

There are two ways to apply the manual workaround on affected machines. Each of the provided options offers a complete solution. You'll need to apply the workaround on each of the affected machines **before** the next update is offered, which is anticipated in the next "patch Tuesday" date of **November 11, 2025**. Applying the workaround requires a reboot, so plan accordingly.

After you apply either workaround, hotpatch updates released in November and December 2025 will install without requiring a reboot.

### Option 1: Use Local or Group Policy to enable the remediation

1. Download and install the Windows 11 24H2, Windows 11 25H2, and Windows Server 2025 KB5062660 251028_18301 Feature Preview package. This installs the Local or Group Policy template (`ADMX` file) for this specific remediation.

2. Select **Start**, type **gpedit**, then select **Edit group policy**. Navigate to **Computer Configuration\Administrative Templates\KB5062660 251028_18301 Feature Preview\Windows 11, version 24H2, 25H2\KB5062660 251028_18301 Feature Preview**.

   For more information on deploying and configuring this special Group Policy, see Use Group Policy to enable an update that is disabled by default.

3. In the right window pane, open **KB5062660 251028_18301 Feature Preview**, select **Enabled**, then select **OK**.

4. Reboot the affected machine.

5. Run the following command to remove the **DeviceLicensingServiceCommandMutex** entry from the registry. If this entry isn't present on the affected device, removal is ignored.

   PowerShell

   ```
   try {
   Remove-ItemProperty -Path 'HKLM:\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Subscriptions' -Name 'DeviceLicensingServiceCommandMutex' -ErrorAction Stop
   } catch {
   Write-Host "DeviceLicensingServiceCommandMutex entry not present, skipping removal."
   }
   ```

   Alternatively, use your preferred registry editing tool or automation solution to delete the same value. Don't delete the whole registry key.

## Option 2: Use a script to enable the remediation

Open an elevated PowerShell window on each of the affected machines and run the following commands. The last command prompts you to restart your device. The mitigation isn't complete until the machine is rebooted, and it's recommended that you restart immediately after running this script.

PowerShell

```
Stop-Service -Name 'HIMDS'
New-Item -Path 'HKLM:\SYSTEM\CurrentControlSet\Policies\Microsoft\FeatureManagement\Overrides' -Force
New-ItemProperty -Path 'HKLM:\SYSTEM\CurrentControlSet\Policies\Microsoft\FeatureManagement\Overrides' -PropertyType 'dword' -Name '4264695439' -Value 1 -Force
try {
```

```
  Remove-ItemProperty -Path 'HKLM:\SOFTWARE\Microsoft\Windows
NT\CurrentVersion\Subscriptions' -Name 'DeviceLicensingServiceCommandMutex' -
ErrorAction Stop
} catch {
  Write-Host "DeviceLicensingServiceCommandMutex entry not present, skipping
removal."
}
Restart-Computer -Confirm
```

# Next steps

Now that Hotpatch is enabled, here are some articles that might help you with updating your computer.

- Hotpatch for Windows Server
- Patch a Server Core installation
- Automatic VM Guest Patching
- Azure Update Manager

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Last updated on 10/31/2025

# Windows release health

Quickly find official information on Windows updates and servicing milestones. Access resources, tools, and news about known issues and safeguards to help you plan your next update. Want the latest Windows release health updates? Follow @WindowsUpdate on X.



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DEPLOY
**Secure Boot client playbook for certificate...**



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**Windows 11 Roadmap: new and upcomin...**



WHAT'S NEW
**An IT pro's guide to Windows 11,...**



DEPLOY
**Windows Server Secure Boot playboo...**



OVERVIEW
**Understanding Windows monthly...**

## Message center

- Take action: NTLMv1 hardening changes might affect network connections on devices without Credential Guard
- Windows Server 2022 has reached end of mainstream support
- 90-Day Reminder: Windows Server 2016 will reach end of updates on January 12, 2027

See more >

## Windows 11, version 26H2

- Known issues
- Resolved issues
- Release notes
- Windows 11 release information
- How to get Windows 11, version 26H2

## Windows 11, version 26H1

- Known issues
- Resolved issues
- Release notes
- Windows 11 release information

## Windows 11, version 25H2

- Known issues
- Resolved issues
- Release notes
- Windows 11 release information

What to know about Windows 11, version 26H1

How to get Windows 11, version 25H2

## Windows 10, version 22H2

- Known issues
- Resolved issues
- Release notes
- Windows 10 release information
- How to get Windows 10, version 22H2

## Windows Server 2025

- Known issues
- Resolved issues
- Release notes
- Windows Server release information
- What's new in Windows Server 2025

## Windows Server 2022

- Known issues
- Resolved issues
- Release notes
- Windows Server release information
- What's new in Windows Server 2022

## Additional versions

See details on known and resolved issues for other supported versions of Windows and Windows Server.

- Known issues: earlier versions

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Open the Get Help app in your Windows device to find resources to troubleshoot common issues.

# Windows Server - License Terms

Article • 03/31/2022

Review our Windows Server-related license terms.

- Additional software for Windows Server 2016
- Windows Server Technical Preview Expiration
- Windows Server 2016 Technical Preview License Terms
- Microsoft Software License Terms - MICROSOFT.WINDOWSSERVER.SYSTEMINSIGHTS
- Microsoft Software License Terms - MICROSOFT.WINDOWSSERVER.SYSTEMINSIGHTS.CAPABILITIES
- Windows Admin Center - License Terms

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## Feedback

Was this page helpful? Yes No